FOREWORD

I am happy to write the foreward for this book, because it is borne out of the intelligent and focused work of the writer. I have known the writer; Adegoke Abiola Oluseyi both private and official capacities, and his involvement in teaching and preparing students for the Unified Tertiary Matriculation Examination, Post-UTME, Pre-degree, School of Nursing entrance examinations and other special programmes with the Legend Tutorial College, Ile-Ife, Osun State for some time. Mr. Oluseyi is a brilliant and dedicated coordinator of this College, and they have had a very good record of successes in all the examinations.

This material (book) on OAU Post-UTME in all the main subjects involved in the Obafemi Awolowo University, Ile-Ife is a product of the intelligent preparation of students for Post-UTME,Pre-Degree, UTME and other related examinations by Mr. Oluseyi. The solutions are provided by seasoned academics in the University, and the scope spans most entrance examinations into the Nigerian tertiary institutions. I wholeheartedly recommend the book for everyone preparing (or preparing candidates) for Post-UTME, Pre-degree and related entrance examinations into any tertiary institution in Nigeria.

Thank you

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PREFACE

This series is borne out of enormous passion to salvage an appalling rate of failure in OAU Post-UTME by prospective candidates. Having written this same exam and passed through the programme myself, I find it imperative to make available a material equipped with all necessary recipes to scale through the first and the most critical hurdle of OAU Admission processing. This material is essentially prepared in a surgical and comprehensive analysis of answers to past OAU Post-UTME entrance examination. It is axiomatic that after an extensive and intensive digestion of this material candidates would be confident and aware of the modus operandi of the entrance examination questions. However, it is not to be used as a substitute for recommended textbooks. You can consult other materials. All suggestions, observations, recommendations or constructive criticisms should be sent to seyilegend@gmail.com or any of the numbers listed below. I wish you the best of God.

...ADEGOKE ABIOLA OLUSEYI
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OBAFEMI ALWOLOWO UNIVERSITY, ILE-IFE
2015/2016 ADMISSION EXERCISE
CUT-OFF MARKS

FACULTY/PROGRAMME	MERIT	CATCHMENT AREAS						
		EKITI	**LAGOS**	**OGUN**	**ONDO**	**OSUN**	**OYO**	**ELDS**
ADMINISTRATION								
Accounting	262	253	248	254.5	258	258.5	255	255
International Relations	235	215	210.5	229.5	230	231.5	224.5	234.5
Public Administration	236.5	201.5	216	219.5	212.5	232	223	222
Local Government Studies	202	202	202	202	202	202	202	
AGRICULTURE		**EKITI**	**LAGOS**	**OGUN**	**ONDO**	**OSUN**	**OYO**	**ELDS**
Agric Economics	200	200	200	200	200	200	200	200
Animal Science	200	200	200	200	200	200	200	200
Agric Extension & Rural Sociology	200	200	200	200	200	200	200	200
Family Health & Protection	200	200	200	200	200	200	200	200
Crop Production & Protection	200	200	200	200	200	200	200	200
Soil Science	200	200	200	200	200	200	200	200
ARTS		**EKITI**	**LAGOS**	**OGUN**	**ONDO**	**OSUN**	**OYO**	**ELDS**
English Language	256	242	229	242	242	252	251	220
Literature-in-English	246	221	206	234	206	240	209	206
Dramatics Arts	250	219	203	238	237	248	243	201
History	207	207	207	207	207	207	207	207
French	206	206	206	206	206	206	206	206
German	211	211	211	211	211	211	211	211
Portuguese	222	222	222	222	222	222	222	222
Music	203	203	203	203	203	203	203	203
Linguistics	256	241	242	236	246	250	240	219
Philosophy	244	221	204	228	223	229	232	216
Religious Studies	206	206	206	206	206	206	206	206
Yoruba	211	211	211	211	211	211	211	211
EDUCATION		**EKITI**	**LAGOS**	**OGUN**	**ONDO**	**OSUN**	**OYO**	**ELDS**
Physical & Health Education	200	200	200	200	200	200	200	200
Int. Science	200	200	200	200	200	200	200	200
Language Arts	210	205	205	205	205	205	205	205
Social Studies	208	205	205	205	205	205	205	200
Yoruba	200	200	200	200	200	200	200	200
Fine Arts	200	200	200	200	200	200	200	200
Music	200	200	200	200	200	200	200	200
Religious Studies	200	200	200	200	200	200	200	200
History	200	200	200	200	200	200	200	200
English	280	247	254	239.5	238	238.5	227	-
French	200	200	200	200	200	200	200	200
Economics	232.5	206.5	225	229	213	201.5	202	-
Geography	200	200	200	200	200	200	200	200
Political Science	200	200	200	200	200	200	200	200
Physics	200	200	200	200	200	200	200	200
Chemistry	200	200	200	200	200	200	200	200
Biology	200	200	200	200	200	200	200	200
Mathematics	200	200	200	200	200	200	200	200
Guid/CLN	200	200	200	200	200	200	200	200

FACULTY/PROGRAMME	MERIT	CATCHMENT AREAS						
		EKITI	**LAGOS**	**OGUN**	**ONDO**	**OSUN**	**OYO**	**ELDS**
ENV. DESIGN & MANGMT.								
Architecture	237.5	227	206.5	236.5	219	230.5	242	219.5
Building	241	200	200	200	200	200	200	200
Urban & Reg. Planning	204	200	200	200	200	200	200	200
Quantity Surveying	205.5	200	200	200	200	200	200	200
Fine Arts	249.5	225	228	235	220.5	242.5	230	209.5
Estate Management	201	200	200	200	200	200	200	200
LAW		**EKITI**	**LAGOS**	**OGUN**	**ONDO**	**OSUN**	**OYO**	**ELDS**
Law	284.5	276.5	274.5	277.5	282.5	280.5	276.5	233
PHARMACY		**EKITI**	**LAGOS**	**OGUN**	**ONDO**	**OSUN**	**OYO**	**ELDS**
Pharmacy	256.5	248.5	238.5	250	248	252.5	249.5	Various
SCIENCE		**EKITI**	**LAGOS**	**OGUN**	**ONDO**	**OSUN**	**OYO**	**ELDS**
Applied Geophysics	200	200	200	200	200	200	200	200
Biochemistry	227	229	225.5	234.5	232.5	235.5	228	213
Botany	200	200	200	200	200	200	200	200
Chemistry	200	200	200	200	200	200	200	200
Engineering Physics	239.5	-	235.5	-	-	231.5	-	200
Geology	247	236	229	243.5	241	233.5	233	200
Industrial Chemistry	223	216	200	216.5	214.5	221	212.5	204.5
Mathematics	200	200	200	200	200	200	200	200
Microbiology	245.5	230.5	230	239.5	236	235	235.5	211.5
Physics	237.5	-	-	-	-	227.5	-	209
Statistics	200	200	200	200	200	200	200	200
Zoology	200	200	200	200	200	200	200	200
SOCIAL SCIENCE		**EKITI**	**LAGOS**	**OGUN**	**ONDO**	**OSUN**	**OYO**	**ELDS**
Demography & Soc. Statistics	208	208.5	208.5	208.5	208.5	208.5	208.5	208.5
Economics	272	260	249	260	264	270.5	266	250
Geography	213	213	213	213	213	213	213	213
Political Science	236.5	228.5	226	226	226	235	233	200
P.P.E	236.5	228.5	226	226	226	235	233	200
Psychology	237	213.5	223.5	228.5	229	225	223	202
Sociology & Anthropology	236	236	236	230	233	235	211	202
COLLEGE OF HEALTH SCIENCE		**EKITI**	**LAGOS**	**OGUN**	**ONDO**	**OSUN**	**OYO**	**ELDS**
Medicine	287	286.5	286.5	286.5	286.5	285	286.5	281
Dentistry	270	264.5	264.5	265	268.5	264.5	264.5	262
Nursing	256	253.5	253.5	253.5	253.5	253.5	253.5	250
Medical rehabilitation	254	253	252	252	252	252	253	250
TECHNOLOGY		**EKITI**	**LAGOS**	**OGUN**	**ONDO**	**OSUN**	**OYO**	**ELDS**
Agricultural Engineering	216	200	200	200	200	200	200	200
Chemical Engineering	272	260	256	267	270	260	266	238
Civil Engineering	259	250	254	250	256	257	253	219
Computer Science & Engineering (Single & Combined Honour)	264	247	247	254	261	263	258	236
Computer Science with Econs	259	237	250	250	254	237	237	200
Comp. Science with Math.	271	256	256	256	256	256	259	233
Electronic and Electrical Engr.	268	262	251	263	250	264	262	227
Food Engineering	200	200	200	200	200	200	200	200
Food Science & Technology	227	222	200	219	224	221	216	214
Mat. Sc. & Engineering	247	200	200	200	200	200	200	200
Mechanical Engineering	266	259	241	251	257	259	261	240

2015 ENGLISH LANGUAGE QUESTIONS

Fill out the gaps in the passage below with the correct option from the list provided in brackets in front of the gaps.

1 —-——-- **(a. Jargon b. Vernacular c. Dialects d. Wazobia**) is specialized language that appears in a non-specialized context thus giving a 2. -------— (**a. Generalized b. Technical c. Restricted d. Straightforward**) flavour to statements that would be better 3. —--------- (**a. Addressed b. Written c. Expressed d. Spoken**) in everyday words. When you are writing a 4. ------------- (**a. Paper b. Document c. Treatise d. Pamphlet**) in, say, economics, anthropology, or psychology, you can and should use terms that are meaningful within the 5. ——----: (**a. Pool b. Conundrum c. Register d. Field**) cash flow, kinship structure, paranoid, and so forth. But those same terms become jargon when used out of 6. —-—— (**a. Place b. Meaning c. Contest d. Context**).

Choose the option nearest in meaning to the italicized words in the following sentences.

7. He cannot hide his *aversion* for Kemi's unrepentant behaviour. a. abhorrence b. sadness c. ignominy d. moodiness
8. The *investments* in stocks seemed to have gone down the drain with this meltdown in the banking sector a. businesses b. capital c. surpluses d. dividends
9. My efforts at making her to see my point were *rebuffed*. a. embraced b antagonized c. snubbed d. successful
10 He is a *lout* and can't be relied upon at all. a. vagabond b. thug c. unserious d liar
11. The undisputed boxer was quite a *mouthful* for his opponent. a. not a match b. undefeated c. evenly match d. boastful
12. Such stories are difficult to believe because they are *make-belief*. a. lies b. vituperations c. genuine d. fantasy

Choose the most appropriate option to complete the gaps in the following sentences.

13. I am sure all those who mistook President Buhari's -—-——--- demeanour for cowardice will soon know him for who he actually is. a. stealthy b. sloppy c. snaky d. succulent
14. The new Inspector General of Police alleged that terrorists had ——- the rank and file of the force. a. Proliferated b conquered c. infiltrated d. insulated
15. He lay awake, his whole body — sleep. a. acting for b. looking for c. drumming for d. aching for
16. Please think —---- everything and let me have your answer tomorrow. a. thoroughly b. through c. around d. on
17. I asked you a. when you are going to get marry b the time when you are going to get marry c. at what point you are getting married d. when you were going to get married

Choose from the options A-D the word opposites in meaning to the underlined word.

18. I guess he is <u>indifferent</u> to our plans to rid Nigeria of societal ills. a. interested in b. opposed to c. bothered d not interested in
19. The increase in transportation cost <u>imperiled</u> my sister's plan to travel this month. a. propelled b. restricted c. disturbed d. hoodwinked
20. The criminal's answers to the questions during interrogation were <u>evasive</u> a. harsh b. outspoken c. clever d direct

Read the passage below and answer the questions under it.

Among his papers, there is the farewell lecture given in 1925 when he retired from his Copenhagen chair at the age of 65 — protesting himself 'an old fogey', though English studies were fortunate that so youthfully creative a 'fogey' was to go on writing for almost a further two decades. In this splendid apologia, he explained that for him linguistic investigation' involved primarily understanding the texts… to penetrate into the innermost thoughts of the best men and women'. 'Speech is the noblest instrument to bindman to man, and it is by speech as by literature, or best by both combined, that one comes to understanding the people from whom they emanated.' First and foremost, of course, a student of language, he insisted on studying language at its best, and ii that v he hoped, he says, 'to have imparted to my hearers some of my own enthusiasm for the great poets.

My greatest enjoyment, and no doubt that of my hearers as well, has been in my Chaucer classes, partly because Chaucer t such a wonderful power of describing human beings.' So far from confining himself to expounding linguistic history for its own sake, he sees his work as 'combating the ghastly malady of our time, nationalism', 'the. Essential mark' of which 'is antipathy, disdain, finally hatred'. 'Especially now since the World- war this is a task of the greatest importance, since it is necessary that the wounds of this gruesome time should be healed.' Thus he spoke to his students in 1925. Sadly, this noble friend of mankind was to see a still more gruesome manifestation of nationalism and to die in 1943 when his country had already suffered for some years the horrors of the Nazi occupation, when there was little opportunity 'to diffuse knowledge and love oF what is best in other peoples'.

Growth and Structure of the English language by Otto Jespersen

21. Another word used in the passage that can serve as a synonym to 'ghastly' is a. expounding b. essential c. splendid d. gruesome
22. The figure of speech used by the writer iii 'though English studies were fortunate that so youthfully creative a 'fogey' was to go on writing for almost a further two decades' in describing the person being talked about is _______ a. hyperbole b. irony c. innuendo d. sarcasm

23. The writer believes that __________ a. Human beings are best understood by what they say and or by what is written about them. b. Human beings are very difficult to penetrate in a linguistic investigation. c. The innermost thoughts of all men and women are the major preoccupations of linguists d. Literature and language are combined in any worthwhile linguistic enquiry to understand human beings.
24. According to the writer, he derived greatest enjoyment in a. Chaucer, the great personality he befriended when he was in school b. the profound ideas expounded by his teachers while he was in school c. series of Lectures he received about Chaucer arid his writings d. the wonderful ways his teacher described human beings in many of his lectures
25. We can categorically pinpoint from the passage that the writer was talking about a. a former student of his ho is intelligent b. a disturbing trend in linguistic study c a distinguished scholar who had impacted positively on the field of discussion d. the ghastly malady of our time, nationalism', 'the essential mark' of which 'is antipathy, disdain, finally hatred

2015 ENGLISH LANGUAGE SOLUTIONS

1. **A – Jargon**
2. **B – Technical**
3. **D – Spoken**
4. **C – Treatise**
5. **D – Field**
6. **D – Context**
7. **A – Abhorrence**
8. **D – Dividends**
9. **C – Snubbed** means disregarded, dishonoured, disparaged
10. **B –**
11. **A – Not a Match**
12. **D – Fantasy**
13. **A – Stealthy** means trying to avoid being noticed or done quietly, slowly and cautiously in order to escape notice. The other meanings are: furtive, silent, surreptitious, quiet, cautious, covert.
14. **C – Infiltrated** means to enter an organization to spy on it. It also means to become part of an organization, or enter a place surreptitiously in order to gather information or influence events. The synonyms for this are: Penetrate, Permeate, Intrude, gain access to, creep into etc.
15. **D – Aching for**
16. **B – Through** - Think through means to consider all aspects of something. It also means to consider or reflect on something carefully, especially in order to reach a decision. It's synonyms are: consider, contemplate, ponder, mull over, weigh up.
17. -
18. **A – Interested in**
19. **A – Propelled**
20. **D – Direct**
21. **D - Gruesome**
22. **–**
23. **D –**
24. **D –**
25. **C -**

English Language Question 2014

Choose the most appropriate option from a — d to complete the gaps in the following sentences.

1. The high-profile witness has been discredited having been accused of being ________ with the truth in his testimony. a. biological b. geographical c. circumventing d. economical
2. The jailed businessman has left his family in ________due to the confiscation of his property by the government. a. bliss b. quandary c. opulence d. Eldorado
3. While the host community was condemned roundly for their hostility, the visiting contingent was applauded for their________ a. friendliness b. hospitality c. dexterity d. methodology
4. It is hard to quantify the that________ the abducted Chibok girls would have gone through since they were taken away by the dreaded Boko Haram insurgents. a. rape b. trauma c. isolation d. insecurity
5. The Federal Government has expressed the fear that the violent activities of the insurgents may ________ if powerful foreign assistance is not received soon. a. extrapolate b. proliferate c. metamorphose d. escalate

Fill in the blank spaces with the most appropriate option from options lettered a — d.

6. The driver was reckless, the road was slippery due to the early morning downpour, and ________ there was a crash that claimed two lives. a. frantically b. subsequently c. consequently d. unequivocally
7. Did you know we were very fortunate to run ________for the newly elected president of the country when he was campaigning for the office? a. errant b. errands c. an errand d. around
8. I am sure that if you probe further, the accused person will reveal where the ________money is kept. a. pilfered b. missed c. lost d. robbed
9. I was waving frantically but you drove ________me. a. past b. pass c. passed d. passing

Choose the option nearest in meaning to the italicized words in the sentences below.

10. You need to go and study the *etymology* of the underlined words in the returned essay,' the lecturer told the student. a. meaning b. technicality c. originality d. origin
11. The *erratic* power supply these days has caused a lot of damage to household items that use

electricity. a. lackadaisical b. regular c. uneven d. high-voltage

Choose the correct option from letters a — d to fill the blank spaces in the sentences below.

12. He won the elections but many people were killed by his thugs, and thus many said his was a ________ a. Philistine victory b. pyrrhic victory c. crocodile victory d. pseudo victory
13. The aspirant for the highest office in the coming elections has started distributing live cows to each electorate in his ward; but some rejected it because they considered it a ________ a. an Eldorado gift b. Grievous gift c. Greece gift d. Greek gift
14. The workers expressed their heartfelt thanks ________the management for the notable improvement to their conditions of service. a. toward b. with c. towards d. to
15. The man told him point blank that his argument was bereft ________sound reasoning. a. with b. of c. in d. off

Read the passage and use it to answer questions 16 — 19.

As we both fed our eyes wistfully at the used under-wear section, Vivian noticed a heap of women's underwear, a mixture of braziers of various sizes and designs, panties, gstrings and tongs, underskirts, lingerie of different colours, all heaped and scattered on a big bedspread like a pile of rubbish. Some were quite ancient and threadbare, while a few appeared not to have suffered much oppression in the hands and private parts of their previous owners. It was apparent that the international businessman who imported such inglorious assortment had agents with prongs, long enough to dig deep into the deepest and farthermost refuse bins and dumps of Europe to be able to meet the demand back home.

16. We can infer from the passage above that a. the effort of international businessmen was commended by the writer for their contribution to Nigeria's economy b. the writer believes that Nigerian government has not been doing enough to encourage importers of goods to remain in business and recoup money they have invested in their business c. the writer makes comic comments on the porous Nigerian borders through which contraband goods are imported into the country d. the writer subtly castigates and derides the patronage of substandard goods that dots the Nigerian markets in the name of imported goods
17. '________all heaped and scattered on a big bedspread like a pile of rubbish' The figure of speech used in that quotation is a. metaphor b. simile c. hyperbole d. pun
18. *It was apparent that the international businessman who imported such inglorious assortment had agents with prongs, long enough to dig deep into the deepest and farthermost refuse bins and dumps of Europe to be able to meet the demand back home*.

    From the quotation above, it is apparent that the writer is being a. metaphorical c. sarcastic d. specific b. categorical
19. Another word that means the same as the word 'threadbare' as it is used in the passage is________ a. worn-out b. dirty c. expensive d. cheap

Fill out the gaps in the passage below with the correct option from the list provided in brackets in front of the gaps.

20 ________ (a. Jargon b. Vernacular c. Dialects d. Wazobia) is specialized language that appears in a nonspecialized context, thus giving a 21. ________ (a. Generalized b. Technical c. Restricted d. Straightforward) flavour to statements that would be better 22. ________ (a. Addressed b. Written c. Expressed d. Spoken) in everyday words. When you are writing a 23. ________ (a. Paper b. Document c. Treatise d. Pamphlet) in, say, economics, anthropology, or psychology, you can and should use terms that are meaningful within the 24. ________ (a. Pool b. Conundrum c. Register d. Field) cash flow, kinship structure, paranoid, and so forth. But those same terms become jargon when used out of 25. ________ (a. Place b. Meaning c. Contest d. Context).

ENGLISH LANGUAGE SOLUTION 2014

1. (a) circumventing – to find a way of avoiding restrictions imposed by a rule or law without actually breaking it.
   (b) Geographical – relating to geography or to the geography of a specific region.
   (c) Biological – related to living organisms
   (d) Economical – resourcefully frugal
   The correct option is D – It is an idiomatic expression which means a way of saying that somebody has left out some important facts especially when you do not want to say they are lying.

2. (a) bliss – complete happiness
   (b) quandary – a state of uncertainty or indecision as to what to do in a difficult situation.
   (c) opulence – characterized by an obvious or lavish display of wealth or affluence.
   (d) Eldorado –
   The correct option is B

3. Friendliness
   The correct option is B

4. Trauma means emotional shock or bodily injury.
   The correct option is B

5. Extrapolate means to estimate a value that falls outside a range of known values
   - Proliferate – is an intransitive verb which means "to increase greatly in number"
   - Metamorphose – to change physical form
   - Escalate – to become or cause something to become greater, more serious, or more intense.

   The synonyms of escalate are: intensify/worsen/deteriorate

6. – frantically means anxiously, agitatedly, desperately
   - Subsequently means successively, next, afterword, after
   - Consequently means as a result, therefore, thus
   - Unequivocally means clearly, unambiguously, explicitly

   The correct option is C
   The accident happened "as a result" of the recklessness of the driver, slippery road and early morning downpour.

7. "run errands" . **The correct option is B**

8. Pilfer means to steal small items of little value, especially habitually.
   The correct option is A
   Please note that when you use "rob", it means to take something illegally from a person or place, especially by using FORCE, THREATS, or VIOLENCE.

9. **B –** When something/an action, is performed in quick tempro especially when it can't be done in slowmotion, the verb will remain in its base form e.g. take a salute, smash an egg, leap over the fence
   The correct option is B

10. Etymology – is the study of history or origins of words.
    The correct option is D

11. When something is erratic, it means it is not consistent, regular or predictable. Uneven means varying and inconsistent especially inequality, thoroughness, or duration.
    The correct option is C

12. Pyrrhic Victory means a victory that is not worth winning because the winner has suffered or lost so much in winning it. This word "PYRRHIC" actually originated from PYRRHUS, the King of Epirus who defeated the Romans in 279BC but lost many of his men.
    The correct option is B;

13. **The correct option is D;** Greek gift

14. Thanks takes the preposition "to"

15. Bereft takes the preposition "of"
    The correct option is B
16. **The correct option is D**
17. **The correct option is B**
18. **The correct option is C**
19. **The correct option is A**
20. **The correct option is A**
21. **The correct option is B**
22. **The correct option is D**
23. **The correct option is C**
24. **The correct option is D**
25. **The correct option is D**

ENGLISH LANGUAGE QUESTION 2013

In each of **questions 1 - 3**, choose the option **nearest in meaning** to the word(s) or phrase in **italics**.

1. Nobody could say precisely when the landlord became a *recluse*
   A. loner B. drunkard C. nincompoop D. cantankerous
2. If I had known his delicate state of mind, I would not have *broached* the matter
   A. told them B. divulged C. brought up D. cancelled
3. Don't talk like that; you know the Professor will not entertain such *vituperative* remarks.
   A. irresponsible B. insulting C. angry D. illiterate

In each of questions 4— 8, fill each gap with the most appropriate option from the list provided.

4. My daughter would become ………if I paid no attention to her behaviour.
   A. rhascally B. rhapsodically C. rascally D. rascality

5. When the soldiers saw that resistance was ………they stopped fighting.
   A. inadvertent B. futile C. inappropriate D. insurmountable

6. The last time the man saw his ex-wife, she………
   A. was thinking of a proposal of starting a new business.
   B. was intending to start a new business.
   C. intended to start a new business
   D. was going to start a new business.
7. Two days before the execution, the robber was taken to the place where he would be ………With doleful eyes, helooked at the spot where his execution………
   A. hunged/was to be taken place B. hanging/shall take placeC. hung/would take place D. hanged/was to take place

8. 'Don't ………', said the leader, 1 want a decision now'.
   A. prevaricate B. predicate C. precipitate D. be pejorative

In each of questions 9- 11, choose the most appropriate option opposite in meaning to the word or phrases in italics.

9. He has been advised to beware of political *jobbers* if he hopes to succeed.
A. neophytes B. masquerades C. stockbrokers D. masterminds

10. What a *tangled* web we weave, when we try to deceive!
A. complicated B. crooked C. simple D. Loose

11. She devoted too much time to the *peripheral* aspects.
A. superficial B. minor C. main D. Real

For question 12, choose the letter which contains the correct phonetic Symbol in the underlined sounds below.

12. Women
A. /l/ B. /e/ C. /ou/ D. /u/

For question 13, choose the word which contains the correct sound as given in each of the sounds below.

13. /v/
A.off B.fan C. of D. Four

In each of questions 14 and 15, choose the option that best completes the gap(s)

14. The car owner does not think about the — of his vehicle and other payments involved in owning it.
A. transportation B. depreciation C. calculation D. Appreciation

15. We shall offer a good job to a — to register guests in the Central Hotel.
A. waiter B. watchman C. cashier D.receptionist

In questions 16 and 17, fill in the right word or phrase

16. There is not — sense in what that politician has just said.
A. many B. plenty C. lot of D. much

17. I'm sorry I can't give you any of the oranges, I have ....left.
A. few B. little C. only a little D. a few

Fill in the gaps in the following sentences with appropriate preposition

18. Lawrence did not win the contract — the long run
A. at, B. in, C. on, D. to

19. Memuna was careful not to fall — Ameen's tricks
A. into, B. for, C. in, D. with

20. The Commander had placed his troop — alert.
A. in, B.on, C. at, D.over

PASSAGE

All over the world till lately, and in most of the world still today, mankind has been following the course of nature that is to say, it has been breeding up to the maximum. To let nature take her extravagant course in the reproduction of the human race may have made sense in an age in which we were also letting her take her course in decimating mankind by the casualties of war, pestilence, and famine. Being human, we have at least revolted against that senseless waste. We have started to impose on nature's heartless place a humane new order of our own. But when once man has begun to interfere with nature, he cannot afford to stop half way. We cannot, with impunity, cut down the death rate, and at the same time allow the birth-rate to go on taking nature's course. We must consciously try to establish an equilibrium or, sooner or later, famine will stalk abroad again.

Now answer the following questions.

21. 'humane' as used in the passage means
A. wise B. human C. benevolent D. sensible

22. 'We must consciously try to establish an equilibrium' in the passage implies that mankind must
A. strive not to be wasteful B. realistically find an equation C. purposely find a balance D. deliberately try to tight nature

23. The main idea of this passage is that
A. nature is heartless.B. man should control the birth-rateC. man should change nature's course graduallyD. pestilence causes more deaths than war

24. The author observes that
A. war, pestilence and famine were caused by the extravagance of natureB. it was wise at a time when mankind did not interfere with normal production.C. nature is heartless and senseless. D. there was a time when uncontrolled birth made sense.

25. Which of these statements does not express the opinion of the author?
A. many people had died in the past through want and disease.B. mankind should not have the maximum number of children possible.C. mankind has started to interfere with the work of nature.D. Man's present relationship withnature in matters of birth and death is a happy one.

ENGLISH LANGUAGE 2013

1. **A –** Recluse is a noun and it means a person who like being alone and it means also a loner.

2. **C –** Broach means to begin a discussion which people disagree about especially because it is embarrassing.
3. **C –** The word vituperative is an adjective which means cruel and angry remarks or criticism.
4. **C –** A rascally child is one who does not respect other people and enjoys playing tricks on them.
5. **B –** the word futile is an adjective which means pointless or having no point, because there is no chance of success.
6. **B –**
7. **D –** The word hang can be used for both human beings or animate and nominate objects, but when it is used for objects, its past tense form is hung while when it is used for a human being, its past tense form will be hanged.
8. **A –** Prevaricate is a verb which means to beat about the bush and not giving a direct answer to a question because you want to hide the truth.
9. **B –** A jobber as the Italised words is an expert who specialises in buying and selling of shares for people. Therefore, its opposite will be masquerades which means people who pretend to be what they are not.
10. The word tangled means twisted together in an untidy way or complicated and not too easy to understand and its opposite will be simple.
11. **C -** The word peripheral in this content means not too important like the main part of something and it opposite will be main.
12. The word women is transcribed in this form/wimin/, the "O" is the long vowel sound /i/ - **A**
13. The world "of" is transcribed thus /av/ or /∧∨/ or /D∨/ and among all the options, it is only "of" that has the sound /∨/ - **C** is the answer.
14. **B –** Depreciation means to lose value over a period of time and car does not appreciate, but depreciates.
15. **D –** A receptionist is a person who deals with people arriving at a hotel or in an office in a company.
16. **D –** Sense is an uncountable noun and cannot be used within any as means is used for countable nouns such as human beings.
17. **A –** Few as a word shows not enough to be shared with another person and is also used for countable nouns.
18. **B –** Long run goes with preposition in as an idiom and in the long run means after a very long period of time.
19. **B –** Fall collocates or goes with the preposition for. Fall for something therefore, means to be tricked into believing something that is not true.
20. **B –** Alert also goes with the preposition on and to be on the alert means to be ever ready or prepared for something.
21. **C –** The word humane is being kind to people or animals by making sure they do not suffer more than is necessary and benevolent means to be kind, helpful and generous.
22. **C –** It means to strike a balance and not overdo things.
23. **C -** From the last three lines of the passage, it is evident that man has been advised to go in a gradual way to achieve its aim in order to strike a balance.
24. **A –** War, pestilence and famine are embedded in nature which if left uncontrolled, will exalate to maximum.
25. **D –** The above three options express the author's mind.

ENGLISH LANGUAGE 2012

Choose the correct options to fill the gaps in the three questions below.

1. I don't understand *what exactly you were saying.*what is the name of the subordinate clause in this sentence? (a) adverbial clause (b) nominal clause (c) adjectival clause (d) interrogative clause
2. Are you going to Lagos _________ your car? (a) with (b) in (c) through (d) by
3. Another name for relative clause is __________ (a) adverbial clause (b) infinitive clause (c) adjectival clause (d) relational clause

Choose from the option a-d the word or phrase that is nearest in meaning in meaning to the word underlined

4. My efforts at making her to see my point were rebuffed. (a) embraced (b) antagonized (c) snubbed (d) successful
5. He is a lout and can't be relied upon at all. (a) shifty (b) thug (c) unserious (d) liar
6. The speech was delivered with great trepidation. (a) fear (b) dexterity (c) power (d) creativity
7. He cannot hide his aversion for Kemi's unrepentant behaviour. (a) Abhorrence (b) sadness (c) ignominy (d) moodiness
8. My sister is known by everybody to be scurrilous . (a) pleasant (b) vituperative (c) active (d) inactive

Choose from the options a-d the word opposite in meaning to the underlined word.

9. The criminal's answers to the questions during interrogation were evasive. (a) harsh (b) outspoken (c) clever (d) direct
10. I guess he is indifferent to our plans to rid Nigeria of societal ills. (a) interested in (b) opposed to (c) bothered (d) not interested in
11. The girl is very somber in her style of dressing. (a) solemn (b) pleasant (c) provocative (d) exposing

Select from the options a-d the correct meaning of the idiom underlined

12. The manager behaved as if he had bats in the belfry. (a) he was pleasant (b) he had strange ideas (c) he was speechless' (d) he was angry
13. I learnt Agnes was off colour this morning. she was not in class. (a) busy somewhere else (b) not able to wake up early enough (c) not in her right mind (d) not in good health

Fill in the spaces in the passage below with the appropriate words

One needs to observe the __14___ of the __15___ to appreciate how interesting it can be. I witnessed a land __16___ many years back. The ___17___ had ___18___ those who sold him a piece of land to court for ___19___ on his land. Appearing for the defendants the __20___ argued that the land had been lying untouched for over thirty years and had become a hideout for social miscreants.

	A	B	C	D
14	Work	proceedings	hearing	working
15	Law	document	war	ruling
16	quarrel	law	matter	case
17	plaintiff	defendant	accused	criminal
18	arrested	sued	fought	prosecuted
19	trespassing	claiming	building	working
20	legal luminary	plaintiff	Defence counsel	criminal

Choose from the options a-d the appropriate verb that best completes the following sentences.

21. Although she suffered a lot of hardship she still _______ (a) gloated (b) glowed (c) flowed (d) splashed
22. Our plane__________down at Aminu Kano International Airport at exactly 12 midnight. (a) landed (b) descended (c) got (d) touched
23. The disease was ________ very rapidly in the community. (a) widening (b) catching people (c) spreading (d) raging

Choose from the options a-d the correct spelling.

24. A. ebmbaras B. embarass C. embarras D. embarrass
25. A. conterfit B. conterfiet C. counterfeit D. counterfeet

2012 ENGLISH LANGUAGE SOLUTION

1. B: If an expression that has an adjectival marker (what, who e.t.c.) comes after a verb such expression is a nominal clause / phrase.
2. D: You travel by car, train, aeroplane, road, air, railway, e.t.c.
3. C: Adjectival clause is the same as relative clause
4. C: Rebuffed means to turn down a request or suggestion.
5. A:
6. A: Trepidation means great fear.
7. A: Aversion means hatred. Abhorrence means the same.
8. B
9. D: To be evasive means to be indirect.
10. A: To be indifferent means not to be interested in something.
11. C: Somber means dull, sad and serious likewise solemn.
12. B: For someone to have bats in the belfry means for the person to behave strangely especially in a strange way.
13. D: "Off colour" means to be ill, sick, funny, smutty, dirty, crude, coarse e.t.c.
14. C: Hearing in law registers is the proceeding of the court in a law case
15. D: Ruling means judgment. It can also be called verdict.
16. D
17. A: A plaintiff is the complainant in a court case
18. B
19. A
20. C: The defence council is the legal team standing for the accused in a law case.
21. B: It means to be happy and satisfied.
22. D: Aeroplane touches down and not lands.
23. C 24. D D 25. C

2011 POST-UTME SCREENING EXERCISE

Passage I

The privilege of blackening one's stool is not granted to every dead chief or queen-mother without conditions. The honour is merited only on the fulfillment of certain conditions on the part of the occupant of a stool. The blackening of a king's stool is regarded as the greatest honour that could be conferred on a ruler; thus in many Akan states only the stools of kings who proved to be true leaders are blackened.

No royal person's stool is reserved unless he died while still a ruler. A destooled chief is the last person whose memory anybody wants to keep fresh. He must have broken a taboo or committed a serious crime to merit his degradation. He may have committed adultery with his servants' wives; he may have bought and sold slaves, who are considered as heirlooms to the stool; he may have used the oath unreasonably; he may have cursed people. All these crimes can deprive a chief of his regal powers. Once this happens, he becomes, in the eyes of the people, more insignificant than a commoner who has no right whatever to be a chief.

However, a chief may die on the stool', and yet not have his stool blackened. This is so because one must die a good death'. Sudden death through an accident destroys the right to have one's stool blackened. So does death through an unusual disease like leprosy, lunacy, epilepsy and dropsy — which, if discovered in time, are causes for destoolment. The only exception here is death in war which magnifies one's fame and dignity. But even here, if it is found out that one fell when retreating. or running away, from the enemy, one is regarded as at treacherous and infamous leader who should be erased from all historical memory. A chief who suffered from an unclean disease, but got cured before dying, is said to have been engaged in a personal difficult war with the disease and emerged triumphant. Such a chief is worthy of respect. Suicide is, perhaps, one of the worst deaths a chief could undergo. Under no condition whatever will the stool of a ruler who takes his own life, or is killed by a fetish' be consecrated.

Answer the following questions on the passage

1. Which of the following is true according to the passage? **A.** it is entirely up, to the chief whether or not his stool will eventually be blackened B. it is partly up to him, partly due to circumstances beyond his control. C. it is entirely due to circumstances beyond his control D. it depends entirely on people's opinion of him during his lifetime.
2. A destooled chief can be correctly defined A. a chief who has committed crimes B. a chief who was removed during hisreign C. a chief who has broken taboos D. a chief who is more insignificant than a commoner
3. What is meant by die on the stool'? A. dying a miserable and unworthy death B. dying as a reigning ruler C. dying while on stool in the palace D. dying after a disease of stooling
4. Which of the following is the most suitable title for the passage? A. reasons for destoolment B. how to live a worthy life by an Akan ruler C. an aspect of traditional custom of the Akan people D. stool blackening by the royal personages.
5. The most basic condition that qualifies anybody for stool destoolment is A. dying a worthy death B. having being crowned as a ruler C. being a ruler with tangible achievements D. having respect for traditional customs

In the following sentences, choose the option that is most nearly opposite to the underlined word.

6. In any group, there are people who display <u>apathy</u> A. enthusiasm B. patience C. respect D. tolerance.
7. Ferni was very <u>open</u> about his ambition .A. silent B. withdrawn C. closed D. secretive.
8. Angela is very <u>indolent.</u> A. perfect B. devoted C. diligent D. trustworthy
9. Rather than <u>support</u> the chairman, Olu slept off. A. deny B. oppose C. doubt D. back
10. Ngozi's beauty is <u>natural</u> A. unnatural B. artificial C. awkward D. fake

Passage 2

The passage below has gaps numbered 1 to 10. Immediately following each gap, four options are provided. Choose the most appropriate option for each gap.

Coach Samson Siasia has asked that Heartland be paid their __11__ salaries and _12__ so as to ___13__ them to victory against Al-Ahly of Egypt in Sunday's all-important CAF Championships League match. Heartland are third in Group B with four points __14__ as many matches and need to __15 __defeat at second-placed Al-Ahly to stay in the __16 ___for a place in the semi-final of Africa's most prestigious club competition. Siasia told MTN Football.com that the team would be better motivated if they at least receive their August salaries before the Al-Athly __17___. We have to make sure that they are paid their salaries __I 8__ so that they could play the game of their lives. The government has done very well, but it will be a big morale __19__ to get paid for them to go out there and play. Siasia informed MTNFootbafl.com that
Heartland plan to employ the counter __20__to get a result in Cairo.

11. A. left over B. outstanding C. owed D. late
12. A. match bonuses B. match payments C. match wages D. match fees
13. A. push B. instigate C. spur D. move
14. A. from B. in C. at D. with
15. A .afford B. annul C. avoid D. afford
16. A. focus B. centre C. running D. front
17. A. show B. showdown C. show up D. blow out
18. A. to time B. for time C. as due D. as and when due
19. A. boomer B. inspirer C. booster D. pusher
20. A. attack B. attacker C. getter D. goal

For questions 21 -23, choose the best options from letters a — d that best summarises the information contained in the underlined sentence.

21. In an answer to the question as to how life is treating him, the politician said <u>it never rains but it pours</u>. A. things are getting decidedly worse .B. his financial status is deteriorating. C. the blessings of life shower on him like a heavy rain. D. he is contented with improved fortunes.
22. <u>Camilla waited for her friend</u> in the library for a good hour. A. Camilla enjoyed the sixty minutes she waited for her friend .B. When Camilla was waiting; she spent the time in a profitable way. C. Camilla waited for her friend rather more than sixty minutes. D. It was good for Camilla to wait an hour for her friend.
23. <u>This is your instruction</u> and I have had no hand in it. From this sentence we know that the writer A. does not support the instruction B. is refusing to obey the instruction C. dislikes the person that issues the instruction D. is somehow happy with the instruction

For questions 24 and 25, choose the letter which contains the correct phonetic symbol in the underlined sounds below.

24. Pl<u>um</u>b A. /m/ B. /b/ C. /ph/ D. /p/
25. W<u>o</u>men A. /l/ B. /e/ C. /ou/ D. /u/

SOLUTION

Passage I

1. A 2. D 3.B 4. C
5. A

Antonyms refer to words that are opposite in meanings.

6. Apathy means lack of enthusiasm, interest or emotionally emptiness.

The correct answer is A - **Enthusiasm**

7. Open means not sealed, apart, wide or allowing access.

The correct answer is D - **Secretive**

8. Indolent means lazy, lethargic, slow to change etc.

The correct answer is C —. **Diligent**

9. Support means give assistance, tolerance, encouragement etc.

The correct answer is B — **Oppose**

10. Natural derived from nature or existing in; not made, caused by or processed by humans.

The correct answer option is B — **Artificial**

Passage 2

The passage has to do with registered words i.e words peculiar to certain fields (Sport, Law etc.)

11. -B — Outstanding12. A — Match bonuses 13.C — Spur14. A — From
15. C — Avoid16. C — Running 17. B — Showdown 18. A — To time
19. C — Booste r 20. A — Attack

Numbers 21— 23 are based on idiomatic expression; words that have fixed expressions with nonliteral meaning. It is the natural way of using language or stylistic expression.

21.A 22.C 23.A

Questions 24-25 are on English sound:

24. Plumb-/p/Am/
    Whenever /m/ consonant is followed by /b/, the /b/ sound is silent. In the transcription letter "b" will not appear. The sound contained in letters "mb" is /m/ **The correct answer is A**
25. Women - /wlmln/
    The sound contained in letter "0" is /i/. **The correct answer is A**.

2010 POST-UTME SCREENING EXERCISE

Choose the words that are closer in meaning to the words in initial positions.

1. **Futile**: (a) worthless (b) vain (c) dangerous (d) useless
2. **Halt:** (a) wait (b) fault (c) stop (d) stay
3. **Virtuous:** (a) seeing (b) good (c) upright (d) religious
4. **Renowned** (a) famous (b) popular (c) well read (d) familiar
5. **Solitary** (a) private (b) sultry (c) alone (d) lonely

In each of questions 6 and 7, choose the option that best completes the gap(s)

6. The car owner does not think about the ______ of his vehicle and other payments involved in owing it.
   (a)Transportation (b) depreciation (c) calculation (d) appreciation
7. We shall offer a good job to a ______ to register guests in the Central Hotel.
   (a) Waiter (b) watchman (c) cashier (d) receptionist

In each of the questions 8 and 9, choose the option opposite in meaning to the word in italics.

8. Lola was *agitated* when the sad news of her mother's accident was broken to her. (a)excited (b) calm (c) uncontrollable (d) unreasonable
9. The president took exception to the *ignoble* matter. (a) honourable (b) embarrassing (c) dishonourable (d) extraordinary

In each of questions 10-12, select the option that best explains the information conveyed in the sentence.

10. The crowd in the hall is *intimidating* (a) The crowd is frightening (b) The crowd is angry (c) The crowd is overwhelming (d) The crowd is riotous
11. The event of last Friday show that there is *no love lost* between the principal and the Vice-Principal.
    (a) They like each other (b) They work independently (c) They cannot part company (d) They dislike each other
12. Adawo is an *imp.* (a) Adawo behaves badly (b) Adawo behave decently (c)Adawo behaves differently (d) Adawo behave queerly

In questions 13 and 14, select from the options to fill in the gaps

13. There is no _______ sense in what that politician has just said. (a) many (b) plenty (c) lot of (d) much
14. The candidate made _________ at the village square a day before the elections (a) a sermon (b) an address (c) an eulogy (d) a lecture (e) a speech

In each of Questions 15 to 17, choose the option that best complete the gap(s)

15. The city ____ as a federal capital only _____ the last twenty years. (a) existed/over (b) has existed/for (c) was existing/from (d) is existing/in
16. He is _____ Kaduna _____ an official assignment. (a) at/in (b) at/for (c) in/on (d) for/in
17. The members of the other team agree _____ all the terms of contract. (a) on (b) by (c) to (d) with

In each of Questions 18 and 19, choose the word(s) or phrases which best fill(s) the gap(s)

18. After Jerry had made the bed, he ____ on it. (a) layed (b) laid (c) lied (d) lay
19. The buildings damaged by the rainstorm _____ schools, hospitals and private houses. (a) included (b) include (c) were included (d) was including

In each of the Questions 20 and 21, fill the gap(s) with the most appropriate option.

20. _____ any problems, I shall travel to London tomorrow on a business trip. (a) In spite of (b) Given (c) Barring (d) In case
21. 'I can't stand people prying into my private life', Ladi said '_____' agreed Agbemu. (a) Me either (b) Me too (c) I also (d) Likewise myself

In each of Questions 22 to 25, choose the option that has the same consonant sound as the one represented by the letter(s) underlined.

22. Cheap: (a) machine (b) sheep (c) chip (d) chemist
23. School: (a) cool (b) chart (c) itch (d) leech
24. Pharmacy: (a) every (b) rough (c) plough (d) wave
25. Happy: (a) our (b) eyes (c) honour (d) behind

SOLUTION

Synonyms are words that are closer in meanings:

1. **Futile** means Fruitless, unproductive, etc. **The correct answer is B**
2. **Halt** means top or pull up etc. **The correct answer is C.**
3. **Virtuous** mea*ns* good, moral, ethical, decent, honest etc. **The correct answer is B**
4. **Renowned** means famous, prominent etc. **The correct answer is A**
5. **Solitary** means secluded, shunning company, companionless, antisocial, reclusive etc.
   The correct answer is C
6. **Depreciation** means the process of becoming less valuable over a period of time.
   The answer is Depreciation.
   The correct answer is B
7. **Receptionist** – a person who works in a place such as hotel, office, or hospital which welcomes and helps visitors.
   The answer is receptionist.
   The correct answer is D
8. **Agitated** *means* upset, disturb, distressed, nervous, etc. **The correct answer is B**
9. **Ignoble** means dishonourable, unworthy, shameful, etc. **The correct answer is A**
10. Intimidating means overwhelming. **The correct answer is C**
11. **"No love lost"** means dislike, ill will, hate-when it takes the word between.
    The correct answer is D
12. **Imp** means a little devil.
    The correct answer is A
13. **The correct answer is D**

14.	**E**	15.	**B**	16.	**C**
17.	**A**	18.	**D**	19.	**A**
20.	**A**	21.	**C**	22.	**C**
23.	**A**	24.	**B**	25.	**D**

2009 POST-UTME SCREENING EXERCISE
USE OF ENGLISH
COMPREHENSION

Read the following passages carefully and answer the questions that follow

Passage 1

The best acceptable definition of history is that it is a record of the past actions of mankind, based on surviving evidence. It is this evidence that is historian employs to chronicles and correlate events, by which he arrives at conclusions which he believes to be valid. Hence, the historian is referred to as an interpreter of the development of mankind.

It should be understood that there is more than one way of treating the past. For example, in trying to deal with the revolutions in Nigeria, past and present, the historian may describe the events in a narrative order or, he may choose to concentrate on analysis of the general causes, comparing their stages of evolution in other countries.

The historian does not seek to attain the same kind of results as the scientist, who can verify his conclusions by repeating his experiment under controlled conditions. Whilst he also attempts to classify the phenomena, the historian is more likely to consider events in terms of their uniqueness.

Added to this is the fact that history is concerned, fundamentally, with the lives and actions of men, and as such, the historian's search for causes is bound to be relatively subjective as compared to that by the scientist. In essence, however, historians are agreed and insist that history should be written as scientifically as possible and that evidence should be analyzed with the same objective attitude employed by the scientist when he examines certain phenomena of nature.

1. History could be defined as (a) a record of evolution of a country (b) a record of development of mankind (c) a record of present actions of mankind based on surviving evidence (d) a record of the past action of mankind based on surviving evidence.
2. According to the passage, one of the duties of a historian is (a) to predict the future (b) to analyze the past and future (c) to explain the significance of the past events (d) to interpret the development of mankind
3. How can history be scientifically recorded? (a) by examining available evidence and analyzing unusual occurrences (b) by falsifying and fabricating available facts (c) by speculating on what was and ought to have been (d) by concealing some of the evidence.
4. The scientist tends to be more reliable than a historian because (a) he works in a laboratory (b) he is better qualified (c) he can cross check his results several times (d) he has more time to work at his experiments
5. According to the passage, a historian should try to examine a material (a)scientifically (b) subjectively (c) accurately (d) objectively.

Passage 2

From the apex of the Niger Delta southwards, dry land, overgrown with dense forests still virginal in various spots, gives way to seasonally inundated zones. Here, sweet water swamps with strands of raffia palms gradually merge into tidal swamps of brackish ooze, where mud skippers thrive under the arching roots of mangroves. The Niger, fingering through a thousand creeks, meets the sea in a dozen estuaries. Strong rivers current drifts and mud across the rivers mouths, sealing them again and again to navigation.

6. According to the passage, how would you describe a seasonally inundated zone? (a) a zone always covered with mud (b) a zone always covered with shallow water (c) a zone under water at certain times of the year (d) a zone subject to heavy rain every season.
7. What is brackish ooze? (a) a strong river current (b) a mixture of fresh water and mud (c) a mixture of fresh water and salt water (d) fresh and clear water
8. Where do mudskippers thrive? (a) in the creeks (b)in the swamps (c) in the mangroves (d) in the roots
9. Where does the Niger meet the sea? (a) in the creeks (b) in the Delta (c) in the swamps (d) in the forest
10. 'Fingering through' as used in the passage means (a) cutting across (b)passing through (c) cutting between (d) passing between

Lexis and Structure

In each of the following sentences, there is one word underlined and one gap. From the list of words lettered (a) to (e), choose the word that is mostly nearly opposite in meaning to the word underlined and which will appropriately fill the gap in the sentence. (11 – 15).

11. She was very proficient hairdresser but had little aptitude for sewing in which she was ____ (a) new (b) unskilled (c) unlearned (d) ignorant (e) awkward
12. A metal will expand when it is heated and ____ when it cools (a) shorten (b) lesser (c) contract (d) congeal (e) curtail
13. Athletes wishing to get rid of their ____ and get more energy should take more exercise (a) fat (b) oxygen (c) lethargy (d) trainer (e) espots
14. If you do not accept the offer of a job in the secretariat within the next one week, we shall assume you have _____ it. (a) denied (b)refused (c)deprived (d) lost
15. The political aspirant asked the villagers to support him and not to _____ (a) deny (b) undermine (c) defy (d) despite (e) attack

From the words lettered a-e below each of the following sentences, choose the one which is nearest in meaning to the underlined word, as it is shown the sentence (Nos. 16 – 20)

16. After finishing the 800 metres race, he fell asleep from exhaustion (a) weakness (b) fatigue (c) overwork (d) eagerness (e) sloth
17. The footballers went back to their camp sullenly (a) cheekily (b) quickly (c) stubbornly (d) resentfully (e) silently
18. After Warri, on our way to Benin, we passed through a dense forest (a) crowded (b) close (c) thick (d) heavy (e) wooded
19. Last night, there was a very fierce rain storm (a) raging (b) storming (c) angry (d) violent (e) ferocious
20. The examiner said that the candidate's performance in the examination was not good enough (a) failure (b) achievement (c) E-marks (d) presentation (e) marks

In each of the following question, fill each gap with the appropriate option from the list. Following exercises express different times by using different tenses. From the options suggested, choose any one that best suits each context

21. The editor was not happy that the Nigeria press was hemmed ___ (a) up (b) across (c) in (d) over (e) sideway
22. More _____ to your elbow as you campaign for press freedom (a) energy (b) power (c)effort (d) grease (e) kinetic
23. A child that shows mature characteristics at an early age may be described as _____ (a) precocious (b) ingenious (c) premature (d) preconceived
24. That is a very terrible woman; everyday she makes a lot of noise about one thing or the other. I'm not surprised, that's what her sisters _____ too (a) are used to doing (b) do (c) always used to do (d) are doing
25. Sir, I'm not lying about the matter, I know nothing of it. If I knew, _______ (a) I must tell you (b) I can tell you (c) I would tell you (d) I shall tell you

SOLUTION

Comprehension – Passage 1

1. C 2. D 3. A 4. C 5. D

Passage 2

6.B 7.B 8. C 9. A 10. B

Lexis and Structure

Antonyms are words that are opposite in meaning

11 Proficient means having a high degree of skill in something; very skilled. The answer is unskilled. **The correct answer is B**

12 Expand means make or become larger and greater in size, number or importance etc. The answer is shorten. **The correct answer is B.**

13 Energy means vigour, effort and enthusiasm into an activity, work, etc. Antonyms are: fatigue, lethargy, weakness, tiredness, etc. the answer is lethargy

The correct answer is C.

14 Accept means respond to action, to take willingly something that is offered. Antonym is refuse. **The correct answer is B.**

15 Support means encouragement, boost, help, assist somebody or something etc. antonym are: discredit, disapprove, refuse, under-rate, etc. The closest option is undermine
The correct answer is B.

16 Exhaustion means the state of being very tired due to fatigue. Synonyms are: fatigue, wearing etc. The answer is Fatigue. **The correct answer is B.**

17 Sullenly means hostilely silent and badly tempered etc. The answer is resentful.
The correct option is D

18 Dense means tightly packed, very thick. The answer is thick
The correct answer is C

19 Fierce means aggressive, violent or intense etc. The answer is angry. **The correct answer is C.**

20 Performance means how well or badly you do something. Antonyms are: inaction, failure, challenges, etc. The answer is marks. **The correct answer is E**

21 Hemmed means to surround somebody or something so that it cannot grow easily. The preposition that goes with the verb hem is 'in' so.**The correct option is C**

22 **B 23. A 24.B 25.C**

2008 POST-UME SCREENING EXERCISE
USE OF ENGLISH

Read the following passage carefully and answer the questions based on it.

Why should an artist attempt to concentrate his experience of life in a unique work of art? No. final answer can be given, but two possible reasons suggest themselves. Man seems always to have preferred order to disorder. His whole progress on earth has been a struggle to this end Everything he has done, from the creation of vast empires to the growing of small garden, has been a triumph, in greater or lesser degree, of order over chaos. To help control his own thought,6.the sudden surprises of his limitless mind, he has had to invent language. As each new thing appears, whether it be an idea or an object, he gives it a name and thus brings it into line with the things he already understands. And he has invented for himself more than one kind of language. There is a language of painting, a language of architecture, or mathematics – to name but three each has its own special symbols, its own form of logic; and each enables him to express some of the myriad thoughts that crowd his mind. High among the languages of man is the language of music.

1. Through his struggles man has achieved (a) the return of a state of utter confusion (b) the complete destruction of vast empires (c) the growth of disorder from order (d) the transaction of order out of chaos.
2. Man invent the language because (a) it helped to organize his thoughts and unceasing ideas (b) there was nothing he could do at the time to diversify his talents (c) he already had control over his mind and his countless ideas (d) it was a method or realizing his position as a Supreme Being.
3. By naming objects or ideas, man was able to (a) comprehend less and less the things around and about him (b) clarify things and correlate them with facts already known (c) allow an area of complete confusion to develop in language (d) make visual impressions for more important than ever before.
4. The various language can be identified by (a) their use of the same marks or signs and systems of logic (b) the manner in which their logic agrees and their symbolism is similar (c) the endings of the various symbols and their simplified logic (d) the own science of reasoning and their peculiar marks or signs.
5. The work "myriad" (line 11) as used in the context means (a)terrible (b) mysterious (c) frequent (d) multitude

In each of question 6 – 10, there is a gap. Complete the gaps with appropriate item from the option A – D under sentence.

6. If you try to write without having a clear idea, you often end up just ____ without saying anything very meaningful (a) drooling (b) boasting (c) gambling (d) rambling
7. The four of you should share the remainder _____ you. (a) among (b) around (c) between (d)within
8. I have no doubt that Enyimba will _____ Oaks next Saturday. (a) flog (b) whip (c) win (d) beat
9. "You need not go ____ down the road before you notice a huge white building on the road", the man said: (a) inside (b) farther (c) further (d) deep
10. If your writing lacks coherence, your reader will just find something else to read or ____ the television. (a) tune in (b) turn on (c) switch up (d) open.

Choose the appropriate option to complete the following:

11. The president promised a higher allocation to the education sector in this year's budget, _____ (a) isn't he? (b) did he? (c) didn't he? (d) doesn't he?
12. The picture is ascribed to Leonardo da Vinci. This means that ______ (a) Leonardo da vinci painted it (b) Leonardo da Vinci might have painted it (c) Leonardo da Vinci definitely painted it (d) Leonardo da Vinci did part of the painting.

13. The Principal's reference to the cane ______ the boy with much mental uneasiness. (a) inflicted (b) assaulted (c) afflicted (d) insulted
14. The government's envoy had left the country again in his latest round of <u>trouble shooting.</u> The underlined expression means (a) trip marring efforts (b) troublesome efforts (c) peace making efforts (d) trouble making efforts
15. His three sons, Sanmi, Chukwu and Collins are eleven, nine and seven _______ (a) respectively (b) respectedly (c) succeedingly (d) successively
16. As the examination progressed, it was observed that more candidate started into space. This means many candidates (a) looked into the sky (b) looked straight for long but to nothing in particular (c) looked through the window for would-be helpers (d) tried to ensure that the spaces, between them were well maintained.
17. Hundreds of car went ______ us before we were given a ride to the campus. (a) pasted (b) past (c) passes (d) by
18. You told me that Johnson is your trusted friend, why did you not <u>stand up for</u> him during his trial? (a) defend (b) ridicule (c) pity (d) disown
19. When you pronounce the word <u>university</u>, how many sounds could, how many sounds could you perceive? (a) 5 (b) 4 (c) 10 (d)

SOLUTION

1. **D. 2.A 3.A 4.C 5.D 6.D 7.A 8.C 9.B 10.B 11.C. 12.A 13.A**

<u>14.Trouble shooter</u> is a person that settles trouble is an effort to settle trouble and bring peace. The answer is peacemaking efforts. **The correct answer is C**

15 **A**

16 'Stared into space' means to look straight in front of you without looking at a particular thing, usually because you are thinking about something. **The correct answer is B**

17 **C**

18 'Stand up for' is a phrasal verb meaning to support a course or defend somebody or something you believe. **The correct answer is A**

19 University- u/ni/ver/si/ty – 5 syllable. **The correct answer is A**

2007 POST-UME SCREENING EXERCISE
ENGLISH LANGUAGE

Fill in the blanks in the following sentences making use of one of the four options in the letters A – D.

1. They ______ arrived Lagos by now, all things being equal (a) had (b) must (c)might have (d) would have
2. The pupils _____ so much noise that the teacher had to tell them to stand up and raise up their hands (a) had been making (b)should have made (c) were making (d) had made
3. I ______ that he was insincere all along (a) could know (b) must know (c) should have known (d) may have known
4. If I had gone to Lokoja, I ______ the opportunity of seeing the president (a) should not have had (b) must not have had (c)would not have had (d) should not have

Choose the preposition that best fills the gaps in the following sentences

5. My sister does not have flair ____ Mathematics (a) at (b) at (c) with (d) for
6. When I got to her house, she was still ____ bed (a) in (b) on (c) on the (d) in the
7. During the demonstration, the anti-riot policemen were instructed to break ____ the students' defense line (a) off (b) open (c) through (d) down
8. I was ____ hearing distance of the speaker 9a) at (b) in (c) on (d) within

From the words labeled A – D in numbers 9 to 20, choose the one that best completes each of the following sentences

9. The electricity cable had to be _____ enough to be laid along the bend in the road (a) elastic (b) taut (c) compliment (d) flexible
10. We didn't have a lot of money, so I had to live quite ______ (a) niggardly (b) frugally (c) wastefully (d) grudgingly
11. I acted too impetuously, I do not know what _____ me (a) came along (b) came over (c) came at (d) came on
12. The hurricane raged for several days, learning a trial of _____ across the land (a) destruction (d) depoilation (c) desperation (d) demolition
13. If a player breaks the rule during a match, one point will be ______ to his opponent (a) adjourned (b) credited (c) debited (d) allied
14. Mouse-traps are not always very effective, as some mice prove to be remarkably ______ (a) inaccessible (b) perpetual (c) inconspicuous (d) elusive
15. At the frontier he hid the watches in his pocket in order to ____ customs duty (a) evade (b) incur (c) repel (d) deceive
16. The driver was short of petrol, so he _____ down all the hills with the engine switched off (a) glided (b) cut (c) wheeled (d) coasted
17. Everyone in my family has a job. My mother is a teacher, my father is an engineer, and my granny _____ (a) used to sell roast chicken (b) is selling roast chicken (c) has roast chicken (d) sells roast chicken
18. One curious thing about my uncle is that he wishes _____ (a) he is having eight wives (b) he had eight wives (c) he can have eight wives (d) he can be allowed to have eight wives.
19. I'm afraid, you know. My father has been sleeping since 4:00 pm yesterday. It's about time _____ (a) to wake up (b) he wakes (c) he woke up (d) he's awake

20. Are you deaf? I asked you ______ (a) how old you were (b) how old are you (c) how old is your age (d) what is your age
21. What is a sentence? (a) it is made up of words (b) it is made up of phrases and clauses (c) it is a group of words giving a complete sense (d) it can be simple or complex
22. What is a clause? (a) it is made up of words (b) it is made up sentences (c) it is made up of phrases (d) it is a group of words containing a finite verb
23. What is a phrase? (a) it is a group of words containing a finite verb (b) it is a group of words giving a complete sense (c) it is a group of words not containing a finite verb (d) it usually begins with a preposition of a participle.
24. An example of a finite verb is (a) going to school every day (b) given his position as the principal (c) while going to school (d) while they were going to school
25. What is tense? (a) it has to do with present, past and future times (b) it is a correspondence between the form of the verb and the concept of time (c) it is derived from the latin word 'transpire' (d) it is a controversial topic in linguistics
26. What is aspect? (a) it is a manner in which the verbal action is experienced or regarded (b) it reflects the attitude or mood of the speaker (c) it is made up of progressive and perfective forms (d) it is normally joined together with tense
27. What are minor sentences? (a) they are complete sentences (b) they are incomplete statements (c) they are incomplete statements but normally function as sentences (d) they are used by writers for economical purposes.
28. In measuring one's linguistics competence in a particular language, itemize four sentence types that one need to master (a) simple, complex-multiple, difficult and more difficult (b) simple, complex, compound and compound-complex (c) simple, more-simple, difficult, more difficult (d) rational, more rational, logical and more logical.

SOLUTION

1. **D**
2. A present continuous tense takes a verb with '-ing'. The correct answer is 'were making'. **The correct answer is C 3. C 4.C 5.D 6.A 7.A 8.D 9.D.**

10 The answer is 'frugally' which means using only as much money or food as is necessary. **The correct answer is B**

11 **The correct answer is B**

12 The answer is 'destruction'. Destruction is the act of damaging something so badly that it no longer exists. **The correct answer is A**

13 Credited means praised or approved because you are responsible for something great that happened. **The correct answer is B**

14 .**D** 15.A 16.D 17.D 18. B

19.Anytime 'it's high time' or 'it's about time' is used, it is followed with a past tense. The answer is 'he woke up'. **The correct answer is C**

20 'The correct answer is A

21 The correct option is B

22.A clause is a group of words that includes a subject and a finite verb. **The correct option is D**

23 A phrase is a group of words without a finite verb. Hence, not making a complete sense.**The correct option is C**

24 A finite verb is a verb that reflects tense, number, and person. Eg, eats, sleeps, slept, was, were, etc. The answer is 'while they were going to school' **The correct answer is D**

25 A tense is a form of verb that shows the time of the performance of an action..**The correct answer is B**

26 An aspect shows if the action in a sentence is in progress or has been completed, hence, we have progressive and perfective aspect. The answer is 'it is made up of progressive and perfective forms'. **The correct answer is C**

27 Minor sentence is a sentence fragment. E.g., I suppose, last month, or so, I think, etc. the answer is 'they are incomplete statements but normally function as sentences. **The correct answer is C**

28 The answer is simple, complex, compound and compound-complex. **The correct answer is B**

2006 POST-UME SCREENING EXERCISE
ENGLISH LANGUAGE

From the words lettered A – E, choose the word that is alike in meaning to the word underlined

1. Andrew made some bellicose statement about his strength to other boys in the street. This means that Andrew (a) is a brave man (b) wishes to fight (c) is a coward (d) loves to help others with his power (e) has a lot of power
2. The excuse that he forgot about the meaning was a flimsy one. This means the excuse was (a) very bad (b) a complete lie (c) difficult to believe (d)not important (e) a very good one
3. Scrupulous politicians do not have a place in the Nigerian polities. Scrupulous politician are: (a) honest (b) dishonest (c) corrupt (d) good natured (e) insincere
4. The woman was happy because her gorgeous dressing made her quit obtrusive. The woman was very (a) appreciated (b) proud (c) good (d) noticeable (e) excellent
5. The man is known to be a sly. I won't trust him with anything. This means the man is known as a (a) deceiver (b) thief (c) kidnapper (d) rogue (e) burglar

From the alternative suggested, select the answer that best expresses the same meaning as the expression italicized in each exercise.

6. Don't take the plate away; *it is possible for the owner to ask for it* (a) the owner might (b) the owner can (c) the owner is going to (d) the owner will come to (e) none of them
7. I wonder *if you would allow me* to put out the fire (a) I might (b) I can (c) I should (d) all of them
8. When you great-grandmother was in Vietnam, *did she have the ability* to speak *Chinese?* (a) has she been able (b) was she enabled (c) could she (d) how possible was it for her (e) none of them
9. I know a carpenter that *knows how to* make that kind of wardrobe (a) could (b) has the know-how (c) can (d) can be able (e) may be able to
10. Frances, where is your male visitor? Don't lie to me, *it is not possible that he has gone through the high window* (a) he couldn't have (b) he can't have (c) he shouldn't (d)he mustn't have (e) none of them
11. My father *made no bones about* telling his friend how he felt about his behavior. This means that my father (a) spoke well to his friend about his behavior (b) spoke honestly to his friend about his behavior (c) spoke in the open to his friend about his behavior (e) spoke with all his might to his friend about his behavior
12. After much talk, my brother taught it was time to *hit the hay*. This mean my brother taught it was time to (a) make hay while the sun shine (b) burn the collection of hay (c) go to bed (d) keep quiet (e) tell the others off
13. The housemaster was *foaming in the mouth* when he discovered that some students had sneaked out of the hostel. This means the housemaster (a) was very sad (b) had epilepsy (c) became silent and calculative (d) was uncontrollably furious (e) was jittery
14. Who told Mabel she could sing? She really *laid an egg* at the talent show. This means (a) Mabel's performance was very embarrassing (b) Mabel's performance was very interesting (c) Mabel's performance was very impressive (d) Mabel's performance was not very bad (e) Mabel's performance was like that of a hen laying an egg
15. Mr. Johnson is *on the warpath because* his car broke down again. This means Mr. Johnson is (a) ready to fight his mechanic (b) started fighting the government because the road was bad (c) very infuriated (d) fighting a war with his family in the car (e) drawing a battle in between him and his mechanic

For questions 16 – 20, choose among the options A – E the word that is nearest in meaning to the italicized words in each of the sentences

16. The president announced that all political prisoners have been *pardoned.* (a) condemned (b) severely rebuked (c) banished (d) reprieved (e) released
17. He *resented* been criticized every time by his boss (a) preferred (b) abhorred (c) ignored (d) carefully considered (e) enjoyed
18. The most striking thing about the just-concluded world cup finals in Germany was the complete *eclipse* of the defending champion- Brazil (a) sudden disappearance (b) defeat (c) failure (d) brilliant performance (e) arrogance
19. As he watched the winning film, his face remain *inscrutable* (a) unreadable (b) pale (c) unfriendly (d) impossible to please (e) bright
20. His latest album has done much to *boost* his reputation as a writer (a) increase (b) establish (c) nourish (d) destroy (e) decrease

For questions 21 – 25, choose from the option A –E the word or phrase opposite in meaning to the underlined word.

21. The doctor certified the tumour <u>malignant</u> (a) benign (b) ripe (c) painless (d) dangerous (e) slow
22. Andrew is too <u>garrulous</u> for my liking (a) dull (b) apathetic (c) laconic (d) easy going (e) dumb
23. The man hold <u>parochial</u> views on almost every issue (a) rational (b) realistic (c) popular (d) broad minded (e) sensible
24. Your idea on this issue seem to me quite <u>novel</u> (a) bookish (b)dangerous (c) archaic (d) genuine (e) good
25. The people appreciated the chairman for his <u>invaluable</u> contributions to the community's development (a) worthless (b) costly (c) unrecognized (d) incalculable (e) meaningless
26. Which of the following statement is true with regard to summary writing? (a) details are more important than main ideas (b) main ideas are more important than examples (c) illustration are more important than main ideas (d)elaboration, exemplification and details are more important than main idea (e) none of the above

Choose the appropriate option to complete the following:

27. At the crusade, we prayed to God to _____ this on us (a) breath His breathe (b)breathe His breath (c) breathe His breathe (d) breath His breath
28. The chairman, committee of Deans needs to see your friend Dele urgently, do you know his _______? (a) where and about (b) whereabouts (c)whereabout (d) where and abouts
29. "As from now, this university will have zero tolerance for any form of malpractice", said the Vice-chancellor. The vice-chancellor said that (a) as from then, the university will have zero tolerance for any form of malpractice (b) as from now, this university would have zero tolerance for any form of malpractice (c)as from the, his university will have zero tolerance for any form of malpractice (d) as from then, that university will have zero tolerance for any form of malpractice.
30. The teacher took me for one of those students who could not spell such word as (a) 'miscellaneous and maintenance' (b) 'miscellaneous and maintenance' (c) 'miscellaneous and maintenance' (d) 'miscellaneous and maintenance'

31. God should take control of the heart of the organizers of this Post-UTME screening exercise, they should not make this test ______ than UTME (a) more tough (b) more tougher (c) much tougher (d) more much tougher

SOLUTION

1. Bellicose means warlike, ready or inclined to quarrel or fight. **The correct answer is B**
2. Flimsy means unconvincing, weak etc. **The correct answer is C**
3. Scrupulous means having moral integrity, careful about paying attention to every details. **The correct answer is A**
4. Obtrusive means annoying, highly noticeable in an unpleasant way. **The correct answer is D**
5. Sly means crafty, evasive, mischievous etc. **The correct answer is A**
6. Possible – able to happen, maybe real or true, might happen. **The correct answer is A**
7. Wonder – amazed admiration or awe. **The correct answer is B**
8. Ability – be able, competence or intelligence. **The correct answer is C**
9. Know-how – knowledge or experience of doing something. **The correct answer is B**
10. Not possible – can't be done or exist. **The correct answer is A**
11. Made no bones – **The correct answer is B**
12. Hit the hay –**The correct answer is C**
13. Foaming at the mouth – **The correct answer is D**
14. Lay an egg – **The correct answer is C**
15. On the warpath- **The correct answer is C**
16. Pardoned – **The correct answer is D**
17. Resented – **The correct answer is B**
18. Complete eclipse – **The correct answer is C**
19. Inscrutable – **The correct answer is A**
20. Boost – to increase or improve. **The correct answer is A**
21. Malignant – impossible to be controlled easily and likely to cause death. **The correct answer is A**
22. Garrulous – talking too much, wordy. **The correct answer is C**
23. Parochial – narrow- minded. **The correct answer is D**
24. Novel – new, interesting and often seeming slightly strange. **The correct answer is C**
25. Invaluable – extremely or very useful. **The correct answer is A**
26. **B 27.B 28.C 29.A 30.C 31.A**

2015 OAU Post UTME: Mathematics Questions

1. Simplify $\frac{3^{n+3} - 3^{n+2}}{3^{n+1} - 3^{n}}$ (a) -9 (b) 9 (c) 10 (d) -10
2. Find the value of p which satisfies the equation $\sqrt{p} - \frac{6}{\sqrt{p}} = 1$ (a) 4 (b) -4 (c) 9 (d) -9
3. Find the area of the circle $4x^2 + 4y^2 - 400 = 0$ (a) 10π sq. units (b) 40π sq. units (c) 400π sq. units (d) 100π sq. units
4. Let the mean of x, y^{-1}, z^5 be 6. Find the mean of 10, y^{-1}, 12, x, z^5 (a) 7 (b) 8 (c) 9 (d) 10
5. What is the addition of y and x – intercepts of the line $\frac{2}{3}x + \frac{3}{2}y + 9 = 0$? (a) -19.5 (b) 19.5 (c) 20.5 (d) -20.5
6. Given that $h(x) = 3 + 2x$ and $f(x) = 1 - x$, find $h(-f(x))$ (a) $1 - 2x$ (b) $1 + 2x$ (c) $2x - 1$ (d) $-1 - 2x$
7. Noting that $\cos\alpha = \sin(90 - \alpha)$, find y in term of x in the equation $\cos\left(1 + \frac{1}{2}x\right) = \sin\left(\frac{3}{2}y\right)$ (a) $y = \frac{178+x}{3}$ (b) $y = \frac{x-178}{3}$ (c) $y = \frac{178-x}{3}$ (d) $y = \frac{-(178+x)}{3}$
8. For what values of x is $x^{-1} < -1$? (a) $0 < x < 1$ (b) $x < -1,\ x > 0$ (c) $x > 1,\ x < 0$ (d) $-1 < x < 0$
9. In how many ways can the letters of the word NWAFOR be permuted? (a) 7200 (b) 72 (c) 720 (d) 72000
10. If α, β are the roots of equation $18 + 15x - 3x^2 = 0$, find $\alpha\beta - \alpha - \beta$ (a) 11 (b) -11 (c) 10 (d) -10
11. Resolve $\frac{1}{(1-x^2)}$ into partial fractions (a) $\frac{1}{2(1+x)} - \frac{1}{2(1-x)}$ (b) $\frac{1}{2(1+x)} + \frac{1}{2(x-1)}$ (c) $\frac{1}{2(x+1)} + \frac{1}{2(1-x)}$ (d) $\frac{1}{2(1-x^2)}$
12. Given that the sum of infinity $S_\infty = a + ar + ar^2 + \ldots = \frac{a}{1-r}$, to what sum does the infinite series $1 - \frac{2}{3} + \frac{4}{9} - \frac{8}{27} + \ldots$ converge? (a) $-\frac{3}{5}$ (b) $\frac{5}{3}$ (c) $-\frac{5}{3}$ (d) $\frac{3}{5}$
13. What is the value of x for which $x^2 - 5x + 6$ is minimum? (a) $\frac{5}{2}$ (b) $-\frac{5}{2}$ (c) 3 (d) -3
14. Integrate $5x^4 + e^{-x}$ with respect to x (a) $-e^{-x} + x^5 + K$ (b) $e^{-x} + x^5 + K$ (c) $-e^{-x} - x^{-5} + K$ (d) $-e^{-x} + x^4 + K$
15. If X = {2, 3, 6, 7, 8} and Y = {6, 7, 10, 3, 17}, find $Y - (X \cap Y)$ (a) { } (b) {10, 17} (c) {2, 3, 6, 7, 8, 10, 17} (d) {6, 3, 7}
16. Find the angle the line $\frac{1}{\sqrt{3}}y - x = 0$ makes with positive y – axis (a) 30° (b) 60° (c) 0° (d) 45°
17. Find the value of k in the equation $\frac{5}{2\sqrt{2}} - \sqrt{8} = k\sqrt{2}$ (a) $\frac{4}{3}$ (b) $\frac{3}{4}$ (c) $-\frac{3}{4}$ (d) $-\frac{4}{3}$
18. Evaluate $\int_0^1 3^x log3dx$ (a) 3 (b) 4 (c) 1 (d) 2
19. The probability of an event A given by P(A) is a number between (a) -1 and 1 (b) 0 and ½ (c) 0 and 1 (d) -1 and 0
20. Noting that $\sin^2\theta + \cos^2\theta = 1$, simplify $\frac{1-cos\theta}{sin^2\theta}$ (a) $\frac{1}{1+cos\theta}$ (b) $\frac{1}{1-cos\theta}$ (c) $\frac{1}{1+sin\theta}$ (d) $\frac{1}{1-sin\theta}$

21. A circle has an eccentricity (a) < 1 (b) 1 (c) > 1 (d) 0
22. If two element A and B are independent then P(A and B) is (a) P ($A \cap B$ (b) P ($A \cup B$) (c) P(A) (d) P(B)

2015 OAU Post UTME: Mathematics Solution

1. $\frac{3^{n+3} - 3^{n+2}}{3^{n+1} - 3^n}$

Recall from the law of indices: $a^{x+y} = a^x \times a^y$. Therefore, this question can be written as

$\frac{(3^n \times 3^3) - (3^n \times 3^2)}{(3^n \times 3^1) - 3^n}$

Since 3^n is common to both numerator and the denominator, factorization can be made. Then we have:

$\frac{3^n(3^3 - 3^2)}{3^n(3^1 - 1)} => \frac{27-9}{3-1} = \frac{18}{2} = 9$ **The Correct Option is B**

Alternatively, we can make 3^n equal to a variable, say x.

Then,

$\frac{3^n \times 27 - 3^n \times 9}{3^n \times 3 - 3^n} = \frac{27x - 9x}{3x - x} = \frac{18x}{2x} = 9$ **The Correct Option is B**

2. $\sqrt{P} - \frac{6}{\sqrt{P}} = 1$

Multiplying both sides by $\sqrt{P}$, we have

$P - 6 = 1 \times \sqrt{P}$

$P - 6 = \sqrt{P}$

Squaring both sides

$(P-6)^2 = (\sqrt{P})^2$

$P^2 - 12P + 36 = P$

$P^2 - 13P + 26 = 0$

$P^2 - 4p - 9P + 36 = 0$

P(P – 4) – P(P – 4) = 0

(P – 9)(P – 4) = 0

P – 9 = 0 or P – 4 = 0

P = 9 or 4

The correct option is C

Please take note that the two roots of the resulting quadratic equation i.e. 9 and 4 are both in the options. It is therefore essential to test the one that satisfies the equation by simply substituting

$\sqrt{4} = \frac{6}{\sqrt{4}} \neq 1$ But $\sqrt{9} - \frac{6}{\sqrt{9}} = 1$

Therefore, the right value is 9 as already said.

3. $4x^2 + 4y^2 - 400 = 0$

Dividing both sides by 4, we have:

$x^2 + y^2 - 100 = 0$

$x^2 + y^2 = 100$

$x^2 + y^2 = r^2$

Therefore,

$r^2 = 100$, $r = \sqrt{100} = 10$

Area = $\pi r^2 = \pi(10^2) = 100\pi$

The correct option is D

4. Mean, $\bar{x} = \frac{x + y^{-1} + z^5}{3} = 6$

$x + y^{-1} + z^5 = 18$…………… (i)

for 10, y^{-1}, 12, x, z^5; the mean,

$\bar{x} = \frac{10 + y^{-1} + 12 + x + z^5}{5} = \frac{x + y^{-1} + z^5 + 22}{5}$ ..(ii)

Substituting equation (i) into equation (ii), we have:

$\bar{x} = \frac{18+22}{5} = \frac{40}{5} = 8$

The correct option is B

5. $\frac{2}{3}x + \frac{3}{2}y + 9 = 0$

$\frac{3}{2}y = -\frac{2}{3}x - 9$

Multiply both sides by $\frac{2}{3}$

$\frac{2}{3} \times \frac{3}{2}y = \frac{2}{3} \times -\frac{2}{3}x - \left(\frac{2}{3} \times 9\right)$

$\therefore y = -\frac{4}{9}x - 6$

Therefore, the intercept on the y – axis = -6

To find the intercept on the x – axis, put y = 0

$0 = \frac{-4x}{9} - 6$

$0 = \frac{-4x - 54}{9} - 54$

Then,

-4x – 54 = 0, -4x = 54, $x = \frac{54}{-4} = -13.5$

y + x intercept = -6 + (-13.5) = -6 – 13.5

= -19.5

The correct option is A

6. *h(x)* = 3 + 2*x*

f(x) = 1 – *x*

h(-f(x)) = *h*(-(1 – *x*))

= *h*(*x* – 1)

Since *h(x)* = 3 + 2*x*,

h(*x* – 1) = 3 + 2(*x* – 1)

= 3 + 2x – 2

= 1 + 2x

The correct option is B

7. $\cos\left(1 + \frac{1}{2}x\right) = \sin\left(\frac{3}{2}y\right)$

Cos α = Sin(90 – α), Also Sin α = Cos(90–α)

Therefore, $Sin\ \left(\frac{3}{2}y\right) = Cos\ \left(90 - \frac{3}{2}y\right)$

Taking this into the original equation, we have

$Cos\ \left(1 + \frac{1}{2}x\right) = Cos\ \left(90 - \frac{3}{2}y\right)$

Then,

$1 + \frac{1}{2}x = 90 - \frac{3}{2}y$

$\frac{1}{2}x + \frac{3}{2}y = 90 - 1$

$\frac{x}{2} + \frac{3y}{2} = 89$

$\frac{x+3y}{2} = 89$

$x + 3y = 178$

$3y = 178 - x, y = \frac{178-x}{3}$

The correct option is C

8. **IF** the correct question is: $\boldsymbol{x^{-1}} < 1$, then

$\frac{1}{x} < 1$

Are we to multiply both sides by x? No! it is not known whether x is negative or positive. If x is negative, you already known we are to reverse the sign upon multiplication. But if positive, we are to keep the sign. Since the squares of negative and positive numbers are always positive, we multiply both sides by x^2, then keep the sign as it is.

$x^2 . \frac{1}{x} < 1 . x^2$

$x < x^2$

$x - x^2 < 0$
$x(1-x) < -0$
Either
$x < 0$ or $1 - x < 0$
$x < 0$ or $1 < x$
i.e.
$x < 0$ or $x > 1$
The correct option is C
Assuming the question dictates that $\mathbf{x^{-1} < -1}$ as in the question.
We multiply both sides by x^2 as we did above and then continue in the usual way
$\frac{1}{x} < -1$
$x^2 . \frac{1}{x} < -1 . x^2$
$x < -x^2$
$x + x^2 < 0$
$x(1+x) < 0$
$x < 0$ or $1 + x < 0$
$x < 0$ or $x < -1$

9. NWAFOR has a total of 6 letters, so it can be permuted in 6! Ways
$6! = 6 \times 5 \times 4 \times 3 \times 2 = 720$ ways
The Correct option is C

10. $18 + 15x - 3x^2 = 0$
There are two ways to approach this problem; one is to find the roots of the equation and carry out the operation as demanded, the other is to employ the knowledge of the relationship between roots of a quadratic equation.
$18 + 15x - 3x^2 = 3x^2 - 15x + 18 = 0$
Then,
$3x^2 - 15x - 18 = 0$
Solving this gives x = -1 or 6
$3x^2 - 15x - 1\ 8 = 0$
$3x^2 + 3x - 18x - 18 = 0$
$3x(x + 1) - 18(x + 1) = 0$
$(3x - 18)(x + 1) = 0$
$3x - 18 = 0$ or $x + 1 = 0$
$3x = 18$ or $x = -1$
$x = \frac{18}{3} = 6$or - 1
Therefore, $\alpha\beta - \alpha - \beta = (-1)(6) - (-1) - 6$
$= -6 + 1 - 6$
$= - 11$
Let us use the relationships that exist between the roots.
In $18 + 15x - 3x^2$,
The coefficient of x^2, -3 = a, that of x = b while the constant, 18 = c
The sum of the two roots i.e. $\alpha + \beta = \frac{-b}{a} = \frac{-15}{-3} = 5$
The product of the two roots i.e. $\alpha\beta = \frac{c}{a} = \frac{18}{-3} = -6$
$A\beta - \alpha - \beta = \alpha\beta - (\alpha + \beta)$
$= -6 - 5 = - 11$ as before
The correct option is B

11. $\frac{1}{1 - x^2}$
$1 - x^2 = 1^2 - x^2$
Using difference of two squares:
$a^2 - b^2 = (a - b)(a + b)$
$1^2 - x^2 = (1 - x)(1 + x)$
Then,
$\frac{1}{1 - x^2} = \frac{1}{(1-x)(1+x)} \equiv \frac{A}{1-x} + \frac{B}{1+x}$
$= \frac{A(1+x) + B(1-x)}{(1-x)(1+x)}$
Therefore,
$A(1 + x) + B(1 - x) = 1$
To find A, one employs a value of x that will eliminate B; x = 1 does this
$A(1 + 1) + B(1 - 1) = 1$
$2A + 0 = 1, 2A = 1, A = \frac{1}{2}$
To find B, one uses a value of x that ensures the elimination of A; x = -1 is suitable for this.
$A(1 + (-1)) + B(1 - (-1)) = 1, A(1 - 1) + B(1 + 1) = 1$
$0 + 2B = 1$
$\therefore B = \frac{1}{2}$
$\therefore \frac{1}{x - x^2} = \frac{\frac{1}{2}}{1-x} + \frac{\frac{1}{2}}{1+x}$
$= \frac{1}{2(x+1)} + \frac{1}{2(1-x)}$
The correct option C

12. $S_\infty = \frac{a}{1-r}$ (given)
$a = 1, r = 1 \div \left(-\frac{2}{3}\right) = \frac{1 \times 3}{-2} = -\frac{3}{2}$
$\therefore S_\infty = \frac{1}{1-\left(-\frac{3}{2}\right)} = \frac{1}{1+\frac{3}{2}} = \frac{1}{\frac{5}{2}} = \frac{2}{5}$

13. $x^2 - 5x + 6$ (given)
$\frac{dy}{dx} = 2x - 5$
For a minimum value, $\frac{dy}{dx}$ must be zero
$2x - 5 = 0, 2x = 5$
$x = \frac{5}{2}$
The correct option is A

14. $\int 5x^4 + e^{-x} dx$
$\frac{5x^5}{5} - e^{-x} + K$
$= x^5 - e^{-x} + K$
The correct option is A

15. X = {2, 3, 6, 7, 8}
Y = {6, 7, 10, 3, 17}
$X \cap Y$ = {3, 6, 7}
$Y - (X \cap Y)$ represents elements in Y that are not present in $X \cap Y$
Therefore, $Y - (X \cap Y)$ = {10, 17}
The correct option is B

16. $\frac{1}{\sqrt{3}} y - x = 0$
$\frac{y}{\sqrt{3}} = x$
So, $y = x\sqrt{3} = \sqrt{3}(x)$ (From y = mx + c)
Hence, the gradient is $\sqrt{3}$
$m = \tan\theta$
$\theta = \tan^{-1} m = \tan^{-1}\left(\sqrt{3}\right)$
$\theta = 60^o$
Hence, the line makes an angle of 30^o with the positive y = axis
The correct option is A

17. $\frac{5}{2\sqrt{2}} - \sqrt{8} = K\sqrt{2}$
Multiply through by $2\sqrt{2}$

$5 - 2\sqrt{2} \times \sqrt{8} = K\sqrt{2} \times 2\sqrt{2}$
$5 - 2\sqrt{16} = 2K\sqrt{4}$
$5 - 8 = 4K$
$-3 = 4K,\ K = \frac{-3}{4}$
The correct option is C

18. $\int_0^1 3^x log3dx$
Integrating $3^x log3dx$ gives 3^x
Therefore,
$\int_0^1 3^x log3dx = [3^x]_0^1$
$= 3^1 - 3^0 = 3 - 1 = 2$
The correct option is D

19. The probability of an event occurring is always a value between 0 and 1
The correct option is C

20. $\frac{1-\cos\theta}{sin^2\theta}$
Since $\sin^2\theta + \cos^2\theta = 1$
$\sin^2\theta = 1 - \cos^2\theta$
Then,
$\frac{1-cos\theta}{sin^2\theta} = \frac{1-cos\theta}{1-cos^2\theta} = \frac{1}{1+cos\theta}$
The correct option is A

21. A circle has an eccentricity of 0.
The correct option is B

22. Two elements A and B are independent if the chance or probability of occurrence of A does not affect that of B, then P(A and B) = P(A ∩ B) = P(A) × P(B)
The Correct option is A

MATHEMATICS QUESTION 2014

1. Find the angle in degree which the line $x - \sqrt{3y} = 0$ makes with the positive y-axis A: 30 B: 90 C: 60 D: 180
2. If equation 6 – kx + $2x^2$ = 0 has equal roots, find k^2 + 4. A: 48 B8: 52 C: 44 D: 96
3. Simplify $log_{100}\sqrt{10^{-1}}$. A: $-\frac{1}{8}$ B: $-\frac{1}{4}$ C. $\frac{1}{4}$ D: $\frac{1}{8}$
4. Evaluate $\left(\frac{1}{25}\right)^{-\frac{1}{2}} + \left(\frac{1}{8}\right)^{-\frac{2}{3}}$ A: 8 B:10 C: 9 D: 6
5. Find the remainder when x^4 - 11x + 2 is divided by x A: 2 B: 6 C: -2 D: 5
6. If α and β are the roots of equation cx^2 + ax + b - 0, find αβ A: $-\frac{b}{a}$ B: $-\frac{a}{c}$C: $\frac{b}{c}$ D: $\frac{c}{a}$
7. The binary operation ⊗ is defined by a ⊗ b = 2a - l. Find 3 ⊗ (2 ⊗ 1) A: 3 B: 4 C: 5 D:6
8. Two coins are tossed; find the probability of having at least two heads A: $\frac{1}{2}$ B: $\frac{3}{4}$C: $\frac{1}{4}$ D: 1
9. Solve the equation $\frac{1}{1+\frac{1}{x^3}} = 0$A: 0 B: -1 C: +1 D: ∞
10. If x is a real number and x + 11 < 0 evaluate $\frac{|x|}{x}$A: 0 B: -1 C: 1 D: 2
11. Simplify: $\frac{30}{\sqrt{2}} + \sqrt{50}$A: $4\sqrt{5}$ B: $20\sqrt{2}$C: $5\sqrt{5}$ D: $10\sqrt{2}$
12. If m is the gradient of the line pq – px – qy = 0 and q ≠ 0, find $\frac{1}{m}$A: $\frac{q}{p}$ B: $\frac{p}{q}$C: $-\frac{q}{p}$ D: $-\frac{p}{q}$
13. If P is directly proportional to $\sqrt{Q}$; P = 20 when Q = 4. Find Q when P = 100 A: 200 B: 300 C: 100 D: 400
14. Obtain the centre of the circle $7(y^2 + 10y) + 7x^2 = 1$. A: (0, 5) B: (-5, 0) C: (0, -5) D: (5, 0)
15. Given $\int_{-a}^{a} 15x^2dx$ = 3430, find the value of the constant a A: 8 B: 6 C: 7 D: 9
16. Evaluate $\frac{d}{dx}$(InSin3x) A: 3cot3x B: 3tan3x C. $\frac{1}{sin3x}$ D: 3sin3x
17. Find the equation of a line which passes through a point (-2, 3) and makes an angle of 45° with positive x-axis A: y – x – 5 = 0 B: y + x + 5 = 0 C: x – y – 5 = 0 D: y – x + 5 = 0
18. If cosA = $\frac{12}{13}$ and A is an acute angle, find (1 + tan^2A) A: $\frac{144}{25}$B: $\frac{25}{144}$ C: $\frac{169}{25}$ D: $\frac{169}{144}$
19. Integrate the function 1 — 2x. A: x – x^2 + K B: x + x^2 + K C: -x – x^2 + K D: x - x^2 + K
20. Find the sum to infinity of the sequence 1, -1, 1, -1, 1, -1, ... A: 2 B: $-\frac{1}{2}$ C: 1 D: $\frac{1}{2}$
21. Differentiate 2 – sin(2 - ax) with respect to x A: acos(2 – ax) B: -asin(2 – ax) C: -acos(2 – ax) D: -asin(2 – ax)
22. Simplify $\left(\frac{8\sqrt{n}}{\frac{3}{m^{\frac{3}{2}}}}\right)^2 \left(\frac{4^{-1}m^2}{2n^{-2}}\right)$. A: $128n^3m^{-1}$ B: $8n^3m^{-1}$ C: $8n^3m$ D: $8n^4m$

MATHEMATICS 2014 SOLUTION

1. $x - \sqrt{3}y = 0$
$\sqrt{3}y = x + 0$
$y = \frac{1}{\sqrt{3}}x + 0$
Compare with y = mx + c
$m = \frac{1}{\sqrt{3}}$
$Tan\theta = m = \frac{1}{\sqrt{3}}$
$\therefore\ \theta = Tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$
$\theta = 30^o$
(A)

2. For 6 – kx + $2x^2$ to have equal roots, it means it is a perfect square
∴ $b^2 = 4ac$ holds
a = 2, b = -k, c = 6
$(-k)^2 = 4 \times 2 \times 6$
$k^2 = 48$
$k = \sqrt{48}$
∴ $k^2 + 4 = 48 + 4$
= 52
(B)

3. $log_{100}\sqrt{10^{-1}}$
$= log_{100}10^{-\frac{1}{2}}$
$= log_{10^2}10^{-\frac{1}{2}}$
$= \left(\frac{1}{2} \times -\frac{1}{2}\right) log_{10}10$
$= -\frac{1}{4}$
(B)

4. $\left(\frac{1}{25}\right)^{-\frac{1}{2}} + \left(\frac{1}{8}\right)^{-\frac{2}{3}}$
$= (25)^{-1 \times -\frac{1}{2}} + (8)^{-1 \times -\frac{1}{2}}$
$= 25^{\frac{1}{2}} + 8^{\frac{2}{3}}$
$= 5 + 2^{3 \times \frac{2}{3}}$
$= 5 + 2^2$
$= 9$
(C)

5. $x^4 - 11x + 2.$
The reminder when divided by x is known by equating x to zero
x = 0 and substitute
$0^4 - 11(0) + 2 = 2$
$\therefore$ Remainder = 2
(A)

6. $cx^2 + ax + b = 0$
$\alpha\beta = \frac{a}{c}$
Where c = b, a = c
$\therefore \alpha\beta = \frac{b}{c}$

7. a ⊗ b = 2a – 1
3 ⊗ (2 ⊗ 1)
= 3 ⊗ (2(2) – 1)
= 3 ⊗ 3
= 2 (3) – 1
= 6 – 1
= 5
(C)

8. The probability of having at least two heads is
P(HH) = $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$
(C)

9. $\frac{1}{1+\frac{1}{x^3}} = 0$
$\frac{1}{1} + \frac{1}{x^3}$
$\frac{x^3+1}{x^3}$
Then $1 \div \frac{x^3+1}{x^3} = 0$
$\frac{1}{1} \times \frac{x^3}{x^3+1} = 0$
$\frac{x^3+1}{x^3} = \frac{0}{1}$
$x^3 = 0$ then $x = 0$
(A)

10. x + 11 < 0
x < - 11
$\frac{1x1}{x} = \frac{\pm 11}{11}$
= -1
(B)

11. $\frac{30}{\sqrt{2}} + \sqrt{50}$
$= \frac{30+\sqrt{50 \times 2}}{\sqrt{2}}$
$= \frac{30+10}{\sqrt{2}}$
$= \frac{40}{\sqrt{2}}$
$= \frac{40 \times \sqrt{2}}{\sqrt{2} \times \sqrt{2}} = \frac{40\sqrt{2}}{2}$
$= 20\sqrt{2}$
(B)

12. pq – px – qy = 0
pq– px = qy
Divide both side by q
$y = \frac{pq}{q} - \frac{p}{q}x$
$y = p - \frac{p}{q}x$
$y = \frac{p}{q}x + p$
y = mx + c
$\therefore m = -\frac{p}{q}$
$\therefore \frac{1}{m} = -\frac{q}{p}$
(C)

13. $P \propto \sqrt{Q}$
$P = K\sqrt{Q}$
$10 = K\sqrt{4}$
$\therefore K = \frac{20}{2}$
K = 10
When P = 100
$100 = 10\sqrt{Q}$
$\sqrt{Q} = 10$
Q = 100
(C)

14. $7(y^2 + 10y) + 7x^2 = 1$
$7[(y + 5)^2 - 25] + 7(x - 0)^2 = 1$
$7[(y + 5)^2 + (x - 0)^2 - 25] = 1$
$(x - 0)^2 + (y + 5)^2 - 25 = \frac{1}{7}$
$(x - 0)^2 + (y + 5)^2 = \frac{1}{7} + 25$
$(x - 0)^2 + (y + 5)^2 = \frac{176}{7}$
Compare to the equation of circle
$(x - h)^2 + (y - k)^2 = \gamma^2$
Where h and k are the centre of the circle:
$\therefore$ centre of the circle is (0, -5)
(C)

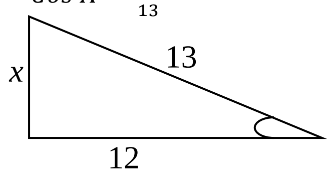
15. $\int_{-a}^{a} 15x^2 dx = 3430$
$\left[\frac{15x^3}{3}\right]_{-a}^{a} = 3430$
$[5x^3]_{-a}^{a} = 3430$
$5(a)^3 - 5(-a)^3 = 3430$
$5a^3 + 5a^3 = 3430$
$10a^3 = 3430$
$a^3 = 343$
$a = \sqrt[3]{343}$
a = 7
(C)

16. $\frac{d}{dx}(In\ Sin\ 3x)$
Let u = Sin 3x
$\frac{du}{dx} = 3\ Cos\ 3x$
$\frac{d}{du}(In\ u) = \frac{1}{u}$
$\therefore \frac{d}{dx}(In\ Sin\ 3x) = \frac{1}{u} \times Cos\ 3x$
$= \frac{3\ Cos\ 3x}{u}$
$= \frac{3\ Cos\ 3x}{Sin\ 3x}$
$= 3\ Cot\ 3x$
(A)

17. $\theta = 45^o$
$m = Tan\theta = Tan\ 45^o$
m = 1

at point (-2, 3)
$m = \frac{y-3}{x-(-2)}$
$1 = \frac{y-3}{x+2}$
$x + 2 = y - 3$
$y - x - 3 - 2 = 0$
$y - x - 5 = 0$
(A)

18. $Cos\ A = \frac{12}{13}$



$13^2 = x^2 + 12^2$
$x^2 = 13^2 - 12^2$
$x^2 = 25$
x = 5
$Tan\ A = \frac{5}{12}$
$\therefore$ 1 + $\tan^2$ A
$= 1 + \left(\frac{5}{12}\right)^2$
$= 1 + \frac{24}{144}$
$= \frac{144+25}{144}$
$= \frac{169}{144}$
(D)

19. $\int 1 - 2x$
$= x - \frac{2x^2}{2} + k$
$= x - x^2 + k$
(A)

20. 1, -1, 1, -1, 1, -1
$S_\infty = \frac{a}{1-r}$
$r = \frac{-1}{1} = -1, a = 1$
$S_\infty = \frac{1}{1-(-1)} = \frac{1}{2}$
$S_\infty = \frac{1}{2}$
(D)

21. In any ramification, the expression contains two terms which are to be treated separately.
The differential form of 2 is 0. Let us now differentiate the expression sin (2 – ax) and combine the values obtained.
Sin(2 – ax) is a composite function with (2 – ax) as the "core". The differential form of the core is (-a). The differential form of sin is Cos. Sin(2 – ax) can therefore be represented differentially as the product of Cos(2 – ax) and –a = -aCos(2 – ax). Let us now combine the form:
0 – (-acos(2 – ax) = a cos(2 – ax)
Option A
Method 2
2 – sin(2 – ax) contains two terms: "2" whose differential form is 0 and sin(2 – ax) which is to be carefully treated.
Let **y = sin(2 – ax)**, also let **u = 2 – ax**
Then **y = sinU**
$\frac{dy}{dx}\ cosu;\ \frac{du}{dx} = -a$
We can say that $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$
= cosux – a => -a cos u
But u = 2 – ax, therefore $\frac{dy}{dx} = -a\cos u = -a\cos(2-ax)$
Putting the two together, we have:
0 – (-acos(2 – ax) = **acos(2 – ax)**
The correct options is A

22. $\left[\frac{8\sqrt{n}}{m^{\frac{3}{2}}}\right]^2 \left[\frac{4^{-1}m^2}{2n^{-2}}\right]$
$= \frac{\left(8n^{\frac{1}{2}}\right)^2}{m^{\frac{3}{2}\times 2}} \times \frac{\left(\frac{1}{4}m^2\right)}{\left(\frac{2}{n^2}\right)}$
$= \frac{64n}{m^3} \times \frac{m^2n^2}{4\times 2}$
$= \frac{64n \times m^2 \times n^2}{8m^3}$
$= \frac{8n^2}{8m}$
$= 8n^3m^{-1}$
(B)

MATHEMATICS 2013 QUESTION

1. If the probability of success in an even is $\frac{y}{x}$. What is the possibility of failure? A $\frac{x-y}{x}$ B. $\frac{y-x}{x}$ C. $\frac{x-y}{y}$ D.$\frac{y-x}{y}$
2. What is the circumference of the circle $x^2 + y^2 = \left(\frac{7}{\pi}\right)^2$? A. 16units B. 14 units C. 15 unit D. 15 units
3. Resolve $\frac{1}{x(1+x)}$ into partial fractions. A. $\frac{1}{x} + \frac{1}{1+x}$ B. $\frac{1}{1+x} - \frac{1}{x}$ C. $\frac{-1}{x} - \frac{1}{1+x}$ D. $\frac{1}{x} - \frac{1}{1+x}$
4. Solve the equation $5^{x^2} = 25^{x+4}$ A. -4, 2 B. -4, -2 C. 4, - 2 D. 4, 2
5. Evaluate $\sum_{n=1}^{4}(2^n + 1)$ A. 28 B. 31 C. 29 D. 32
6. Find point of intersection of the lines 3x – 2y = 5, 2x + 5y = -7 A. $x = \frac{11}{19}, y = -\frac{31}{19}$ B. $x = -\frac{11}{19}, y = \frac{31}{19}$ C. $x = -\frac{11}{19}, y = -\frac{31}{19}$ D. $x = \frac{11}{19}, y = \frac{31}{19}$
7. Solve $4x^2 + 20x - 24 = 0$ A. 1, 6 B. -1, -6 C. 6, -1 D. -6, 1
8. What is the 15th term of the sequence -3, 2, 7,...? A. 65 B. 66 C. 68 D. 67
9. Evaluate $\frac{log\sqrt{27} - log\sqrt{8}}{log3 - log2}$ A. $\frac{2}{3}$ B. $-\frac{2}{3}$ C. $\frac{3}{2}$ D. $-\frac{3}{2}$
10. Integrate $4x^3 + \frac{1}{x}$ with respect to x. A. In x + x^4 + K B. $x^{-1} + x^4$ + K C. $12x^2 - x^{-2}$ + K D. $\frac{1}{5}x^5 + x^{-2}$ + K
11. Find the diameter of the circle $2x^2 + 2y^2 - 50 = 0$. A. -10units B. 10 units C. 25 units D. -25units
12. What is the coordinate of centre of the circle $x^2 + y^2 + 2x - 4y = 10$? A. (-1, -2) B. (1, 2) C. (-1, 2) D. (1, -2)

13. Simplify $log_x x^4 + log_x 4^x$ A. 4x B. $\frac{4}{x}$ C. 4 + x D. $4xlog_{4x}^{4x}$

14. Solve the equation $3^{x+1} = 27^{1-x}$ A. $\frac{1}{2}$ B. $-\frac{1}{2}$ C. $\frac{3}{4}$ D. $-\frac{3}{4}$

15. Given $f(x) = 3 + x$ and $g(x) = 3 - x$, find $g(f((x))) = 3 - f(x)$
A. 6 B. x C. -x D. 0

16. What is the reminder when $x^3 + 5x^2 - 6x + 1$ is divided by $x - 1$? A. -1 B. 2 C. -2 D. 1

17. If α, β are the roots of equation $6 + 5x - x^2 = 0$, find αβ + α + β. A. 11 B. -11 C. 1 D. -1

18. What is the value of y for which the function $\frac{y-1}{y+1}$ is undefined? A. -1 B. 1 C. 0 D. 2

19. What is the distance between the points (-1, 5) and (-7, -3)? A. 9 B. 10 C. 11 D. 12

20. If X = {2, 3, 6, 7, 8}, and Y = {6, 7, 10, 3, 17}, find X ∩Y A. { } B. 3, 6, 7 C. {2, 3, 6, 7, 8, 10, 17} D. {6, 3, 7)

21. Differentiate sin(2x – 5) with respect to x. A. cos(2x – 5) B. –cos(2x – 5) C. 2cos(2x – 5) D. -2cos(2x – 5)

22. If δ, λ are the roots of equation $x^2 - 5x + 7 = 0$, find the value of $\delta^2 + \lambda^2$ A. 25 B. -25 C. -11 D. 11

MATHEMATICS SOLUTION 2013

1. Pr (success) $= \frac{y}{x}$
Pr(failure) $= 1 - \frac{y}{1}$
$= \frac{x-y}{x}$ **"C"**

2. Circumference = 2πr
$x^2 + y^2 = \left(\frac{7}{\pi}\right)^2$
from $x^2 + y^2 + 2gx + 2fy + c = 0$
$(x - a)^2 + (y - b)^2 = r^2$
$(x - 0)^2 + (y - 0)^2 = r^2$
$(x - 0)^2 + (y - 0)^2 = \left(\frac{7}{\pi}\right)^2$
$r = \frac{7}{\pi}$
Circumference $= 2\pi \times \frac{7}{\pi}$
= 14unit **"B"**

3. $\frac{1}{x(1+x)} = \frac{A}{x} + \frac{B}{1+x}$
$= \frac{A(1+x)+Bx}{x(1+x)}$
Where x = 0
1 = A(1 + 0) + 0
A = 1
When x = 1
1 = A(1 – 1) + B(- 1)
1 = -B
$= \frac{1}{x} - \frac{1}{1+x}$ **"D"**

4. $5^{x2} = 25^{x+4}$
$5^{x2} = 5^{2(x+4)}$
$x^2 = 2x + 8$
$x^2 - 2x - 8 = 0$
$x^2 - 4x + 2x - 8 = 0$
$x(x - 4) + 2(x - 4) = 0$
$(x - 4)(x + 2) = 0$
x = 4 or – 2 **"C"**

5. $\sum_{n=1}^{4}(2^n + 1)$
This means sum up by making n from 2 to 4
$2^2 + 1 + 2^3 + 1 + 2^4 + 1 = 5 + 9 + 17 = 31$
"D"

6. 3x – 2y = 5 ....(i)
2x + 5y = -7......(ii)
Equ (i) ×2: 6x – 4y = 10
Equ (ii) ×3: 6x + 15y = -21
19y = -31
$y = \frac{-31}{19}$
from equ (ii)
2x + 5y = -7
$2x + 5\left(\frac{-31}{19}\right) = -7$
$2x = -7 + \frac{155}{19}$
$2x = \frac{-133+155}{19}$
$2x = \frac{22}{19}$
$x = \frac{22}{19} \times \frac{1}{2} = \frac{11}{19}$
:. $x = \frac{11}{19}$, $y = \frac{-31}{19}$ **"C"**

7. $4x^2 + 20x - 24 = 0$
Divide through by 4
$x^2 + 5x - 6 = 0$
$x^2 + 6x - x - 6 = 0$
x(x + 6) – 1(x + 6) = 0
(x + 6) (x – 1) = 0
:. x = -6 or 1 **"D"**

8. -3, 2, 7
d = 2 – (- 3) = 5
$T_5 = a + (n - 1)d$
= -3 + (5 – 1)5
= -3 + 14(5) = -3 + 70 = 67 **"D"**

9. $\frac{log\sqrt{27} - log\sqrt{8}}{log3 - log2}$
$\frac{log\sqrt{\frac{27}{8}}}{log3}$
$\frac{log\sqrt{\frac{2^3}{2^3}}}{log\frac{3}{2}}$
$\frac{log\frac{3^{\frac{3}{2}}}{2^{\frac{3}{2}}}}{log\frac{3}{2}} = \frac{\log\left(\frac{3}{2}\right)^{\frac{3}{2}}}{log\frac{3}{2}} = \frac{3}{2}\frac{log\frac{3}{2}}{log\frac{3}{2}}$
$= \frac{3}{2}$ **"C"**

10. $\int 4x^3 + \frac{1}{x}\, dx$
$\int 4x^3 dx + \int \frac{1}{x}\, dx$
$\frac{4x^{3+1}}{3+1} + inx + k$
$\frac{4x^4}{4} + inx + k$
$x^4 + inx + k$ **"A"**

11. $2x^2 + 2y^2 - 50 = 0$
Divide throughout by 2
$x^2 + y^2 - 25 = 0$
$(x - 0)^2 + (y - 0)^2 = 5^2$
r = 5 units :. D = 2r = 2(5) = 10units
"B"

12. $x^2 + y^2 + 2x - 4y = 10$
Using completing square method
$x^2 + 2x + y^2 - 4y = 10$
$x^2 + 2x + 1 - 1 + y^2 - 4y + 4 = 10 + 1 + 4$
$(x + 1)^2 + (y - 2)^2 = 15$
$(x + 1)^2 + (y - 2)^2 = (\sqrt{15})^2$
$(x - a)^2 + (y - b)^2 = r^2$
:. a = -1 b = 2 **"C"**

13. $log_x x^4 + log_x 4^x$
$= 4log_x x + x10_4 4$
= 4 × 1 + x × 1
= 4 + x **"C"**

14. $3^{x+1} = 27^{1-x}$
$3^{x+1} = 3^{3(1-x)}$
$3^{x+1} + 3^{3-3x}$
x + 1= 3 – 1
4x = 2
$x = \frac{2}{4} = \frac{1}{2}$ **"A"**

15. $f(x)$ = 3 + x and $g(x)$ = 3 – x
$g\big(f((x)\big) = 3 - f(x)$
= 3 – 3 – x
= -x **"C"**

16. $x^3 + 5x^2 - 6x + 1$ by x – 1
x – 1 = 0; x = 1
Rem = $1^3 + 5(1)^2 - 6(1) + 1$
= 1 + 5 – 6 + 1
= 1 **"D"**

17. $6 + 5x - x^2 = 0$
$x^2 - 5x - 6 = 0$
a = 1, b = 5, c = -6
$\alpha\beta = \frac{c}{a}$ $\alpha + \beta = \frac{-b}{a}$
$\alpha\beta = \frac{-6}{1}$ $\alpha + \beta = \frac{-5}{1}$
αβ = - 6 α + β = -5
αβ + α + β = -6 – 5 = -11 **"B"**

18. $\frac{y-1}{y+1}$
y + 1 = 0
y = -1 **"A"**

19. $D = \sqrt{(y_2 - y_1)^2 + (x_2 - x_1)^2}$
(-1, 5) and (-7, 3)
$D = \sqrt{(-3 - 5)^2 + (-7 - (-1))^2}$
$= \sqrt{(-8)^2 + (-6)^2}$
$= \sqrt{64 + 36}$
$= \sqrt{100} = 10$ **"B"**

20. X = {2, 3, 6, 7, 8}
Y = {6, 7, 10, 3, 17}
X∩Y = {3, 6, 7} = {6, 3, 7} **"D"**
Option B has no bracket, so its automatically wrong

21. Differentiate
Sin(2x – 5)
$= \frac{d(2x-5)}{dx} = 2$
:. Sin(2x – 5)
= 2cos(2x – 5) **"C"**

22. $x^2 - 5x + 7 = 0$
a = +1, b = -5, c = +7
$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
$\alpha + \beta = \frac{-b}{a} = \frac{-5}{1} = -5$
$\alpha\beta = \frac{c}{a} = \frac{7}{1} = 7$
$\alpha^2 + \beta^2 = (-5)^2 - 2(7)$
= 11 **"D"**

MATHEMATICS 2012 QUESTIONS

1. What is the highest possible value of $\frac{8}{1+x^2}$ if $0 \le x \le 3$? (a) 8 (b) 4 (c) 2 (d) 16
2. The fifth term in the progression 9, 27, 81, . . . is (a) 243 (b) 3^7(c) 729 (d) 3^8
3. The interior angles of an hexagon are 120^0, 100^0, 80^0, 150^0, x and 130^0. The value of x is (a) 170^0 (b) 20^0 (c) 120^0 (d) 140^0
4. Obtain the product of 1100_2 and 101_2 (a) 111100_2 (b) 110100_2 (c) 2220_2 (d) 1144_7
5. Simplify$\left(\frac{8\sqrt{n}}{m^{\frac{3}{2}}}\right)^2 \left(\frac{4^{-1}m^2}{2n^{-2}}\right)$ (a) $128n^3m^{-1}$ (b) $8n^3m^{-1}$ (c) $8n^4m$ (d) $8n^3m$

The universal set U consists of all integers. Subsets of U are defined as:
A = {y : y ≤ 3}
B = {y : 5 < y < 12}
C = {y : -2 ≤ y < 5}
Use the information above to answer question

6. A ∩ (B ∪ C)' is (a) {y < - 4} (b) ∅ (c) {y < 0} (d) {-4 ≤ y ≤ 3}
7. Make k the subject of the formula $m = \frac{2nk}{p} + \frac{k}{2p}$ (a) $k = \frac{2mp}{2n+1}$ (b) $k = \frac{2mp}{4n+1}$ (c) $k = \frac{mp}{2n+1}$ (d) $k = \frac{2n+1}{2mp}$
8. Evaluate $\log_8 128 + \log_3 9$ (a) 19 (b) 48 (c) $\frac{13}{3}$(d) 6
9. Find the value of y if $\frac{1}{2}\log_3 y = 2$ (a) 9 (b) 18 (c) $\frac{9}{2}$(d) 81
10. Which of the following is a perfect square? (a) $x^2 - 3x - 4$ (b) $x^2 + 9x + 9$(c) $2x^2 + 2x + 2$(d) $x^2 + 2x + 1$
11. Integrate $\sqrt{2x+1}$ (a) $\frac{1}{3}(2x+1)^{\frac{3}{2}} + K$ (b) $\frac{1}{3}(2x+1)^{-\frac{3}{2}} + K$ (c) $-\frac{1}{3}(2x+1)^{\frac{3}{2}} + K$ (d) $-\frac{1}{3}(2x+1)^{-\frac{3}{2}} + K$
12. Obtain the centre of the circle $3y^2 + 3(x+5)^2 = 17$ (a) 0,5) (b) (-5,0) (c) (0,-5) (d) 5,0)
13. If $f(x+1) = \frac{x^2+1}{x^3}$, find $f(2)$ (a) $\frac{5}{8}$ (b) 2 (c) $\frac{1}{4}$ (d) 1
14. The quadratic equation whose roots are $(x - 3)$ and $\left(x + \frac{1}{3}\right)$ is (a) $x^2 + \frac{8}{3}x - 1 = 0$ (b) $x^2 - 2x - 3 = 0$ (c) $x^2 - \frac{8}{3}x - 1 = 0$ (d) $x^2 - 9 = 0$
15. If the bearing of a town B from A is 145^0, the bearing of A from B is (a) 305^0 (b) 325^0 (c) 35^0 (d) 145^0

16. A number is selected randomly from the set of integers 1 to 30 inclusive. The probability that he number is prime is (a) $\frac{4}{15}$ (b) $\frac{1}{3}$ (c) $\frac{3}{15}$ (d) $\frac{7}{30}$

17. Differentiate $cos\, ax$ with respect to x (a) $a\, sin\, ax$ (b) $\frac{1}{a} sin\, ax$ (c) $-a\, sin\, ax$ (d) $-\frac{1}{a} sin\, ax$

18. Obtain the values of x in $|x - 9| = 16$ (a) 25 (b) -7 (c) 25, -7 (d) -25, 7

19. Which of these numbers is an irrational number (a) sin 0^0 (b) sin 30^0 (c) sin 60^0 (d) sin 90^0

20. Give $\int_{-a}^{a} 15x^2 dx = 3430$, find the value constant a (a) 8 (b) 6 (c) 7 (d) 9

21. Which of these lines is at right angle with the line $x = -7$ (a) $2x + y = -1$ (b) $2x - y = 1$ (c) $y = 0$ (d) $x = 49$

The scores of students in a class test are shown in the table below: Use the information to answer Question 13.

Scores	1	2	3	4	5	6	7	8
No. of Students	0	1	3	5	3	4	2	0

22. The modal score is (a) 5 (b) 4 (c) 6 (d) 8

MATHEMATICS 2012 SOLUTION

1. $\frac{8}{1+x^2}, 0 \le x \le 3$

When estimating a fraction, it will be connected that the bigger the numerator, the bigger the fraction.

Conversely, the bigger the denominator, the smaller the fraction.

To get the highest possible value for $\frac{8}{1+x^2}$, the denominator must be the smallest possible; this value occurs if x = 0

Then, the expression $\frac{8}{1+x^2} = \frac{8}{1+0^2} = \frac{8}{1} = 8$

But wait! It seems we are obsessed with making the dominator as small as possible; **why can't we employ negative numbers** (Since they are smaller than 0)?

No! Below are the reasons:

(i) The range states $0 \le x \le 3$, which implies that x must not be less than zero and must not be greater than 3.

(ii) Even if we are to disregard the range given, a negative number is surely make the denominator larger. **But how?**

Recall that the squares of all number (both positive and negative) are positive. If I choose a negative number, say – 2, the fraction becomes $\frac{8}{1+(-2)^2} = \frac{8}{1+4} = \frac{8}{5} = 1.6$

Hence, the right choice of x = 0, which makes the fraction equal 8.

The correct option is A

2. 9, 27, 81, . . .

This is a GP whose first term, a = 9 and common ratio, $r = \frac{27}{9}$ or $\frac{81}{27} = 3$

The nth term = ar^{n-1}

The fifth term = $9(3^{5-1}) = 9(3^4) = 9(81) = 729$

The correct option is C

3. The sum of interior angles of a polygon is given by (n – 2)180, where n is the number of sides.

For hexagon, the sum is (6 – 2)180° = 720°

This means the sum of the stated angles must be 720°

120° + 100° + 80° + 150° + x + 130° = 720°

580° + x = 720°

x = 720° – 580° = 140°

The correct option is D

4. $1100_2 \times 101_2$

Method 1

This entails doing the multiplication in the usual ways, maintaining the base

```
      1 1 0 0
    × 1 0 1
    -------
      1 1 0 0
    0 0 0 0
  1 1 0 0
  -----------
  1 1 1 1 0 0
  -----------
```

The correct option is A

Method 2

This method is particularly appreciated when division is involved. With this, you can be rescued from the horrific jaws of long division.

The very first step is to convert the numbers to base ten

$1100_2 = 1^3\, 1^2\, 0^1\, 0^0{}_2 = 1 \times 2^3 + 1 \times 2^2 + 0 + 0$

$101_2 = 1^2\, 0^1\, 1^0{}_2 = 1\times2^2 + 0 + 1\times2^0$

$= 5_{ten}$

The next step is to carry out the operation as demanded. For division, divide the numbers in base ten. Here, we are to multiply: $12_{ten} \times 5_{ten} = 60_{ten}$.

Since our answer is to be left in base two, the last step is to convert the number (the result) to base two

2	60
2	30 r 0
2	15 r 0
2	7 r 1
2	3 r 1
2	1 r 1
	0 r 1

Thus, $60_{ten} = 11100_2$

The correct option is A as before

5. $\left(\frac{8\sqrt{n}}{m^{\frac{3}{2}}}\right)^2 \left(\frac{4^{-1}m^2}{2n^{-2}}\right)$

$= \left(\frac{8n^{\frac{1}{2}}}{m^{\frac{3}{2}}}\right)^2 \left(\frac{4^{-1}m^2}{2n^{-2}}\right)$

$= \frac{8^2 n}{m^3} \times \left(\frac{4^{-1}m^2}{2n^{-2}}\right)$

$= \frac{64n}{m^3} \times \frac{\frac{1}{4}m^2}{2n^{-2}}$

This can be written as

$\left(\frac{64 \times \frac{1}{4}}{2}\right)\left(\frac{nm^2}{m^3 n^{-2}}\right)$

$= (8)(n^{1-(-2)} \times m^{2-3})$

$= (8)(n^3 m^{-1}) = 8n^3 m^{-1}$

The correct option is B

6. A = {y : y ≤ 3} = {. . . -4, -3, -2, -1, 0, 1, 2, 3}
B = {y : 5 < y < 12} = {6, 7, 8, 9. 10, 11}
C = {y : -2 ≤ y < 5} = {-2, -1, 0, 1, 2, 3, 4}
B∪C = {-2, -1, 0, 1, 2, 3, 4, 6, 7, 8, 9, 10, 11}
(B∪C)′ = {… -4, -3, 5, 12, 13, 14…}
A∩(B∪C)′ = { = {… -4, -3}
The correct option is D

7. $m = \frac{2nK}{P} + \frac{K}{2P}$
Then,
$m = \frac{2(2nK)+K}{2P} = \frac{4nK+K}{2P}$
Thus,
$4nK + K = 2mP$
$K(4n+1) = 2mP$
$K = \frac{2mP}{4n+1}$
The correct option is B

8. $Log_8 128 + log_3 9$
Let us solve each term separately and add the results
$log_8 128 = log_{2^3} 2^7 = \frac{7}{3} log_2 2 = \frac{7}{3} \times 1 = \frac{7}{3}$
$log_3 9 = log_3 3^2 = 2log_3 3 = 2 \times 1^3 = 2$
Thus,
$log_8 128 + log_3 9 = \frac{7}{3} + 2$
$= \frac{7}{3} + \frac{2}{1} = \frac{7+6}{3} = \frac{13}{3}$
The correct option is C

9. $\frac{1}{2} log_3 y = 2$
$\boldsymbol{Plog_x a = log_x a^P}$
Then, we can write
$log_3 y^{\frac{1}{2}} = 2$
If $\boldsymbol{log_x a = m, a = x^m}$
Then, we can write:
$y^{\frac{1}{2}} = 3^2$
By squaring both sides, we can write
$\left(y^{\frac{1}{2}}\right)^2 = (3^2)^2$
$y = 3^4 = 81$
The correct option is D

10. For a perfect square, $b^2 = 4ac$
Recall that a is the coefficient of x^2, b is that of x while c is the constant.
Now, let us check the expression whose b^2 equals 4ac
(i) $x^2 - 3x - 4$
$b^2 = (-3)^2 = 9$, $4ac = 4 \times 1 \times -4 = -6$
Hence, $x^2 - 3x - 4$ is not a perfect square
(ii) $x^2 + 9x + 9$
$b^2 = 9^2 = 81$, $4ac = 4 \times 1 \times 9 = 36$
Thus, x^2 9x + 9 is not a perfect square
(iii) $2x^2 + 2x + 2$
$B^2 = 2^2 = 4$, $4ac = 4 \times 2 \times 2 = 16$
Therefore, it is not a perfect square
(iv) $x^2 + 2x + 1$
$b^2 = 2^2 = 4$, $4ac = 4 \times 1 \times 1 = 4$
Then, $x^2 + 2x + 1$ is a perfect square
The correct option is D

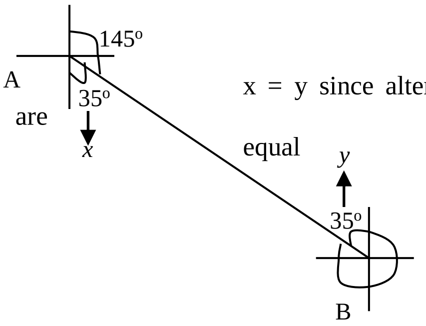
11. $\sqrt{2x+1} = (2x+1)^{\frac{1}{2}}$
We need $\int (2x+1)^{\frac{1}{2}} dx$
Let u = 2x + 1
$\frac{du}{dx} = 2$
$dx = \frac{du}{2}$
Then, $\int (2x+1)^{\frac{1}{2}} dx = \int u^{\frac{1}{2}} . \frac{du}{2}$
$= \frac{1}{2} \int u^{\frac{1}{2}} . du$
$= \frac{1}{2} \times \frac{u^{\frac{1}{2}-1}}{\frac{1}{2}+1} + K$
$= \frac{1}{2} \times \frac{u^{\frac{3}{2}}}{\frac{3}{2}} + K$
$= \left(\frac{1}{2} \div \frac{3}{2}\right) u^{\frac{3}{2}} + K$
$= \left(\frac{1}{2} \times \frac{2}{3}\right) u^{\frac{3}{2}} + K$
$= \frac{1}{3} u^{\frac{3}{2}} + K$ put U = 2x+1 to have:
$\frac{1}{3}(2x+1)^{\frac{3}{2}} + K$
The correct option is A

12. For a circle with p and q as centre, the equation of the circle is:
$x^2 + y^2 - 2p_x - 2qy = c$
All we need to do is to express the given equation in the form above. We can then figure out the values of p and q
Given: $3y^2 + 3(x+5)^2 = 17$
$3y^2 + 3(x^2 + 10x + 25) = 17$
$3y^2 + 3x^2 + 30x + 75 = 17$
$3y^2 + 3x^2 + 30x = 17 - 75$
$3y^2 + 3x^2 + 30x = -58$
Dividing through by 3, we have $y^2 + x^2 + 10x = -\frac{58}{3}$
As can be seen, this new expression does not contain a term that can match 2qy, in the standard equation.
This is rectified by adding 0y to the expression, knowing fully well that this does not affect its value in any way.
We have: $x^2 + y^2 + 10x - 0y = -\frac{58}{3}$
Good! Let us now compare this with the standard equation.
$10x = -2Px, P = \frac{10x}{-2x} = -5$
$-0y = -2qy, q = \frac{-0y}{-2y} = 0$
Thus, (p, q) = (-5, 0)
The correct option is B

13. $f(x+1) = \frac{x^2+1}{x^3}$
To find $f(2), x+1 = 2, x = 1$
Thus, $\frac{x^2+1}{x^3} = \frac{1^2+1}{1^3} = \frac{2}{1} = 2$
The correct option is B

14. The roots are $(x - 3)$ and $\left(x + \frac{1}{3}\right)$
To get the equation, the two modified roots are to be multiplied
$(x-3)\left(x + \frac{1}{3}\right) = 0$
$x^2 + \frac{x}{3} - 3x - 1 = 0$
$x^2 + \frac{x - 9x}{3} - 1 = 0$

$x^2 - \frac{8x}{3} - 1 = 0$
The correct option is C

15. 

x = y since alternate angle are equal

The bearing of A from B = 360° – 35° = 325°
The correct option is B

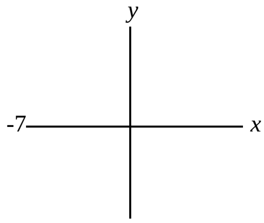
16. The prime numbers between 1 – 30 are 2, 3, 5, 7, 11, 13, 17, 19, 23 and 29
This gives a total of 10 numbers
Thus, P(prime) = $\frac{10}{30} = \frac{1}{3}$
The correct option is B

17. y = cosax, Needed: $\frac{dy}{dx}$
Let u = ax
Then, y = cosu
Differentiate the two expression
$\frac{du}{dx} = a$ and $\frac{dy}{du} = -sinu$
$\frac{dy}{dx} = \frac{du}{dx} \times \frac{dy}{du} = ax - sin\ U = -asin\ U$
$-$ a sin $a \times -$sin U
The correct option is C

18. $|x - 9| = 16$
It is highly important to note that x – 9 = 16 is not the same as |x – 9| = 16
The latter implies that the positive value of the expression on the L.H.S gives 16.
This means that x – 9 = 16 or – 16
If x – 9 = 16
x = 16 + 9 = 25
if x – 9 = -16
x = -16 + 9 = -7
The two values are 25 and -7
The correct option is C

19. Irrational numbers are those that can not be expressed as a ratio
Sin 60° = 0.866025403 cannot be expressed as a ratio. Unlike values like sin90° (which equals 1) and sin30° (which equal $\frac{1}{2}$).
The correct option is

20. $\int_{-a}^{a} 15x^2\, dx = 3430$
Then,
$\left[\frac{15x^3}{3}\right]_{-a}^{a} = 3430$
Normally, the value of an integral expression is obtained by subtracting the value obtained using the lower limit from the one calculated using the upper limit. i.e.
$3430 = \frac{15a^3}{3} - \frac{15(-a)^3}{3}$
$= \frac{15a^3}{3} + \frac{15a^3}{3}$
$= \frac{30a^3}{3} = 10a^3$
Then,
$10a^3 = 3430$
$a^3 = \frac{3430}{10} = 343$
Thus,
$a^3 = 7^3, a = 7$
The correct option is C

21. 

Therefore, x = -7 is perpendicular to y = 0
The correct option is C

22. The mode is the value with the highest frequency or occurrence. Hence, the modal score is 4
The correct option is B

MATHEMATICS 2011 QUESTIONS

1. If the universal set U = {1,2,3,4,5,6,7,8,9,10}, M = (1,3,5,7,9) and N = (2,4,6,8,10) which of the following is equal to (M U N)'? A. (M ∩ N)' , B. M'UN', C. M'∩ N', D. M ∩ N.
2. Cos (180 - θ) is equivalent to A. cos (θ - 180), B. cosθ, C. –cosθ, D. –cos (180 + θ).
3. Find the equation of the circle with centre (-1, 3) and radius 4. A. $x^2 + y^2 - 6x + 2y = 6$, B. $X^2 + y^2 + 2x - 6y = 16$, C. $X^2 + y^2 - 6x + 2y = 16$, D. $X^2 + y^2 + 2x - 6y = 6$
4. Find $\frac{dy}{dx}$, if $y = \frac{3}{\sqrt{x}}$ A. $-\frac{3}{2}x^{-\frac{3}{2}}$ B. $3x^{\frac{-3}{2}}$ C. $\frac{3}{2}x^{\frac{-3}{2}}$ D. $\frac{3}{4}x^{\frac{-3}{2}}$
5. Integrate $\frac{1}{2x}$ A. Not defined B. 0, C. $\frac{1}{2}Inx + C$ D. $\frac{1}{4}x^2 + C$
6. A die is tossed twice. What is the probability of obtaining a total of 6 if both numbers are odd? A. $\frac{1}{12}$ B. $\frac{1}{18}$ C. $\frac{5}{36}$ D. $\frac{1}{6}$
7. If the mean of the numbers a, b, c, d, e is x, find the mean of numbers $a + k, b + 2k, c - k, d - 2k, e$. A. x B. $x + k$ C. $x - k$ D. $2x$
8. Factroize: $a^2 - b^2 + (a + b)^2$. A. $2a^2$ B. 2a (a – b) C. 2a (a + b) D. 2b (b– a)
9. Let α and β be roots of quadratic equation $x^2 + 2x - 3 = 0$, then αβ is A. -3 B. -2 C. 2 D. 6
10. Covert 69_{10} to a number in base two A. 1001101 B. 10100001 C.1000101 D. 100101
11. The reciprocal of $\frac{\frac{3}{4}}{\frac{1}{4}+\frac{1}{3}}$ is A. $1\frac{2}{7}$ B. $\frac{7}{9}$ C. $-1\frac{2}{7}$ D. $-\frac{7}{9}$
12. The speed of 30 kilometers per minute, expressed in centimeters per second is A. 5 B. 50 C. 500 D. 5000

13. Evaluate x if, $\log_4$ (x + 3) (x- 3) = 2 A. 3 or -3 B. 5 or -5 C. 5 or -3D. 3 or -5

14. Given that $a = \frac{1}{2-\sqrt{3}}$, $b = \frac{1}{2+\sqrt{3}}$,find the value of$a^2 + b^2$A. $\frac{14}{37}$ B.7 C. 14 $+ 2\sqrt{3}$ D. 14

15. If the binary operation * is defined as x * y =2, find 2 * (4 * 5) A. 4 B. 5 C. -5 D. 2

16. Find the value of $\sqrt{6+\sqrt{6+\sqrt{6+\sqrt{6+\sqrt{6}}}}}$A. -2 B. 2 C.6 D. 3

17. Find the value of$\int_0^{\frac{\pi}{2}}(2\pi + 2cos2x)dx$A. π^2 + 1 B. π^2 C. π^2 - 4 D. π^2 + 3

18. The circle $2x^2 + 2\left(y - \frac{3}{2}\right)^2 = 2$has centre and radius respectively as A. $\left(0,\frac{3}{2}\right)$ and 2 B. $\left(0,\frac{-3}{2}\right)$and 1 C. $\left(\frac{3}{2},0\right)$ and 2 D. $\left(0,\frac{3}{2}\right)$and 1

19. The line perpendicular to the straight line $y + \frac{3}{2}x - 1 = 0$ has the gradient (a) $-\frac{2}{3}$ (b) $\frac{3}{2}$ (c) 3 (d) $\frac{2}{3}$

20. Find x if $2^{x2} = 4^{(x+4)}$ (a) -2 or 4 (b) 4-2 or 2 (c) -4 or 4 (d) -4 or 2

21. Express in partial fraction$\frac{3x}{x^2-1} \equiv \frac{A}{x-1} + \frac{B}{x+1}$. Then A and B respectively is (a) -3, 3 (b) $\frac{2}{3},\frac{2}{3}$ (c) $-\frac{3}{2}, -\frac{3}{2}$(d) $\frac{3}{2},\frac{3}{2}$

22. A square has a perimeter of 40cm. What is its area in cm square?(a) 80 (b) 1600 (c) 100 (d) 160

MATHEMATICS 2011 SOLUTION

1. U = {1,2,3,4,5,6,7,8,9,10},
M = {1,3,5,7,9}
N = {2,4,6,8,10}
M∪N = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}
(M∪N)′ = { } or ϕ (Since M∪N contains all the elements in the universal set)
Also,
M′ = {2, 4, 6, 8, 10}, N′ = {1, 3, 5, 7, 9}
M′∩N′ = { } or ϕ (Since M′ and N′ have nothing in common)
Thus,
(M∪N)′ = M′∩N′
The correct option is C

2. Using the knowledge of compound angles,
cos(A + B) = cosAcosB – sinAsinB
cos(A – B) = cosAcosB + sinAsinB
Then,
Cos(180 - θ) = cos180 × cosθ + sin180 × sinθ
Sin180 = 0, cos 180 = -1
Thus,
Cos(180 - θ) = -1cosθ + 0sinθ
= -cosθ + 0
= -cosθ
The correct option is C

3. For a circle with radius r and centre (p, q), the equation is:
$(x - p)^2 + (y - q)^2 = r^2$
Here, p = -1 and q = 3. Also, r = 4
$(x - (-1))^2 + (y - 3)^2 = 4^2$
$(x + 1)^2 (y - 3)^2 = 16$
$x^2 + 2x + 1 + y^2 - 6y + 9 = 16$
Then,
$x^2 + y^2 + 2x - 6y = 6$
The correct option is D

4. $y = \frac{3}{\sqrt{x}} = \frac{3}{x^{\frac{1}{2}}} = 3x^{-\frac{1}{2}}$
Then, $\frac{dy}{dx}$ can be obtained by simply differentiating $3x^{-\frac{1}{2}}$
$y = 3x^{-\frac{1}{2}}$
$\frac{dy}{dx} = \left(-\frac{1}{2} \times 3\right)\left(x^{\frac{1}{2}-1}\right)$
$= -\frac{3}{2}x^{-\frac{3}{2}}$
The correct option is A

5. We are to integrate $\frac{1}{2x}$
$\frac{1}{2x} = \frac{1}{2} \times \frac{1}{x}$
Then,
$\int \frac{1}{2x}dx = \int\left(\frac{1}{2} \times \frac{1}{x}\right)dx = \frac{1}{2}\int\frac{1}{x}dx$
$= \frac{1}{2}In\ x + c$
The correct option is C

6. Let us first construct the table

+	1	2	3	4	5	6
1	1, 1	1, 2	1, 3	1, 4	1, 5	1, 6
2	2, 1	2, 2	2, 3	2, 4	2, 5	2, 6
3	3, 1	3, 2	3, 3	3, 4	3, 5	3, 6
4	4, 1	4, 2	4, 3	4, 4	4, 5	4, 6
5	5, 1	5, 2	5, 3	5, 4	5, 5	5, 6
6	6, 1	6, 2	6, 3	6, 4	6, 5	6, 6

Out of 36, there are three chances of getting a total of 6 from two odd numbers.
The chances are with 5, 1; 3, 3; and 1, 5 as can be seen from the table
Hence, the probability = $\frac{3}{36} = \frac{1}{12}$
The correct option is A

7. For numbers a, b, c, d, e, the mean is
$\frac{a+b+c+d+e}{5} = x$
Thus, a + b + c + d + e = 5x
For numbers a + k, b + 2k, c – k, d – 2k, e, the mean is
$\frac{(a+k)+(b+2k)+(c-k)+(d-2k)+e}{5}$
$= \frac{a+b+c+d+e+k+2k-k-2k}{5}$
$= \frac{a+b+c+d+e}{5}$
But a + b + c + d + e = 5x. Then, the mean
$= \frac{5x}{5} = x$
The correct option is A

8. $(a^2 - b^2) + (a + b)^2$
From the knowledge of difference of two squares,
$a^2 - b^2 = (a + b)(a - b)$. Therefore,
$(a^2 - b^2) + (a + b) = [(a + b)(a - b)] + (a + b)^2$
= (a + b)[(a – b) + (a + b]
= (a + b)[a – b + a + b]

$= (a + b)[a + a – b + b]$
$= (a + b)[2a]$
$= (2a)(a + b)$

The correct option is C

9. For a quadratic equation whose roots are α and β, the product $\alpha\beta = \frac{c}{a}$ (where c is the constant and a is the coefficient of x^2)
For $x^2 + 2x – 3 = 0$
a = 1, c = -3
Thus, $\alpha\beta = \frac{-3}{1} = -3$
The correct option is A
<u>Method</u> 2
We can also find the two roots and then multiply them
$x^2 + 2x – 3 = 0$
$x^2 + x – 3x – 3 = 0$
$x(x + 1) – 3(x + 1) = 0$
$(x – 3)(x + 1) = 0$, Either $x – 3 = 0$ or $x + 1 = 0$
x = -1 or + 3
If α = -1, β = 3
αβ = -1 × 3 = -3 as before
The correct option is A

10.

2	69	↑
2	34 r 1	
2	17 r 0	
2	8 r 1	
2	4 r 0	
2	2 r 0	
2	1 r 0	
	0 r 1	

Thus, $69_{ten} = 100010_{two}$
The correct option is C

11. $\frac{\frac{3}{4}}{\frac{1}{4}+\frac{1}{3}} = \frac{\frac{3}{4}}{\frac{3+4}{12}} = \frac{\frac{3}{4}}{\frac{7}{12}} = \frac{3}{4} \div \frac{7}{12} = \frac{3}{4} \times \frac{12}{7} = \frac{9}{7}$
For any number x, the reciprocal is $\frac{1}{x}$. Hence, the reciprocal of $\frac{9}{7}$ is $\frac{1}{\frac{9}{7}} = \frac{7}{9}$
The correct option is B

12. Speed = 30km/min
To convert the speed to centimeters per second, we convert the distance to centimeter and the time to seconds
1km = 100,000cm
Then, 30km = 30 × 100,000cm
= 3000000cm
1min = 60s
30km/min = $\frac{3000000cm}{60s}$
= 50000cm/s

13. $Log_4(x + 3)(x – 3) = 2$
If $\log_a P = x$, $P = a^x$
Then, $(x = 3)(x – 3) = 4^2$
$(x + 3)(x – 3) = 16$
$x^2 – 3x + 3x – 9 = 16$
$x^2 – 9 = 16$
$x^2 = 16 + 9 = 25$
$x = \pm\sqrt{25} = \pm 5$
The correct option is B

14. $a = \frac{1}{2-\sqrt{3}}, b = \frac{1}{2+\sqrt{3}}$
$a^2 = \left(\frac{1}{2-\sqrt{3}}\right)^2 = \frac{1}{2-\sqrt{3}} \times \frac{1}{2-\sqrt{3}}$
$= \frac{1}{4-2\sqrt{3}-2\sqrt{3}+3}$
$= \frac{1}{4-4\sqrt{3}+3}$
$= \frac{1}{7-4\sqrt{3}}$
Let us now conjugate it
$\frac{1}{7-4\sqrt{3}} = \frac{1}{7-4\sqrt{3}} \times \frac{7+4\sqrt{3}}{7+4\sqrt{3}}$
$= \frac{7+4\sqrt{3}}{7^2-(4\sqrt{3})^2} = \frac{7+4\sqrt{3}}{49-48}$
$= \frac{7+4\sqrt{3}}{1}$
$= 7 + 4\sqrt{3}$
Deductively,
$b^2 = \left(\frac{1}{2+\sqrt{3}}\right)^2 = 7 - 4\sqrt{3}$
Then,
$a^2 + b^2 = 7 + 4\sqrt{3} + 7 - 4\sqrt{3}$
$= 7 + 7 + 4\sqrt{3} - 4\sqrt{3}$
$= 14$
The correct option is D

15. x * y = 2
Then, 4 * 5 = 2
Thus,
2 * (4 * 5) = 2 * 2
Since x * y = 2
2 * 2 = 2
The correct option is D

16. Let $\sqrt{6+\sqrt{6+\sqrt{6+\sqrt{6+\sqrt{6}}}}} = x$
By squaring both sides, we have:
$6 + \sqrt{6+\sqrt{6+\sqrt{6+\sqrt{6}}}} = x^2$
Then,
$6 + x = x^2$
$x^2 - x = 6$
$x^2 - x - 6 = 0$
$x^2 - 3x = 2x - 6 = 0$
$x(x - 3) + (x - 3) = 0$
$(x + 2)(x - 3) = 0$
Either
$x + 2 = 0 \text{ or } x - 3 = 0$
$\therefore x = -2 \text{ or } + 3$
But the square root of a number cannot be negative. Hence, x = 3
The correct option is D

17. $\int_0^{\frac{\pi}{2}} (2\pi + 2cos2x)\, dx$
$\int 2\pi\, dx = 2\pi x, \int 2cos2x\, dx$
$= 2\int cos2x\, dx$
$= 2\left(\frac{sin2x}{2}\right)$
Then, $2\int cos2x\, dx$
$= 2\pi x + \frac{2sin2x}{2}$
$= 2\pi x + \sin 2x$
Introducing the limits, we have:
${}_0^{\frac{\pi}{2}}[2\pi x + sin2x]$

Then we have
$\left[2\pi\left(\frac{\pi}{2}\right)+\ sin2\left(\frac{\pi}{2}\right)\right]-\ [2\pi(0)+\ sin2(0)]$
$=\ \pi^2+\ sin\pi-\sin 0$
(Both sinπ and sin0 equal 0)
Then, π^2 = sin π - sin 0 = π^2
The correct option is B

18. For a circle whose centre is (p, q) and the radius is r
$(x-p)^2 = (y-q)^2 = r^2$
For the circle $2x^2+\ 2\left(\frac{y-2}{2}\right)^2=2$
$x^2+\ \left(\frac{y-2}{2}\right)^2=1$
This means the same as:
$(x+0)^2+\ \left(\frac{y-2}{2}\right)^2=\ 1^2$
Then,
$(x-p)^2 = (x+0)^2$
x – p = x + 0
Thus, p = 0
Also,
$(y-q)^2 = \left(\frac{y-2}{2}\right)^2$
Then, $q\ =\ \frac{3}{2}$
Finally, $r^2 = 1^2$, r = 1
Therefore, the centre is $\left(0,\frac{3}{2}\right)$ and the radius is 1
The correct option is D

19. $y=\ \frac{3}{2}x-1=0$
For a straight line, y = mx + c, the gradient is m and the intercept is c.
Arranging the given equation to match the standard equation, we have:
$y=\ \frac{-3}{2}x+1$
Thus, the gradient is $\frac{-3}{2}$
Also, when two lines are perpendicular, the product of their gradients =-1. Let the gradient of the unknown line be m_2. Then,
$\frac{-3}{2}\times\ m_2=-1, m_2=\ \frac{-1}{\frac{-3}{2}}=\ \frac{2}{3}$
The correct option is D

20. $2^{x^2}=\ 4^{(x+4)}$
$2^{x^2}=(2^2)^{(x+4)}$
$2^{x^2}=\ 2^{2x+8}$
Then,
$x^2=2x+8$
$x^2-\ 2x-8=0$
$x^2+\ 2x-4x-8=0$
$x(x+2)-4(x+2)=\ 0$
$(x-4)(x+2)=\ 0$
$x=4\ or-2$
The correct option is A

21. $\frac{3x}{x^2-1}\ \equiv\ \frac{A}{x-1}+\ \frac{B}{x+1}$
$=\ \frac{A(x+1)+B(x-1)}{(x-1)(x+1)}$
Then,
3x = A(x + 1) +B(x – 1)
To find A, we must employ a value of x, that will eliminate B. x = 1 will do this.
Then,
3(1) = A(1 + 1) + B(1 – 1)
3 = 2A, $A\ =\ \frac{3}{2}$
To find B, a value of x that will eliminate A must be employed, x = -1 will do this.
Then,
3(-1) = A(-1 + 1) + B(-1 – 1)
-3 = -2B, $B\ =\ \frac{-3}{-2}=\ \frac{3}{2}$
The correct option is D

22. Since the perimeter of the square is 40cm, each side will measure
$\frac{40cm}{4}=>10cm$
The area of a square = L^2
Where L = length of a side
Then,
Area of the square = $(10cm)^2 = 100cm^2$
The correct option is C

MATHEMATICS 2010 QUESTIONS

1. Find x if $x^2-\ 2x\ -\ 15\ =\ 0$ A. 3, -5 B. -3, 5 C. 1, 15 D. -2, -15
2. A father leaves a legacy of N45 million for his children – Peter, David, and Paul – to be shared in the ratio 7:5:3. What amount in million Naira would each receive respectively? A. ₦14, ₦7, ₦3 B. ₦15, ₦5, ₦3 C. ₦21, ₦15, ₦9 D. ₦20, ₦16, ₦10
3. As ∅ tends to zero, what does cosθ tend to? A. sin θ B. 0 C. $\frac{1}{2}$ D. 1
4. The expression 2cos θ + $\sin^2$θ has the numerical value A. 1 B. 2 C. 4 D. 0
5. If tan$x\ =\ \frac{\sin x}{\cos x}$, find $tan(90^o\ +\ x)$ for acute value of x. A. –cot x B. – tan x C.cot x D. tan x
6. Evaluate the length of perpendicular from A to BC A. $\frac{\sqrt{52}}{12}$ cm B. $\frac{12}{\sqrt{52}}$ cm C. $\frac{24}{\sqrt{52}}$ cm^2 D. $\frac{24}{\sqrt{52}}$ cm
7. The indefinite integral of xe^x, for any real constant c is A. c B. $x\ +e^x\ +\ c$ C. $x^2\ +\ e^x\ +\ c$ D. $e^x(x-1)+c$
8. Find the area under the curve y(x) = sin x between x = 0 and x = π. A. 2 B. 1 C. -2 D. π
9. Let the letter P, Q, R and S denote parallelogram, quadrilateral, rectangle and square respectively. Using subset notation, which of these inclusions is correct? A. Q ⊂ R ⊂ P ⊂ S B. R ⊂ 9 ⊂ P ⊂ S C. S ⊂ P ⊂ R ⊂ Q D. S ⊂ R ⊂ P ⊂ Q
10. In a convex polygon with n sides, the sum of interior angles is A. (n-2)πB. 2(n-1)π C.4(n-1)π D. (2n-4)π
11. Find the equation of the line perpendicular to the line $y\ =\ 2x\ +\ 1$ and passing through a point (3, 1). A. $y=\ \frac{1}{2}x+\ \frac{5}{2}$B. $y=\ -\frac{1}{2}x+\ \frac{5}{2}$ C. $y\ =\ x\ +\ 5$ D. $2y\ =\ x\ +\ 5$
12. What is the distance between points (1,2) and (4,5) on a plane? A. $3\sqrt{2}$ B. $2\sqrt{3}$ C. 3 D. 9

13. Integrate $\int 2tan\,(2x + \pi)dx$ A. $2cos\,(2x + \pi) + k$ B. $log\,(cos(2x + \pi) + k$ C. $-log(cos(2x + \pi) + k$ D. $4cot(2x + \pi) + k$

14. Find the values of x for which $5 + 2x - 3x^2 = 0$ A. -2 and $\frac{6}{5}$ B. -1 and $\frac{5}{3}$ C. -2 and -1 D. $\frac{6}{5}$ and $\frac{5}{3}$

15. If $\left(\frac{3}{4}\right)^x \left(\frac{2}{3}\right)^y = \frac{32}{27}$, find the value of $3y - 2x$ A. -1 B. 7 C. 1 D. -7

16. The integral values of y which satisfy the inequality 1 < 5 – 2y ≤ 7 are A. -1, 0, 1, 2 B. 0, 1, 2, 3 C. -1, 0, 1, 2, 3 D. -1, 0, 2, 3

17. If $x^2 - 5x + 6 = (x - a)^2 + b$, the value of b is A. $-\frac{1}{4}$ B. $\frac{5}{2}$ C. 2 D. 3

18. The scores of 16 students in a mathematics test are 65, 65, 55, 60, 60, 65, 60, 70, 75, 70, 65, 70, 60, 65, 65, 70. What is the sum of the median and modal scores? A. 125 B. 130 C. 140 D. 150

19. A businessman invested a total of ₦200,000 in two companies which paid dividends of 5% and 7% respectively. If he received a total of ₦11,600, how much did he invest at 7%. A. ₦140,000 B. ₦160,000 C. ₦80,000 D. ₦100,000

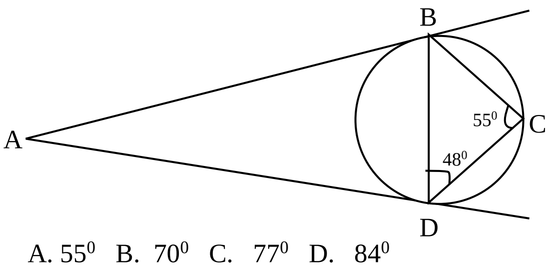
20. If $a\sqrt{5} + b\sqrt{2}$ is the square root of $95 - 30\sqrt{10}$, the values of a and b are respectively. A. 5,2 B. 2, -5 C. -5, 3 D. 3, -5

21. If $\frac{x}{y} = \frac{z}{w} = c$, find the value of $\frac{3x^2 - xz + z^2}{3y^2 - yw + w^2}$ in terms of c A. $3c^2$ B. $\frac{17c^2}{4}$ C. $2c - c^2$ D. c^2

22. Express $\frac{5y-12}{(y-2)(y-3)}$ in partial fractions A. $\frac{2}{y-2} - \frac{3}{y-3}$ B. $\frac{2}{y-2} + \frac{3}{y-3}$ C. $\frac{2}{y-3} - \frac{3}{y-2}$ D. $\frac{5}{y-3} - \frac{4}{y-2}$

23. The second terms of an infinite geometric series is $-\frac{1}{2}$ and the third term is $\frac{1}{4}$. Find the sum of the series. A. 2 B. 1 C. $\frac{3}{2}$ D. $\frac{2}{3}$

24. In the figure AB and AD are tangents to the circle. If BCS = 55^0 and BDC = 48^0, find BAD.



A. 55^0 B. 70^0 C. 77^0 D. 84^0

25. Find the area of the triangle. A. 24cm B. 24cm^2 C. 12cm^2 D. 12cm

MATHEMATICS 2010 SOLUTIONS

1. $x^2 - 2x - 15 = 0$
$x^2 + 3x - 5x - 15 = 0$
$x(x + 3) - 5(x + 3) = 0$
$(x - 5)(x + 3) = 0$
Either $x - 5 = 0$ or $x + 3 = 0$
$x = 5$ or -3
Method 2: Formula method
$x^2 - 2x - 15 = 0$
a = 1, b = -2, c = -15
$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
$\frac{-(-2) \pm \sqrt{(-2)^2 - (4 \times 1 \times -15)}}{2 \times 1}$
$= \frac{2 \pm \sqrt{4+60}}{2} = \frac{2 \pm 8}{2}$
$= \frac{2+8}{2}$ or $\frac{2-8}{2}$
$= 5$ or -3
The correct option is B

2. Total amount = ₦45million
Peter: David : Paul = 7 : 5 : 3
Total ratio = 15
Peter's share = $\left(\frac{₦45}{15} \times 7\right)$million
₦21 million
David's share = $\left(\frac{₦45}{15} \times 5\right)$million
= ₦15 million
Paul's share = $\left(\frac{₦45}{15} \times 3\right)$million
= ₦9 million
The correct option is C

3. When ϕ = 0, cosϕ = 1
Then, when ϕ tends toward 0,
cosϕ tends toward 1
The correct option is D

4. This equation is incomplete. The numerical value of 2cosθ + sin^{2}θ cannot be obtained until the value of θ is stated.

5. $Tanx = \frac{Sinx}{Cosx}$
Then, $\tan(90 + x) = \frac{\sin(90+x)}{\cos(90+x)}$
From compound angles,
Sin(A + B) = sinAcosB + cosAsinB
Cos(A + B) = cosAcosB – sinA sin B
Thus,
$\frac{Sin(90+x)}{\cos(90+x)} = \frac{sin90cosx + cos90\,sinx}{cos90cosx - sin90\,sinx}$
$= \frac{1 \times cosx + 0 \times sinx}{0 \times cosx - 1 \times sinx}$
$= \frac{cosx}{-sinx}$
$But\ \frac{cos\theta}{sin\theta} = cot\theta$
Thus, $\frac{-cosx}{sinx} = -cotx$
The correct option is A

6. The diagram is missing

7. We are to find $\int xe^x\,dx$
Let u = x, $\frac{du}{dx} = 1, du = dx$
Recall that $\int udv = uv - \int vdu$
$dv = e^x dx$
$v = \int e^x\,dx = e^x$
Now, we can say:
$\int udv = x\,.\,e^x - \int e^x\,dx$
$= xe^x - e^x + c$
$= e^x(x - 1) + c$
The correct option is D

8. $y(x) = sinx$
Finding the area under the curve between $x = 0$ and $x = \pi$ as lower and upper limits
$Area = \int_0^\pi sinx\,dx$
$[-cosx]_0^\pi$
$= -cos\pi - (-cos0)$

$= +1 - (-1)$
$= 1 + 1$
$= 2$
The correct option is A

9. Parallelogram, square and rectangle are all quadrilaterals
Then, PCQ, SCQ and RCQ
Also,
A rectangle can be formed from parallelogram. Thus RCP.
Finally, a square can be formed from a rectangle.
Then, SCR
Then, we can write: SCRCPCQ
The correct option is D

10. The sum of the interior angles of a polygon = (n = 2)180 = (2n – 4)90
(n – 2)180 = (n – 2)π
The correct option is A

11. The gradient of the line $y = 2x + 1$ is 2 (by comparing with $y = mx + c$. Where m is the gradient
Since the unknown line is perpendicular to $y = 2x + 1$. The products of their gradient must equal – 1.
Let the gradient of this second line be m_2. Then,
$m_2 \times 2 = -1$, $m_2 = \frac{-1}{2}$
It is said that the line pass through point (3, 1). Then
$m_2 = \frac{y-1}{x-3}$
$\frac{-1}{2} = \frac{y-1}{x-3}$
$2y - 2 = -x + 3$
$2y = -x + 3$
$y = \frac{-x+5}{2} = \frac{-1}{2}x + \frac{5}{2}$
The correct option is B

12. The distance between two points (x_1, y_1) and (x_2, y_2) is giving by
$d = \sqrt{(y_2 - y_1)^2 + (x_2 - x_1)^2}$
$= \sqrt{(5-1)^2 + (4-1)^2}$
$\sqrt{9+9} = \sqrt{18} = 3\sqrt{2}$
The correct option is A

13. $\int 2\tan(2x + \pi)\,dx = 2\int \tan(2x + \pi)\,dx$
$= 2\int \frac{\sin(2x+\pi)dx}{\cos(2x+\pi)}$
$= \int \frac{-\sin(2x+\pi)}{\cos(2x+\pi)}dx$
$= -\log[cos(2x + \pi)] + c$
The correct option is C

14. $5 + 2x - 3x^2 = 0$
$-3x^2 + 2x + 5 = 0$
Multiplying through by – 1, we have:
$3x^2 - 2x - 5 = 0$
$3x^2 + 3x - 5x - 5 = 0$
$3x(x + 1) - 5(x + 1) = 0$
$(3x - 5)(x + 1) = 0$
Either $3x - 5 = 0$ or $x + 1 = 0$
$x = \frac{5}{3}\ or - 1\ 0$
The correct option is B

15. $\left(\frac{3}{4}\right)^x\left(\frac{2}{3}\right)^y = \frac{32}{27}5+$
$\frac{3^x}{4^x} \times \frac{2^y}{3^y} = \frac{32}{27}$
$\frac{3^x}{2^{2x}} \times \frac{2^y}{3^y} = \frac{2^5}{3^3}$
$\left(\frac{3^x \times 1}{2^{2x}}\right) \times \left(\frac{2^y \times 1}{3^y}\right) = \frac{2^5 \times 1}{3^3}$
$3^x.2^{-2x}.2^y.3^{-y} = 3^{-3} \times 3^{-3}$
$a^x \times a^y = a^{-2x+y} = 3^{-3}.2^5$
Thus,
$x - y = -3$ ……………………………(i)
$-2x + y = 5$ ………………………………(ii)
Adding equations (i) and (ii), we have
$x - 2x = 2$
$-x = 2, x = -2$
Substitute $x = -2\ in\ equation\ (i)$
$x - y = -3$
$-2 - y = -3$
$y = -2 + 3 = 1$
Then,
$3y - 2x = 3(1) - 2(-2)$
$= 3 + 4 = 7$
The correct option is B

16. – 1 < 5 – 2y ≤ 7. This can be broken into – 1 < 5 – 2y and 5 – 2y ≤ 7
– 1< 5 – 2y
5 – 2y > 1
–2y > - 1 – 5
–2y > - 6
$y < \frac{-6}{-2}$ (Note the reversal of sign)
$y < 3$
Also,
5 – 2y ≤ 7
–2y ≤ 7 – 5
–2y ≤ 2
$y \geq \frac{2}{-2}$ (Please note the reversal of sign)
$y \geq -1$
If ≥ -1 and $y < 3, -1 \leq y < 3$
The values in this range are – 1, 0, 1 and 2
The correct option is A

17. $x^2 - 5x + 6 = (x - a)^2 + b$
It is simple, we just need to expand the term on the RHS
$x^2 - 5x + 6 = (x - a)(x - a) + b$
$x^2 - 5x + 6 = x^2 - 2ax + a^2 + b$
It can be deduced that
(i) $-2ax = -5x$
(ii) $a^2 + b = 6$
From (i)
$a = \frac{-5x}{-2x} = \frac{5}{2}$
From (ii)
$a^2 + b = 6$
$\left(\frac{5}{2}\right)^2 + b = 6, \frac{25}{4} + b = 6$
$b = 6 - \frac{25}{4}$
$= \frac{24-25}{4} = \frac{-1}{4}$
The correct option is A

18. Let us first arrange the scores in an order, say ascending order: 55, 60, 60, 60, 60, 65, 65, 65, 65, 65, 65, 70, 70, 70, 70, 75

The score that has the highest frequency is 65. Hence, modal score = 65

$Median = \frac{65+65}{2} = 65$

Sum of median and modal score = 65 + 65 = 130

The correct option is B

19. Total amount = ₦200,000

$I = \frac{PRT}{100}$

Let the interest at 5% be I, and that at 7% be I_2

$I_1 + I_2 =$ ₦11,600

Let the amount invested at 7% be x.

Then,

$I_2 = \frac{x \times 7 \times 1}{100}, I_2 = \frac{(200{,}000-x)\times 5 \times 1}{100}$

$I_1 + I_2 = \frac{x \times 7 \times 1}{100} + \frac{(200{,}000-x)\times 5 \times 1}{100} = 11{,}600$

$\frac{5(200{,}000-x)}{100} + \frac{7x}{100} = 11{,}600$

$\frac{200{,}000-x}{20} + \frac{7x}{100} = 11{,}600$

$\frac{1{,}000{,}000-5x+7x}{100} = 11{,}600$

$1000000 + 2x = 1160000$

$2x = 1160000 - 1000000$

$2x = 160{,}000$

$x = \frac{160{,}000}{2} = 80{,}000$

₦80,000

The correct option C

20. $a\sqrt{5} + b\sqrt{2} = \sqrt{95 - 30\sqrt{10}}$

Square both sides

$\left(a\sqrt{5} + b\sqrt{2}\right)^2 = \left(\sqrt{95 - 30\sqrt{10}}\right)^2$

$\left(a\sqrt{5} + b\sqrt{2}\right)^2 = 95 - 20\sqrt{10}$

$\left(a\sqrt{5} + b\sqrt{2}\right)\left(a\sqrt{5} + b\sqrt{2}\right) = 95 - 30\sqrt{10}$

$5a^2 + ab\sqrt{10} + ab\sqrt{10} + 2b^2 = 95 - 30\sqrt{10}$

$5a^2 + 2b^2 + 2ab\sqrt{10} = 95 - 30\sqrt{10}$

Then, it can be deduced that:

(i) $5a^2 + 2b^2 = 95$ and (ii) $2ab\sqrt{10} = -30\sqrt{10}$

From (ii)

$2ab = -30$

$ab = \frac{-30}{2} = -15$

$b = \frac{-15}{a}$ ………………………….(iii)

Substituting (iii) into (i), we have:

$5a^2 + 2\left(\frac{-15}{a}\right)^2 = 95$

$5a^2 + 2\left(\frac{225}{a^2}\right) = 95$

$5a^2 + \frac{450}{a^2} = 95$

Let $a^2 = x$

$5x + \frac{450}{x} = 95$

$\frac{5x^2=450}{x} = 95$

$5x^2 + 450 = 95x$

$5x^2 - 95x + 450 = 0$

$x^2 - 91x + 90 = 0$

$x^2 - 9x - 10x + 90 = 0$

$x(x-9) - 10(x-9) = 0$

$(x-10)(x-9) = 0$

Either $x - 10 = 0$ or $x - 9 = 0$

$x = 10$ or 9

When $x = 0$

$a^2 = 10$

When $x = 9$

$a^2 = 9$

$a = 3$

From (iii)

$b = \frac{-15}{a} = \frac{-15}{3} = -5$

Then, a and b are 3 and – 5 respectively

The correct option is D

21. $\frac{x}{y} = \frac{z}{w} = c$

With this, the following modifications are possible:

(i) $\frac{x}{z} = c$ (ii) $\frac{y}{w} = c$

We are required to express $\frac{3x^2 - xz + z^2}{3y^2 - yw + w^2}$

In term of c

From (i), $x = zc$

From (ii), $y = wc$

Then,

$\frac{3x^2 - xz + z^2}{3y^2 - yw + w^2} = \frac{3(zc)^2 - z^2c + z^2}{3(wc)^2 - w^2c + w^2}$

$= \frac{3z^2c^2 - z^2c + z^2}{3w^2c^2 - w^2 + w^2}$

$= \frac{z^2(3c^2 - c + 1)}{w^2(3c^2 - c + 1)}$

$= \frac{z^2}{w^2} = \left(\frac{z}{w}\right)^2$

But $\frac{z}{w} = c$

Thus, $\left(\frac{z}{w}\right)^2 = c^2$

The correct option is D

22. $\frac{5y-12}{(y-2)(y-3)} = \frac{A}{(y-2)} + \frac{B}{(y-3)}$

$= \frac{A(y-3) + B(y-2)}{(y-2(y-3)}$

Then,

A(y – 3) + B(y – 2) = 5y – 12

To find A, a value of y which eliminates B(y – 2) must be employed, y = 2 does this.

A(2 – 3) + B(2 – 2) = 5(2) – 12

-A = 10 – 12

-A = -2, A = 2

To get B, a value of y which eliminates A(y – 3) must be employed; y = 3 does this

A(3 – 3) + B(3 – 2) = 5(3) – 12

B = 15 – 12

= 3

Then, $\frac{5y-12}{(y-2)(y-3)} = \frac{2}{y-2} + \frac{3}{y-3}$

The correct option is B

23. $ar^{n-1} = Un$

For the second term

$-\frac{1}{2} = ar^{2-1} = ar$

For the third term

$\frac{1}{4} = ar^{3-1} = ar^2$

$ar = -\frac{1}{2}$ ……………………………(i)

$ar^2 = \frac{1}{4}$ ……………………………(ii)

Dividing equation (ii) by (i), we get:

$r = \frac{1}{4} \div -\frac{1}{2} = \frac{1}{4} \times \frac{-2}{1}$

$= -\frac{1}{2}$

From equation (i)

$ar = -\frac{1}{2}$

$a \times -\frac{1}{2} = -\frac{1}{2}$

$a = 1$

Sum to infinity = $\frac{a}{1-r}$

$= \frac{1}{1-\left(-\frac{1}{2}\right)}$

$= \frac{1}{1+\frac{1}{2}} = \frac{1}{\frac{3}{2}} = \frac{2}{3}$

The correct option is D

24. $B\hat{C}D = A\hat{D}B$ (angles in alternate segment)
Also, $B\hat{C}D = A\hat{B}D$ (same reason)
Then, $A\hat{D}B = A\hat{B}D = B\hat{C}D = 55^o$
$A\hat{D}B + A\hat{B}D + B\hat{A}D = 180$
$110^o + B\hat{A}D = 180^o$, $B\hat{A}D = 180^o - 110^o = 70^o$
The correct option is B

25. Incomplete question

MATHEMATICS 2009 QUESTIONS

1. Factorize $6x^2 - 14x - 12$. A. $2(x+3)(3x-2)$ B. $6(x-2)(x+1)$ C. $2(x-3)(3x+2)$ D. $6(x+2)(x-1)$E. $(3x+4)(2x+3)$
2. What is the product of $\frac{27}{5} - (3)^{-3}$ and $\left(\frac{1}{5}\right)$ A. 5 B. 3 C. 2 D. 1 E. $\frac{1}{25}$
3. If the length of the sides of a right angled triangle are $(3x+1)$cm, $(3x-1)$cm and xcm, what is x? A. 2 B. 6 C. 18 D. 12 E. 0
4. Evaluate$\frac{(xy^2-x^2y)}{(x^2-xy)}$ A. -3 B.$\frac{3}{5}$ C. $\frac{4}{5}$ D. 3 E. 4
5. Two fair dice are rolled. What is the probability that both show up the same number of point? A. $\frac{1}{36}$ B. $\frac{7}{36}$ C. $\frac{1}{2}$ D. $\frac{1}{3}$ E. $\frac{1}{6}$
6. In 1984, Tolu was 24 years old and his father 45 years. In what year was Tolu exactly half his father's age? A.1982 B.1981 C. 1983 D. 1979 E. 1978
7. Find the probability that a number selected at random 40 to 50 is a prime A.$\frac{3}{11}$ B. $\frac{5}{11}$ C. $\frac{5}{10}$ D. $\frac{4}{10}$ E. $\frac{7}{12}$
8. A man kept 6 black, 5 brown and 7 purple shirts in a drawer. What is the probability of his picking a purple shirt with his eyes closedA.$\frac{1}{7}$B.$\frac{7}{18}$ C.$\frac{11}{18}$ D.$\frac{7}{11}$ E.$\frac{5}{10}$
9. An $(n-2)^2$ sided figure has n diagonals, find the number n of diagonals for a 25 sided figure A. 8 B. 7 C. 6 D. 9 E. 10
10. Find the probability of selecting a figure which is parallelogram from a square, a rectangle, a rhombus, a kite and a trapezium A.$\frac{3}{5}$ B. $\frac{2}{5}$C. $\frac{4}{5}$ D. $\frac{1}{5}$ E. $\frac{5}{6}$
11. If P varies inversely as V and V varies directly as R^2, find the relationship between P and R given that R = 7 when P = 2 A. $P = 98R^2$ B. $PR^2 = 98$ C. $P^2R = 89$ D. $P = \frac{1}{98R}$ E. $P = \frac{R^2}{98}$
12. If 7 and 189 are the first and fourth terms of a geometric progression respectively find the sum for the first three terms of the progression A.182 B. 180 C. 91 D. 63 E. 28
13. Find the positive number n such that thrice its square is equal to twelve times the number A. 1 B. 4 C. 2 D. 5 E. 9
14. A sector of a circle of radius7.2cm which subtends an angle of 300^0 at the centre is used to form a cone. What is the radius of the base the cone? A. 6cm B.7cm C. 8cm D. 9cm D. 5cm
15. If $pq + 1 = q^2$ and $t = \frac{1}{p} - \frac{1}{pq}$ express t in terms of q A. $\frac{1}{pq}$ B. $\frac{1}{q+1}$ C. $\frac{1}{q+2}$D. $\frac{1}{p+1}$ E.$\frac{1}{1-q}$
16. Given a regular hexagon, calculate each interior angle of the hexagon A. 60^0 B. 30^0 C. 120^0 D. 15^0E. 135^0
17. Find n if $\log_2 4 + \log_2 7 \log_2 n = -1$ A.10 B. 14 C. 12 D. 27 E. 26
18. If x = 1 is root of the equation $x^3 - 2x^2 - 5x + 6$, fin d the other roots. A. -3 and 2 B. 2 and 2 C. 3 and -2 D. 1 and 3 E. -3 and 1
19. The value of $(0.303)^3 - (0.02)^3$ is A. 0.019 B. 0.0019 C.0.00019D. 0.000019 E. 000035
20. List all integers satisfying the inequality -2 < 2x – 6 < 4 A. 2,3,4,5 B. 2,3,4 C. 2,5 D. 3,4,5 E. 4,5
21. If $3^{2y} - 6(3^y)=27$, find y. A. 3 B.1 C. 2 D.-3 E. 1
22. A number of pencils were shared out among Peter, Paul and Audu in the ratio 2:3:5 respectively. If Peter got 5, how many were share? A. 15 B. 25 C. 30 D. 50 E. 55
23. A sum of money was invested at 8% per annum simple interest. If after 4 years, the money became ₦330.00, what is the amount originally invested? A. ₦180 B. ₦165 C. ₦150 D. ₦200 E. ₦250

MATHEMATICS 2009 SOLUTION

1. $6x^2 - 14x - 12$
$2(3x^2 - 7x - 6)$
$2(3x^2 - 9x + 2x - 6)$
$2(3x(x-3) + 2(x-3)$
$2(3x+2)(x-3)$
The correct option is C

2. $\left(\frac{27}{5} - 3^{-3}\right)\frac{1}{5}$
$= \left(\frac{27}{5} - \frac{1}{3^3}\right)\frac{1}{5}$
$= \left(\frac{27}{5} - \frac{1}{27}\right)\frac{1}{5}$
$= \frac{1}{5}\left(\frac{27}{5}\right) - \frac{1}{5}\left(\frac{1}{27}\right)$
$= \frac{27}{25} - \frac{1}{135} = 1.0726 = 1$
The correct option is D

3. For a right angled triangle, the square of the hypotenuse side equals the sum of the squares of the two other sides. Don't forget that the longest side is the hypotenuse. Of the three sides, 3x + 1, 3x – 1, and x, 3x + 1 is the longest. Then, we can write
$(3x+1)^2 = (3x-1)^2 + x^2$

$9x^2 + 6x + 1 = 9x^2 - 6x + 1 + x^2$
$9x^2 + 6x + 1 = 10x^2 - 6x + 1$
$10x^2 - 6x + 1 - 9x^2 - 6x - 1 = 0$
$x^2 - 12x = 0$
$x(x - 12) = 0$
Either x = 0 or x – 12 = 0
Thus, x = 0 or 12
However, employing x as 0 will make the side 3x – 1 negative and the side x of the triangle non-existent.
Therefore, x can only be 12
The correct option is D

4. $\frac{xy^2 - x^2y}{x^2 - xy} = \frac{xy(y-x)}{x(x-y)}$
$= \frac{y(y-x)}{(x-y)}$
This can also be written as $\frac{-y(-y+x)}{(x-y)} = \frac{-y(x-y)}{(x-y)} = -y$

5.

+	1	2	3	4	5	6
1	1, 1	1, 2	1, 3	1, 4	1, 5	1, 6
2	2, 1	2, 2	2, 3	2, 4	2, 5	2, 6
3	3, 1	3, 2	3, 3	3, 4	3, 5	3, 6
4	4,1	4, 2	4, 3	4, 4	4, 5	4, 6
5	5, 1	5, 2	5, 3	5, 4	5, 5	5, 6
6	6, 1	6, 2	6, 3	6, 4	6, 5	6,6

Out of 36, there are 6 chances of getting the same number, as can be seen from the table above.
Then,
P(the same number) = $\frac{6}{36} = \frac{1}{6}$
The correct option is E

6. Let the present year be 1984
Tolu's present age = 24
His fathers age = 45
Some years ago, Tolu's age was 24 – x while his father's age would be 45 – x.
As at then, Tolu's age was half his father's age
i.e. 24 – x = $\frac{45-x}{2}$
2(24 – x) = 45 – x
48 – 2x = 45 – x
x = 3
i.e. Tolu's age was half his father's age three years ago
Year = 1984 – 3 = 1981
The correct option is B

7. Sample space = 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50
Event space = 41, 43, 47
n(sample space) = 11
n(event space) = 3
$P = \frac{n(event\ space)}{n(sample\ space)} = \frac{3}{11}$
The correct option A

8. Black shirts = 6
Brown shirts = 5
Purple shirts = 7
Total = 18
P(purple) = $\frac{7}{18}$
The correct option is B

9. $(n - 2)^2 = 25$
n – 2 = 5
n = 5 + 2 = 7
The correct option is B

10. Sample space = square rectangle, rhombus, kite and trapezium
n(sample space) = 5
n(event space) = 3
P = $\frac{3}{5}$
The correct option is A

11. $P \propto \frac{1}{v}, v \propto R^2$
$P = \frac{k}{v}, v = cR^2, Then, P = \frac{k}{cR^2}$
Let $\frac{k}{c}$ $be\ w$
Then, $P = \frac{w}{R^2}$
$PR^2 = w$
P = 2, R = 7
W = 2 × 7^2= 2 × 49 = 98
Thus,
$P = \frac{98}{R^2}$
$PR^2 = 98$
The correct option is B

12. a = 8 …………………(1)
Fourth term = $ar^{4-1} = ar^3$ = 189 ………(ii)
Divide (ii) by (i)
$\frac{ar^3}{a} = \frac{189}{7}$
$r^3 = 27$
$r^3 = 3^3$
r = 3
Sum of terms of GP = $\frac{a(r^n - 1)}{r-1}$
If r > 1 or $\frac{a\,(1 - r^n)}{1-r}$, if r < 1
Since r > 1
Sum = $\frac{a(r^n - 1)}{r-1} = \frac{7(3^3 - 1)}{3-1} = \frac{7(27-1)}{2} = \frac{7 \times 26}{2} = 91$
The correct option is C

13. Let number be x
If three times the square of x equals twelve times x, we can write $3x^2 = 12x$
Then,
$3x^2 - 12x = 0$
$3x(x - 4) = 0$
3x = 0 or x – 4 = 0
x = 0 or 4
The correct option is B

14. In this arrangement, the area of the sector equals the curved surface area of the cone.
Area of the sector = $\frac{\theta}{360} \times \pi r^2 = \frac{300}{360} \times 3.142 \times 7.2^2$
= 135.734cm^2
Curved surface area of a cone = $\pi r l$
r = radius of its base, l = slant height
N.B: The radius of the sector equals the slant height of the cone.
Curved surface area of cone = 3.142 × r × 7.2 = 22.6224r
Equating the two, we have
22.6224r = 135.734

$r = \frac{135.734}{22.6224}$ = 6cm
The correct option is A

15. pq + 1 = q^2, $t = \frac{1}{p} - \frac{1}{pq}$
pq = $q^2 - 1$
$p = \frac{q^2-1}{q}$
$t = \frac{1}{\frac{q^2-1}{q}} - \frac{1}{q^2-1} = \frac{q}{q^2-1} - \frac{1}{q^2-1}$
$= \frac{q-1}{(q^2-1)} = \frac{q-1}{(q^2-1^2)}$
Using difference of two squares on the denominator
$t = \frac{q-1}{(q-1)(q+1)} = \frac{1}{q+1}$
The correct option is B

16. The sum of the interior angles of a regular polygon with n sides is given by (n – 2)180. Thus each interior angle
$= \frac{(n-2)180}{n}$
For a hexagon, n = 6
Each angle $= \frac{(6-2)180}{6} = \frac{720}{6}$ = 120°
The correct option is C

17. $\log_2 4 + \log_2 7 - \log_2 n = 1$
$\mathbf{\log_p a + \log_p b - \log_p c = \log_p\left(\frac{a\times b}{c}\right)}$
Thus,
$log_2\left(\frac{4\times 7}{n}\right) = 1, log_2\left(\frac{28}{n}\right) = 1$
If $\log_a P = x$, $p = a^x$
Then, $\frac{28}{n} = 2^1 = 2$
$n = \frac{28}{2} = 14$
The correct option is B

18. Since x = 1, is a root of the equation, the expression can be divided by x – 1 without any remainder. We can, then easily obtain the other factor, which will unarguably be the quotient of the division

$$\begin{array}{r l}
 & x^2 - x - 6 \\
x-1 & \overline{)\,x^3 - 2x^2 - 5x + 6} \\
 & \underline{x^3 - x^2} \\
 & -x^2 - 5x \\
 & \underline{-x^2 + x} \\
 & -6x + 6 \\
 & \underline{-6x + 6} \\
 & 0 \quad 0
\end{array}$$

Thus, $x^2 - x - 6$ is another factor
$x^2 - x - 6 = x^2 - 3x + 2x - 6$
x(x – 3) + 2(x – 3)
= (x + 2) (x – 3)
Hence,
x + 2 = 0 and x – 3 = 0
x = - 2 and x = 3
Thus, the other two factors are – 2 and 3
The correct option is C

19. $0.303^3 - 0.02^3$
You can just subtract directly. This gives **0.02781**
We can also employ the knowledge of difference of two cubes; $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$
Let a = 0.303 and b = 0.02
$0.303^2 - 0.02^3 = (0.303-0.02)(0.303^2+0.303(0.02)+0.02^2)$
= 0.283(0.091809 + 0.00606 + 0.0004)
= 0.283(0.098269)
= **0.02781** as before
No suitable option

20. -2 < 2x – 6 < 4

-2 < 2x - 6	and	2x – 6 < 4
2x – 6 > - 2		2x < 10
2x > 4		x < 5
x > 2		

The values are 3 and 4
No suitable option

21. $3^{2y} - 6(3^y) = 27$
$3^{2y} - 6(3^y) - 27 = 0$
$(3^y)^2 - 6(3^y) - 27 = 0$
Let $3^y = a$
$a^2 - 6a - 27 = 0$
$a^2 = 3a - 9a - 27 = 0$
$a(a+3) - 9(a+3) = 0$
$(a-9)(a+3) = 0$
a = 9 or – 3
since $3^y = a$
$3^y = 9$, $3^y = 3^2$, y = 2
The correct option is C

22. Let the total number of pencils be x
Sum of ration = 2 + 3 + 5 = 10
Peter's share = $\frac{x}{10} \times 2$
But Peter's share = 5
Then,
$\frac{x}{10} \times 2 = 5$
$\frac{2x}{10} = 5, 2x = 50$
$x = \frac{50}{2} = 25$
The correct option is B

23. Let the principal be P
Amount = ₦330
Interest 330 – P
Interest, $I = \frac{PRT}{100}$
$330 - P = \frac{P\times 8\times 4}{100}$
$330 - P = \frac{32P}{100}$
$33000 - 100P = 32P$
132P = 33000
$P = \frac{33000}{132}$ = ₦250
The correct option is E

MATHEMATICS 2008 QUESTIONS

1. The expression $a^3 + b^3$ is equal to A. $(a^2+b)(a-ab+b^2)$ B. $(a+b)(a^2-ab+b^2)$ C. $(a-b^2)(a^2-ab+b)$ D. $(a-b)(a^2+ab+b^2)$
2. Factorize $16(3x+2y)^2 - 25(a+2b)^2$. A. (12x + 8y – 5a – 10b)(12x+8y-5a-10b) B.20(3x+2y-a-10b)(3x+2y+a+1-2b) C. (12x + 8y +5a + 10b)(12x+8y-5a-10b) D. 20(3x+2y+a+2b)(3x+2y+a+2b)
3. A cone has base radius 4cm and height 3cm. The area of its curved surface is A. $251\pi cm^2$ B. $20\pi cm^2$ C. $24\pi cm^2$ D. $15\pi cm^2$

4. A cylinder has height 4cm and base radius 5cm, its volume to 3 significant figure is A. 314.2cm^3 B. 31.42cm^3 C. 251.4cm^3 D. 251cm^3
5. Let $3logx + logy = 3$. Then, y is A.$\left(\frac{x}{10}\right)^3$ B.$\left(\frac{x}{10}\right)^{-3}$C.$\left(\frac{10}{x}\right)^{-\frac{1}{3}}$ D.$\left(\frac{10}{x}\right)^3$
6. If$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, then y is A.$\pm\frac{b}{a}\sqrt{a^2 - x^2}$ B.$\pm\frac{b}{a}\sqrt{x^2 - a^2}$ C. $\frac{a}{b}\sqrt{a^2 - x^2}$D. $\pm\frac{a}{b}\sqrt{x^2 - a^2}$
7. A cyclist rode for 30 minutes at xkm/hr and due to a breakdown, he had to push the bike for 2hrs at (x-5)km/hr. If the total distance covered is less than 60km, what is the range of values for x? A. x<14 B. x<29 C. x<28 D. x<20
8. The expression $ax^2 + bx$ takes the value 6 when x = 1 and 10 when x = 2. Find its value when x = 5 A.10 B. 12 C. 6 D. -10
9. Dividing $2x^3 - x^2 - 5x + 1$ by $x + 3$ gives the remainder A. -3 B. 47 C. 61 D. -47
10. Let $f(x) = 2x3 - 3x^2$ – 5x + 6. If x – 1 divides $f(x)$ find the poles of the function. A. $1, 2, \frac{3}{2}$B. $1, 2, -\frac{3}{2}$C. -1,2,3 D. 1,-2,$-\frac{3}{2}$
11. The difference of two numbers is 10, while their product is 39. Find these numbers A. -3 and 10 or 13 and 10 B. 3 and -10 or 3 and 13 C. 3 and -3 or 3 and 13 D. -3 and -13 or 13 and 3
12. The average age of x pupils in a class is 14 years 2 months. A pupil of 15 years 2 months joins the class and the average age is increased by one month. Find x. A. 12 B. 6 C. 11 D. 14

All the 120 pupils in a school learn Yoruba or Igbo. Given that 75 learn Yoruba and 60 learn Igbo.

13. How many learn both languages? A. 60 B. 45 C. 15 D. 120
14. How many learn Igbo only? A. 45 B. 30 C. 15 D. 60

Suppose we have matrices $A = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 2 \\ 4 & 3 \end{pmatrix}$

15. Find $A^2 + AB - 2A$ A. $\begin{pmatrix} -5 & -9 \\ 12 & 14 \end{pmatrix}$ B. $\begin{pmatrix} -1 & -4 \\ 8 & 7 \end{pmatrix}$ C. $\begin{pmatrix} -4 & -4 \\ 12 & 13 \end{pmatrix}$ D. $\begin{pmatrix} 0 & -4 \\ -8 & -6 \end{pmatrix}$
16. Evaluate the integral$\int_1^2 \left(x^2 + \frac{1}{x}\right) dx$A. $\frac{8}{3} + In2$ B. $\frac{7}{3} + In2$ C. $\frac{7}{3} - In37/3$ – In3 D. $\frac{8}{3}$
17. If the distance covered by a body in time t seconds is s = t^2 – 6t^2 + 5t, what is its initial velocity? A. 0ms^{-1} B. -4ms^{-1} C. (3t^2 – 12t + 5)ms^{-1} D. 5ms^{-1}

Suppose D, E and P are subset of a universal ∪. Let U be the set of natural numbers not greater than 10, while D, E and P are respectively the set of odd numbers, even number and prime numbers. For any set X, its compliment is denoted by X^1 and ∅ denote the empty set.

18. Display the set D∩P A. {3,5,7}B. {2} C. {4,6,8,10} D. {2,3,5,7}
19. Find D∩E A. {2} B. {3,5}C. {1,2,3,4,5,6,7,8,9,10} D. ∅
20. The trigonometric expression cos2A + Sin2A can be written as A. cosA (CosA-sinA) B. cos2A + sin2A–2sinAcosA C. 2sinAcosA+ cos^{2}A + sin^{2}A D. cos^{2}A + sin^{2}A – 2sinAcosA

A bag contains 10 balls of which 3 are red and 7 are white. Two balls are drawn at random. Find the probability of none of the balls is red if the draw is

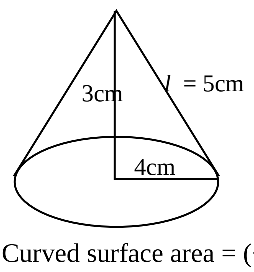
21. With replacement A. 0.9 B. 1 C. 0.4 D. 0.49
22. Without replacement A. 0.1 B. 0.47 C. 0.42 D. 0.21
23. In a throw of a fair die, the probability of obtaining an even number is A. 1 B. $\frac{3}{2}$ C. $\frac{1}{6}$ D. $\frac{2}{3}$
24. Two fair coins are tossed simultaneously. What is the probability of obtaining at least 1 tail turns up? A. $\frac{1}{4}$ B. $\frac{3}{4}$ C.$\frac{1}{2}$ D. 1

Let α and β be the roots of the equation $x^2 - 5x + 4 = 0$

25. Find the values of $\frac{1}{\alpha} - \frac{1}{\beta}$ A. $\pm\frac{4}{3}$ B. $\frac{3}{4}$ C. $\pm\frac{3}{4}$ D. $\frac{1}{5}$
26. A regular polygon has each of it angles as 160^0. What is the number of sides of the polygon? A. 18 B. 36 C. 9 D. 20
27. One angle of an octagon is 100^0 while the other sides are equal. Find each of these exterior angles. A. 80^0 B. 60^0 C. 140^0 D. 40^0

MATHEMATICS 2008 SOLUTION

1. It is important to know the following identities:
(i) a^2 – b^2 = (a + b)(a – b)
(ii) a^3 – b^3 = (a – b)(a^2 + ab + b^2)
(iii) a^3 + b^3 (a + b)(a^2 – ab + b^2)
The correct option is B
2. 16(3x + 2y)^2 – 25(a + 2b)^2
4^2(3x + 2y)^2 – 5^2(a + 2b)^2
[4(3x + 2y)]^2 – [5(a + 2b)]^2
At this point, the knowledge of the difference of two squares is employed: a^2 – b^2 = (a + b) (a – b)
∴ [4(3x + 2y)]^2 – [5(a + 2b)]^2
= [4(3x + 2y)] + [5(a + b)](4(3x + 2y) – 5(a + 2b)]
(12x + 8y + 5a + 10b)(12x + 8y – 5a – 10b)
The correct option is C
3. r = 4m, h = 3m
The curved surface area = π r*l*
Where *l* = slant height
The slant height must be found first



$l^2 = 3^2 + 4^2$
$l^2 = 9 + 16 = 25$
$l = 5$cm

Curved surface area = (π × 4 × 5)cm^2 = 20πcm^2
The correct option is B
4. Volume of a cylinder = πr^{2}h
π = 3.142, r = 5cm, h = 4cm

$V = 3.142 \times 5^2 \times 4$
$= 314.2cm^3$
The correct option is A

5. $3\log x + \log y = 3$
$3\log_{10}x + \log_{10}y = 3$
$\log_{10}x^3 + \log_{10}y = 3$
$\log_{10}(x^3y) = 3$
Then,
$x^3y = 10^3$
$y = \frac{10^3}{x^3} = \left(\frac{10}{x}\right)^3$
The correct option is D

6. $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$
$\frac{y^2}{b^2} = \frac{x^2}{a^2} - 1$
$\frac{y^2}{b^2} = \frac{x^2 - a^2}{a^2}$
$y^2 = \frac{b^2(x^2 - a^2)}{a^2}$
$y = \pm\sqrt{\frac{b^2(x^2 - a^2)}{a^2}}$
$y = \pm\sqrt{\frac{b^2}{a^2}(x^2 - a^2)}$
$y = \pm\frac{b}{a}\sqrt{x^2 - a^2}$
The correct option is B

7. Distance = speed × time
While riding, distance, $d_1 = \left(x \times \frac{1}{2}\right)$km
While pushing the bike, distance, $d_2 = (x - 5) \times 2 = 2x - 10$
Total distance = $d_1 + d_2$
$= \frac{x}{2} + 2x - 10$
$= \frac{x + 4x - 20}{2} = \frac{5x - 20}{2}$
Since the total distance is less than 60km,
$\frac{5x-20}{2} < 60$
$5x - 20 < 120$
$5x < 140$
$x < \frac{140}{5}$
$\therefore\ x < 28$
The correct option is C

8. $ax^2 + bx$
When x = 1, the expression equals 6
Then,
$a(1)^2 + b(1) = 6$
$a + b = 6$ ………………………… (i)
When x = 2, its value is 10
Then,
$a(2)^2 + b(2) = 10$
$4a + 2b = 10$
$2a + b = 5$ ………………………… (ii)
Subtract equation (i) from equation (ii)
$2a – a = 5 – 6$
$a = - 1$
Put a = -1 in equation (i) to get b
$- 1 + b = 6$, $b = 6 + 1 = 7$
We can now evaluate the expression when x = 5
$ax^2 + bx = (-1)(5)^2 + 7(5)$
$= - 25 + 35$
$= 10$
The correct option is A

9. By remainder theorem, the remainder obtained when $2x^2 – x^2 – 5x + 1$ is divided by x + 3 is $f(-3)$
Let $f(x) = 2x^3 – x^2 – 5x + 1$
Then,
$f(-3) = 2(-3)^2 – (-3)^2 – 5(- 3) + 1$
$= -54 – 9 + 15 + 1$
$= -47$
Method 2
$2x^2 - 7x + 16$
$x + 3\sqrt{2x^3 - x^2 - 5x + 1}$
$(-)\ \underline{2x^3 + 6x^2}$
$(-)\ -7x^2 - 5x$
$\underline{-7x^2 - 21x}$
$(-)\ 16x + 1$
$\underline{16x + 48}$
-47
The correct option is D

10. $f(x) = 2x^3 - 3x^2 - 5x + 6$
Since x – 1 is a factor, the other factors can be obtained by dividing the function by x – 1
$2x^2 - x + 6$
$x + 1\sqrt{2x^3 - x^2 - 5x + 1}$
$(-)\ \underline{2x^3 + 2x^2}$
$(-)\ -x^2 - 5x$
$\underline{-x^2 - x}$
$(-)\ -6x + 6$
$\underline{-x + 6}$
$0 \quad 0$
Thus, $2x^3 – 3x^2 – 5x + 6 = (x – 1)(2x^2 – x – 6)$
$2x^2 – x – 6 = (x – 2)(2x = 3)$
Then,
$2x^3 – 3x^2 – 5x + 6 = (x – 1)(x – 2)(2x + 3)$
The three factors x – 1, x – 2 and 2x + 3 poles of the function are 1, 2, and $\frac{-3}{2}$ (obtained by equating each factor to zero)
The correct option is B

11. Let the two numbers be x and y
$x – y = 10$ …………….. (i)
$xy = 39$ ……………….. (ii)
$x = 10 + y$ …………….. (iii)
Substitute (iii) into (ii)
$(10 + y)y = 39$
$10y + y^2 = 39$
$y^2 + 10y – 39 = 0$
$y^2 – 3y + 13y – 39 = 0$
$y(y – 3) + 13(y – 3) = 0$
$(y + 13)(y – 3) = 0$
Either y + 13 = 0 or y – 3 = 0
y = - 3 or 3
When y = -13
x = 10 + (-13) = -3
When y = 3
x = 10 + 3 = 13
Thus, the two numbers are either – 13 and – 3 or 3 and 13
The correct option is D

12. Average age = 14years, 2 months = 170month
$\frac{Total\ age}{x} = 170$
Total age = 170x

When a pupil of 15years, 20months (182 months) join, the total age will be 170x + 182

New average = $\frac{170x+182}{x+1}$

It is said that new average age increased by 1. This means that

$\frac{170x+182}{x+1} = 170 + 1$

$\frac{170x+182}{x+1} = 171$

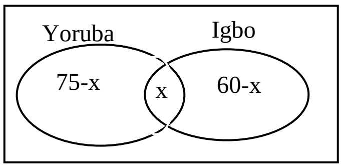
170x + 182 = 171x + 171

171x - 170x = 182 – 171

x = 11

The correct option is C

13.



75 – x + x + 60 – x = 120

135 – x = 120

x = 135 – 120 = 15

The correct option is C

14. Igbo only = 60 – x

= 60 – 15 = 45

The correct option is A

15. $A = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix} \quad B = \begin{pmatrix} 0 & 2 \\ 4 & 3 \end{pmatrix}$

$A^2 = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix}\begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} -1 & -4 \\ 8 & 7 \end{pmatrix}$

$AB = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix}\begin{pmatrix} 0 & 2 \\ 4 & 3 \end{pmatrix} = \begin{pmatrix} -4 & -1 \\ 12 & 13 \end{pmatrix}$

$2A = 2\begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} 2 & -2 \\ 4 & 6 \end{pmatrix}$

$A^2 + AB - 2A = \begin{pmatrix} -1 & -4 \\ 8 & 7 \end{pmatrix} + \begin{pmatrix} -4 & -1 \\ 12 & 13 \end{pmatrix} - \begin{pmatrix} 2 & -2 \\ 4 & 6 \end{pmatrix} = \begin{pmatrix} -7 & -3 \\ 16 & 14 \end{pmatrix}$

No Suitable option

16. $\int_1^2 \left(x^2 + \frac{1}{x}\right) dx$

$= \left[\frac{x^3}{3} + In x + c\right]_1^2$

$= \left[\frac{2^3}{3} + In 2 + c\right] - \left[\frac{1^3}{3} + In\ 1 + c\right]$

$= \left(\frac{8}{3} + In 2 + c\right) - \left(\frac{1}{3} + 0 + c\right)$

$= \frac{8}{3} - \frac{1}{3} + In\ 2$

$= \frac{7}{3} + In\ 2$

The correct option is B

17. S = t^2 – $6t^2$ + 5t

$v = \frac{ds}{dt} = 2t - 12t + 5$

The velocity at t = 0 is the initial velocity

= 2(0) – 12(0) + 5

$V_{initial}$ = (0 – 0 + 5)m/s

= 5m/s

The correct option is D

18. U = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}

D = {1, 3, 5, 7, 9}

E = {2, 4, 6, 8, 10}

P = {2, 3, 5, 7}

D ∩ P = {3, 5, 7}

The correct option is A

19. D ∩ E = { } or ϕ

The correct option is D

20. Cos2A = cos(A + A) = $2\cos^2$A – 1

Sin2A = sin(A + A) = 2sinAcosA

Cos(A + A) + sin(A + A) = $\cos^2$A – $\sin^2$A + 2sinAcosA

No suitable option

21. Probability of none of the ball is red = probability of both is white (since there are only two colours involved)

P(white)(with replacement) = $\frac{7}{10} \times \frac{7}{10} = 0.49$

The correct option is D

22. P(white)(without replacement) = $\frac{7}{10} \times \frac{6}{10} = 0.47$

The correct option is B

23. Chances: 1, 2, 3, 4, 5, or 6 = 6

Even chances: 2, 4, or 6 = 3

P(even) = $\frac{n(even)}{n(total)} = \frac{3}{6} = \frac{1}{2}$

No suitable option

24.

	T	H
T	TT	TH
H	HT	HH

At least one tail = TT, TH, HT

P(at least one tail) = $\frac{3}{4}$

The correct option is B

25. x^2 – 5x + 4 = 0

You can employ the knowledge of the relationship between the roots of a quadratic equation.

More easily, you can find the two roots and carryout the operation as demanded.

x^2 – 5x + 4 = 0

x^2 – x – 4x + 4 = 0

x(x – 1) – 4(x – 1) = 0

(x – 4)(x – 1) = 0

x – 4 = 0 or x – 1 = 0

x = 4 or 1

If α = 4 and β = 1

$\frac{1}{\alpha} - \frac{1}{\beta} = \frac{1}{4} - \frac{1}{1} = \frac{1-4}{4} = \frac{-3}{4}$

If α = 1 and β = 4

$\frac{1}{\alpha} - \frac{1}{\beta} = \frac{1}{1} - \frac{1}{4} = \frac{4-1}{4} = \frac{3}{4}$

Hence, $\frac{1}{\alpha} - \frac{1}{\beta} = \pm\frac{3}{4}$

The correct option is C

26. Each interior angle of a regular polygon is $\frac{(n-2)180}{n}$, (where n is the number of sides)

Thus,

$\frac{(n-2)180}{n} = 160$

180n – 360 = 160n

180n – 160n = 360

20n = 360

$n = \frac{360}{20}$ = 18

The correct option is A

27. Sum of the interior angles = (n – 2)180

For octagon, n = 8

Sum = (8 – 2)180 = 6 × 180 = 1080°

Since one of the angles is 100°, the seven remaining angles = 1080° – 100° = 980°

Each of the seven remaining angles = $\frac{980}{7}$ = 140°

Interior angle + exterior angle = 180°

Each exterior angle = 180° – 140° = 40°

The correct option is D

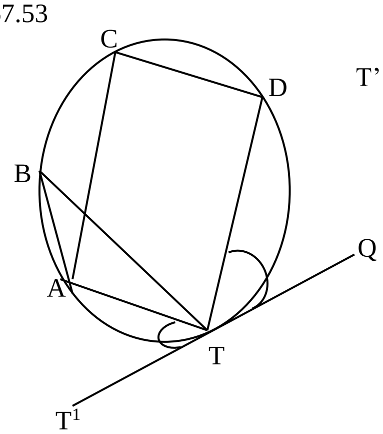
MATHEMATICS 2007 QUESTIONS

1. The interior angles of a pentagon are: 180^0, 80^0, $78^0$and x. The value of x is: A. 75^0 B. 108^0 C. 120^0D. 134^0
2. All the vertices of an isosceles triangle lie on a circle and each of the base angles of the triangle is 65^0. The angle subtended at the centre of the circle by the base of the triangle is? A. 130^0 B. 115^0 C. 100^0D. 65^0
3. A square tile measures 20cm by 20cm. How many of such tiles will cover a floor measuring 5m by 4m? A. 500 B. 400 C. 320 D. 250
4. The volume of a certain sphere is numerically equal to twice its surface area. The diameter of the sphere is: A. 6 B. 9 C. 12 D. $\sqrt{6}$
5. A bearing of 310^0, expressed as a compass bearing is: A. $N50^0W$ B. $N40^0W$ C. $S40^0W$ D. $S50^0W$.
6. Which of the following specified sets of data is not necessarily sufficient for the construction of a triangle? A. three angles B. two sides and a right angle C. two sides and an included angle D. three sides
7. The average age of the three children in a family is 9 years. If the average age of their parent is 39 years, the average age of the whole family is: A. 20 years B. 21 years C. 24 years D. 27 years.
8. Simplify$1 + \frac{2}{3} - 3 \div \left(1 + \frac{2}{3}\ of\ \frac{6}{7}\right)$ A. $-\frac{8}{33}$B. $\frac{21}{11}$ C. $\frac{33}{21}$D. $-\frac{21}{8}$
9. If $1 + \frac{1}{1+\frac{1}{1+\frac{1}{x}}} = 5$ find xA. $\frac{3}{7}$ B. $\frac{7}{3}$ C. $-\frac{3}{7}$ D. $-\frac{7}{3}$
10. Evaluate x in base 3 if $41_x - 22_x = 17_x$. A. 11 B. 8 C. 12 D. 22
11. A woman buys 4 bags of rice for Ł56 per bag and 3 bags of beans for Ł26 per bag using the currency “LONI” (Ł) in base 7. What is the total cost of the items in another currency “MONI” (M̶) in base 8? A. M224 B. M114 C. M340 D. M440
12. When the price of egg was raised by ₦2 an egg, the number of eggs which can be bought for ₦120 is reduced by 5. The present price of an egg is A. ₦6 B. ₦7 C. ₦8 D. ₦10
13. How long will it take a sum of money invested at 8% simple interest to double the original sum? A. 8 years B.10.5 years C. 12 years D. 12.5 years
14. The journey from Lagos to Ibadan usually takes a motorist 1 hour 30 minutes. By increasing his average speed by 20km/hr, the motorist saves 15 minutes. His usual speed, in km/hr is A. 100 B. 90 C. 85 D. 80
15. The smallest section of a rod which can be cut into exactly equal sections, each of either 30cm or 36cm in length is A. 90cm B. 180cm C. 360cm D. 540cm
16. If x = 0.0012 + 0.00074 + 0.003174, what is the difference between x to 2 decimal places and x to 1 significant figure? A. 0.01 B. 0.0051 C. 0.1 D. 0.005
17. The angle of depression of two points A and B on a plane field from the top of a mast erected between A and B are 30^0 and 45^0 respectively. If A is westward of B, find /AB/ if the height of the mast is 15m from the field A. $15\sqrt{3}m$ B. $5(3 + \sqrt{3})m$ C. $15(1 + \sqrt{3})m$ D. $15(\sqrt{3} - 1)$
18. The radius of a circle is given as 10cm subject to an error of 0.2cm. The error in the area of the circle is A. $\frac{1}{4}$ %B. $\frac{1}{50}$% C. 2% D. 4%
19. If θ is acute, evaluate$\frac{cos(90-\theta)+sin(180-\theta)}{Cos(180-\theta) - sin(90-\theta)}$A. tanθ B. -tanθ C. cotθ D. -cotθ
20. In a survey of 100 students in an institution, 80 students speak Yoruba, 22 speak Igbo, while 6 speak neither Igbo nor Yoruba. How many students speak Yoruba and Igbo? A. 96 B. 8 C. 64 D. 12
21. A bag contains 5 yellow balls, 6 green balls and 9 black balls. A ball is drawn from the bag. What is the probability that it is a black or yellow ball? A.$\frac{37}{160}$ B.$\frac{133}{400}$ C.$\frac{77}{800}$D. $\frac{233}{180}$

The table below shows the distribution of weight measure for 100 students.

Weight (kg)	60-62	63-65	66-68	69-71	72-74
F	5	18	42	27	8

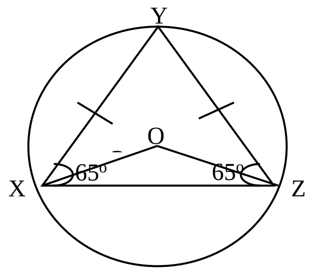
22. Calculate the mean of the distribution to two decimal places A. 64.45 B. 62.45 C. 67.45 D. 65.45
23. Calculate the mode of the distribution to two decimal places A. 67.33 B. 65.33 C. 65.53 D. 67.53
24.



T^1 Q is a tangent to the circle ABCDT, and DTQ = 40^0, angle ATT^1 = 30^0, then angle ATD is A. 70^0B. 90^0 C. 250^0 D. 110^0

MATHEMATICS 2007 SOLUTION

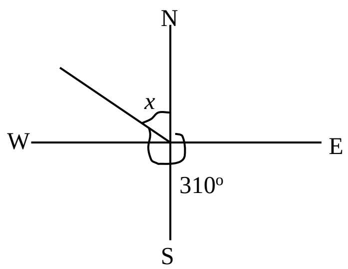
1. Sum of the interior angles = $180^o + 118^o + 78^o + x + 80^o$
$= 456^o + x$
The sum of the interior angles of a pentagon (5 sided polygon) is $(5 - 2)180 = 3 \times 180 = 540^o$
Hence,
$456^o + x = 540^o$
$x = 540^o - 456^o$
$= 84^o$
No suitable option

2.


YXZ + XZY + ZYX = 180°
(sum of the angles in a triangle)
$65^o + 65^o + ZYX = 180^o$
$130^o + ZYX = 180^o$
$ZYX = 180^o - 130^o = 50^o$
ZOX = 2 ×ZYX (angle at the centre of a circle is twice the angle at the circumference)
$= 2 \times 50^o$
$= 100^o$
The correct option is C

3. Side of tile = 20cm = 0.2m
Area of tile = $(0.2m)^2 = 0.04m^2$
Area of floor = 5m × 4m = $20m^2$
Number of tiles = $\frac{Area\ of\ floor}{Area\ of\ tile} = \frac{20m^2}{0.04m^2} = 500$
The correct option is A

4. Volume of a sphere = $\frac{4}{3}\pi r^3$
Area of a sphere = $4\pi r^2$
Since the volume is twice the area, we can write
$\frac{4}{3}\pi r^3 = 2 \times \pi r^2$
$\frac{4}{3}\pi r^3 = 8\pi r^2$
$4\pi r^3 = 24\pi r^2$
Dividing both sides by πr^2, we have
$4r = 24$
$r = \frac{24}{4}$ = 6units
$Diameter = 2r = 2 \times 6units = 12\ units$
The correct option is C

5.


The bearing is Nx^oW
What is x?
$310^o + x = 360^o$ (sum of angles at a point)
$x = 360^o - 310^o = 50^o$
Thus, the bearing is N50^oW
The correct option is A

6. A triangle cannot be constructed (with details known) if only the angles are given
The correct option is A

7. Let the ages of the children be x, y, and z
Average age = $\frac{x+y+z}{3} = 9$
Then, $x + y + z = 3 \times 9 = 27$
Let the ages of the parents be v and w
Average age of parents = $\frac{v+w}{2} = 39$
$v + w = 2 \times 39 = 78$
Average age of the family = $\frac{v+w+x+y+z}{5} = \frac{78+27}{5}$
= 21 years
The correct option is B

8. $1 + \frac{2}{3} - 3 \div \left(1 + \frac{2}{3}\ of\ \frac{6}{7}\right)$
$= \frac{3+2}{3} - 3 \div \left(1 + \frac{2}{3} \times \frac{6}{7}\right)$
$= \frac{5}{3} - 3 \div \left(1 + \frac{4}{7}\right)$
$= \frac{5}{3} - 3 \div \left(\frac{11}{7}\right)$
$= \frac{5}{3} - 3 \div \frac{11}{7}$
$= \frac{5}{3} - 3 \times \frac{7}{11}$
$= \frac{5}{3} - \frac{21}{11}$
$= \frac{-8}{33}$
The correct option is A

9. $1 + \frac{1}{1+\frac{1}{1+\frac{1}{x}}} = 5$
$1 + \frac{1}{1+\frac{1}{\frac{x+1}{x}}} = 5$
$1 + \frac{1}{1+\frac{x}{x+1}} = 5$
$1 + \frac{1}{\frac{2x+1}{x+1}} = 5$
$1 + \frac{x+1}{2x+1} = 5$
$= \frac{2x+1+x+1}{2x+1} = 5$
$= \frac{3x+2}{2x+1} = 5$
$3x + 2 = 5(2x + 1)$
$3x + 2 = 10x + 5$
$7x = -3$, $x = \frac{-3}{7}$
The correct option is C

10. $41_x - 22_x = 17_x$
Let us carry out the operation in base 10
$4^11^0{}_x = 4 \times x^1 + 4 \times x^0 = 4x + 1$
$2^12^0{}_x = 2 \times x^1 + 2 \times x^0 = 2x + 2$
$1^17^0{}_x = 1 \times x^1 + 7 \times x^o = x + 7$
Hence,
$41_x - 22_x = 17_x$ can be written as
$(4x + 1) - (2x + 2) = x + 7$
$4x + 1 - 2x - 2 = x + 7$
$2x - 1 = x + 7$
$2x - x = 7 + 1$
$x = 8$
Is this our final answer?
No! The value of x is required in base 3. Let's quickly do the conversion

3	8
3	2 r 2
	0 r 2

$= 22_3$

The correct option is D

11. We are to first get the total cost in LONI. We can then covert the amount to MONI. Please keep the base in mind.
Cost of rice = $56_7 \times 4 = 323_7$
Cost of beans = $26_7 \times 3 = 114_7$
Total cost = $323_7 + 114_7 = 440_7$
This is the cost in LONI.
But we are to convert to MONI which is the base 8.
This simply means we should convert 440_7 to a number in base 8. We first need to get the number in base 10
$4^2 4^1 0^0{}_7 = 4 \times 7^2 + 4 \times 7^1 + 0 \times 7^0$
$= 196 + 28 + 0 = 224_{10}$
Good! We can now convert this to base 8

8	224
8	28 r 0
8	3 r 4
	0 r 3

$= 340_8$

Then, the amount can be written as M340
The correct option is C

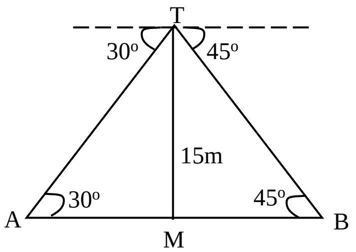
12. Let the original price of egg be x
As at then, number of eggs that can be bought with ₦120 equaled $\frac{120}{x}$
The new price of egg = ₦(x + 2)
The number of egg that can be bought now equals $\frac{120}{x} - 5$
Total price = price per one × number of eggs
$120 = x + 2 \times \left(\frac{120}{x} - 5\right)$
$120 = x + 2 \left(\frac{120-5x}{x}\right)$
$120 = \frac{(x+2)(120-5x)}{x}$
$120x = 120x - 5x^2 + 240 - 10x$
Then,
$0 = -5x^2 + 240 - 10x$
$0 = -5x^2 - 10x + 240$
Multiplying both sides by -1, we have:
$5x^2 + 10x - 240 = 0$
Dividing through by 5, we have
$x^2 + 2x - 48 = 0$
$x^2 + 8x - 6x - 48 = 0$
$x(x + 8) - 6(x + 8) = 0$
$(x - 6)(x + 8) = 0$
$x - 6 = 0$ or $x + 8 = 0$
$x = 6$ or -8
Since price cannot be negative, x = ₦6
Is this the final answer? No! Recall that x is not the present price but the original price
New price = x + 2 = ₦(6 + 2) = ₦8
The correct option is C

13. If the original principal is doubled, it means that the interest equals the principal i.e. I = P
$I = \frac{PRT}{100} = \frac{P \times 8 \times T}{100}$
$I = \frac{8PT}{100}$
But I = P, Hence:
$P = \frac{8PT}{100}$
8PT = 100P
$T = \frac{100P}{8P} = 12.5$years
The correct option is D

14. Distance = speed × time
Let the usual speed be x
Time taken at x = 1hr, 30mins = 1.5hr
New speed = x + 20
New time taken = 1hr, 30mins minus 15minues = 1hr, 15min = 1.25hr.
Since the distance is the same, we can write:
Usual speed × usual time = New speed × new time
i.e. $1.5x = 1.25(x + 20)$
$1.5x = 1.25x = 25$
$1.5x - 1.25x = 25$
$0.25x = 25$
$x = \frac{25}{0.25} = 100$
Therefore, the usual speed is 100km/hr
The correct option is A

15. This is a periphrastic way of demanding the LCM of 30 and 36
$30 = 2 \times 3 \times 5$
$36 = 2 \times 3 \times 3$
LCM = $2 \times 2 \times 3 \times 5 \times 3 = 180$
Hence, the smallest section of a rod which can be cut into exactly equal sections, each of either 30cm or 36cm is 180cm
The correct option is B

16. x = 0.0012 + 0.0074 + 0.003174 = 0.005114
To 2 d.p, x = 0.01
To 1 s.f, x = 0.005
Difference = 0.01 – 0.005 = 0.005
The correct option is D

17.


From the diagram,
$\tan 30^o = \frac{15}{AM}$
$AM = \frac{15}{tan30^o} = \frac{15}{\frac{1}{\sqrt{3}}} = 15\sqrt{3}$
$tan45^0 = \frac{15}{MB}$
$MB = \frac{15}{tan45^o} = \frac{15}{1} = 15$
AB = AM + MB
$= 15\sqrt{3} + 15$
$= 15(\sqrt{3} + 1)$
$= 15(1 + \sqrt{3})$
The correct option is C

18. r = 10cm, error = 0.2cm
Actual area = $\pi r^2 = \pi \times 10^2 = 100\pi cm^2$

With an error of 0.2cm, the radius is either (10cm + 0.2cm) or (10cm – 0.2cm) = 10.2cm or 9.8cm

Error area = $\pi(10.2)^2cm^2 = 104.04\pi cm^2$

Or $\pi(9.8)^2cm^2 = 96.04\pi cm^2$

Error = $(104.04 - 100)\pi cm^2 = 4.04\pi cm^2$

Or $(100 - 96.04)\pi cm^2 = 3.96\pi cm^3$

% error = $\frac{error}{actual\ value} \times 100\%$

$= \frac{4.0\pi cm^2}{100\pi cm^2} \times 100\% = 4.04\%$

Or $\frac{3.96\pi cm^2}{100\pi cm^2} \times 100\% = 3.96\%$

The correct option is D

19. $\frac{\cos(90-\theta)+\sin(180-\theta}{\cos(180-\theta)-\sin(90-\theta)}$

Since θ is acute, sin(180 - θ) = sinθ

Also, cosα = sin(90 - α)

Therefore, cos(90 - θ) = sin θ

Cos180 - θ = -cosθ

Sin(90 - θ) = cosθ

Then, we can write

$\frac{\sin\theta+\sin\theta}{-\cos\theta-\cos\theta} = \frac{2\sin\theta}{-2\cos\theta} = \left(\frac{2}{-2}\right)\frac{\sin\theta}{\cos\theta}$

- 1 × tanθ = -tanθ

The correct option is B

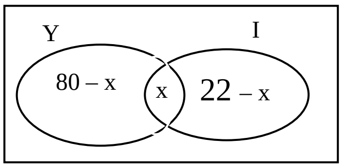
20. $\in = 100$

$\cap(Y) = 80$

$\cap(I) = 22$

$\cap(I \cup Y)' = 6$

Let $(I \cup Y) = x$



80 – x + x + 22 – x + 6 = 100

108 – x = 100

x = 8

The correct option is B

21. Total number of balls = 20

P(black or yellow) = P(black) + P(yellow)

$P(black) = \frac{Number\ of\ black\ ball}{total} = \frac{9}{20}$

$P(yellow) = \frac{Number\ of\ yellow\ ball}{total} = \frac{5}{20}$

$P(black\ or\ yellow) = \frac{9}{20} + \frac{5}{20} = \frac{14}{20} = \frac{7}{10}$

22.

Weight in kg	Average (X)	Frequency (f)	fx
60 – 62	61	5	305
63 – 65	64	18	1152
66 – 68	67	42	2814
69 – 71	70	27	1890
72 – 74	73	8	584

$Mean\ (\bar{X}) = \frac{\sum fx}{\sum f}$

$\sum fx$ = 305 + 1152 + 2814 + 1890 + 584 = 6745

$\sum f$ = 5 + 18 + 42 + 27 + 8 = 100

$\bar{X} = \frac{6745}{100} = 67.45$

The correct option is C

23. The modal class is 66 – 68

Since it has the highest frequency. Don't forget that this is a range and not the mode itself. Whenever group data is involved, the mode is given by $L_1 + \left(\frac{f_x}{f_x + f_y}\right)c$

Where

L_1 = Lower boundary of the modal class

f_x = Difference between the frequency of the modal class and that of the class before it.

f_y = Difference between the frequency of the modal class and that of the class after it.

C = Interval of the modal class the

The boundary of the 66 – 68 class is 65.5 – 68.5

Then,

L_1 = 65.5

C = 68.5 – 65.5 = 3

From the table,

f_x = 42 – 18 = 24

f_y = 42 – 27 = 15

Then,

Mode = $65.5 + \left(\frac{14}{24+15}\right)3$

= 65.5 + 1.8461 = 67.35

The correct option is A

24. ATT[1] + DTQ + ATD = 180° (angles in a straight line)

30° + 40° + ATD = 180°

70° + ATD = 180°

ATD = 180° – 70° = 110°

The correct option is D

POST UME EXAM MATHEMATICS 2006 QUESTIONS

1. Solve for p in the following equation given in base two 11(p + 110) = 1001p A. 10 B. 11 C. 110 D. 111
2. Factorize $16(3x+2y)^2 - 25(a+2b)^2$A. (12x+8y +5a+10b((12x+8y-5a-10b)B. (12x+8y-5a-10b)(12x+8y-5a-10b)C. 20(3x+2y – a-2b) (3x+2y+a+2b) D. 20(3x+2y+a+2b)(3x+2y+a+2b)
3. A cone has base radius 4cm and height 3cm, the area of its curved surface is A. $12\pi cm^2$ B. $24\pi cm^2$ C. $20\pi cm^2$D. $15\pi cm^2$
4. Let log y + log 3 = 3. Then y is (a) $\left(\frac{10}{x}\right)^3(b)\left(\frac{x}{10}\right)^3$ (c)$\left(\frac{x}{10}\right)^{-3}$ (d)$\left(\frac{10}{x}\right)^{-\frac{1}{3}}$
5. If$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, then y is A. $\pm\frac{b}{a}\sqrt{x^2 - a^2}$ B. $\frac{a}{b}\sqrt{a^2 - x^2}$C. $\pm\frac{a}{b}\sqrt{x^2 - a^2}$ D. $\pm\frac{b}{a}\sqrt{a^2 - x^2}$
6. A cyclist rode for 30 minutes at xkm/hr, if the total distance covered is less than 60km, what is the range of values for x? A. x < 14 B. x < 20 C. x < 29D. x < 28
7. A business invested a total of ₦200,000 in two companies which paid dividends of 5% and 7% respectively. If he received a total of ₦11,600 as dividend, how much did he invest at 5%? A. ₦160,000 B. ₦140,000 C. ₦120,000 D. ₦80,000

8. In a class, 37 students take at least one of Chemistry, Economics and Government, 8 students take Chemistry, 19 takes Economics and 25 take Government. 12 students take Economics and Government but nobody take Chemistry and Economics. How many students take both Chemistry and Government? A. 3 B. 4 C. 5 D. 6

9. Z is partly constant and partly varies inversely as the square of d, when d = 1, z = 11 and when d = 2, z = 5. Find the value of z when d = 4. A. 2 B. 3.5 C. 5 D. 5.5

10. Expand the expression $(x^2-2x-3)(x^2+x+1)$ A. x^4-4x^2-5x-3 B. $-x^3-4x^2+5x-3$ C. $x^4-x^3-4x^2-5x-3$ D. x^4-4x^2-5x-3

Suppose we have matrices $A = \begin{pmatrix}1 & -1\\ 2 & 3\end{pmatrix}$ and $B = \begin{pmatrix}0 & 2\\ 4 & 3\end{pmatrix}$

11. Find $A^2+AB-2A$ A. $\begin{pmatrix}-5 & -9\\ 12 & 14\end{pmatrix}$ B. $\begin{pmatrix}-1 & -4\\ 8 & 7\end{pmatrix}$ C. $\begin{pmatrix}-4 & -4\\ 12 & 13\end{pmatrix}$ D. $\begin{pmatrix}0 & -4\\ -8 & -6\end{pmatrix}$

12. The inverse of matrix B isA.$\frac{1}{8}\begin{pmatrix}-3 & 2\\ 4 & 0\end{pmatrix}$ B. $\begin{pmatrix}-3 & 2\\ 4 & 0\end{pmatrix}$ C. $\frac{1}{8}\begin{pmatrix}3 & -4\\ -2 & 0\end{pmatrix}$ D. $\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$

13. The indefinite integral of the function f(x) = x cos x for any constant k, is (a) -x cos x + sin x + k (b) x sin x – cos x (c) x sin x + cos x + k (d) x + sin x + cos + k

14. Evaluate the integral$\int_1^2 \left(x^2 + \frac{1}{x}\right)dx$ A. $\frac{8}{3} + In2$B.$\frac{7}{3} + In2$C.$\frac{7}{3} - In3$ D.$\frac{8}{3}$

15. The trigonometric expression cos2A + sin2A can be written as A. cosA(cosA-sinA) B. $\cos^2A+\sin^2A$ – 2sinAcosA C. 2sinAcosA + $\cos^2A + \sin^2A$ D. $\cos^2A + \sin^2A$ – 2 sin AcosA

Suppose D, E and P are subsets of a universal set U. Let U be the set of natural numbers not greater than 10, while D, E and P are respectively the set of odd numbers, even number and prime number. For any set X, its complement is denoted by X′ and ϕ denote the empty set

16. Display the set D ∩P A. {3,5,7} B. {2} C. {4,6,8,10} D. {2,3,5,7}

17. Find D∩E, A. {2} B. {3,5}C. {1,2,3,4,5,6,7,8,9,10} D. ϕ

A bag contains 10 balls of which 3 are red and 7 are white. Two balls are drawn at random. Find the probability of none of the balls is red, if the draw is

18. With replacement: A. 0.49 B. 1 C. 0.47 D. 0.21

19. Without replacement A. 0.1 B. 047 C. 0.47 D. 0.21

20. A regular polygon has each of it angles as 160^0. What is the number of sides of the polygon? A. 36 B. 9 C. 18 D. 20

21. A girl walks 30m from a point P on a bearing of 40^0 to a point θ, she then walks 30m on a bearing of 140^0 to a point R. The bearing R from O is A. 90^0 B. 50^0 C. 45^0 D. 40^0

22. How many different three digit numbers can be formed using the integers 1 to 6 if no integer occurs twice in a number? A. 720 B. 1260 C. 2520 D. 5040

23. In how many different ways can be letters of the word GEOLOGY be arranged in order? A. 720 B. 1260 C. 2520 D. 5040

MATHEMATICS 2006 SOLUTION

1. 11(P + 110) = 1001P

Since the numbers are in base two, all our operations must be done in base two.

$11 \times P = 11P_2$

		1	1	0
		×	1	1
		1	1	0
	1	1	0	
1	0	0	0	0_2

Then, 11(P + 110) = 11P + 10010

Thus,

11P + 10010 = 1001P

10010 = 1001P – 11P (The subtraction must be done in base two)

10010 = 110P

$P = 10010_2 \div 110_2$

At this juncture, two alternatives are available

We can use long division to get the value of P. We can also convert the two numbers to base ten, divide them and then get the answer down to base two.

Let us do it the second way

$1^40^30^31^10^0{}_2 = 1 \times 2^4 + 0\times2^3+0\times2^2+1\times2^1+0\times2^0$

$= 16 + 0 + 0 + 2 + 0 = 18_{ten}$

$1^21^10^0{}_2 = 1\times2^2 + 1\times2^1 + 0\times2^0$

$= 4 + 2 + 0 = 6_{ten}$

$\frac{10010_2}{110_2} = \frac{18_{ten}}{6_{ten}} = 3_{ten}$

Let us now convert to base 2

2	6
2	3 r 0
2	1 r 1
	0 r 1

$= 110_2$

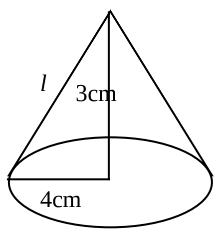
Thus P = 110

The correct option is C

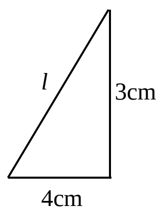
2. This is the same as question 2, 2008.

The correct option is A

3.

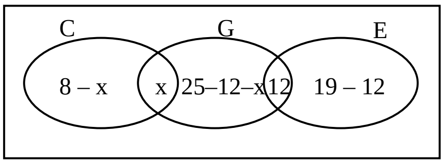


From the diagram



Using pythagora's theorem
$l^2 = 4^2 + 3^2$
= 16 + 9 = 25
$l = \sqrt{25} = 5cm$
Curved surface area of a cone
= $\pi r l$ = $(\pi \times 4 \times 5)cm^2 = 20\pi cm^2$
The correct option is C

4. This is the same as question 5, 2008.
The correct options is A
5. This is the same as question 6, 2008.
The correct option is A
6. This is the same as question 7, 2008.
The correct option is D
7. This is the same as question 19, 2010.
The correct option is C
8. $\in = 37$
$\cap$(C) = 8
$\cap$(E) = 19
$\cap$(G) = 25
$\cap$(E $\cap$ G) = 12
$\cap$(C $\cap$ G) = 0
Let $\cap$(C $\cap$ G) = x
Let us now put all these together in a venn diagram. Caution must be taken while drawing as Economies and Chemistry must not overlap, since nobody is taking both



8 –x + x + 25 – 12 – x + 12 + 19 – 12 = 37
40 – x = 37
x = 40 – 37 = 3
The correct option is A

9. $z = c + \frac{k}{d^2}$
When d = 1 and z = 11
$11 = c + \frac{k}{1^2} = c + \frac{k}{1}$
Then,
$11 = \frac{c+k}{1}$
$c + k = 11$ ……………………….(i)
When d = 2 and z = 5
$5 = c + \frac{k}{2^2} = c + \frac{k}{4}$
$5 = c + \frac{k}{4}$
$5 = \frac{4c+k}{4}$
Then,
4c + k = 20……………………….(ii)
Let us take equations (i) and (ii) together
c + k = 11 ………… (i)
4c + k = 20……….. (ii)
Let us subtract equation (i) from (ii)
4c – c) + (k – k) = 20 – 11
3c = 9
$c = \frac{9}{3} = 3$
Let us substitute c = 3 into equation (i)
3 + k = 11, k = 11 – 3 = 8
Good! Let us now find z, which d = 4
$z = c + \frac{k}{d^2} = 3 + \frac{8}{4^2} = 3 + \frac{8}{16} = 3 + \frac{1}{2} => 3.5$
The correct option is B

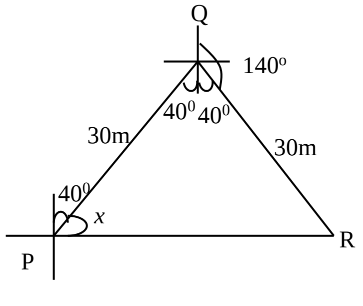
10. $(x^2 - 2x - x)(x^2 + x + 1)$
$= x^4 + x^3 + x^2 - 2x^3 - 2x^2 - 2x - 3x^2 - 3x - 3$
$= x^4 - x^3 = 4x^2 - 5x - 3$
The correct option is C

11. $A = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix} \quad B = \begin{pmatrix} 0 & 2 \\ 4 & 3 \end{pmatrix}$
$A^2 = A \times A = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix}\begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} -1 & -4 \\ 8 & 7 \end{pmatrix}$
$AB = A \times B \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix}\begin{pmatrix} 0 & 2 \\ 4 & 3 \end{pmatrix} = \begin{pmatrix} -4 & -1 \\ 12 & 13 \end{pmatrix}$
$2A = 2 \times \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} 2 & -2 \\ 4 & 6 \end{pmatrix}$
$A^2 + AB - 2A = \begin{pmatrix} -1 & -4 \\ 8 & 7 \end{pmatrix} + \begin{pmatrix} -4 & -1 \\ 12 & 13 \end{pmatrix} - \begin{pmatrix} 2 & -2 \\ 4 & 6 \end{pmatrix} = \begin{pmatrix} -7 & -3 \\ 16 & 14 \end{pmatrix}$

12. $B^{-1} = \frac{AdjB}{|B|}$
$B = \begin{pmatrix} 0 & 2 \\ 4 & 3 \end{pmatrix}$
$AdjB = \begin{pmatrix} 3 & -2 \\ -4 & 0 \end{pmatrix}$
$|B| = (0 \times 3) - (4 \times 2) = 0 - 8 = -8$
$B^{-1} = \frac{AdjB}{|B|} = \frac{\begin{bmatrix} 3 & -2 \\ -4 & 0 \end{bmatrix}}{-8} = \frac{-1}{8}\begin{bmatrix} 3 & -2 \\ -4 & 0 \end{bmatrix}$
$= \frac{1}{8}\begin{bmatrix} -3 & 2 \\ 4 & 0 \end{bmatrix}$
The correct option is A

13. We need $\int xcosx\, dx$
Integrating by part
$\int udv = uv - \int vdu$
Let u = x, $\frac{du}{dx} = 1$
dv = cosx, v = sinx
$\int xcosx\, dx = xsinx - \int sinx\, dx = xsinx + cosx + k$
The correct option is C

14. This is the same as question 16, 2008.
The correct option is B
15. This is the same as question 20, 2008.
16. This is the same as question 18, 2008.
The correct option is A
17. This is the same as question 19, 2008.
The correct option is D
18. This is the same as question 21, 2008.
The correct option is A
19. This is the same as question 22, 2008.
The correct option is B
20. This is the same as question 26, 2008.
The correct option is C
21.



<PQR = 40º + 40º = 80º

Let us obtain the length PR using cosine rule

$PR^2 = PQ^2 + QR^2 - 2(PQ \,.\, QR)\cos P\hat{Q}R$

$= 30^2 + 30^2 - 2(30 \times 30)\cos 80^o$

= 900 + 900 – 312.5667

= 1800 – 312.5667

= 1487.433

$PR = \sqrt{1487.433}$

$= 38.567\text{m}$

We can now get x by using sine rule

$\frac{30}{sinx} = \frac{38.567}{sin80^0}$

$sinx = \frac{30\, sin80^O}{38.567}$

$= 0.766049$

$x = sin^{-1}0.766049 = 50^o$

The bearing of R from Q = 40º + 50º = 90º

The correct option is A

22. The number of three digit numbers that can be formed using integers 1 to 6 is $6P_3$

$= \frac{6!}{(6-3)!} = \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 1} = 120$

The correct option is B

23. GEOLOGY has 7 letters.

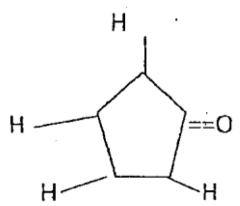
Since G and O occur twice, the number of ways the word can be arranged $= \frac{7!}{2!2!} = \frac{7 \times 6 \times 5 \times 4 \times 3 \times 2}{2 \times 2} = 1260$ways

The correct option is B

CHEMISTRY 2015

1. Blackboard chalk is a mixture of limestone and gypsum. A piece of blackboard chalk weighing 5.42g was immersed in HCI. The CO_2 produced weighed 1.11g. What is the percentage of limestone in the chalk? A 67.0% B 53.3% C 50.8% D 46.5%
2. In the Lassaine's test for the halogens, the sodium fusion extract is acidified and then treated with silver nitrate. A pale yellow precipitate soluble in ammonium hydroxide shows the presence of A fluorine B chlorine C bromine D iodine
3. Given the half-cell reaction, 2Br → Br2, how many moles of electron will be required to produce 0.56 dm^3 of bromine at s.t.p.? [molar volume of gas at s.t.p= 22.4 dm^3]. A 0.05 B 0.10 C 0.20 D 1.00
4. One mole of an alkanol and one mole of an alkanal, each containing n moles of carbon atoms in its molecule, was separately combusted in excess oxygen. The volumes of steam produced by the compounds were in ratio 5:4. What are the respective formulae of the compounds? A C_4H_9OH and C_4H_7CHO B C_4H_9OH and C_3H_7CHO C C_3H_7OH and C_3H_7CHO D C_4H_9OH and C_4H_9CHO
5. The basic tenet of valence bond electron pair repulsiontheory is that the pairs of electrons making the sigma bonds dictate the shape of molecules. The pi-bonds often encounter in some molecules serve to A distort the shape of molecules B after the angle between the atoms in molecules C. shorten the sigma bonds in molecules D. explain the shape of molecules.
6. The equill6rium constant at constant pressure, Kp for the reaction,

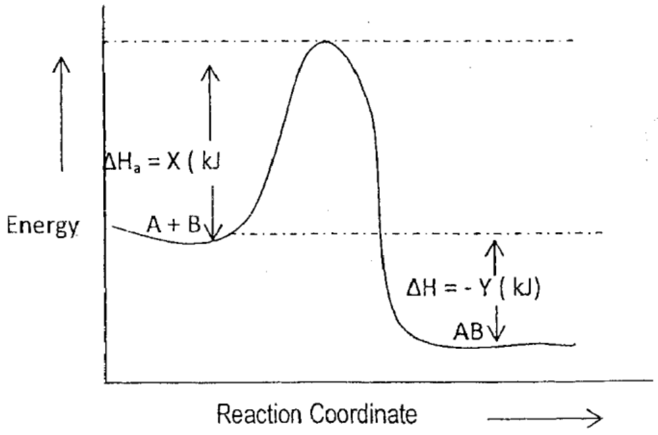
   $2NO_{(g)} + Cl2_{(g)} \leftrightarrow$ 2NOCI(g) is 800. The partial pressures of the gas mixture at equilibrium are 0.02 atm each for NO and Cl_2. What is the partial pressure of NOCI in the mixture? A 0.02 atm. B 6.41×10^{-4} atm. C 0.04 atm. D 0.08 atm.
7. How many sigma (Ѡ) bonds are there in the molecule below:



   A 4 B5 C 8 D 10
8. The enthalpy changes for the atomization of methane and ethane are 1648 and 2820 kJ mol^{-1} respectively. The C — C mean bond dissociation energy in kJ mol^{-1} is therefore A 348 B --348 C 1176 D –1176
9. What volume of 0.750 mol dm^{-3} Na_2CO_3 solution could be diluted to 250 cm^3 to reduce the concentration to 0.025 mol dm^{-3}? A 16.0cm3 B 14.2cm3 C 10.4cm2 D 8.3cm3
10. An organic compound with the molecular of C_4H_80 has the Isomers (I) butanal (ii) cyclobutanol (iii) cyclobutanone(iv)2 –methyipropanal A (i) and (ii) B (i), (ii) and (iv) C (I), (ii) and (iii) D (ii) and (iv)
11. 0.125 mol of magnesium is dropped into 25cm^3 of 4.0 moldrn^{-3} HCI. how many moles of hydrogen gas are evolved? [Mg 24] A 0.050 B 0250 C 0.125 D 0.063
12. Which of the followings is thermally stable?

    A $CuCO_3$ B $Pb(NO_3)_2$ C Na_2NO_3D Na2CO3
13. An atom has a core and outside the core anelectron occupies an orbital for which the principal quantum number, n = 4, I = 0, m 0 and s $+\frac{1}{2}$% or $-\frac{1}{2}$%. The atom is Likely to be A boron B sodium C potassium D fluorine
14. An element B, has two isotopes Band B present in ratio 2:3. The relative atomic mass of B is, A 20.5 B 21.2 C 23.4D 25.0
15. A chloroform solution of pure organic compound was spotted at a distance 0.80 cm from the base of a 20 cm long chromatoplate. If the compound has the Rt value of 0.50 and moves half way up the 20 cm long plate, what is the distance of the solvent front from the top of the plate upon elution? A 0.80cm B 1.0cm C 1.2cm D 1.4cm
16. Platinum electrodes are dipped into copper sulphate solution in a voltameter. The solution left after electrolysis is A clear B blue C puple D pale green

17. The equilibrium constant, K_c for the reaction, $NO_{(g)}$ + $\frac{1}{2}O2_{(g)} \rightarrow NO_{(g)}$, is 35.2. What is the value of K_c for the reaction, $N0_{2(g)}$ $NO_{(g)}$ + $\frac{l}{2}O2_{(g)}$? A 35.2 B 17.6 C $2.84x10^{-2}$D $1.24x10^3$
18. $CH_3CH_2CI + NaOH \rightarrow CH_3CH_2OH + NaCl$
The above reaction is best described will proceed to form the products by a pcocess known as A. nucleophilic addition B. nucleophilic substitution C. electrophilic addition D. electrophilic substitution
19. What quantity of current is required to deposit 2.4g of copper in a period of 750 seconds during an electrolytic deposition process? [Cu = 64, 1F = 96500 coul mol^{-1}] A. 9.65A B. 10.81A C. 12.33A D 15.54 A
20. The standard reduction potentials for the following half-cell reactions are,
$2H_20_{(l)} = O2_{(g)} + 4H^+_{(aq)} + 4e$ E° = -1.23V $2H_20_{2(i)}$ = $2O_{2(g)}$ $+4H^+_{(aq)}$ + 4e E° — 0.68 V
What is the E° for the reaction, $2H_2O_{2(i)}$ = $2H_2O_{2(g)}$ + $0_{2(g)}$ at 298 K? A -0.68 V B. - 1.23 V C + 0.55 V 0 + 1.91 V
21. The energy for the dissociation of molecule AB in kJ in the diagram of energy against the reaction coordinate shown below is,



AX B -Y CX-(-Y) D -(X+Y)
22. Bonding in ammonium chloride is A. ionic, covalent and dative B ionic and covalent C covalent and dative D ionic

2015 OAU POST UTME CHEMISTRY

1. $CaCO_{3(s)} + 2HCl_{(aq)} \rightarrow CaCl_{2(aq)} + H_2O_{(l)} + CO_{2(g)}$
Molar mass of CO_2 = 44gmol^{-1}, Molar mass of $CaCO_3$ = 100gmol^{-1}
Mass of CO_2 = $\frac{1.11g}{44gmol^{-1}}$ 0.02523
From the balanced equation of reaction, 1mol of CO_2 is produced from 1mol of $CaCO_3$. Hence, 0.02523mol of CO_2 will be produced from 0.02523mol of $CaCO_3$
$\therefore$Mole of $CaCO_3$ = (0.02523 × 100)g => 2.523g
Percentage of limestone ($CaCO_3$) in the original mixture = $\frac{2.523g}{5.42g}$ × 100% = 46.55%
The correct option is D
2. Lassaigne's sodium fusion test is commonly employed for identifying elements such as nitrogen, sulphur and the halogens.
In the case of the halogens, detection is carried out by adding dilute trioxonitrate (v) acid (HNO_3) and silver trioxonitrate (v) to the filtrate. If the halogen present is **chlorine**, the precipitate will be **white** and readily **soluble** in ammonium hydroxide.
If it is **bromine**, the precipitate will be **pale yellow** and **soluble** in ammonium hydroxide.
If it is **iodine**, it will be **yellow** and **insoluble** in ammonium hydroxide.
The correct option is C
3. $2Br^- \rightarrow Br_2 + 2e^-$
1mol of Br_2 (22.4m^3) requires two moles of electron. Therefore 0.56dm^3 will require $\left(\frac{2}{22.4} \times 0.56\right)$mol = 0.05mol
The correct option is A
4. The general combustion equation of an alkanol is: $C_nH_{2n} + 10H + \left(\frac{3n}{2}\right)O_2 \rightarrow nCO_2 + (n + 1)H_2O$
That of an alkanal is:
$C_nH_{2n+1}CHO + \left(\frac{3n+2}{2}\right)O_2 \rightarrow (n+1)CO_2 + (n+1)H_2O$
Here, great care must be taken because if the alkanol and the alkanal will have the same number of carbon atom, 'n' as featured in the equations above will not be the same. Why?
All the carbon atoms in an alkanol are found in the alkyl group whereas one carbon atom is still found in the functional group for an alkanal. This implies that if "n" in the alkanol is n_x, that in the alkanal will be $n_x - 1$. Let us unite the two equations by inserting n_x and $n_x - 1$ as appropriate in the equations.
For the Alkanol
$C_{nx}H_{2n_x+1} + OH + \left(\frac{3n_x}{2}\right)O_2 \rightarrow n_xCO_2 + (n_x + 1)H_2O$
For the alkanal
$C_{nx-1}H_{2(n_x-1)+1}\,CHO + \left(\frac{3(n_{x-1})+2}{2}\right)O_2 \rightarrow (n_{x-1} + 1)CO_2 + (n_{x-1} + 1)H_2O$
The combustion equation of alkanal then becomes
$C_{n_{x-1}}H_{2n_x-1}CHO + \left(\frac{3n_{x-1}}{2}\right)O_2 \rightarrow n_xCO_2 + n_xH_2O$
It is said in the question that the values of steam produced by the alkanol and that produced by the alkanal were in the ratio 5 : 4; therefore
$\frac{n_x+1}{n_x} = \frac{5}{4}$
$4n_x + 4 = 5n_x,\ n_x = 4$
To get the formula of the alkanol, put $n_x = 4$;
$C_{n_x}H_{2n_x-1}$OH => C_4H_9OH
To get that of the alkanal, put $n_x = 4$ into
$C_{n_x-1}H_{n_x-1}CHO$ => C_3H_7CHO
Respectively, the alkanol and the alkanal are therefore C_4H_9OH and C_3H_7CHO

The correct option is B

Method 2

One can quickly eliminate option A, C and D as the alkanol and the alkanal in each option do not have the same number of carbon atoms as indicated in the question. also, C_4H_9OH contains **ten** hydrogen atoms while C_3H_7CHO contains **eight** hydrogen atoms

Ratio of steam = ratio of hydrogen = 10 : 8 = 5 : 4 **Option B** has therefore met the two criteria required.

But wait, why did I have to pass through such a rigour in method 1, when I could simply handle it the easy way as in method 2?

I will answer this by referring to a statement from Albert Einstein (a German – born U.S. physicist): "*By academic freedom, I understand the right to search for truth and to publish and teach what one holds to be true. This also implies a duty: one must not conceal any part of what one has recognized to be true*"

This is why Legend Tutorial College is truly Legendary: we teach and publish it all!

5. Generally, bounds are classified into Pi – bonds and sigma bonds. Sigma bonds are more common than Pi – bonds. Whenever there is Pi – bond, the bond angle is altered. The bond angle in ethane is 120º while it is 180º in ethyne. This is because ethane has one Pi-bond while ethyne has two.

The correct option is B

6. $2NO_{(g)} + Cl_{2(g)} \rightleftharpoons 2NOCl_{(g)}$

$K_p = 800$

But $K_p = \frac{P_{NOCl}^2}{P_{NO}^2 \cdot P_{Cl_2}} = 800$

$P_{NO} = 0.02$atm, $P_{cl} = 0.02$atm

Then,

$\frac{P_{NOCl}}{(0.02)^2 \cdot 0.02} = 800$

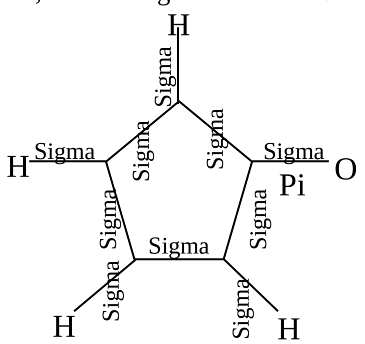
$P_{NOCl}^2 = 800 \times (0.02)^2 \times 0.02$

$= 800 \times (0.02)^2 = 6.4\times10^{-3}$

$\therefore P_{NOCl} = \sqrt{6.4 \times 10^{-3}} = 0.08$atm

The correct option is D

7. When the bonds present in a compound are clearly indicated, the number of Pi and sigma bonds can be known. Every single bond should be taken as sigma. Whenever a double bond is present, it is one sigma and one Pi. If it is a triple bond, it is one sigma and two Pi.



The molecule above contains one Pi and ten sigma bonds.

The correct option is D

8. The enthalpy of atomization represents the energy required to convert one mole of a compound into gaseous atoms under standard conditions.

In order to convert

```
        H
        |
   H –  C – H
        |
        H
```

to atoms, four C – H bond must be broken. Therefore, 4(C – H) = 1648kJmol^{-1}

C – H = $\left(\frac{1648}{4}\right)$KJmol^{-1} = 412KJmol^{-1}

Ethane, C_2H_6 can be represented as

```
      H   H
      |   |
  H – C – C – H
      |   |
      H   H
```

As can be clearly seen; ethane contains one C – C and six C – H bonds. Therefore, 6(C – H) + C – C = 2820

But 1C – H takes 412KJmol^{-1}, Then,

6(412) + C – C = 2820

C – C = 2820 – 6(412) = 2820 – 2472 = 348KJmol^{-1}

The correct option is A

9. Dilution law can be employed here. Essentially, it says $C_1V_1 = C_2V_2$, where C and V represent concentration and volume respectively

$C_1 = 0.750$, $V_1 = ?$, $C_2 = 0.025$moldm^{-2}, $V_2 = 250$cm^3

Then,

0.750moldm^{-3} × V_1 = 0.025moldm^{-3} × 250cm^3

$V_1 = \left(\frac{0.025 \times 250}{0.750}\right)$cm^3 = 8.33cm^3

The correct option is D

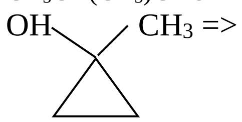
10. Let me give you some powerful tips here, when the molecular formula conforms to $C_nH_{2n}O$ and there is no carbon to carbon double bond, you should expect the isomers to be alkanals, alkanones and cycloalkanols.

When the formula is $C_nH_{2n+2}O$, the groups to be suspected are alkanols and alkoxyalkanes C_4H_8O conforms to $C_nH_{2n}O$. therefore, the possible isomers are alkanals, alkanones and cycloalkanols. The isomers, of C_4H_8O are:

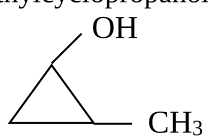
(i) C_3H_7CHO – Butanal

(ii) $CH_3CoC_2H_5$ – Butanone

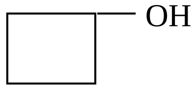
(iii) $CH_3CH(CH_3)CHO$ – 2-methy/propanal

(iv) 

methylcyclopropanol

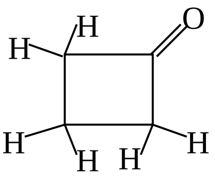
(v) 

2-Methylcyclopropanol

(vi) 

Cyclobutanol

Please note carefully that cyclobutanone is not an isomer of C_4H_8O. We can draw its formula structurally to deduce its molecular formula



The correct option is B

11. $Mg + 2HCl \rightarrow MgCl_2 + H_2$
Mole of Mg = 0.125mol,
mole of HCl = $\left[\left(\frac{25}{1000}\right) \times 4\right]$ = 0.1mol
comparing these values with what the equation dictates, it will be clearly seen that HCl is the limiting reagent.
From the equation, **2**mol of HCl produces **1** mol H_2. Therefore, 0.1mol of HCl will produce 0.05mol of H_2.
The correct option is A

12. Na_2CO_3 and K_2CO_3 are not decomposed on heating. They are therefore said to be thermally stable.
The correct option is D

13. Since the principal quantum number of the outermost shell is 4. There must be four shells in the electronic structure. Potassium is the only element (in the list given) that satisfies this criterion.
The correct option is C

14. R.A.M = $\frac{(Mass_1 \times Ratio_1) + (Mass_2 \times Ratio_2)\ldots}{Ratio_1 + Ratio_1\ldots}$
$= \frac{(20 \times 2) + (22 \times 3)}{2+3} = 21.2$
The correct option is B

15. Origin = 0.80cm
Length of plate = 20cm, R_f = 0.50
$R_f = \frac{Distance\ Travelled\ by\ Solute\ from\ the\ Origin}{Distance\ Travelled\ by\ Solvent\ from\ the\ Origin}$
$0.5 = \frac{(10-0.8)cm}{x}, x = \frac{9.2}{0.5}$ = 18.4cm
Note that x represents the distance of the solvent from the origin. Distance from the bottom of the tank = 18.5cm + 0.80cm = 19.2cm
Distance from the top = 20cm – 19.2cm = 0.80cm
The correct option is A

16. The electrolysis of copper sulphate rids the solution of its deep blue colour. The final colour will be pale blue.
The correct option is B

17. The K_c for a backward reaction equals $\frac{1}{K_c}$ for the forward reaction. If the K_c for the reaction NO + ½O_2 is $\frac{1}{35.2}$ = 2.84×10^{-2}
The correct option is C

18. $CH_3CH_2Cl + NaOH \rightarrow CH_3CH_2OH + NaCl$
The **reaction** essentially involves **interchange** of Cl^- and OH^-. Cl^- and OH^- are **nucleophiles**. The reaction can therefore be described as **nucleophilic substitution reaction.**
The correct option is B

19. 64g of copper requires 2 × 96500C (copper is divalent)
2.4g will require $\left(\frac{2 \times 95600}{64}\right)$ × 2.4 = 7237.5C
Q = It, $I = \frac{Q}{t} = \left(\frac{7237.5}{750}\right)A$ = 9.65A
The correct option is A

20. $2H_2O \rightarrow O_2 + 4H^+ + 4e^-$, E^0 = -1.23v, $2H_2O_2 = 2O_2 + 4H^+ + 4e^-$, E^0 = -0.68
To obtain the desired equation, one reverses the first equation and then adds the resulting equation to the second equation. Please note that reversing the first equation will change E^0 from –1.23V to +1.23V.
$2H_2O_2 \rightarrow 2H_2O + O_2$, E^0 = +1.23V + (- 0.68V) = 0.55V
The correct option is C

21. The enthalpy change is the difference in enthalpy between product(s) and reactant(s). ΔH = -YY has clearly indicated this. Please note that ΔH_a = X is only tantamount to the activation energy.
The correct option is B

22. NH_4Cl has ionic bond between NH_4^+ and Cl^-, covalent in NH_3 and dative bond between NH_3 and H^+
The correct option is A

CHEMISTRY QUESTION 2014

1. When $CuSO_4$ solution is treated with ammonia solution drop by drop till it is added in excess, a precipitate is first formed which then dissolves in excess to give a deep blue solution. The deep blue solution is (a) $Cu(OH)_2$ (b) $[Cu(NH_3)_4](OH)_2$ (c) $CuSO_4$ (d) $[Cu(NH_3)_2]SO_4$
2. If 7.0g of ethane at s.t.p occupy 5.6dm^3, what volume will 7.5g of ethane t the same condition occupy? [C = 12; H = 1; GMV at s.t.p = 22.4dm^3] (a) 6.0dm^3 (b) 5.6dm^3 (c) 5.2dm^3 (d) 9.4dm^3
3. Which of the alcohols below is likely to be oxidized to give the acid
CH_3 – C – H – COOH?
          |
        CH_3
(a) Butan-1-ol (b) 2-Methylpropan-2-ol (c) 2-Methylpropan-1-ol (d) Propan-2-ol
4. The name of CH_3 – $OCOC_2H_5$ is (a) methoxyethane (b) methylpropanoate (c) ethyl ethanoate (d) propylmethanoate
5. Which of these elements has the highest first ionization energy? (a) Rb (b) Li (c) Na (d) K
6. Which of the followings is responsible for the conduction of electricity in a gas enclosed in a glass tube containing two electrodes at a reduced pressure and to which a high voltage is

applied? (a) Cations and anions (b) Cations (c) Electrons (d) Cations and electrons

7. The type of reaction an alkanoic acid cannot udergo is (a) oxidation (b) combustion (c) decomposition (d) esterification

8. What is the pH of 2.5×10^{-3} $moldm^3$ barium hydroxide solution? (a) 11.5 (b) 11.6 (c) 11.7 (d) 11.8

9. The complete oxidation of propan-1-ol yields (a) CH_3CH_2CHO (b) CH_3COCH_3 (c) CH_3COOCH_3 (d) CH_3CH_2COOH

10. Which of the followings will change the equilibrium constant of the reaction
$CO_{(g)} + H_2O_{(l)} \Leftrightarrow CO_{2(g)} + H_{2(g)}$? (a) Increase of temperature (b) Increase of concentration of CO (c) Removal of CO_2 from the mixture (d) Decrease of pressure

11. $M_{(s)} + xH_2SO_{4(aq)} \rightarrow M(SO_4)_{x(aq)} + xH_{2(g)}$ Which of the following elements will not undergo the above reaction? (a) Zn (b) Na (c) Cu (d) Ca

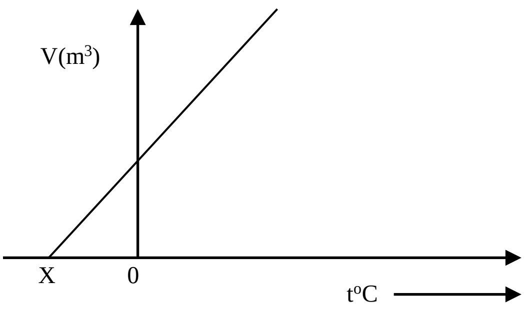
12. A physical change is exemplified by the (a) burning of bush (b) rusting of a metal (c) dissolution of calcium in water (d) heating of ammonium chloride

13. The number of neutrons in the deuterium atom is/are (a) 0 (b) 1 (c) 2 (d) 3

14. Which of these is correct about methyl orange? (a) Yellow in excess aqueous hydrogen ions (b) Pink in excess aqueous hydrogen ions (c) Orange in excess aqueous hydrogen ions (d) Colourless in excess aqueous hydrogen ions

15. Which of these does not affect the rate of a particular chemical reaction? (a) The order of the reaction (b) The size of the particles of the reaction (c) The temperature of the reaction (d) The concentration of the reactants

16. An ideal gas changing volume as temperature raises can be represented by the diagram below:



The point marked X on the graph is (a) -273K (b) 273K (c) 273°C (d) -100K

17. How many isometric dichlorobenzenes are obtainable? (a) 1 (b) 2 (c) 3 (d) 4

18. By accurate description, ozone in the reaction,
$O_{3(g)} + H_2O_{2(l)} \rightarrow H_2O_{(l)} + 2O_{2(g)}$ is (a)displaced to form oxygen (b) decomposed to form oxygen (c) oxidized to oxygen (d) reduced to oxygen

19. $150cm^3$ of nitrogen II oxide were sparked with $100cm^3$ of oxygen, what volume of nitrogen IV oxide will be produced at s.t.p? (a) $100cm^3$ (b) $75cm^3$ (c) $50cm^3$ (d) $150cm^3$

20. Which of the following ions will interact with water to give a solution of pH < 7? (a) Na^+ (b) NH_4^+ (c)

21. In the electrolysis of acidified water using platinum electrodes, $1dm^3$ of oxygen was collected at s.t.p by passing 4.5 amperes into the electrolytic cell. Calculate the time required to set his amount of gas free. [Molar volume of gas at s.t.p = $22.4dm^3$; 1 Faraday = 9650 coulombs] (a) 40.4min (b) 63.8min (c) 55.5min (d) 5.68 min

22. Two plugs of glass wool soaked in concentrated ammonia and concentrated hydrochloric acid are placed at the opposite ends of a long tube such that the distance between them is $100cm^3$. The ratio of the distance covered by ammonia to that covered by hydrochloric acid when a white smoke is first noticed is (a) 2.140 (b) 1.465 (c) 0.982 (d) 0.476

2014 CHEMISTRY SOLUTION

1. When ammonia solution is added in drops to any solution containing Cu^{2+} (not only $CuSO_4$), a light blue precipitate of $Cu(OH)_2$ is formed.
$Cu^{2+}_{(aq)} + 2NH_3 + 2H_2O \rightarrow 2NH_{4(aq)}^{+} + Cu(OH)_{2(s)}$
When excess ammonia solution is added, a deep blue solution is formed.
$Cu(OH)_2 + 4NH_4OH \rightarrow Cu(NH_3)_4(OH_2 + 4H_2O$
The correct option is B

2. Avogadro's law states that equal volumes of gases under the same conditions of temperature and pressure contain the same mole. A corollary to this is that if the moles are equal, the volume will be equal at constant temperature and pressure.
Molar mass of ethane, $C_2H_4 = 28gmol^{-1}$
Molar mass of ethane, $C_2H_6 = 30gmol^{-1}$
Mole of ethane = $\frac{7.0g}{28gmol^{-1}}$ = 0.25mol
They have equal moles and it is also stated that the two gases are measured at s.t.p. Therefore, 7.5g of ethane occupies the same volume as 7.0g of ethane, which is $5.6dm^3$
The correct option is B

3. It is worthy of notice that it is the oxidation of primary alkanols that can yield alkanoic acids. This has sufficiently ruled out 2-methylpropan-2-ol (which is a tertiary alkanol) and propan-2-ol (which is a secondary alkanol). We are now left with options A and C. In order to know which is which, consider the general equation for the oxidation of primary alkanols into alkanoic acid:
RCH_2OH + oxidizing agent $\xrightarrow{H_2SO_4}$ RCOOH
The logical deduction that can be made from this is that it is the carbon atom containing the hydroxyl group that is slightly modified to yield the alkanoic acid, other chains and branches are undisturbed. The acid is 2-methylpropan-1-oic

acid. Therefore, the alkanol that gives this is 2-methylpropan-1-ol

The correct option is C

4. The conventional way of naming esters is to name the alkanol component before the alkanoic acid component. It must be noted that the formula RCOOR for esters can also be written as ROCOR. $CH_3OCOC_2H_5$ is therefore the same as $CH_3COOC_2H_5$ which is ethyl ethanoate

The correct option is C

5. Ionization energy increases across the periods and decreases down the groups. It will therefore decrease from Lithium (Li) to Francium (Fr) in group 1. This means that Lithium has the highest.

The correct options is B

6. When an electrolyte conducts electricity, it utilizes free mobile ions (cations and anions) in doing so. When a gas conducts, it does with electrons.

The correct option is C

7. Alkanoic acids undergo combustion reaction for example:

$CH_3COOH + 2O_2 \rightarrow 2CO_2 + 2H_2O$

They undergo decomposition reaction, for example:

COOH $\underset{\Delta}{\rightarrow}$ HCOOH + CO_2
|
COOH

They undergo esterification reaction, for example:

$CH_3COOH + CH_3OH \rightarrow CH_3COOCH_3 + H_2O$

The correct option is A

8. $Ba(OH)_2 \rightarrow Ba^{2+} + 2OH^-$

x x 3x

2.5×10^3M $2\times2.5\times10^{-3}$M

Since every mole of $Ba(OH)_2$ furnishes 2 moles of OH^-, 2.5×10^{-3} moldm^{-3} of $Ba(OH)_2$ will contain 5×10^{-3}moldm^{-3} of hydroxide ion

$[OH^-]\ [H^+] = 10^{-14}$, $[H^+] = \frac{10^{-14}}{[OH^-]} = \frac{10^{-14}}{[5\times10^{-3}]} = 2\times10^{-12}$moldm^{-3}

$pH = -\log_{10}[H^+] = -\log_{10}(2\times10^{-12}) = 11.7$

The correct option is C

9. The partial oxidation of propan-1-ol yields propanal (Option A) while the complete oxidation yields propanoic acid, CH_3CH_2COOH.

The correct option is D

10. It is only a change in temperature that affects the value of the equilibrium constant.

The correct option is A

11. The equation demonstrates the action of some metals on dilute acids. Metals that are below hydrogen in the electrochemical series (Cu, Hg, Ag, Au, and Pt) will not displace hydrogen from an acid.

The Correct option is C

12. I tis asserted that ammonium chloride sublimes. This means that it undergoes a physical transition from the solid form into the gaseous form without contrary to what is known as ordinary dissolution, calcium reacts with water to produce an alkaline solution and liberate hydrogen gas.

$Ca + 2H_2O \rightarrow Ca(OH)_2 + H_2O$

No one needs to be told that burning of bush and rusting of a metal are chemical changes.

The correct option is D

13. Deuterium, an isotope of hydrogen is represented symbolically as ${}^2_1D$ or ${}^2_1H$. The mass number (2 in this case) is a sum of proton number (1 in this case) and the neutron number, i.e. P + N = mass number

1 + N = 2, N = 2 – 1 = 1

The correct option is B

14. Methylorange is red/pink in excess hydrogen ions (i.e. in acidic solutions), organge in neutral and yellow in excess hydroxyl ions (i.e. in alkaline solutions).

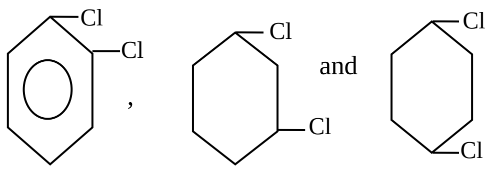
The correct option is A

15. The order of a reaction expresses the dependence of the reaction rate on concentration. It is not a factor that affects the rates of chemical reactions.

The correct option is A

16. The curve shows the relationship between the volume of a gas and its temperature (Charles' Law). The point marked "X" is known as the absolute zero temperature. It is the point at which the volume of the gas theoretically becomes zero. Numerically, it is Okelvin or -273ºC. Continence yourself that these are equal in magnitudes.

17. The possible isomeric dichlorobenzenes are:



The correct option is C

18. The accurate description of this reaction is decomposition of ozone into oxygen. An oxygen atom from hydrogen peroxide is utilized in producing two moles of oxygen molecules

$H_2O_{2(l)} \rightarrow H_2O + O$

$O + O_3 \rightarrow 2O_2$

The correct option is B

19. $2NO_{(g)} + O_{2(g)} \rightarrow 2NO_{2(g)}$

From the equation, 2 moles of NO react with 1 mole of O_2 to produce 2 moles of NO_2. Therefore, 150cm^3 of NO will react with 75cm^3 of O_2 to produce 150cm^3 of NO_2. The remaining 25cm^3 of O_2 will be left unreacted with.

The correct option is D

20. Na^+ interacts with water to produce an alkaline solution (i.e. a solution with pH > 7). The same thing goes with NH_4+. $RCOO^-$ interacts with water to produce an acidic solution (i.e. a solution with pH < 7)

$Na^+ + OH^-$ (from H_2O) $\rightarrow$ NaOH

$RCOO^- + H^+$ (from H_2O) → RCOOH

21. The equation for the electrolysis of acidified water is:

$4OH^- \rightarrow 2H_2O + O_2 + 4e^-$

This implies that 4F of electricity is required to produce 1 mole of O_2 (don't forget that $1e^- \equiv$ 1F).

4F = 1mole of O_2 = 22.4dm^3 at s.t.p

1dm^3 of oxygen will therefore be produced by $\left(\frac{4}{22.4}\right) F$ = 0.17857F

1F = 96500C

0.17857F is therefore (0.17857 × 96500)C = 17232.143C

Quantity of electricity = Current (ampere) × Time (second)

17232.143C = 4.5A × Time

∴ Time = $\left(\frac{17232.143}{4.5}\right) s$ = 3829.36s

= 63.82 minutes

The correct option is B

22. Graham's law of diffusion states that the diffusion rate of a gas is inversely proportional to the square root of its density. The diffusion rates and densities of two goes A and B can be compared using a mathematical representation of the law $\frac{rA}{rB} = \sqrt{\frac{\rho_B}{\rho_A}} = \sqrt{\frac{M_B}{M_A}}$

Where r_A and r_B are the diffusion rates of the concerned gases, ρ_A and ρ_B are their densities and M_A and M_B their relative molecular masses. Since diffusion rate is directly proportional to the distance travelled, one can write $\frac{d_A}{d_B} = \sqrt{\frac{M_B}{M_A}}$

Where d_A and d_B are the distances travelled

R.M.M of NH_3 = 17 and that of HCl = 36.5

Therefore, $\frac{d_{NH_3}}{d_{HCl}} = \sqrt{\frac{M_{HCl}}{M_{NH_3}}} = \sqrt{\frac{36.5}{17}}$ = 1.4653

The correct option is B

CHEMISTRY QUESTION 2013

1. A motor truck releases an average of 5.0g CO into air for every km covered. How many molecules of CO will be emitted into the air if the truck travels 8km?[C = 12;O = 16; N_A=6.02 × 10^{23}] A. 4.32 × 10^{22} B. 2.48 × 10^{23}C. 8.6 × 10^{23} D. 6.82 × 10^{21}
2. A sample of an organic compound was weighed to be 0.250g and subjected to Kjeldahl treatment. The ammonia produced was neutralized by 27.0 cm^3 of 0.100 mol dm^{-3}HCl. What is the percentage of nitrogen in the compound? [H = 1; N = 14]

   A. 18.4% B. 17.8%C. 15.1% D. 13.3%
3. Given the half -redox reaction $O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$, how many moles of electron will be required toproduce 3.0 × 10^{22} molecules of water? A. 0.05 B. 0.10C. 0.15 D. 2.0
4. The quantum number L in an atom defines

   A. the shell K, L, M B. orbitalsC. multiplicity D. degeneracy of orbitals
5. The hybridization of the central atom in a molecule

   A. dictates the shape of the molecule B. shortens the sigma bond in the molecule C. distorts the shape of the molecule D. serves to explain the shape of the molecule
6. Lithium with atomic number of 3 is a

   A. Strong reducing agent B. Strong oxidizing agent C. Weak reducing agent D. Weak oxidizing agent
7. The correct name for $HCOOC_2H_5$ is

   A. methylethanoate B. ethylethanoate C. ethylmethanoate D. Propylethanoate
8. When CaC_2 reacts with water, the organic product forms is

   A. ethanol B. ethanoic acid C. Ethane D. ethyne
9. 100 cm^3 of ethyne was mixed with 240 cm^3 of oxygen in a combustion chamber. What volume of carbon (iv) oxide is produced?

   A. 100cm^3B. 24cm^3C. 138cm^3 D. 192cm^3
10. Uranium-235 explodes when bombarded with a slow moving neutron according to the equation below:

    $^{235}_{92}U + \, ^{1}_{0}n \rightarrow \, ^{94}_{36}Kr + Ba + 3^{1}_{0}n$

    The atomic number and mass of Ba respectively are

    A. 46and 126 B. 36 and 116 C. 56 and 139 D. 66 and 146
11. The reduction potential of two electrodes are

    X^{2+} + 2e- → X, E° = 0.042V

    Y^++e- → Y, E°=0.012V

    Calculate the free energy change for the cell that ismade up of the electrodes [F = 96500 Coulomb mol^{-1}]

    A. 4.20kJ B. 5.79kJC. 6.86 kJ D. 10.55 kJ
12. Which of SF_4, SiH_4, CO_2, ICI, CH_2Cl_2, SO_2 and XeO_3will not show the property of permanent dipole?

    A. CO_2 and SiH_4 only B. SF_4 and SiH_4 only C. CO_2, SiH_4 and XeO_3 only D. SF_4, SiH_4, CO_2 and ICI
13. A sample of water weighs 200.00 g at 298 K. What is the volume of this quantity of water in cubic meters given that the density of water at 298 K is 0.98gcm^{-3}?

    A. 2.04 × 10^{-3} m^3 B. 2.04 × 10^{-6} m^3C. 2.04 × 10^{-9} m^3 D. 2.04 × 10^{-4} m^3
14. What is the pH of 500 cm3 of 0.02 mol dm^{-3}tetraoxosulphur (VI) acid?

    A. 1.456 B. 1.333C. 1.455 D. 1.699
15. The main product of electrophilic addition of HCl to 2-methylpropene is

A. 2-chloro-2-methylbutane B. 2-chloro-2-methylbutene C. 2-methyl-2-chloropropene D. 2-chloro-2-methylpropane

16. Which of the following compounds would you expect to show positive iodoform test?
(i) Batanone (ii) Propanoic acid(iii) Ethanol (iv) Benzaldehyde(v) But-2-one
(i) and (ii) B. (i) and (iii)C. (iv) and (v) D. (ii) and (iii)

17. The complete combustion of one mole of an alkanol is shown below
$CH_{2n+1}CHO + xO_2 \rightarrow yCO_2 + 2H_2O$
What is the value of x in terms of n?
A. $\frac{3n+1}{2}$ B. $\frac{3n-1}{2}$ C. $\frac{3n}{2}$ D. $\frac{3n+3}{2}$

18. An ion has a charge of +3. The nucleus of the ion has a mass of 120. The number of neutrons in the nucleus is 1.50 times that of the number of protons. How many electrons are in the ion?
A. 55 B. 48 C. 45 D. 42

19. Consider the following reactions
(i) $LiOH + CO_2 \rightarrow Li_2CO_3 + H_2O$
(ii) $2H_2 + O_2 \rightarrow 2H_2O$
(iii) $2Cu + O_2 \rightarrow 2CuO$
(iv) $HCl + AgNO_3 \rightarrow AgCl + HNO_3$
Which of these reactions are redox reactions?
A. (I) and (iii) only B. (i), (ii) and (iii) only C. (ii) and (iv) only D. (ii) and (iii) only

20. Which of the following metals cannot displacehydrogen from steam?
A. Copper B. Iron C. Strontium D. Lithium

21. Consider the exothermic reaction
$2SO_{2(g)} + O_{2(g)} \leftrightarrow 2SO_{2(g)}$
If the temperature of the reaction is reduced from 600 °C to 300 °C and no other changes take place then,
A. the reaction rate increases B. concentration of SO_3 decreases C. concentration of SO_3 increases D. SO2 gas becomes unreactive

22. The molarity of 5% by weight of aqueous solution of tetraoxosulphate (VI) acid [molecular weight 98] is
A. 0.537 mol dm^{-3} 8. 0.208 mol dm^{-3} C. 0.551 mol dm^{-3} D. 0.333 mol dm^{-3}

CHEMISTRY SOLUTION (OAU POST UTME 2013)

1. 5g of CO every km covered 8km = (8 × 5)g
= 40g
$n = \frac{Mass}{Molar\ Mass} = \frac{40}{12+16} = \frac{40}{28} = 1.429$
$n = \frac{No\ of\ particles}{6.02 \times 10^{23}}$
No of particle = $1.429 \times 6.02 \times 10^{23} = 8.6 \times 10^{23}$
"C"

2. "C"

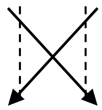
3. $O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$
From the equation above,
$4e^- \rightarrow 2H_2O$
4moles of electron → $2 \times 6.02 \times 10^{23}$ water
x moles → 3.0×10^{22}
$x = \frac{4 \times 3.0 \times 10^{22}}{2 \times 6.02 \times 10^{23}}$
= 0.10 **"B"**

4. The quantum number *l* in an atom defines orbital and also the shape of the orbital principal quantum number n, depicts the effective distance of electron to the nucleus. m = degeneracy **"B"**

5. The hybridization dictates the shape of the molecule. **"A"**

6. Li $1s^2 2s^1$
Group = 1; period = 2
Group 1 have the highest/ strongest reducing ability because they are highly electropositive **"A"**

7. H C O O C_2 H_5



Ethyl methaloate **"C"**

8. $CaC_2 + 2H_2O \rightarrow C_2H_2 + Ca(OH)_2$
Ethyne **"D"**

9. $C_2H_2 + O_2 \rightarrow CO_2 + H_2O$
$C_xH_y + \left(x + \frac{y}{4}\right)O_2 \rightarrow xCO_2 + \frac{y}{2}H_2O$
$C_2H_2 + \left(2 + \frac{2}{4}\right)O_2 \rightarrow 2CO_2 + H_2O$
$C_2H_2 + \left(\frac{5}{2}\right) O_2 \rightarrow 2CO_2 + H_2O$
Multiply throughout by 2
$2C_2H_2 + 5O_2 \rightarrow 4CO_2 + 2H_2O$
2 : 5 4 : 2
$100cm^3$ $240cm^3$ - -
$\frac{2}{5} \times 240$ $240cm^3$ $\frac{2}{5} \times 240$ $\frac{2}{5} \times 240$
$96cm^3$ $240cm^3$ $192cm^3$ $96cm^3$
$4cm^3$ - $192cm^3$ $96cm^3$
:. Volume of oxide produced = $192cm^3$
"D"

10. ${}^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{94}_{36}Kr + {}^{x}_{y}Ba + 3{}^{1}_{0}n$
235 + 1 = 9 + x + 3(1)
236 = 9 + x + 3
236 = 12 + x
x = 236 – 12 = 224
236 = 97 + x
x = 236 – 97
= 139
92 + 0 = 36 + y + 0
y = 92 – 36
= 56 **"C"**

11. $\Delta G = -nFE_\Theta$
$E_\Theta = 0.042 - 0.012$
= 0.030
$\Delta G = -2 \times 96500 \times 0.030$
= 5790
= 5.79kJ **"B"**

12. **D –**

13. m = 200g at 298k
$\rho = \frac{Mass}{Volume}$
$0.98 = \frac{200}{Volume}$

Volume = $\frac{200}{0.98}$
= 204.08cm^3
100
100cm =
$(100cm^3) = (1m)^3$
$1000000cm^3 = 1m^3$
204.08 = x
x = $\frac{204.08}{1000000}$
= $2.04\times10^2\times10^{-4}m^3$ **"D"**

14. $n = \left|\frac{CV}{1000}\right| = \frac{0.02\times500}{1000}$
$H_2SO_4 \leftrightarrow 2H^+ + SO_4^{2-}$
$0.02moldm^{-3}$ 2 × $0.02moldm^{-3}$
:. $[H^+] = 0.04moldm^{-3}$
pH = $-\log_{10}[H^+] = \log_{10}[0.04]$

15.
```
      H   H    CH3
      |   |     |
H – C – C  =  C – H  +  HCl
      |
      H
          ↓
   H   Cl    H
   |    |     |
H – C – C  =  C – H
   |    |     |
   H  CH3  H
```
2 – chloro – 2 – methyl propane **"D"**

16. **C –**

17. For C from the equation
n + 1 = y .... (1)
For H for the equation
2n + 1 + 1 = 4 ...... (2)
1 + 2x = 2y + 2 .... (3)
From (1) y = n + 1
1 + 2x = 2(n+1) + 2
1 + 2x = 2n + 2 + 2
2x = 2n + 4 – 1
2x = 2n + 3
$x = \frac{2n+3}{2}$ (No answer)

18. $^{120}X^{3+}$
Neutron = 1.50 proton
n = 1.5p
n + p = 120
1.5p + p = 120
2.5p = 120
$p = \frac{120}{2.5}$
P = 48
Since proton number = 48
Electron number = 48 – 3
= 45 **"C"**

19. **"B"**

20. **"C"**

21. For an exothermic reaction, an increase in temperature will favour the backward reaction while a decrease will favour the forward reaction. Hence Ans = **"C"**

22. $\frac{5}{100}\times98$
= 4.9
98 – 4.9 = 93.1
$\frac{4.9}{93.1}$ Nil

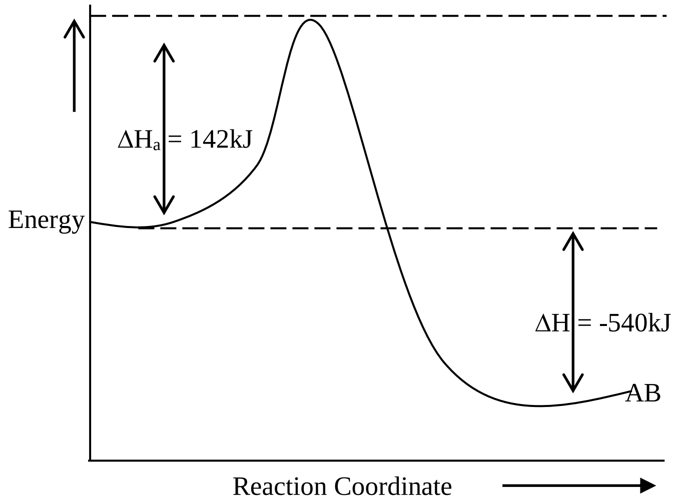
CHEMISTRY 2012 QUESTIONS

1. During the electrolysis of dilute tetraoxosulphate (VI) acid solution, 0.05 mole of electrons were passed. What volume of gas was produced at the anode? (a) $2.24dm^3$ (b) $0.560dm^3$ (c) $0.280dm^3$ (d) $0.224dm^3$
2. What volume of $0.750moldm^{-3}$ Na_2CO_3 solution could be diluted to $250cm^3$ to reduce the concentration to $0.025moldm^{-3}$? (a) $16.8cm^3$ (b) $14.2cm^3$ (c) $10.4cm^3$ (d) $8.3cm^3$
3. When $70cm^3$ of $3.0moldm^3$ Na_2CO_3 is added to $30cm^3$ of $1.0moldm^3$ $NaHCO_3$, the concentration of Na^+ ions in $moldm^{-3}$ in the solution is (a) 1.5 (b) 4.5 (c) 2.0 (d) 3.5
4. The reaction, Y → Product is of first order with the initial concentration of Y = 3.55×10^3 $moldm^3$ and the rate constant of $5.25 \times 10^{-3}s^{-1}$. What is the half-live of the reaction? (a) 350s (b) 215s (c) 132s (d) 615s
5. Given the half-cell reaction, 2Br → Br_2, how many moles of electron will be required to produce $0.56dm^3$ of bromine at s.t.p? [molar volume of gas at s.t.p. $22.4dm^3$] (a) 0.05 (b) 0.10 (c) 0.20 (d) 1.00
6. The equilibrium constant, Kc for the reaction, $NO_{(g)} + \frac{1}{2}O_{2(g)} \rightarrow NO_{2(g)}$, is 35.2 What is the value of K for the reaction, $NO_{(g)} \rightarrow NO_{2(g)} + \frac{1}{2}O_{2(g)}$ (a) 35.2 (b) 17.6 (c) 2.84×10^{-2} (d) 1.24×10^3
7. An atom has a core and outside the core an electron occupies an orbital for which the principal quantum number n = 4, l = 0, m = 0 and s = $+\frac{1}{2}$ or $-\frac{1}{2}$. The atom is likely to be (a) boron (b) sodium (c) potassium (d) fluorine
8. An element B, has two isotopes $^{20}_{10}B$ and $^{22}_{10}B$ present in ratio 2:3. The relative atomic mass of B is (a) 20.5 (b) 21.2 (c) 23.4 (d) 25.0
9. What quantity of current is required to deposit 2.4g of copper in a period of 750 seconds during an electrolytic deposition process? [Cu = 64, IF = $9650Cmol^{-1}$] (a) 9.65A (b) 10.81A (c) 12.33A (d) 15.54a
10. Platinum electrodes are dipped into copper sulphate solution in a voltameter. The solution left after electrolysis is (a) clear (b) blue (c) pale blue (d) sky blue
11. What volume of water is produced when a mixture of $150cm^3$ of hydrogen and $100cm^3$ of oxygen is exploded in a eudiometer? (a) $250cm^3$ (b) $150cm^3$ (c) $100cm^3$ (d) $50cm^3$
12. A chloroform solution of pure organic compound was spotted at a distance 0.80cm from the base of a 20cm long chromatoplate. If the compound has

the r_f–value of 0.505 and moves half way up the 20cm long plate, What is the distance of the solvent front from the top of the plate upon elution? (a) 0.80cm (b) 1.0cm (c) 1.2cm (d) 1.4cm

13. The main organic product named when bromine water is added to but-1-ene is (a) 1-bromobutane (b) 2-bromobutane-1-ol (c) 1-bromobutan-2-ol (d) 2-bromobutan-2-ol
14. The standard reduction potentials for the following half-cell reactions are,

    $2H_2O_{(l)}=C_{2(g)} + 4H^+_{(aq)} + 4e$ $E^0 = -1.23V$
    $2H_2O_{2(l)} = O_{2(g)} + 4H^+_{(aq)} + 4e$ $E^0 = -0.68$ V
    What is the E^0 for the reaction, $2H_2O_{2(l)} = H_2O_{(l)} + O_{2(g)}$ at 298 K? (a) – 0.68 V (b) –1.23 V (c) - 0.55 V (d) + 1.91 V
15. Bonding in ammonium chloride is (a) ionic, covalent and dative (b) ionic and covalent (c) covalent and dative (d) ionic
16. Valence shell electron pair theory through hybridization predicts that boron trichloride is (a) Arrhenius acid (b) Lewis base (c) Lewis acid (d) Lowry-bronsted base
17. The basic tenet of valence bond electron pair repulsion theory is that the pairs of electrons making the sigma bonds dictate the shape of molecules. The pi-bonds often encounter in some molecules serve(a) distort the shape of molecules (b) alter the angle between the atoms in molecules (c) shorten the sigma bonds in molecules (d) explain the shape of molecules
18. A chemical equilibrium is established when (a) concentration of the reactants and products are equal (b) reactants in the system stop forming the products (c) concentrations of the reactants and products remain unchanged (d) reactants in the system are completely transformed to products
19. Oxygen is extracted from water by (a)displacement reaction (b) oxidation reaction (c) reduction reaction (d) decomposition reaction
20. Excess ethanol was sparked with 3g of pure oxygen in a combustion chamber. How many molecules of CO_2 are produced? (N = 6.02×10^{23} molecules mol^{-1}). (a) 6.02×10^{22}(b) 3.01×10^{23} (c) 3.70×10^{21} (d) 2.84×10^{21}
21. Forty (40) grams of sodium nitrate were added to 50 cm^3 of water to give a saturated solution at 298K. If the solubility of the salt is 10.50 $moldm^{-3}$ at the same temperature, what percentage of the salt is left undissolved? (Na = 23, N = 14 and O = 16). (a) 11.56% (b) 2.55%(c) 5.88% (d) 2.45%
22. The energy for the decomposition of molecule AB in kJ in the diagram of energy against the reaction coordinate shown below is (a) 146 (b) -540 (c) 682 (d) 398



CHEMISTRY 2012 SOLUTIONS

1. During the electrolysis of dilute tetraoxosulphate (vi) acid, OH^- is selected at the anode for preferential discharge. The OH^- leads to the liberation of oxygen gas at the anode.
   $4OH^- \rightarrow 2H_2O + O_2 + 4e^-$
   Since 4 moles of electrons are required to produce 1mole of O_2, 0.05mole of electrons will produce $\frac{1mole\ of\ O_2}{4\ moles\ of\ e^-} \times 0.05 mole\ of\ e^-$ = 0.0125mol
   Volume at s.t.p = Molar gas volume × mole
   = $(22.4 \times 0.0125)dm^3$
   = **$0.280dm^3$**
   Alternatively, we can obtain volume directly without having to calculate the mole of oxygen produced.
   Since 1 mole of any gas at s.t.p occupies $22.4dm^3$, we relate the mole of electron and the volume of oxygen produced directly
   Let us do it!
   $4e^- = 22.4dm^3$
   $0.05e^- = x$
   $\therefore x = \frac{(22.4 \times 005)}{4}$ = **$0.280dm^3$**as before
   The correct option is C
2. Since the amount of Na_2CO_3 does not change upon dilution, the conservation of mole can be utilized here
   Mole = Conc. $(moldm^{-3})$ × Volume (dm^3)
   = $0.0250moldm^{-3} \times \left(\frac{250}{1000}\right)dm^3$
   = 0.00625mol
   Conserving the mole, we can write:
   Mole = Conc. $(moldm^{-3})$ × Volume (dm^3)
   $0.00625mol = 0.750moldm^{-3} \times$ Volume (dm^3)
   $\therefore$ Volume $(dm^3) = \frac{0.00625mol}{0.750moldm^{-3}} = 0.00833dm^3$
   = $(0.00833 \times 1000)cm^3$
   = **$8.33cm^3$**
   Alternatively, we can use the dilution equation:
   $C_1V_1 = C_2V_2$
   $0.025 \times 250 = 0.750 \times V_2$

$V_2 = \left(\frac{0.025 \times 250}{0.75}\right)$cm^3
$= \boldsymbol{8.33cm^3}$ as before
The correct option is D

3. Mole of Na_2CO_3 = 3moldm^{-3} × $\left(\frac{70}{1000}\right)$dm^3 = 0.21mol
$Na_2CO_3 \rightarrow 2Na^+ + CO_3^{2-}$
1mole of Na_2CO_3 contains 2 moles of Na^+, then 0.21mole of Na_2CO_3 will contain (2 × 0.21) mole of Na^+ which equals 0.42mole.
Mole of $NaHCO_3$ = 1moldm^{-3} × $\left(\frac{30}{1000}\right)$dm^3 = 0.03mol
$NaHCO_3 \rightarrow Na^+ + HCO_3^-$
1mole of $NaHCO_3$ contains 1 mole of Na^+, then 0.03mole of $NaHCO_3$ will contain 0.03 mole of Na^+.
Total mole of Na^+ = (0.42 + 0.03)mol = 0.45mol
Total Volume = (70 + 30)cm^3 = 100cm^3 = $\left(\frac{100}{1000}\right)$dm^3 = 0.1dm^3
Concentration of Na^+ = $\frac{0.45mol}{0.1dm^3}$ = 4.5moldm^{-3}
The correct option is B

4. The half-life of a first order reaction does not depend on the initial concentration of the reactant. It only depends on the specific rate constant.
$T_{\frac{1}{2}} = \frac{0.693}{K}$, where K = rate constant
$= \frac{0.693}{5.25\times10^{-3}}$ = 132s
The correct option is C

5. $2Br^- \rightarrow Br_2$
The most important thing here is the mole of electrons to balance the equation. Since the overall charges on the left and right side of the reaction are -2 and 0 respectively, two electrons are to be added to the right side. Then:
$2Br^- \rightarrow Br_2 + 2e^-$
∴ 1mole of Br_2 (22.4dm^3 at stp) requires 2 moles of electrons.
Then, 0.56dm^3 will require $\left(\frac{0.56}{22.4} \times 2\right)$mol of e^-
= 0.05
The correct option is A

6. If the equilibrium constant of a reaction A → B is x, that of the reversed reaction, B → A will be $\frac{1}{x}$.
Thus, NO + $\frac{1}{2}O_2 \rightarrow NO_2$, K_C = 35.2
$NO_2 \rightarrow NO + \frac{1}{2}O_2$, $K_c = \frac{1}{35.2}$ = 0.0284 = 2.84×10^{-2}
The correct option is C

7. n = 4, l = 0, m = 0 and s = $+\frac{1}{2}$ or $-\frac{1}{2}$
The Principal Quantum number, n signifies the number of shells to be 4. Of the elements given, only potassium has four shells (K = 2, 8, 8, 1)
The correct option is C

8. R.A.M = $\frac{(Mass_1 \times \%_1) + (Mass_2 \times \%_2)...}{100}$
= (mass$_1$ × α_1) + (mass$_2$ × α_2). . .
Where α represents the fractional abundance.
It can also be given by $\frac{[(Mass_1 \times Ratio_1) + (Mass_2 \times Ratio_2)]}{Sum\ of\ ratios}$
We are to employ this last formula
$$R.A.M = \frac{[(Mass_1 \times Ratio_1) + (Mass_2 \times Ratio_2)]...}{Ratio_1 + Ratio_2 ...}$$
$$= \frac{(20 \times 2) + (22 \times 3)}{2 + 3}$$
$= \left(\frac{40+66}{5}\right)$ = 21.2
The correct option is B

9. $Cu^{2+}_{(aq)} + 2e \rightarrow Cu_{(s)}$
Two Faradays of electricity will be required to produce 1mole of Cu (64g)
Since 64g is produce by 2F, 2.4g will be produced by $\left(\frac{2}{64} \times 2.4\right)F$ = 0.075F
0.075F = (0.075 × 96500)C = 7237.5C
Q = It
$I = \frac{Q}{It} = \left(\frac{7237.5}{750}\right)A$ = 9.65A
The correct option is A

10. Copper sulphate has a white colour in the anhydrous form. When hydrated, the colour becomes blue. This blue colour is due to the presence of Cu^{2+}. The electrolysis of $CuSO_4$ solution using platinum electrodes produces copper at the cathode. This implies that Cu^{2+} is progressively removed from the electrolyte. The colour fades progressively!
The correct option is C

11. $2H_2 + O_2 \rightarrow 2H_2O$
100cm^3 of O_2 requires 200cm^3 of H_2 (as suggested by the equation). Since only 150cm^3 of H_2 is available it is the limiting reagent. H_2 therefore determines the amount of H_2O produced. Since the stoichiometry of H_2 and H_2O is the same (2 : 2), the volume of H_2O produce is also 150cm^3.
The correct option is B

12. Retention factor, R_f = $\frac{Distance\ moved\ by\ the\ solute\ from\ the\ origin}{Distance\ moved\ by\ the\ solvent\ from\ the\ origin}$
It is said that the chloroform is spotted at 0.80cm and it moves half way up the 20cm long plate. Then, the distance moved by the solute = $\left(\frac{20-0.80}{2}\right)$cm = 9.2cm
$$R_f = 0.505 = \frac{9.2cm}{Distance\ travelled\ by\ solvent}$$
$Solvent\ distance = \frac{9.2cm}{0.505}$ = 18.2178cm
From the bottom of the plate, the solvent front = 0.80cm + 18.2178cm = 19.0178cm
Then, from the top of the plate, the solvent front = 20cm – 19.0178cm
= 0.9822cm
= 1.0cm
The correct option is B

13. The main component of bromine water is HOBr – an unsymmetrical reagent. The compound we are using is but-1-ene – an unsymmetrical alkene.

Anytime a reaction involves an unsymmetrical alkene or alkyne with an unsymmetrical reagent, the major product is the one obtained using markovnikov's rule.

It states that the positive part of the unsymmetrical reagent must be added to the carbon atom that has more hydrogen atoms while the negative part of the reagent is added to the second carbon atom.

```
CH3CH2      H
      \    /
       C = C + HOBr→ ?
      /    \
     H      H
```

With the reagent, HOBr, lies another stumbling block! If we are given HCl, HBr, HF or HI the positive part will be hydrogen. But in HOBr the case is different. HOBr is oxobromate (i) acid. This indicates that Br has an oxidation state of +1 while OH is -1 as expected. Br therefore goes to the carbon atom that has more hydrogen atom

```
CH3CH2     H              H   H   H   H
      \    /              |   |   |   |
       C = C + HOBr →  C – C – C – C – H
      /    \              |   |   |   |
     H      H             H   H  OH   Br
```

The name of the compound formed is 1-bromo-butan-2 – ol.

The correct option is C

14. $2H_2O \rightarrow O_2 + 4H^+ + 4e^-$, $E^o = -1.23V$ ……(i)
$2H_2O_2 \rightarrow 2O_2 + 4H^+ + 4e^-$, $E^o = -0.68V$ ….(ii)
The required equation is:
$2H_2O_2 \rightarrow 2H_2O + O_2$
The equations we need to add together are:
(a) The one that has $2H_2O_2$ on the L.H.S and
(b) The one that has $2H_2O$ on the R.H.S
Equation (ii) complete satisfies the first requirement. For the second requirement, we need to reverse equation (i). Note that whenever an equation is reversed, the emf sign changes. Also, note fully well that when an equation is multiplied, the emf value is **NOT** multiplied.
Reversing equation (i) we have:
$O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$, $E^o = +1.23V$ ……(iii)
Adding equation (ii) and (iii) together, we have:
$2H_2O_2 + O_2 + 4H^+ + 4e^- \rightarrow 2H_2O + 2O_2 + 4H^+ + 4e^-$
$E = -0.68V + 1.23V = +0.55V$
Cancelling out common terms on the two sides we have:
$2H_2O_2 \rightarrow 2H_2O + O_2$, $E^o = +0.55V$
The correct option is C

15. Ammonium chloride contains the three bond types – **ionic, covalent** and **dative**.
In ammonia, hydrogen atoms are **covalently** linked to nitrogen atom.
In ammonium ion, ammonia and hydrogen ion are united by **dative** bond.
Now, finally, ammonium chloride comprises ammonium ion (NH_4^+) and chloride ion (Cl^-). The bond between the two **ions** is **ionic** bond.
The correct option is A

16. For majority of atoms, the must have eight electrons at the outermost shell before becoming stable. The structure of BCl_3 is

```
Cl . . B . . Cl
         .
         .
         Cl
```

As can be seen for the central atom, boron, the total number of outer electrons is 6. This indicates that BCl_3 needs two more electrons for octet. It can thus accept a lone pair of electrons.
Lewis defined an acid as a substance that accepts a lone pair of electron while a base is a substance that donates BCl_3 is a Lewis acid.
The correct option is C

17. The presence of Pi-bonds in some compounds alters the bond angles. For instance, the bond angle in ethane is 109.5° – this is because there is no Pi-bond in an alkane. In ethane, the bond angle is 120° – this is due to the presence of one Pi-bond. In ethyne, the bond angle is 180° – this is attributed to the presence of two Pi-bonds
The correct option is B

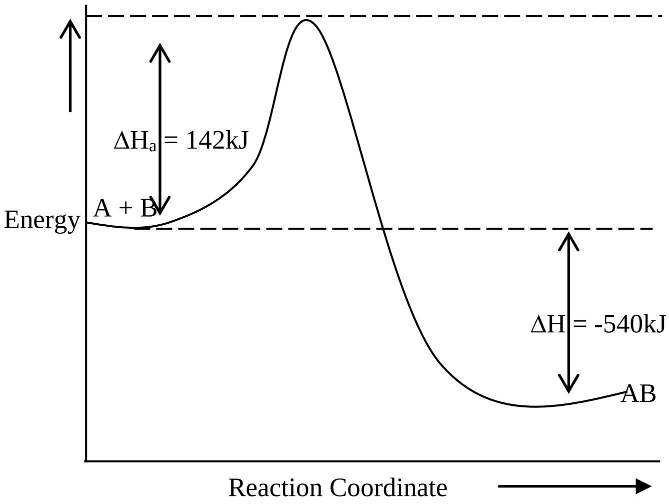
18. Chemical equilibrium is said to be established for a reversible reaction when the rate of the forward reaction equals that of the backward reaction. Then, it appears as if the reaction has stopped. No! it is just that the concentrations of the reactants and products remain unchanged.
The correct option is C

19. Oxygen is extracted from water by electrolysis. The oxygen gas is from the OH^- that migrates to the anode.
Since the reaction at the anode is oxidation, oxygen is obtained from water by oxidation.
The correct option is B

20. $C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$
Mole of oxygen = $\frac{3g}{32gmol^{-1}}$ => 0.09375mol
3 moles of O_2 produces 2 moles of CO_2
Therefore, 0.09375mol of O_2 = $\left(\frac{2}{3} \times 0.09375\right)$mol of CO_2= 0.0625mol
Number of CO_2 molecules = mole of CO_2 × Avogadro's number
= $(0.0625 \times 6.02 \times 10^{23})$
= 3.76×10^{22} molecules

21. Molar mass of $NaNO_3$ = $85gmol^{-1}$
Mole of $NaNO_3$ = $\frac{40g}{85gmol^{-1}}$ = 0.47059mol
If $50cm^3$ contains 0.47059mol, $1000cm^3$ ($1dm^3$) will contain
= $\left(\frac{0.47059}{50} \times 1000\right)$mol = 9.412mol
Since $1dm^3$ will contain 9.412mol, the concentration is $9.412moldm^{-3}$
But the solubility of $10.50moldm^{-3}$ indicates that $1dm^3$ can still hold 10.50mol of the salt. Therefore, no precipitate will be produced at all.

22.



For the dissociation of AB, the energy required is 142KJ –(-540KJ) = 682KJ
The correct option is C

CHEMISTRY 2011 QUESTIONS

1. What condition favours the formation of the product for the endothermic reaction, $N_2O_{4(g)} \leftrightharpoons 2NO_{2(g)}$ (a) a decrease in pressure (b) a decrease in volume (c) an increase in pressure (d) a constant volume
2. What is the percentage yield of water if 0.90g of water is obtained when 29.0g of butane is burned in excess oxygen? (a) 0.02% (b) 0.20% (c) 2.0% (d) 10.0%
3. The order of reactivity of five metals is P > Q > R > S > T. Which of the following reactions can occur spontaneously? (a) $T + P^+ \rightarrow T^+ + P$ (b) $Q + T^+ \rightarrow Q^+ + T$ (c) $R + Q^+ \rightarrow R^+ + Q$ (d) $T + R^+ \rightarrow T^+ + R$
4. An element, Y has the electronic configuration of $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$. (a) Y is a period III element (b) Y contains three electrons in the outer shell (c) Y is a transition metal (d) Y can engage in bonding with the s and p orbitals
5. Which of the following is NOT implicated as a major cause of global warming? (a) NO_2 (b) CO_2 (c) $CFCl_3$ (d) CF_2Cl_2
6. Which of the following shows little or no net reaction when the volume of the system is decreased? (a) $2O_{3(g)} \leftrightharpoons 3O_{2(g)}$ (b) $2NO_{2\,(g)} \leftrightharpoons N_2O_{4(g)}$ (c) $H_{2\,(g)} + I_{2\,(g)} \leftrightharpoons 2HI_{(g)}$ (d) $PCl_{5\,(g)} + Cl_{2\,(g)}$
7. A solution of 0.20 mole of NaBr and 0.20 mole of $MgBr_2$ in 2.0 dm^3 of water is to be analyzed. How many moles of $Pb(NO_3)_2$ must be added to precipitate all the bromide as insoluble $PbBr_2$? (a) 0.30 mol (b) 0.10 mol (c) 0.20 mol (d) 0.40 mol
8. A given volume of methane diffuses in 20s. How long will it take the same volume of sulphur (IV) oxide to diffuse under the same conditions? [C = 12, H = 1, S = 32, O = 16]. (a) 5s (b) 20s (c) 40s (d)60s
9. The reaction, $A + B \rightarrow C$, can be represented by the equation, r = K[A][B], k in this equation is, (a)proportionality constant (b) rate constant (c)equilibrium constant (d)Boltzmann constant
10. The reaction that takes place in Daniel cell is,(a) $Zn/Zn^{2+}//Cu^{2+}/Cu$ (b) $Zn/Zn^{2+}\ //Cu/\ Cu^{2+}$ (c) $Zn^{2+}\ /\ Zn\ //\ Cu^{2+}\ /\ Cu$ (d) $Zn^{2+}\ /\ Zn\ //\ Cu\ /\ Cu^{2+}$
11. Which of the followings is composed of the elements, H, O, Al, and Si? (a) Urea (b) Silica (c) Bauxite (d) clay
12. Which of the following is not a chemical reaction? (a) burning of bush (b) rusting of iron (c) decay of bitter leaves (d) dissolution of potassium hydroxide pellets
13. 100.0g of $KClO_3$ was added to 40.0cm^3 of water to give a saturated solution at 298K. If the solubility of the salt is 20.0$moldm^{-3}$ at 298K, what percentage of the salt is left undisolved? [K = 39, Cl = 35.5, O=16]. (a) 80% (b) 60% (c) 5% (d) 2%
14. A tertiary amine is (a) ethylamine (b) diethylamine(c) triethylamine(d)tetraethylamine
15. Which of the following statements is true when sulphur atom forms its ion? (a) it achieves an inert configuration (b) it transfers two electrons in the process (c) it accepts one electron in the process (d) it gets oxidized in the process
16. An electron described by the quantum number, n = 4, *l* = 3 can be located in what orbital? (a) 4f (b) 3s (c) 3d (d) 4p
17. An aqueous solution of a crystalline salt reacts with dilute HCl to give a yellow precipitate and a gas that tuned dichromate paper green. The crystalline salt may be, (a) $Na_2S_2O_3.5H_2O$ (b) $Na_2CO_3.10H_2O$ (c) Na_2S (d) $NaHCO_3$
18. The oxidation states of nitrogen in ammonium nitrate are, (a) -3, +3 (b) -3, +5 (c) +3, -5 (d) -3, 4
19. Which of these reagents can confirm the presence of a triple bond? (a)hypochlorous acid (b) bromine water (c) acidified $KMnO_4$ (d) Copper chloride
20. An excess 0.10 mol dm^3HCl was poured into a big beaker containing 2g of limestone. The unreacted acid required 25 cm^3 of 0.10 mol dm^{-3} potassium carbonate to neutralize it. What was the original volume of the acid? [Ca = 40, C = 12, O = 16] (a) 250cm^3 (b) 260 cm^3 (c) 400 cm^3 (d) 450 cm^3
21. ${}^{226}_{88}Ra \rightarrow {}^{x}_{86}Rn + \alpha$-particle. What is the value of x in the nuclear reaction? (a) 226 (b) 220 (c) 222 (d) 174
22. In the electrolysis of copper (II) sulphate using copper electrodes, the processes that occur at the anode and cathode respectively are: (a) dissolution and evolution (b) dissolution and

deposition (c) deposition and evolution (d) evolution and deposition

CHEMISTRY 2011 SOLUTIONS

1. $N_2O_{4(g)} \rightarrow 2NO_{2(g)}$
To start with, when a reversible reaction contains gas (es), a change in pressure affects the position of equilibrium.
A decrease in pressure favors the side that has a greater mole of gas (es) while an increase in pressure will favor the side that has smaller mole of gas (es).
The right side has more gas than the left side. Hence, the forward reaction is favored by a decrease in pressure
The correct option is A

2. $2C_4H_{10} + 13O_2 \rightarrow 8CO_2 + 10H_2O$
2(58g) 10(18g)
As already depicted above, 2(58) g (which is 116g) of butane will produce 10(18) g (which is 180g)
Then, 29g of butane is expected to produce $\left(\frac{180 \times 29}{116}\right)$g = 45g
It is said that only 0.90g of water is obtained. The percentage yield
$= \left(\frac{Actual\ yield}{Theoretical\ yield}\right) \times 100\%$
$= \frac{0.90g}{45g} \times 100\%$
= 2%
The correct option is C

3. The P>Q>R>S>T
An ion can be displaced from its Solution only by species that are higher than it in the reactivity series. T is the least reactive of the five metals and as such T cannot displace the ion of other metals. Then, options A and D are automatically wrong.
Since Q>T, it can displace T^+ from its solution.
$Q + T^+ \rightarrow Q^+ + T$
The correct option is B

4. From the electronic of an element a lot of things can be known:
   i. Group
   ii. Period
   iii. Number of unpaired electrons
   iv. Valence electrons
   v. The block the elements belongs.

$Y = 1S^22S^22P^63S^23P^3 = 2, 8, 5$
   i. Since the outermost shell contains five electrons, the element belongs to group (V)!
   ii. Since there are three shells (2, 8, 5) **the element belongs to period 3!**
   iii. $3P^3 = \boxed{\uparrow}\boxed{\uparrow}\boxed{\uparrow}$
   3px 3py 3pz
Since the three electrons are all one – in – a – box, there are three unpaired electrons
   iv. Since, the electronic configuration ends in a shell containing 5 electrons, the valence electron (not valency) is 5!
   v. Since the spdf configuration ends in a p-orbital, the element is a P – blocked element

The correct option is A

5. All the gases listed contribute to global warming. Even water vapor has its own quota of the blame! The most implicated gas is CO_2, next to chlorofluorocarbons, next to CH_4 and then nitrogen oxide (the least implicated).
The correct option is A

6. As already explained, (for question 1), a change of pressure or volume affects the position of equilibrium in reactions where there are equal moles of gases on the two sides.
$H_{2(g)} + I_{2(g)} \leftrightharpoons 2HI$
Since, there are two moles of gases on each side of the equation, a change of pressure or volume does not affect the reaction. The correct option is **The correct option C**

7. $NaBr \rightarrow Na^+ \quad Br^-$
0.20mol 0.20mol 0.20mol
$MgBr2 \rightarrow Mg^{2+} + 2Br^-$
Total Br^- = 0.20mol + 0. 40mol
= 0.60mol
We can now easily obtain the amount of Pb $(NO_3)_2$ required to totally discharge Br – $Pb(NO_3)_2 + 2Br^- \rightarrow PbBr_2 + 2NO_3^-$
1mole is required to discharge 2moles of Br^-. Then, 0.30mol of Pb $(NO_3)_2$ will precipitate 0.6mol of Br^-.
The correct option is A

8. Here, we are to employ Graham's law of diffusion. Please note the difference between the two equations below:
$\frac{r_1}{r_2} = \sqrt{\frac{R.M.M_2}{R.M.M_1}} \qquad \frac{t_1}{t_2} = \sqrt{\frac{R.M.M_1}{R.M.M_2}}$
The first is employed when comparing the diffusion rate directly while the second one, although having the same meaning, is used for time comparison.
t_{CH_4} = 20s, RMM_{CH_4} = 16, RMM_{SO_2} = 64, t_{SO_2}?
$\frac{t_{CH_4}}{t_{SO_2}} = \sqrt{\frac{R.M.M_{CH_4}}{R.M.M_{SO_2}}}$
$\frac{20}{t_{SO_2}} = \sqrt{\frac{16}{64}}$
$\frac{20}{t_{SO_2}} = \frac{1}{2}$
$t_{SO_2} \times 1 = 20 \times 2,\ t_{SO_2} = 40s$
The correct option C

9. $A + B \rightarrow C$
The expression **r= k [A] [B]** is known as the rate law. **The constant K is known as the rate constant.** The rate law/rate equation gives the dependence of reaction rate on the concentration of reacting species. The concentration is represented by [ ]. For instance, the conc. Of A = [A] while the conc. Of B = [B] the exponent of each concentration term (which is one for

both A and B here) is known as the order of the reaction with respect to each reactant.

The correct option is B

10. The shorthand notation for Daniel cell is Zn/Zn2+//CU2+?CU

Don't worry! Anytime you are to identify a shorthand notation of an electrochemical cell, you are to go for the one that has the oxidized forms of the two compartments at the Centre. For instance, in the Daniel cell, the two ions are those at the Centre (separated by the salt bridge). Any one that does not conform to this is unarguably wrong. Also, the compartment where oxidation occurs must be by the left hand side while the one of reduction must be by the right hand side

The correct option is A

11. Urea, $(NH_2)_2CO$ contains nitrogen, hydrogen, carbon and oxygen. As can be seen from the formula.

Bauxite, $Al_2O_3.2H_2O$ contains aluminum, oxygen and hydrogen. Clay is a mixture of different substances such as quart and mica bound together by hydrated aluminum silicate, $Al_2Si_2O_7.2H_2O$. from the formula of the silicate, it can be seen that clay contains H, O, Al and Si (as requested)

The correct option is D

12. Burning of bush, rusting of iron and decay of bitter leaves are chemical reaction as these processes are irreversible. The dissolution of potassium hydroxide is reversible. For instance, we can recover the potassium hydroxide by evaporating the water off

The correct option is D

13. The Solubility of $KClO_3$ quoted as $20moldm^{-3}$ indicates that $1dm^3(1000cm^3)$ of water can hold 20mol of the solute at the given temperature. If this is so, $40cm^3$ of water can only hold $\left(\frac{20mol}{1000cm^3} \times 40cm^3\right)$ => 0.80mol

But how many moles are we given?

Mole of $KClO_3$ given =

Mass = 100g,

Molar mass = 39 + 35.5 + (16 × 3)

= $122.5gmol^{-1}$

As can be seen, the amount is more than can be dissolved.

Excess $KClO_3$ mole

= 0.81633mol – 0.80mol

Mass of excess $KClO_3$ = (0.01633 × 122.5) g

= 2g

Percentage undissolved = $\frac{2g}{100g} \times 100\% = 2\%$

14. A primary amine has just one alkyl group attached to that nitrogen atom.

$CH_3 – N – H$	$CH_3CH_2 – N –H$
Methylamine	Ethylamine

A secondary amine has two alkyl groups attached to the nitrogen atom

$CH_3 – N – CH_3$	$CH_3CH_2 – N – CH_2CH_3$
|	|
CH_3	CH_2CH_3
Dimehylamine	Diethylamine

A tertiary amine has three alkyl groups attached to the nitrogen atom

$CH_3 – N – CH_3$	$CH_3CH_2 – N – CH_2CH_3$
Thrimethylamine	Triethylamine

The correct option is C

15. Sulphur is a non – metal. As such it tends to ionize by electron loss S = 2,8,6

After gaining two electrons, the sulphide ion, S^{2-} has the structure S^{2-} = 2, 8, 8

This is an **inert** configuration! This is because it is the structure of an **inert gas** – argon.

The correct option is A

16. N = 4, *l* = 3

In an electronic configuration, azimuthal quantum number, l describes the shape of the orbital.

L = O describes S – orbital

L = 1 describes P – orbital

17. The gas that turns dichromate green is either SO_2 or H_2S. Of the substances given, the only substance that can generate SO_2 is $Na_2S_2O_3.5H_2O$.

The equation of reaction is: $Na_2S_2O_3 + 2HCl \rightarrow 2NaCl_{(aq)} + 6H_2O_{(1)} + SO_{2(g)} + S_{(s)}$

The correct option A

18. $NH_4NO_3 \rightarrow NH_4^+ + NO_3^-$

In NH_4^+	In NO_3^-
N + (1 × 4) = +1	N + (-2 × 3) = -1
N + 4 = +1	N – 6 = -1
N = + 1 – 4	N – 1 + 6
= -3	= +5

Hence, the values are -3 and +5 respectively.

The correct option is B

19. The presence of terminal alkynes can be detected using $AgNO_3$, CuCl (not $CuCl_2$) and Na in liquid ammonia.

The correct option is D

20. $2HCl + CaCO_3 \rightarrow CaCl_2 + H_2O + CO_2$

$2HCl + K_2CO_3 \rightarrow 2KCl + H_2O + CO_2$

We are simply to get the volumes of HCl required for the two reactions and sum them up.

THE FIRST REACTION

Mole of $CaCO_3 = \frac{2g}{100gmol^{-1}}$ = 0.02mol

Two moles of HCl will be required for each of $CaCO_3$. Then, (0.02 × 2) mole will be required for 0.02mol of $CaCO_3$.

Mole of HCl required = (0.02 × 2)mol = 0.04mol

Mole = concentration x volume

Volume = $\frac{mole}{\text{concentration}}$

= $\frac{0.04mol}{0.1moldm^{-3}} \rightarrow 0.4dm^3$

=> $(0.4 \times 1000)cm^3$

= $400cm^3$

THE SECOND REACTION

Mole of K_2CO_3 = $0.1moldm^{-3} \times \left(\frac{25}{1000}\right)dm^3$ = 0.0025mol

Since two moles of HCl will be required for each mole of K_2CO_3, mole of HCl = (0.0025 × 2) mole = 0.005

Mole of HCl = 0.005mol

Volume = $\frac{mole}{conc.} = \frac{0.005mol}{0.1moldm-3}$

= $0.05dm^3$

= $(0.05 \times 1000)cm^3$

= $50cm^3$

Total volume of HCl for the two reactions

= $400cm^3 + 50cm^3 \rightarrow 450cm^3$

The correct option is D

21. $^{226}_{88}Ra \rightarrow {}^{x}_{86}Rn + \propto -particle$

We just need to set up simple linear equation to solve this

$^{226}_{88}Ra \rightarrow {}^{x}_{86}Rn + ^{4}_{2}\propto$

226 = x + 4, **x** = 226 – 4 = **222**

If the atomic number is not given, we solve in a similar manner. Let us say it is represented by y. then, 88 = y + 2, y =88 – 2 = 86

The correct option is C

22. In the electrolysis of copper (ii) sulphate solution using copper electrodes. The copper anode **dissolves** and get **deposited** on the cathode. Hence, the **respective** processes are **dissolution** and **deposition**.

The correct option is B

CHEMISTRY 2010

1. Whose experiment showed that the atom has a tiny positively charged nucleus? (a) Thompson (b) Rutherford (c) Millikan (d) Dalton
2. Which of the quantum number divides shells into orbitals? (a) principal (b) subsidiary (c) magnetic (d) spin
3. Which of these statements is / are correct of a proton? i. the mass of a proton is one-twelfth the molar mass of carbon ii. The mass of a proton is 1840 times the mass of an electron iii. The mass of a proton is 1.0008g (a) ii only (b) i, ii and iii (c) i only (d)i and ii only
4. Candidate devised the following for the separation of the components of some mixtures. i. components of ink; principles involved is chromatography ii. Components of water and kerosene principle involved is separating funnel iii. Components of iodine and sodium chloride, principles involved is sublimation. In which of the above is the principle used correct? (a)i only (b) ii only (c) iii only (d)i, ii and iii
5. Which of the following procedures will separate a mixture of sand sodium chloride and iodine into its components? A. add water; filter; sublime; evaporate to dryness B. add water sublime filter; evaporate to dryness C. sublime; filter, add water; evaporate to dryness D. sublime; add water; filter; evaporate to dryness
6. The type of bonds in ammonium chloride are A. covalent and electrovalent B. dative and covalent C. dative and electrovalent D. covalent, dative and electrovalent
7. Which of the following types of bonding does not produce a compound? A. ionic bonding B. covalent bonding C. dative bonding D. metallic bonding
8. The combining powers of HCO_3, O, Na, Cl, respectively are A. -2, +1, -1, +1 B. 1 2, 1, 1 C. +1, -2, +1, -1 D. none of these
9. What is the chemical formula of the compound containing: 6.02 x 10^{23} atoms of hydrogen, 35g of chlorine, and 4 moles of oxygen atoms? A. HCl_4O B. HClO C. $HClO_4$D. HCl_2O_4
10. $200cm^3$ of hydrogen were collected over water at 30^0C and 740mm of Hg. Calculate the volume of the gas at s.t.p If the vapour pressure of water at the temperature of the experiment is 14mm of Hg. A. $168.25cm^3$ B. $176.40cm^3$ C. $185.4cm^3$ D. -$172.14cm^3$
11. A given mass of gas occupies a certain volume at 300K. At what temperature will its volume be doubled? A. 400K B. 480K C. 550K D. 600K
12. The basic assumption in the kinetic theory of gas that: "force of attraction and repulsion between gaseous molecules are negligible" implies that: A. molecules will continue their motion indefinitely B. gases will occupy any available space C. gases can be compressed D. none of the above
13. Which of the following is true of a sample of hydrogen gas whose mass is 4.00g under a pressure of 2atm and a temperature of 27^0C? (H = 1, R = 0.082 lit atm. Mol^{-1} deg^{-1}) A. its volume is 24.6 litres B. it contains 6.02 x 10^{23} molecules C. it exists as atoms because of temperature D. none of the above
14. The following are chemical entities identifiable during qualitative analysis. i. SO_4^{2-} ii. H_3O iii. NH_4 iv. OH. Which of them can be detected-by litmus paper? A. ii &iv only B. ii only C. I & iii only D. iv only
15. I. $NaHCO_3$ II. $HaHSO_4$ III. NaCl. Which of these will dissolve in water to give alkaline solution? A. i, ii & iii B. ii only C. i only D. i& ii only
16. Burning of 0.46g of ethanol produced heat that raised the temperature of 100g of water by 30^0C. Calculate the heat of combustion of ethanol, C_2H_5OH. (C = 12, H = 1; O = 16) A. 50 $KJmol^{-1}$ B. $900KJmol^{-1}$C. $1200KJmol^{-1}$ D. $1000KJmol^{-1}$
17. When chlorine is bubbled into potassium iodine solution: A. a white precipitate is seen B. reddish brown colour develops C. solution remains colourless D. blue colour is seen
18. $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$. In the reaction above, an increase in pressure will A. decelerate the reaction B. increase the yield of PCl_3 C. increase the yield of PCl_5 D. accelerate the reaction

19. A saturated solution of silver trioxocarbonate (IV), was found to have concentration of 1.30 x 10^5mol dm^{-3}. The solubility product of trioxocarbonate (IV) is A. 8.79 x 10^{-15} B. 1.69 x 10^{-10} C. 1.82 x 10^{-10}D. 9.84 x 10^{-10}
20. A Zinc half-cell is connected to an iron half-cell through a salt bridge and both are also connected through a copper wire. At which electrode is reduction taking place and which electrode is positively charged? A. zinc, zinc B. iron, iron C. Zinc, iron D. Iron, zinc
21. Which of the following is the difference between an electrolytic cell X and electrochemical cell Y: A. Anode in X is-ve while anode in Y is +ve B. in X, oxidation takes place at the anode while in Y reduction takes place at the anode C. in X, anode is positive while in Y anode is negative D. in X, chemical energy is converted into electrical energy while in Y electrical energy is converted into chemical energy
22. What mass of bromine will saturate complete 6.8g of 3-methylbut-1-yne (H = 1; C = 12; Br = 80). A. 16g B. 32 g C. 12 g D. 24 g
23. 100cm^3 of oxygen and 10cm^3 of butane measured at room temperature and pressure were mixed and exploded. Determine the volume of the mixture when bought back to the original conditions of measurements. A. 125cm^{3}B. 100cm^3 C. 75cm^3 D. none of these
24. Sulphur A. forms two alkaline oxides B. is spontaneously inflammable C. burns with a blue flame D. conducts electricity in the molten state
25. Which of the following combination of reagents will reacts to give chlorine gas? A. sodium chloride, conc H_2SO_4 and Manganese (IV) oxide B. potassium tetraoxomangante C. potassium trioxochloide (V) and conc. H_2SO_4D. potassium tetraoxomangante (VI) and conc. H_2SO_4.

CHEMISTRY 2010 SOLUTION

1. J.J. Thompson discovered the electrons during the famous cathode ray experiment. **Ernest Rutherford discovered the nucleus** during his alpha-scattering experiment. He bombarded a thin gold foil with fast moving alpha particles. He noted that the majority of the particles passed through the foil undeflected while a few of them suffered deflection. He then suggested an atomic model in which the mass is concentrated at the centre (the nucleus) and a large space surrounding the nucleus where electrons are found. **He also related that the deflection of some alpha particles could only be caused by a positive charge at the nucleus** (since the alpha particles themselves are positive). Milikan discovered the charge on an electron through the oil-drop experiment. He obtained a value of -1.6×10C^{-19}. The atomic theory was proposed by John Dalton.
The correct option is B
2. The principal quantum number determines the energy of an electron due to its distance from the nucleus. Orbitals are obtained from shells using azimuthal quantum number. The azimuthal quantum number also gives the shape of an orbital. The magnetic quantum number gives the degeneracy of the orbitals. The spin quantum number describes the spinning of an electron in an orbital. An electron can only spin in the clockwise direction or the anticlockwise direction and never the two together. Does it mean the answer is not in the options? It is there! The **azimuthal quantum number** is also known as the **angular** quantum number, the **secondary** quantum number, the **sommerfield** quantum number or the **subsidiary** quantum number.
The correct option is B
3. (i) The mass of a proton is one-twelfth the molar mass of carbon-12.
(ii) The proton is 1840 times as heavy as an electron.
(iii) The mass of a proton is 1amu (1.67×10^{-24}g) and not 1.008g.
The correct option is D
4. (i) The components of ink are separated by chromatography
(ii) The components of water and kerosene are separated using a separating funnel.
(iii) The components of iodine and sodium chloride are separated by sublimation.
The correct option is D
5. **Sublimation** will remove the iodine component. **Water can then be added**. Since NaCl is soluble in water while sand is not, the sand can be **filtered off**. The filtrate which contains sodium chloride solution can be **evaporated** to obtain the salt in the dried form.
The correct option is D
6. Ammonium chloride has covalent, dative and ionic bonds. Please refer to the solution to question 15, 2012 for detailed explanation.
The correct option is D
7. The metallic bonding is not associated with compound formation. It is the bond holding the atoms of metals together.
The correct option is D
8. Talking about combining power, we are not concerned about the sign (may be negative or positive). Apart from this, it directly relates with the valency or oxidation state.
HCO_3^- has a charge of -1, then, the combining power is 1. Oxygen has a valency of -2, thus, the combining power is 2, sodium has a valency of +1, hence, the combining power is 1. Chlorine has a valency of -1, therefore, the combining power is 1.
The correct option is B
9. Mole of hydrogen atoms = $\frac{Number\ of\ H\ atoms}{Avogadro\ Number}$

$= \frac{6.02\times10^{23}}{6.02\times10^{23}} = 1$

Mole of chlorine atoms = $\frac{Mass\ of\ atoms}{R.A.M}$

$= \frac{35}{35.5} = 0.98 = 1$

Note that the molecular mass, $71gmol^{-1}$ is not to be employed for chlorine. It is just because we are dealing with chlorine atom and not chlorine molecules.

Mole of oxygen atoms is already stated as 4. Putting these together, the formula is $HClO_4$.

The correct option is C

10. Volume of H_2 at 30°C and 740mmHg = V_1 = 200cm³, T_1 = 30°C = 303K, P_1 = 740mmHg – 14mmHg = 726mmHg

Volume at stp = V_2

T_2 = 273K, P_2 = 760mmHg

Using the general gas law

$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

Then, $V_2 = \frac{P_1V_1T_2}{P_2T_1} = \left(\frac{726\times200\times273}{760\times303}\right)$cm³

= 172.14cm³

The correct option is D

Note that when a gas is collected over water, the measured pressure is the sum of gas pressure and the pressure due to water vapour.

Pressure of gas = Total pressure – Vapour pressure

That was why we subtracted the vapour pressure of water from the given pressure in the calculation above.

11. $V_1 = x$, $V_2 = 2x$, T_1 = 300K, T_2= ?

$\frac{V_1}{T_1} = \frac{V_2}{T_2}$

$\frac{x}{300} = \frac{2x}{T_2}$

$T_2 = \frac{300\times2x}{x} => 600K$

The correct option is D

12. Since the forces of attraction and repulsion are negligible, gasses can be compressed easily with no hindrance.

The correct option is C

13. Mass of H_2 = 4.00g

Mole, n of H_2 = $\left(\frac{4}{2}\right)$mol = 2mol

$$PV = nRT$$

$V = \frac{nRT}{P} = \left(\frac{2\times0.082\times300}{2}\right)$litres

= 24.6litres

Number of molecules = Mole × Avogadro number

= 2mol × $6.02\times10^{23}mol^{-1}$

= 1.204×10^{24}molecules

The correct option is A

14. H_3O^+ changes blue litmus paper to red while OH^- changes red litmus paper to blue.

The correct option is A

15. $NaHCO_3$ dissolves to give an alkaline solution $NaHSO_4$ produces an acidic solution upon dissolution in water NaCl dissolves to produce a neutral solution.

The correct option is C

16. Molar mass of ethanol, C_2H_5OH = $46gmol^{-1}$

Mole of C_2H_5OH = $\frac{0.46g}{46gmol^{-1}}$ => 0.01mol

Heat gained by water = mcΔθ = (100g × $4.2Jg^{-1}{}^oC^{-1}$ × 30°C) = 12600J

In essence, 0.01mol of ethanol generates 12600J of heat.

The heat of combustion of a substance is the heat evolved when 1mole of the substance is burnt in excess oxygen.

0.01mol produces 12600J. then, 1mol will produce $\left(\frac{12600}{0.01}\right)$ = 1260,000J = 1260kJ

Then, the heat of combustion is $1260kJmol^{-1}$

The correct option is C

17. This is about the power trend in halogens: F >Cl> Br > I

A halogen can displace another halogen if the former is superior to the latter.

Since chlorine is superior to iodine, it displaces the latter from its solution. The colour seen is the colour of iodine after being displaced by chlorine. So, what colour do you expect? Actually, iodine is a dark blue solid, but when dissolved in water, the colour is reddish brown.

The correct option is B

18. $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$

As already known, an increase in pressure favours the side that has less mole of gas(es) while a decrease in pressure favours the opposite. Then, in this reaction will favour the backward reaction i.e. the formation of PCl_5.

The correct option is C

19. $Ag_2CO_{3(s)} \rightleftharpoons 2Ag^+ + CO_3^{2-}$

$\downarrow \quad \downarrow \quad \downarrow$

$x \quad 2x \quad x$

$K_{SP} = (2x)^2 . x = 4x^2.x = 4x^3$

where x = solubility = conc. = $1.3\times10^{-5}moldm^3$

$K_{SP} = 4x^3 = 4(1.3\times10^{-5})^3 = 8.79\times10^{-5}$

The correct option is A

20. In the construction of an electrochemical cell, the more reactive of the two species is made the anode. Hence, Zn is the anode while Fe is the cathode.

Also, the cathode is the positive electrode while the anode is the negative electrode (Note that reverse is this case for an electrolytic cell). To crown it, oxidation always occurs at the anode while reduction occurs at the cathode.

The correct option is B

21. An electrolytic cell converts electrical energy to chemical energy while an electrochemical cell does the opposite.

- In an electrolytic cell, electrons are forced in from an external source while an electrochemical cell generates electrons.
- The anode is the positive electrode while the cathode is the negative in an electrolytic cell. Opposite is the case for an electrochemical cell.

- In an electrolytic cell, electrodes are in the same compartment while they are in different compartments in an electrochemical cell.
- An electrolytic cell does not require a salt bridge while an electrochemical cell does.
- An electrolytic cell requires only one electrolyte while an electrochemical cell usually takes two.

The correct option is C

22. H H Br Br

| | | |

CH_3–C–C ≡ CH + $2Br_2$ → CH_3 – C – C – C – C –H

| | | |

CH_3 CH_3 Br Br

Molar mass of 3-methylbut-1-yne = $68gmol^{-1}$. 4 bromine atoms (4 × 80g = 320g) will be required for 1mol (68g) of the compound. Thus, 6.8g of the compound will require $\frac{320g}{68g} \times 6.8g$ => 32g

The correct option is B

23. $2C_4H_{10(g)} + 13O_{2(g)} \rightarrow 8CO_{2(g)} + 10H_2O_{(l)}$

2moles of butane require 13 moles of O_2. Thus, $10cm^3$ of butane will take $\left(\frac{13}{2} \times 10\right)$ cm^3 of O_2 = $65cm^3$

This shows that oxygen is in excess. Since the mixture is to be brought back to the original conditions i.e. room temperature and pressure, the water vapour will condense – it ceases from being a gas!

$10cm^3$ of butane will react with $65cm^3$ of oxygen to produce $10cm^3 \times \frac{8}{2}$ of CO_2 = $40cm^3$ of CO_2

Unused O_2 = $100cm^3 - 65cm^3 = 35cm^3$

Residual gas = CO_2 + unused O_2

Volume of residual gas = $40cm^3 + 35cm^3 = 75cm^3$

The correct option is C

24. Sulphur forms acidic and not alkaline oxides. It burns with a blue flame and it is not a conductor of electricity.

The correct option is C

25. Chlorine is prepared by heating sodium chloride with conc. H_2SO_4, using MnO_2

$MnO_2 + 2NaCl + 2H_2SO_4 \rightarrow MnSO_4 + Na_2SO_4 + 2H_2O + Cl_2$

The correct option is A

CHEMISTRY 2009 QUESTIONS

1. The flame used by welders in cutting metals is A. butane gas flame B. Acetylene-hydroflane C. kerosene flame D. oxyacetylene E. oxygen flame
2. Consecutive members of an alkane homogenous series differ by A. CH B. CH_2 C. CH_3 D. C_3H_3 E. C_nH2_{n+2}
3. Which of these will dissolve in HCl? Mg, Fe, Pb and Cu A. all the metals B. Mg, Fe and Cu C. Mg, Fe and Pb D. Mg and Fe only E. Mg only
4. Stainless steel is an alloy of A. carbon, iron and lead B. carbon, iron and chromium C. carbon, iron and copper D. carbon, iron and silver E. carbon and iron only
5. What volume of 0.50 MH_2SO_4 will exactly neutralize $20cm^3$ of 0.1M NaOH solution? A. $2.0cm^3$ B. $5.0cm^3$ C. $6.8cm^3$ D. $8.3cm^3$ E. $10.4cm^3$
6. A gas that can behave as a reducing agent towards chlorine and as an oxidizing agent towards A. O_2 B. NO C. SO_2 D. NH_3 E. CO_2
7. An element that can exist in two or more different structure forms which possess the same chemical properties is said to exhibit A. Polymerism B. Isotropy C. Isomorphism D. Isomerism E. Allotropy
8. The hybridization of the carbon atom in ethyne is A. SP^n B. SP^3 C. SP^2 D. SP E. ES
9. In the Haber process for the manufacturer of ammonia, finely divided iron is used as A. an ionizing agent B. a reducing agent C. a catalyst D. a dehydrating agent E. an oxidizing agent
10. Nitrogen can best be obtained from a mixture of oxygen and nitrogen by passing the mixture over A. potassium hydroxide B. heated gold C. heated magnesium D. heated phosphorus E. calcium chloride
11. At STP, how many litres of hydrogen can be obtained from the reaction of $500cm^3$ of 0.5M H_2SO_4 excess in metal A. $22.4dm^3$ B. $11.2dm^3$ C. $65dm^3$ D. $5.6dm^3$ E. $0.00dm^3$
12. Tetraoxosulphate (VI) ions are final test using A. acidified silver nitrate B. acidified barium chloride C. lime-water D. dilute hydrochloric acid E. acidified had nitrate
13. Which of the following is NOT the correct product formed when the parent metal is heated in air? A. calcium oxide (CaO) B. Sodium Oxide (Na_2O) C. Copper (II) oxide (CuO) D. Tri-iron tetroxide (Fe_3O_4) E. Aluminum Oxide (Al_2O_3)
14. Which of the following roles does sodium chloride play in soap preparation? It A. reacts with glycerol B. purifies the soap C. accelerates the decomposition of the fat and oil D. separates the soap form the glycerol E. coverts the fat acid to its sodium salt
15. The function of sulphur during the vulcanization of rubber is to A. act as catalyst for the polymerization or rubber molecules B. convert rubber from thermosetting thermo plastic polymer C. form chains which bind rubber molecular together D. break down rubber polymer molecule E. shorten the chain length of rubber polymer
16. An element with atomic number twelve is like to be A. electrovalent with a valency of 1 B. electrovalent with a valency of 2 C. covalent

with a valency of 2 D. covalent with a valency of 4

17. Which of the following is an acid salt? A. $NaHSO_4$ B. Na_2SO_4 C. CH_3CO_2Na D. Na_2S E. C_2H_5
18. Which of the following compounds is NOT formed by the action of chlorine on methane? A. CH_3Cl B. C_2H_5Cl C. CH_2Cl_2 D. $CHCl_3$E. CH_4CL
19. Starch can be converted to ethyl alcohol by A. distillation B. fermentation C. isomerization D. cracking E. ozonolysis
20. How many isomers can be formed from organic compounds with formula C_3H_8O A.2 B. 3 C. 4 D. 5 E. 1
21. When platinum electrodes are used during the electrolysis of copper (II) tetraoxoxulphate (V) solution, the solution gets progressively A. acidic B. basic C. Natural D. Atmospheric
22. Which of the following physical properties decreases across the period table A. ionization potential B. electron affinity C. electron negativity D. atomic radius E. electropositive reaction
23. Which of these has the lowest pH value? To be continued A. calcium trioxocarbonates B. sodium trioxocarbonate (IV)C. hydrochloric acid D. ethanoic acid E. hydrocarbon acid
24. Which of the following is used in fire extinguishers? A. carbon (III) oxide B. carbon (IV) oxide C. Sulphur (IV) oxide D. Amonia E. Sulphur (III) oxide
25. Mortar is NOT used for under water construction because A. it hardens B. its hardening does not depend upon evaporation C. it requires concrete to harden D. it will be washed away by the flow of water E. it softens when exposed

CHEMISTRY 2009 SOLUTION

1. Oxyaceytylene is used by welders for welding and for cutting metals
The correct option is D
2. Consecutive members of any homologus series differ in their molecular formula by a CH_2 group and in their relative molecular mass by 14.
The correct option is B
3. Since cu is below hydrogen in the reactivity series, it is clearly understandable why it can not displace hydrogen from HCl. Since metals Mg, Fe and Pb are above hydrogen, one would expect all of them to be able to displace hydrogen from HCl. However, dilute hydrochloric acid does not have an effect on lead.
The correct option is D
4. Steel comprises iron and carbon. Stainless steel comprises iron, carbon and chromium.
The correct option is
5. $H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$
$\frac{C_aV_b}{C_bV_b} = \frac{n_a}{n_b}$
$\frac{0.50 \times V_a}{0.1\ 20} = \frac{1}{2}$
$V_a = \frac{0.1 \times 20 \times 1}{0.5 \times 2}$ = 2.0cm^3
The correct option is A
6. SO_2 is a good reducing agent in the presence of better reducing agent such as H_2S, SO_2 serves as an oxidizing agent.
The correct option is C
7. Allotropy is the ability of an element to exist in two or more different forms. Allotropes of the same element normally differ in their physical properties. Their chemical properties are usually the same.
The correct option is E
8. A carbon – carbon triple bond has SP hybridization. A carbon to carbon double bond has SP^2 hybridization while a carbon to carbon ion bond has SP^3 Hybridization – C ≡ C – H (ethyne)
Since the two carbon atoms are joined by triple bond, ethyne has SP HYBRIDIZATION.
9. Finely divided iron is used as a catalyst in the Haber process for the manufacture of amnonia.
The correct option is C
10. When air is passed over heated phosphorous, phosphorous combines with the oxygen in the air, leaving nitrogen gas as the main substance left
$4P_{(s)} + 5O_{2(g)} \rightarrow 2P_2O_{5(g)}$
The correct option is D
11. $X + H_2SO_4 \rightarrow XSO_4 + H_2$
Mole of H_2SO_4 = 0.5moldm^{-3} × $\left(\frac{500}{1000}\right)$dm^3 = 0.25mol
According to the equation, 1 mole of H_2SO_4 produces 1 mole of H_2, then, 0.25mol of H_2SO_4 will produce 0.25mol of H_2.
∴mole of H_2 = 0.25mol
Volume of H_2 at s.t.p = mole × G.M.V
= 0.25 × 22.4dm^3
= 5.6dm^3
The correct option is B
12. The presence of SO_4^{2-} can be confirmed using $BaCl_2$ and HCl. The acidified $BaCl_2$ is first added. Then, a precipitate is formed. The precipitate must not dissolve when excess HCl is added. If it dissolves, it suggests other anions such as CO_3^{2-} and SO_3^{2-}
The correct option is D
13. Unlike others, sodium reacts when heated in air to form a non-stoichiometric oxide. It forms sodium peroxide, Na_2O_2 instead of the normal sodium oxide, Na_2O.
The correct option is B
14. Sodium chloride is added to soap to separate the glycerol out of it. It also reduces the solubility of the soap.
The correct option is D

15. Sulphur is used in the vulcanization of rubber. It forms cross-linkages that bind the respective layers together.
The correct option is C
16. Since magnesium is an electropositive element, it is said to be electrovalent. It has a valency of +2.
The correct option is B
17. $NaHSO_4$ is an acid salt. Na_2SO_4, Na_2S and CH_3COONa are normal salts.
The correct option is A
18. Methane reacts with chlorine in the presence of sunlight to give a succession of substitution products
$CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$
$CH_3Cl + Cl_2 \rightarrow CH_2Cl_2 + HCl$
$CH_2Cl_2 + Cl_2 \rightarrow CHCl_3 + HCl$
$CHCl_3 + Cl_2 \rightarrow CCl_4 + HCl$
There is no such compound as CH_4Cl
The correct option is E
19. Starch can be converted into ethylalcohol (ethanol) by fermentation. Fermentation is a degradation of large carbohydrate molecules (such as starch) into smaller molecules (such as ethylalcohol and CO_2)
20. For the formula C_3H_8O, alkanols and an alkoxyalkane are obtainable. $CH_3CH_2CH_2OH$ (propan-1-ol), $CH_3CH(OH)CH_3$ (propan-2-ol) and $CH_3 - O - CH_2CH_3$ (methoxyethane).
The correct option is B
21. During this electrolysis, Cu^{2+} is discharged at the cathode while OH^- is discharged at the anode. The progressive removal of OH^- causes the acidity of the electrolyte to increase (since H^+ is now excess).
The Correct option is A
22.

Properties	Across the period	Down the group
Atomic radius	Decrease	Increase
Ionization energy	Increase	Decrease
Electron affinity	Increase	Decrease
Ionic radius	Decrease	Increase
Electronegativity	Increase	Decrease
Metallic character	Decrease	Increase

The correct option is D
23. HCl is the only strong acid. Others are either non-acid or weak acid ($CaCO_3$ and Na_2CO_3 are salts and not acids, ethanoic acid is a weak acid) HCl has the lowest pH.
The correct option is C
24. Being non-flammable, carbon(iv) oxide is used as a fire extinguisher. Its high density also aids this function.
The correct option is B
25. Mortar cannot be used for under water construction because it will be washed away by the flow of water
The correct option is D

CHEMISTRY 2008

1. Silver trioxonitrate (V) on heating, gives A. Ag, N_2O and O_2 B. Ag_2O, N_2 and O_2 C. Ag, O and N, O D. Ag, NO_2 and O_2
2. The most reactive halogen is A. Cl_2B. Br_2 C. F_2 D. 1_2
3. 0.79g of a gas at s.t.p occupied a volume of $250cm^3$. What is the relative molecular mass of the gas? (G.M.V. at s.t.p. = $22.4dm^{-3}$) A. 17 B. 32 C. 64 D. 71
4. The relationship between the density (d) of a gas and the rate (r) at which the gas diffuses is A. r = Kd B. $r = Kd^{-1/2}$ C. $r = Kd^{1/2}$ D. $r = Kd^{-1}$
5. The pressure exerted by a sample of a gas confined in $5.86dm^3$ container at 20^0C is 4.1 atm. what is the number of moles of the gas in the sample? (R = $0.082dm^3atm\ mol^{-1}K^{-1}$) A. 1.00 B. 2.00 C. 3.00 D. 4.00
6. $50cm^3$ of hydrogen are sparked with $100cm^3$ of oxygen at 110^0C and 1 atm. If the whole reaction mixture passes through an alkaline pyrogallol solution, the volume of the residual gas is A. $125cm^3$B. $100cm^3$ C. $75cm^3$ D. $50cm^3$
7. Inter-atomic combinations involves the A. neutrons in the nucleus only B. protons in the nucleus only C. electrons in the outer shell only D. electrons in all the shells.
8. Which of the following is not a characteristic property of ionic compounds? A. solubility in polar solvents B. low melting points C. conduction of electricity in aqueous solution D. fast reactions in solution
9. The compound with the heights ionic character among the following is A. PCl_3 B. CCl_4 C. BCl_3 D. C_SCl
10. Why is the hydrogen gas not found in the atmosphere? It readily reacts with A. carbon (IV) oxide B. oxygen C. nitrogen D. carbon (II) oxide
11. Apart from water, the other product(s) of the neutralization reaction between NaOH solution and nitrogen (IV) oxide is/are A. $NANO_2$ B. $NANO_3$ C. $NANO_3$ and HNO_3 D. $NANO_2$ and $NANO_3$
12. An acid and its conjugate base A. are oppositely charged B. differ only by a hydroxide ion C. differ only by an electron D. differ only a proton
13. A complex salt is A. $KAl(SO_4)_2.12H_2O$B. $Cu(NH_3)_3Cl_2$ C. $K_2S_2O_3.5H_2O$ D. $Mg(OH)Cl$
14. What happens to the conductivity of an electrolyte as its concentration reduces? A. increases B. decreases C. us unaffected D. becomes resistivity
15. If the cost of electricity required to deposit 1g of aluminum is ₦4.00, how much would it cost to deposit 24g of copper? (Al = 27, Cu = 64) A. ₦27.02 B. ₦37.02 C. ₦47.02 D. ₦57.02

16. The overall reaction in an electrochemical cell is $Mg_{(s)} + Cu^{2+}_{(aq)} \rightarrow Mg^{2+}_{(aq)} + Cu_{(s)}$ What is the symbolic representation of the cell? A. $Mg_{(s)}$ + /$Mg^{2+}_{(aq)}$ / $Cu^{2+}_{(aq)}$ / $Cu_{(s)}$ B. $Mg_{(aq)}$ / $Mg^{2+}_{(s)}$ / $Cu^{2+}_{(s)}$ / $Cu_{(aq)}$C. $Cu^{2+}_{(aq)}$ / $Cu_{(s)}$ / $Mg^{2+}_{(aq)}$ D. $Cu_{(s)}$ / $Cu^{2+}_{(aq)}$ / $Mg^{2+}_{(aq)}$ / $Mg_{(s)}$
17. Which of the following metals can be used as sacrificial cathode for preventing corrosion of a length of iron pipe? A. Ag B. Cu C. Mg D. Mn
18. A particle that contains 8 protons, 9 neutrons and 7 electrons could be written as A. $^{16}{}_{8}O$ B. $^{17}{}_{8}O^-$ C. $^{17}{}_{9}O^-$ D. $^{17}{}_{8}O^-$
19. The decreasing order of the magnitude of energy changes isA. phase, chemical, nuclear B. chemical, nuclear, phase C. nuclear, phase, chemical D. nuclear, chemical, phase
20. 0.92g of ethanol raised the temperature of 100g of water from 298K to 312.3K when burned completely. What is the heat of combustion of ethanol? A. +300KJmol^{-1} B. +3000KJmol^{-1}C. -300KJmol^{-1}D. -3000KJmol^{-1}[C = 12; H = 1; O = 16; specific heat capacity of water = 4.2Jg^{-1} K^{-1}]
21. The highest level of molecular disorderliness is found in A. ice at -10^{0}C B. water at 100^{0}C C. steam at 100^{0}C D. ice at 0^{0}C
22. A reaction is spontaneous at all temperature if A. $\Delta G = 0$ B. $\Delta G > 0$ C. $\Delta S < 0$ and $\Delta H >$ o D. $\Delta S > 0$ and $\Delta H < 0$
23. Which of the following reactions of marble with 100cm^3 of 0.100moldm^3. HCl is fastest? A> 5g of marble limp at 50^{0}C B. 5g of marble powder at 50^{0}C C. 5g of marble powder at 25^{0}C D. 5g of marble lump 25^{0}C
24. $A_{(g)} + 2B_{(g)} \rightarrow C_{(g)}$ In the reaction represented by the equation above, the rate of appearance of C is found experimentally to be independent of the concentration of A and to increase four folds when the concentration of B is doubled. The rate law for the reaction is A. Rate = $K(A)^0(B)^4$B. Rate = $K(A)^0(B)^2$ C. Rate = $K(A)(B)^2$ D. Rate = $K(A)^2(B)^0$
25. How is the equilibrium constant for the forward reaction of an equilibrium (K_r) related to that of the reverse reaction (K_r)? A. (K_r) is the additive inverse of (K_r) B. (K_r) is the multiplicative inverse of K_r) C. (K_r) is the same as (K_r) D. the product of (K_r) and (K_r) is zero.
26. What is the concentration of OH^- ions of an aqueous solution of pH 4.4? A. 9.600 x 10^{-10}moldm^{-3} B. 2.512 x 10^{-10}moldm^{-3} C. 9.600 x 10^{-11}moldm^{-3} D. 2.512 x 10^{-11}moldm^{-3}

CHEMISTRY 2008 SOLUTION

1. The trioxonitrates (v) of sodium and potassium are decomposed on heating into the corresponding dioxonitrate (III) and oxygen.
$2NaNO_{3(s)} \xrightarrow[\Delta]{} 2NaNO_{2(s)} + O_{2(g)}$
$2KNO_{3(s)} \xrightarrow[\Delta]{} 2KNO_{2(s)} + O_{2(g)}$
The trioxonitrates (v) of calcium, magnesium, lead, zinc and copper decompose on heating to the corresponding oxides, nitrogen (iv) oxide and oxygen
$2Pb(NO_3)_2 \rightarrow 2PbO_{(s)} + 4NO_{2(g)} + O_{2(g)}$
$2Ca(NO_3)_2 \rightarrow 2CaO_{(s)} + 4NO_{2(g)} + O_{2(g)}$
Those of mercury, silver and gold are decomposed on heating to give the corresponding metals, nitrogen (iv) oxide and oxygen
$2AgNO_{3(s)} \rightarrow 2Ag_{(s)} + 2NO_{2(g)} + O_{2(g)}$
The correct option is D
2. Of the halogens, fluorine is the most reactive. Not only that, fluorine is the most electronegative of all elements.
The correct option is C
3. $Mole = \frac{Volume\ at\ stp}{GMV}$
$= \left(\frac{2500cm^3}{22400cm^3}\right) mol$
= 0.011161mol
$Molar\ mass = \frac{Mass}{Mole} = \frac{0.79g}{0.011161mol}$
= 70.78g/mol
= 71g/mol
Therefore, the relative molecular mass is 71
The correct option is D
4. According to Graham's law of diffusion, the diffusion rate of a gas is inversely proportional to the square root of its density.
$r \propto \frac{1}{\sqrt{d}}$
$r = \frac{k}{\sqrt{d}}$
$= K \times \frac{1}{\sqrt{d}}$
$= K \times \frac{1}{d^{\frac{1}{2}}}$
$= Kd^{\frac{-1}{2}}$
The correct option is B
5. Using the idea gas equation: $PV = nRT$
P = 4.1atm, V = 5.86dm^3, R = 0082atm dm^{3}mol^{-1}k^{-1}, T = 20^{o}C = 20 + 273K = 293K
$n = \frac{PV}{RT} = \left(\frac{4.1 \times 5.86}{0.082 \times 293}\right) mol$
= 1.00mol
The correct option is A
6. $2H_{2(g)} + O_{2(g)} \rightarrow 2H_2O_{(g)}$
Since two moles of hydrogen require 1mole of O_2, 50cm^3 of hydrogen will take just 25cm^3 of oxygen
Excess O_2 => 10cm^3 – 25cm^3 = 75cm^3
Since two moles of hydrogen will produce two moles of $H_2O_{(g)}$ volume of steam is also 50cm^3
Normally, the volume of the residual gas should be the sum of the volume of steam produced and that of the excess oxygen. However, the presence

of alkaline pyrogallol will remove the excess oxygen.

The volume of the residual gas is thus that of steam alone = $50cm^3$

The correct option is D

7. Chemical reactions do not involve protons and neutrons at all. A reaction that involves the nucleons (protons and neutrons) is called a nuclear reaction. Only electrons are involved in chemical reactions. Again, not just electrons, the ones involved are those in the outermost shell alone.

The correct option is C

8. **God forbid!** Ionic compounds do not have low melting points. In the real sense of it, they are characterized by high melting and boiling points.

The correct option is B

9. The highest ionic character is expected of caesium chloride since it is made up of element with substantial electronegativity difference (caesium, a metal and chlorine, a non-metal; each of reputable reactivity).

The correct option is D

10. Hydrogen is not found as a free element in the atmosphere because it readily combines with oxygen.

The correct option is B

11. NO_2 is a mixed anhydride. As such, it produces two acids when dissolved in water – HNO_2 and HNO_3

The sodium hydroxide will react with each of these two acids, producing two different salts.

$NaOH + HNO_2 \rightarrow NaNO_2 + H_2O$

$NaOH + HNO_3 \rightarrow NaNO_3 + H_2O$

The correct option is D

12. A conjugate acid-base pair is a pair of substances which differ in their composition by just one proton (H^+)

The correct option is D

13. A complex salt is a salt that has a complex ion. A complex ion comprises a central metallic ion surrounded by atoms or groups of atoms known as ligands. Examples of complex salts are $K_4[Fe(CN)_6]$, $K_3[Fe(CN)_6]$, $[Cu(NH_3)_4]Cl$ etc.

The correct option is B

14. As concentration reduces, the conductivity of an electrolyte increases. The molar conductivity of the electrolyte is inversely proportional to the square root of its concentration i.e. $\Lambda \propto \frac{1}{\sqrt{Conc}}$, λ = molar conductivity

The correct option is A

15. $Al^{3+} + 3e^- \rightarrow Al$

27g of Al requires 3F of electricity.

1g of Al will require $\left(\frac{3}{27}\right)F = 0.1111F$

$Cu^{2+} + 2e^- \rightarrow Cu$

64g of Cu requires 2F of electricity

24g will require $\left(\frac{2}{64} \times 24\right)F = 0.75F$

We are given ₦4.00 to deposit 1g of aluminum. It simply implies we are buying 0.1111F for ₦4.00. The cost to produce 24g of copper will just mean the price of 0.75F

0.1111F = ₦4.00

0.75F = x

$x = \frac{0.75F \times ₦4.00}{0.1111F}$

= ₦27

The correct option is A

16. $Mg_{(s)} + Cu^{2+}{}_{(aq)} \rightarrow Mg^{2+}{}_{(aq)} + Cu_{(s)}$

The shorthand notation of an electrochemical cell starts with the anode, then the cathode. Anode is where oxidation occurs while reduction occurs at the cathode

$Mg \rightarrow Mg^{2+}$ (oxidation)

$Cu^{2+} \rightarrow Cu$ (reduction)

Then, the short notation will be:

$Mg/Mg^{2+}//Cu^{2+}/Cu$

The correct option is A

17. The metal to be protected must be lower than the protecting metal in the electrochemical series so that the latter goes into solution instead of the former.

Of the elements given, magnesium is the only one that is superior to the metal to be protected – iron

The correct option is C

18. First of all, the number of proton is the atomic number. Also, the atom has undergone the loss of an electron since the protons and electrons are not equal (8 protons and 7 electrons)

The mass number = proton number + neutron number = 8 + 9 = 17

Is the symbol ${}^{17}_{8}O$ or ${}^{17}_{8}O^+$?

Since the atom has lost an electron as already explained, it is ${}^{17}_{8}O^+$.

The correct option is D

19. The greatest energy change is recorded for nuclear reactions, next to phase changes and lastly chemical reactions.

The correct option is C

20. $Mole\ of\ ethanol = \frac{0.92g}{46gmol^{-1}}$ = 0.02mol

Heat gained by water = mcΔθ

= $100g \times 4.2jg^{-1}k^{-1} \times (312.3 - 298)$

= 6006J

The heat of combustion is the heat evolved when 1mole of a substance is burnt in excess oxygen.

The combustion of 0.02mol of ethanol generated 6006J of energy. Thus, 1mol will generate $\left(\frac{6006}{0.02}\right)J$ = 300300J = 300.3KJ

The heat of combustion of ethanol is $300.3Kjmol^{-1}$. Since combustion is always an exothermic reaction, the value should be $-300.3KJmol^{-1}$

The correct option is C

21. The degree of molecular disorderliness increases from solid, through liquid, to gas i.e. gases have the highest degree of disorderliness while solid, have the lowest.

The correct option is C

22. The
23. The greater the surface area in contact with the liquid reaction reactant, the higher the reaction rate. Marble powder has a greater surface area in contact with HCl than marble lump. Also, the higher the temperature the higher the reaction rate. Thus, a reaction is expected to be faster at 50°C than at 25°C
The correct option is B
24. $R = K[A]^0 [B]^x$
Let the [A] = a and [B] = b
$R_1 = K.a^0. b^x$
$R_2 = K.a^0 . (2b)^x$
Also, $R_2 = 4R_1$
Then, $\frac{R_2}{R_1} = 4$
$\frac{K.a^0 .(2b)^x}{K.a^0.b^x} = 4$
Then $\frac{(2b)^x}{b^x} = 4$
$\frac{2^x.b^x}{b^x} = 4$
$2^x = 4$, $2^x = 2^2$
x = 2
The rate law is $R = K[A]^0 [B]^2$
The correct option is B
25. If the equilibrium constant for the forward reaction is x, that of the backward reaction is $\frac{1}{x}$. This implies that one is the multiplicative inverse of the other.
(see: => A multiplicative inverse of a value is what multiplies the value to give 1 e.g. $x \times \frac{1}{x}$)
The correct option is B
26. pH + pOH = 14
4.4 + pOH = 14
pOH = 14 – 4.4 = 9.6
pOH = $-\log_{10}[OH^-]$
9.6 = $-\log_{10}[OH^-]$
$\log_{10}[OH^-] = -9.6$
$[OH^-]$ = antilog (-9.6)
= 2.512×10^{-10}moldm^{-3}
The correct option is B

CHEMISTRY 2007

1. Consider the reaction $2A_{(g)} \rightleftharpoons B_{(g)} + C_{(g)} \Delta H = +25.6KJ$. Which of the following changes will favour the formation of the products of the reaction represented above, at equilibrium? A. decrease in temperature B. increase in pressure C. increase in temperature D. decrease in volume
2. An element $^{226}_{88}X$ undergoes radioactive decay by emitting two alpha particles and a beta radiation. Which of the following nuclei correctly describe the product formed by the reaction? A. $^{222}_{89}T$ B. $^{218}_{84}S$ C. $^{218}_{85}R$ D. $^{222}_{87}Q$
3. Which of the following alkanols will not undergo oxidation reaction by acidified $K_2Cr_2O_7$? A. $CH_3CH_2CH(OH)CH_3$B. $CH_3CH_2CH_2CH_2OH$ C. $CH_3CH_2CH_2C(CH_3)(OH)CH_3$D. none of the above
4. The electronic configuration of the species underlined as in the molecule H_2S is. A. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^4$ B. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$C. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^2$ D. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$
5. The functional group(s) of an amino acid is/are A. NH_2, -COOH B. C_nH2_{n+1}, -COOH C. –COCl, COOH D. COOH, $CONH_2$
6. A student while trying to identify two gases labeled A and B, found that gas A. is acidic to litmus paper and turn acidified potassium dichromate solution green, while gas B. turns red litmus paper blue and forms dense white fume with hydrogen chloride. The correct identity of A and B respectively are A. CO_2 and N_2B. SO_2 and NH_3 C. HCl and NH_3 D. NO_2 and PCl_5
7. Which of the following titrations will have a solution with a pH greater than 7 at the end point (Equivalent point) of the titration?A. titration of sodium hydroxide with tetraoxosulphate(VI) acid B. titration of sodium trioxocarbonate(IV) with hydrochloric acid C. titration of sodium hydroxide with oxalic acid (ethanedioic acid) D. titration of ammonium hydroxide and tiroxonitrate(V) acid
8. 70cm^3 of hydrogen are sparked with 25cm^3 of oxygen at S.T.P. The total volume of the residual gas is A. 20cm^3 B. 35cm^3 C. 45cm^{3}D. 25cm^3
9. A solution X, on mixing with $AgNO_3$ solution, gives a white precipitate soluble in NH_3. A solution Y when added to X, also gives a white precipitate which is soluble on boiling. Solution Y contains A. Ag^+ B. Pb^{2+} C. Pb^{4+} D. Zn^{2+}
10. Which of the following is an exception in the assumptions of the kinetic theory of gases? A. the particles of negligible size B. the particles are in constant random motion C. the particles are of negligible mass D. the particles collide with each other
11. A saturated solution AgCl was found to have a concentration of $1.30x10^{-5}$moldm^{-3}. The solubility product of AgCl therefore is A. 1.30 x 10^{-5} mol^{2}dm^{-6} B. 2.60 x 10^{-12}mol^{2}dm^{-6} C. 1.30 x 10^{-7} mol^{2}dm^{-6} D. 1.69 x 10^{-10}mol^{2}dm^{-6}
12. 0.06g of a hydrocarbon occupies 32cm^3 at S.T.P. Its formula is A. C_3H_6 C. C_2H_2 C. C_2H_4 D. C_2H_6
13. Consider the following compounds (i) $CH_3CH_2(CH_3)_2CH_2CH_3$ (ii) $CH_2CH_2CH_2CH_3$ (iii) $CH_3C\ (CH_3)C(CH_3)_2CH_3$ (iv) $CH_3CH(CH_3)CH_3$. The correct arrangement of the compounds in their increasing order of volatility is A. iv, ii, i, iii B. iii, i, ii, iv C. ii, i, iii, iv D. iii, i, iv, ii
14. Which of the following is NOT a redox reaction? A. $2HNO_2 + 2HI \rightarrow 2H_2O + 2NO + I_2$ B. Zn +

$H_2SO_4 \rightarrow ZnSO_4 + H_2$ C. $BaCl_2 + 2AgNO_3 \rightarrow 2AgCl_2 + Ba(NO_3)_2$D. $4FeO + O_2 \rightarrow 2Fe_2O_3$

15. Which of the following processes leads to increase in entropy? A. mixing of a sample of NaCl and sand B. condensation of water vapour C. Boiling of a sample of water D. cooling a saturated solution
16. Which one of the following statements is true? A. most solids have low densities B. most gases are ionic compounds C. solids, like liquids and gases, do not have fixed shapes and fixed volumesD. gases do not have their own shape but rather expand to fill the shapes of their containers uniformly
17. A liquid begins to boil when A. its volume is slightly decreased B. its vapour pressure is lower than the external pressure C. its vapour pressure equals the external pressure D. its molecules start escaping from the surface
18. An element which exists in more than one crystalline form is said to exhibit. A. polymorphism B. isotopy C. allotropy D. isomerism
19. Which of the following is an example of chemical changes? A. freezing of water B. dissolution of NaCl in water C. rusting of iron D. separating a liquid mixture by distillation
20. The electronic configuration in the ground state of the chloride ion (Cl^-) is A. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$ B. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$ C. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^7$ D. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3d^8$
21. The oxidation number of oxygen in BaO_2 is A. -2 B. -4 C. -1 D. 0
22. Which ONE of the following statements is true for the p – block elements? A. electro negativity increases as we go down a groupB. electro negativity increases regularly from left to right across a period C. electro negativity decreases regularly form left to right across a period D. electro negativity remains almost constant across a period
23. The general formula for carboxylic acids is

A.
```
      O
      ||
R – C – OR¹
```
B.
```
     O     O
     ||    ||
R -C–O–C–R¹
```
C.
```
      O
      ||
R -  C – OH
```
D.
```
      O
      ||
R – C – R¹
```

24. The functional group represented in the compound $CH_3CH_2CH_2CH_2CH{=}O$ is A. Alkanone B. Alkanal C. Aljanol D. Alkanoate

SOLUTION 2007

1. $2A_{(g)} \rightleftharpoons B_{(g)} + C_{(g)} \Delta H = +25.6KJ$

For an endothermic reaction (ΔH +ve), an increase in temperature favours the forward reaction while a decrease in temperature favours the backward reaction. Conversely, for an exothermic reaction (ΔH = -ve), an increase in temperature favours the backward reaction while a decrease in temperature favours the forward reaction.

Since the reaction depicted by the equation is favoured by an increase in temperature

The correct option is C

2. $^{226}_{88}X \longrightarrow {}^{v}_{u}A + 2^{4}_{2}\alpha + {}^{0}_{-1}e$

226 = v + 2(4) + 0

226 = v + 8, v = 226 – 8 = 218

Also,

88 = u + 2(2) – 1

88 = u + 3, u = 88 – 3 = 85

$$^{v}_{u}A = {}^{218}_{85}A => {}^{218}_{85}R$$

The correct option is C

3. Primary alkanols undergo oxidation to give alkanals, further oxidation produce alkanoic acids. Secondary alkanols are oxidized to produces alkanones while tertiary alkanol is not easily oxidized. A primary alkanol has just one alkyl attached to the carbon atom carrying the hydroxyl group. A secondary alkanol has two while a tertiary alkanol has three.

```
        H                  H                 H
        |                  |                 |
H –  C – OH       H–C – OH      R – C – OH
        |                  |                 |
        H                  R                 R
  (Primary )         (Secondary)   (Tertiary)
```

$CH_3CH_2CH_2C(CH_3)(OH)CH_3$ is drawn as below:

```
                        CH₃
                        |
CH₃CH₂CH₂ – C – CH₃
                        |
                        OH
```

Which type of alkanol is this? Good! It is a tertiary alkanol.

The correct option is C

4. In H_2S, sulphur exists as S^{2-}. Then, the number of electrons in the specie will be two greater than the number present in the original sulphur atom.

S contains 16 electrons while S^{2-} contains 18 electrons.

Hence the electronic configuration is:

$1s^2 2s^2 2p^6 3s^2 3p^6$

The correct option is D

5. An amino acid is a bifunctional compound i.e.it has two functional groups. The two are the amino group ($-NH_2$) and the carboxylic group ($-COOH$). The general formula is

```
        H
        |
H₂N – C – COOH
        |
        R
```

The correct option is A

6. The two gases that turn acidified potassium dichromate solution green are SO_2 and H_2S. The

only gas that forms dense white fume with HCl is ammonia.

The correct option is B

7. The titration of a strong acid and a strong base will produce a solution whose pH = 7. That of a strong acid and a weak base will produce a solution with a pH < 7. The one of a weak acid and a strong base will produce a solution with a pH > 7.
Sodium hydroxide is a strong base while oxalic acid is a weak acid. At the end point a solution with pH > 7 is expected.
The correct option is C

8. $2H_2 + O_2 \rightarrow 2H_2O$
$25cm^3$ of oxygen will react with $50cm^3$ of hydrogen to produce $50cm^3$ of H_2O
Unused $H_2 = 70cm^3 - 50cm^3 = 20cm^3$
Volume of residual gas = $50cm^3$ of H_2O + $20cm^3$ of excess $H_2 = 70cm^3$ altogether
No correct option

9. If as indicated, a white precipitate which is insoluble in NH_3 is produced when X is mixed with $AgNO_3$, X contains chloride ion. If Y is added to X (Cl^-) and a white precipitate is formed Y can be a solution containing Ag^+, Hg^{2+} or Pb^{2+}. Of the three, only the precipitate from Pb^{2+} can dissolve on boiling. This is just because $PbCl_2$ is soluble (slightly) in hot water.
The correct option is B

10. The particles of a gas are of negligible size and not mass
The correct option is C

11. $AgCl_{(s)} \rightleftharpoons Ag^+_{(aq)} + Cl^-_{(aq)}$
$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ x & x & x \end{array}$$
where x = molar solubility of AgCl
$K_{sp} = x^1.x^1 \Rightarrow x^2 = (1.3 \times 10^{-5})^2$
$= 1.69\times10^{-10}mol^2dm^{-6}$
The correct option is D

12. Mole of a gas = $\frac{Volume\ at\ stp}{GMV}$
Mole of the hydrocarbon = $\left(\frac{32cm^3}{22400cm^3}\right)mol$
$= 0.001429mol$
Molar mass = $\frac{0.06g}{0.001429mol} = 42gmol^{-1}$
Out of the gases given, only C_3H_6 has a molar mass of $42gmol^{-1}$.
The correct option is A

13. As the molecular force in compounds increases, boiling point increases (volatility decreases). In a given homologous series, volatility decreases as the number of carbon atoms increases. If two or more isomeric compounds (those with the same functional group) are to be compared, the more the branch, the higher the volatility (i) and (iii) have lower volatility than (ii) and (iv) since they have higher number of carbon atoms. (ii) and (iv) contain the same number of carbon atoms – they are isomeric. One has to consider the fact that (ii) is straight chained while (iv) has a branch. Thus, (iv) has the greatest volatility, then (ii)
The order is (iv) > (ii) > (i) > (iii)
The correct option is A

14. In $BaCl_2 + 2AgNO_3 \rightarrow 2AgCl + Ba(NO_3)_2$, the oxidation number of barium is +2 on both sides. That of chlorine is –1 on the two sides. That of silver is +1 on the two sides. The same thing goes for nitrogen and oxygen – their initial and final oxidation states are the same. Thus, the reaction is not a redox reaction.
The correct option is C

15. Entropy is a function that measures the randomness or disorderliness of a system. When water is being boiled, entropy increases cooling water from a particular temperature results in a decrease of entropy.
The correct option is C

16. Gases do not have shape and volume. They assume the volume and shape of the container. Liquids have volume but no shape. Then, they also assume the shape of the container. Solids have definite shape and volume.
The correct option is D

17. A liquid boils when its vapour pressure equals the atmospheric pressure.
The correct option is C

18. Allotropy is the ability of an element to exist in more than one crystalline forms. Allotropes of the same element are alike in chemical properties but differ in physical properties.
The correct option is C

19. The rusting of iron is a chemical reaction as it is not easily reversed. Iron rusts when it comes in contact with oxygen and moisture.
$4Fe + 3O_2 + 2XH_2O \rightarrow 2F_2O.XH_2O$
The correct option is C

20. Cl has 17 electrons, Cl^- has 18 electrons. Cl^- has the electronic configuration:
$1s^22s^22p^63s^23p^6$
The correct option is B

21. BaO_2, barium has an oxidation state of +2.
2 + 2x = 0, where x = 0.N of oxygen
2x = -2
$x = \frac{-2}{2} = -1$
The correct option is C

22. Electronegativity increases across a period and decreases down a group. The same goes for electron affinity and ionization energy. Atomic radius, ionic radius and metallic character increase down the group and decrease across the period.
The correct option is B

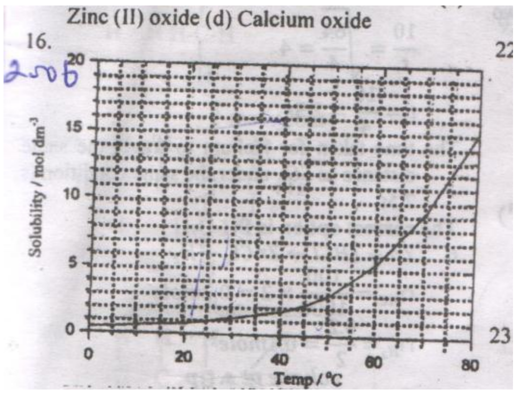
23. The carboxylic acids have the general formula RCOOH where R represents H, alkyl or aryl group. Structurally it is

O
//
R – C
\
OH

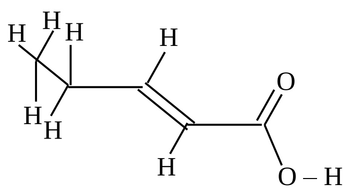
The correct option is C

24. The functional group – CHO belongs to alkanals. Alkanones have – CO – as functional group.
The correct option is B

CHEMISTRY 2006 QUESTIONS

1. A 512cm^3 sample of a gas weighed 1.236g at 20^0C and a pressure of one atmosphere. The reactive molecular mass of the gas is [R = 8.314K^{-1} mol^{-1}; 1 atm = 101,325m^{-3}]. A. 58.07 B. 588,367 C. 5.88 D. 197.9
2. The amount of methane molecules, CH_4, in 8.0 grams of methane is A. 8mol B. 128mol C. 0.5mol D. 3.01 x 10^{25} mol
3. The concentration of a solution obtained by dissolving 0.53g of pure anhydrous Na_2CO_3 in water to make 250cm3 of solution is A. 2.0 x 10^{-5}mol dm^{-3} B. 2.1gdm^{-3} C. 2.0 x 10^{-2} mol dm^{-3} D. 5.0 x $10^{-3}$$moldm^{-3}$
4. What is the maximum volume of CO_2 at s.t.p that can be obtained when dilute hydrochloric acid is added to 10grams of $CaCO_3$? [Ca = 40, C = 12, O = 16] A. 224dm^3B. 22.4dm^3 C. 0.22dm^3D. 22.4dm^3
5. Sulphur (IV) oxide travels a given distance in 10sec. How long will it take equal volume of helium to travel the same distance under the same conditions? [S = 32, O = 16, He = 4] A. 1.6 sec B. 40sec C. 5.0sec D. 2.5 sec
6. The volume of hydrogen gas produced at S.T.P. when 100ml of 2M hydrochloric acid reacts with excess zinc is: A. 2.24dm^3B. 4.48dm^3 C. 1.12dm^3D. 44.8dm^3
7. All the following will liberate a gas when reacted with dilute hydrochloric acid except A. sodium trioxosulphate (IV) salt B. sodium trioxicarbonate (IV) salt C. sodium sulphide D. sodium trioxonitrate (V)
8. The isomer of a compound C_5H_{10} which does NOT decolorize bromine water is A. 2-methylbutane B. 2,2,-dimethylopropane C. 2-methylbut-1-ene D. methylcyclobutane
9. A mixture of nitrogen, oxygen and helium contains 0.25, 0.15 and 0.4mole of these gases respectively. If the pressure contribution due to oxygen was 2.5 atm. The partial pressure of helium is A. 4.0atm B. 0.8 atm C. 3.33 atm D. 6.67 atm
10. Elements P has an atomic number of 12 while element Q has an atomic number 15. Combination of P and Q gave a compound P_mQ_n. the respective values of m and n are A. 2 and 2 B. 2 and 3 C. 3 and 2 D. 2 and 1
11. C_xH_y + $9O_2 \rightarrow 6CO_2 + 6H_2O$. The hydrocarbon, C_xH_y, in the reaction above is most likely A. an alkane B. a benzene C. an alkene D. an alkyne
12. During electrolysis, two cells each containing molten Al_2O_3 and fuse $CaCl_2$ were connected in series A current of 15 Amp was passed through the cells for a given period of time. At the end of the electrolysis 9g of calcium was found to have been deposited at the cathode. What mass of aluminum would be deposited in the second cell. [A = 27, Ca = 40]. A. 8.82g B. 4.44g C. 17.60g D. 4.05g
13. The balanced equation for the reaction of tin(II) salt with potassium heptaoxodichloromate (VI) in an acidic medium can be represented as eSn^{2+} + $fCR_2O^{2-}{}_7$ + $gH^+ \rightarrow hSn^{4-} + iCr^{3+} + jH_2O$. e, f, g, i and j respectively. A. 3, 5, 6, 1 and 4 B. 3, 1, 14, 3, 2 and 7 C. 3, 2, 6, 1, 5 and 6 D. 5, 2, 1, 5, 3 and 2.
14. The pH of a solution containing 0.5 x 10^{-6} M H_2SO_4 is A. 6.3 B. 6.5 C. 6.0 D. 5.0
15. Which of the following oxides will NOT dissolve in both dilute hydrochloric acid and 2M sodium hydroxide solution?A. Lead(II) oxide B. Aluminum oxide C. Zinc(II) oxide D. calcium oxide
16. 

Curve of $KClO_3$. The number of moles of $KClO_3$ crystals produced by cooling 200cm^3 of a saturated solution of the salt from 65^0C to 25^0C is A. 6.3 mole B. 1.26 mole C. 0.63 mole D. 7.30 mole
17. During a compression process involving an ideal gas at pressure P_1, when the volume, V_1 of the gas was halved, the temperature in Kelvin increases by half its initial value. The final pressure P_2 is given by A. $3P_1$ B. $12P_1$ C. $6P_1$ D. $1.5P_1$
18. The I.U.P.A.C. name for the compound.



IsA. pent-3-enoic acid B. pent-4-enoic acid C. pent-2-enoic acid
D. pent-3-ene-1-oic acid

19. Metal salts of long chain fatty acids are known as A. detergent B. double salts C. soaps D. grease

O
||
20. The compound, $CH_3 - C - O - CH_2CH_2CH_3$ when refluxed with dilute HCl, is hydrolyzed to give A. CH_2 COOH and $CH_3CH_2CH_2OH$ B. CH_3 COOH and $CH_3CH_2CH_3$ C. CH_3COOH and CH_3CH_2OH D. CH_3 COOH and $CH_3CH_2CH_2Cl$

O
||
21. The compound $CH_3 - CH_2 - C - CH_3$ is the product of oxidation of A. butan-3-ol B. butan-1-ol C. 3-methylpropan-2-2ol D. butan-2-2ol

22. Which of the following structural formulae is NOT isomeric with the other? A. $CH_3CH_2CH_2CH_2 - OH$ B. $CH_3 - O - CH_2CH_2CH_3$

C. O
/ \
$CH_3HC - CHCH_3$

H
|
D. $H_3C - C - CH_2CH_3$
|
OH

23. Cellulose and starch can be classified as one of the following. A. hydrocarbons B. sugars C. carbohydrates D. alkaloids

CHEMISTRY 2006 SOLUTION

1. Volume = $512cm^3 = 512(10^{-6})m^3 = 5.12 \times 10^{-4}m$
Temperature = 20^oC = (20 + 273)K = 293K
Pressure = 1atm = $101325Nm^{-2}$
R = $8.314Jk^{-1}mol^{-1}$
$PV = nRT$ (ideal gas equation)
$n = \frac{PV}{RT} = \left(\frac{101325 \times 5.12 \times 10^{-4}}{8.314 \times 293}\right)$mol
= 0.021297mol
$Molar\ mass = \frac{Mass}{Mole} = \frac{1.236g}{0.021297mol}$ = $68gmol^{-1}$
The relative molecular mass is 58
The correct option is A

2. Molar mass of CH_4 = 12 + (1 × 4) = $16gmol^{-1}$
Mole of CH_4 = $\frac{8g}{16gmol^{-1}}$ = 0.5mol
Number of molecules = mole × Avogadro's constant
= $0.5 \times 6.02 \times 10^{23}$molecules
= 3.01×10^{23}molecules
The correct option is D

3. Molar mass of Na_2CO_3 = $106gmol^{-1}$
Mole of Na_2CO_3 = $\frac{0.53g}{106gmol^{-1}}$ = $0.25dm^3$
Molarity (conc. In $moldm^{-3}$) = $\frac{Mole}{Volume\ (dm^3)}$
$= \frac{0.005mol}{0.25dm^3}$
= $0.02moldm^{-3}$
= $2 \times 10^{-2}moldm^{-3}$
The correct option is C

4. $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$
Mole of $CaCO_3$ = $\frac{10g}{100gmol^{-1}}$
1 mole of $CaCO_3$ produces 1 mole of CO_2.
Then, 0.1mol of $CaCO_3$ will produce 0.1mole of CO_2.
Volume at s.t.p = Mole of gas × G.mv
= $(0.1 \times 22.4)dm^3$
= $2.24dm^3$
The correct option is A

5. Using Graham's law of diffusion:
$\frac{r_1}{r_2} = \sqrt{\frac{R.m.m_2}{R.m.m_1}}, \frac{t_1}{t_2} = \sqrt{\frac{R.m.m_1}{R.m.m_2}}$
Please, note the difference between the two.
$\frac{t_{SO_2}}{t_{He}} = \sqrt{\frac{R.m.m_{SO_2}}{R.m.m_{He}}}$
$\frac{10}{t_{He}} = \sqrt{\frac{64}{4}} = \sqrt{16} = 4$
$\frac{10}{t_{He}} = 4$, $4t_{He} = 10$, $t_{He} = \frac{10}{4}$ = 2.5s
The correct option is D

6. $Zn_{(s)} + 2HCl_{(aq)} \rightarrow ZnCl_{2(aq)} + H_{2(g)}$
Mole of HCl = $\left(\frac{100}{1000}\right)dm^3 \times 2moldm^{-3}$ = 0.2mol
Mole of H_2 = $\frac{1}{2} \times 0.2mol = 0.1mol$
Volume of H_2 = $(0.1 \times 22.4)dm^3 = 2.24dm^3$
The correct option is A

7. $Na_2SO_{3(s)} + 2HCl_{(aq)} \rightarrow 2NaCl_{(aq)} + H_2O_{(l)} + SO_{2(g)}$
$Na_2CO_{3(s)} + 2HCl_{(aq)} \rightarrow 2NaCl_{(aq)} + H_2O_{(l)} + CO_{2(g)}$
$Na_2S_{(s)} + 2HCl_{(aq)} \rightarrow 2NaCl_{(aq)} + H_2S_{(g)}$
$NaNO_{3(s)}2HCl_{(aq)} \rightarrow 2NaCl_{(aq)} + HNO_{3(aq)}$
It is only $NaNO_3$ that will not liberate a gas on reacting with HCl
The correct option is D

8. The decolorization of bromine water is a test for instauration. Alkenes and alkynes test positive because they are unsaturation. Alkenes conform to the general formula C_nH_{2n}, where n represents the serial number. But this particular formula is also for cycloalkanes.
Don't forget that no alkane is unsaturated. Hence, the cycloalkane – methylcyclobutane (C_5H_{10}) – does not decolourize bromine water.
The correct option is D

9. $X_A = \frac{P_A}{P_T}$, Also $X_A = \frac{n_A}{n_T}$, thus, $\frac{P_A}{P_T} = \frac{n_A}{n_T}$
Where
P_A = pressure due to gas A
P_T = Total pressure
n_A = Mole of A
n_T = Total mole
$\frac{P_{O_2}}{n_T} = \frac{n_{O_2}}{n_T}$
$\frac{2.5atm}{P_T} = \frac{0.15mol}{(0.25+0.15+0.4)mol}$
$\frac{2.5atm}{P_T} = \frac{0.15mol}{0.8mol}$, $P_T = \frac{2.5atm \times 0.8mol}{0.15mol}$
= 13.33atm

We can now use the total pressure to get that of Helium

$X_{He} = \frac{P_{He}}{P_T}$

Mole fraction, $X_{He} = \frac{mole\ of\ He}{Total\ mole} = \frac{0.4}{0.8} = 0.5$

$0.5 = \frac{P_{He}}{13.33atm}$

P_{He} = (0.5 × 13.33)atm = **6.67atm**

Method 2

Since the pressure exerted by a gas increases as its mole increases. We can compare two gases without making any reference to the total mole or pressure

$\frac{P_{He}}{n_{He}} = \frac{P_{O_2}}{n_{O_2}}$

$\frac{P_{He}}{0.4mol} = \frac{2.5atm}{0.15mol}$

$P_{He} = \frac{0.4mol \times 2.5atm}{0.15mol}$ = **6.67atm**

The correct option is D

10. P = 2, 8, 2 Q = 2, 8, 5

The combining power of P is 2 (since its valency is -2) while that of Q is 3 (since its valency is -3)

Then, we can write: P^2Q^3

= $P_3Q_2 = P_mQ_n$

Therefore, m and n are three and two respectively.

The correct option is C

11. $C_xH_y + 9O_2 \rightarrow 6CO_2 + 6H_2O$

The general combustion equation of any hydrocarbon is:

$C_xH_y + \left(\frac{x+y}{4}\right)O_2 \rightarrow xCO_2 + \frac{y}{2}H_2O$

Comparing this standard equation and the one given, we have:

(i) $\frac{x+y}{4} = 9$ (ii) $x = 6$ and (iii) $\frac{y}{2} = 6$

From equation (iii)

$$y = 2 \times 6 = 12$$

Thus, $x = 6$, $y = 12$ in the compound

Then, $C_xH_y = C_6H_{12}$

Since the number of hydrogen atoms is twice that of carbon atoms, it is likely to be an alkene

The correct option is C

12. Using Faraday's second law of electrolysis:

$n_1C_1 = n_2C_2$

n = mole, c = charge on the ion

mole of calcium, $n_{Ca} = \left(\frac{9}{40}\right)$mol = 0.225mol

$C_{Ca} = 2$, $C_{Al} = 3$

$n_{Ca}C_{CA} = n_{Al}C_{Al}$

$0.225 \times 2 = n_{Al} \times 3$

$n_{Al} = \left(\frac{0.225 \times 2}{3}\right)$mol = 0.15mol

Mass = mole × Molar mass

= 0.15mol × $27gmol^{-1}$ => 4.05g

The correct option is D

13. $Sn^{2+} + Cr_2O_7^{2-} + H^+ \rightarrow Sn^{4+} + Cr^{3+} + H_2O$

$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$ (reduction

$Sn^{2+} \rightarrow Sn^{4+} + 2e^-$ (oxidation)

To get equal electrons, multiply the oxidation half equation by 3

$3Sn^{2+} \rightarrow 3Sn^{4+} + 6e^-$

Add the reduction half equation and the modified oxidation half equation to get the overall equation

$3Sn^{3+} + Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 3Sn^{4+} + 2Cr^{3+} + 7H_2O$

Cancel out the common term(s), (only electron in this case)

$3Sn^{2+} + Cr_2O_7^{2-} + 14H^+ \rightarrow 3Sn^{4+} + 2Cr^{3+} + 7H_2O$

Then,

e = 3, f = 1, g = 14, h = 3, i = 2, j = 7

The correct option is B

14. pH = $-\log_{10}[H^+]$

= $-\log_{10}(2 \times 0.5 \times 10^{-6}) = 6$

Why did we multiply the concentration by 2?

The concentration given is for H_2SO_4 and not for H^+. 1 mole of H_2SO_4 produces 2 moles of H^+:

$H_2SO_4 \rightarrow 2H^+ + SO_4^{2-}$

Hence, the hydrogen ion concentration is twice that of the acid.

The correct option is C

15. An oxide that reacts with both acids and bases is known as an **amphoteric oxide**ZnO, Al_2O_3 and PbO are amphoteric oxides. CaO is not amphoteric.

The correct option is D

16. From the curve, the solubility of $KClO_3$ at 65ºC = $7moldm^{-3}$. That at 25ºC = $2moldm^{-3}$

Mole in $200cm^3$ of solution at 65ºC = $\left(\frac{200}{1000}\right) \times 7$ = 1.4mol

Mole at 25ºC = $\left(\frac{200}{1000}\right) \times 2 = 0.4$mol

Preciptate = mole at 65ºC – mole at 25ºC

= 1.4mole – 0.4mole = 1mole

The correct option is B

17. $V_2 = \frac{V_1}{2} = 0.5V_1$

$T_2 = 1.5T_1$

$P_2 = \frac{P_1V_1T_2}{V_2T_1} = \frac{P_1 \times V_1 \times 1.5T_1}{0.5V_1 \times T_1}$

$= \frac{P_1 \times 1.5}{0.5} = 3P_1$

The correct option is A

18. The structure of the compound can be re-written as:

$CH_3CH_2CH = CHCOOH$

The double bond is after the second carbon atom in a compound having five carbon atoms as chain and the carboxylic functional group. Thus, the name is pent-2-enoic acid

The correct option is C

19. Metal salts of long chain fatty acids are known as soaps. The formula is RCOOX, where R = alkyl group and X = metal e.g. $(CH_3(CH_2)_{16}COONa$

The correct option is C

20. The hydrolysis of alkanoate yields alkanoic acids and alkanols

$$CH_3\overset{\overset{\displaystyle O}{\|}}{C} - O - CH_2CH_2CH_3 \xrightarrow[HCl]{OH/H^+} CH_3COOH + CH_3CH_2CH_2OH$$

The correct option is A

21. O

$$
\begin{array}{c}
\quad\quad\;\; \| \\
CH_3 - CH_2 - C - CH_3
\end{array}
$$
is an alkanone (butan-2-one to be precise)

Alkanals result from the oxidation of primary alkanols while alkanones are from secondary alkanols.

You can now easily guess that butan-2-ol is the source. If that sounds too theoretical, simply oxidize the compound and see what you get

$$
\begin{array}{ccc}
\begin{array}{c} \quad\quad O \\ \quad\quad \| \\ CH_3–CH_2–C–CH_3 \end{array} & \text{becomes} & \begin{array}{c} \quad\quad O \\ \quad\quad \| \\ CH_3–CH_2–C–CH_3 \\ \quad\quad | \\ \quad\quad H \end{array}
\end{array}
$$

(Butan-2-ol)

The correct option is D

22. Isomers have the same molecular formula. The compound in option C has a molecular formula C_4H_8O while others have $C_4H_{10}O$. It is, therefore not isomeric with others.

The correct option is C

23. Cellulose and starch are examples of carbohydrates. They are polysaccharides.

The correct option is C

2015 OAU Post UTME: Physics Question

1. Which of the following statements is true of a body moving with simple harmonic oscillation about its equilibrium position? (a) The velocity is maximum when the body is passing through the equilibrium position (b) The velocity is minimum when the body is passing through the equilibrium position (c) Acceleration is maximum when body is passing through the equilibrium position (d) Acceleration is constant throughout the period of motion.
2. A body of mass 1.5kg is fastened to a spring, and lies on a horizontal frictionless table. When the mass is pulled with a force of 32N, the extension in the spring is 0.35m. Calculate the maximum kinetic energy for a simple harmonic oscillation when the initial displacement is 0.2m. (a) 9.80J (b) 9.44J (c) 1.83J (d) 0.45J
3. A man travelled 3km due east in 4 mins by bus on a straight road, and then drove a car 4km due north in 3 mins. Calculate his average velocity (a) 0.71m/s (b) 1.00m/s (c) 11.9m/s (d) 16.7m/s
4. Force of the same magnitude is applied to each of 1kg of lead (Pb), 1kg of iron (Fe) and 1kg of paper. Which of the three masses has the highest acceleration? (a) Paper (b) Iron (c) Lead (d) All accelerations equal
5. A body of mass 8.0kg is connected to a body of mass 5.0kg with an inextensible string passing over a frictionless pulley as shown in Fig. 1. Calculate the force of friction between the 5.0kg mass and the horizontal table if the 8.0kg mass accelerated downwards at $4.5m/s^2$ ($g = 10m/s^2$) (a) 18N (b) 36N (c) 27N (d) 9N

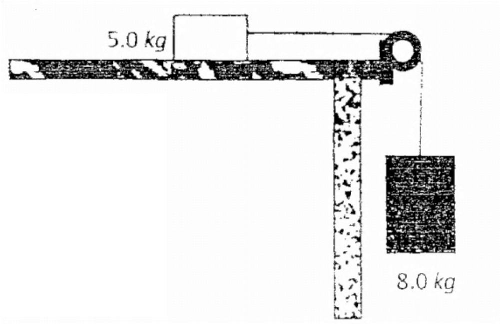
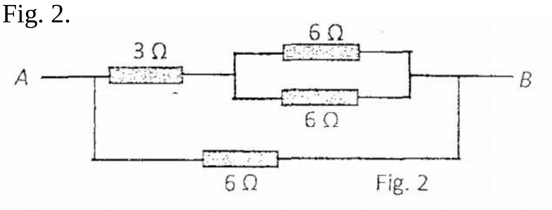


Fig. 1

6. A body of mass M kg is located 0.5m from a body of mass 2M kg. At the same time it is 0.75m from a third body of mass 3M kg and 0.9m from a fourth body of mass 5kg. Which of the three bodies exerts the least gravitational force of attraction on the body of mass M kg? (a) 2M kg (b) 3M kg (c) 5M kg (d) All forces equal
7. The linear coefficient of expansion of aluminium is $23 \times 10^{-6}\,^{o}C^{-1}$, and that of iron is $11 \times 10^{-6o}C^{-1}$. The length of an aluminium rod is 0.24 times the length of an iron rod when both are at 0^oC. The temperature of the aluminium rod is increased to 60^oC while that of the iron rod is increased to 45^oC. Calculate the ratio of the increase in the length of the aluminium rod to that of the iron rod. (a) 1.00 : 0.67 (b) 1.00 : 1.49 (c) 4.00 : 3.00 (d) 2.10 : 1.00
8. Two forces $\vec{A}$ and $\vec{B}$ act on an object. The angle between the force is 60^o. The magnitudes of $\vec{A}$ is 15N while the resultant of the two vectors has a magnitude of 18N. What is the magnitude of vector $\vec{B}$? (a) 3.00N (b) 4.96N (c) 9.95N (d) 39.9N
9. Two force of magnitudes 80N and 60N act one a 5kg object. If the angle between the two forces is 120^o, the acceleration of the body is (a) $14.4m/s^2$ (b) $20.0m/s^2$ (c) $27.1m/s^2$ (d) $121.7m/s^2$
10. Which of the following bodies will lose its heat at the fastest rate? The bodies have the same surface area and initial temperature. (a) A sphere painted black and dull in appearance (b) A cube painted glossy white (c) A toroid painted shiny black (d) A sphere painted shining red.
11. A 10.0kg mass is dragged along a horizontal table by a 60N force. The force is inclined at angle 30^o to the horizontal, and the coefficient of dynamic friction between the mass and the table is 0.15. Calculate the acceleration of the mass. Take $g = 10m/s^2$. (a) $3.7m/s^2$ (b) 4.5m/s (c) 5.0m/s (d) 6.7m/s
12. A man standing at some distance from a sharp cliff, hears the echo of his clap 1.2 second after he claps. How far from the cliff is the man? The speed of sound in air is 340m/s (a) 283.3m (b) 204.0m (c) 408.0m (d) 566.7m
13. A uniform magnetic field lies in the positive x-direction. An electron moving in the negative y-direction enters the region of the magnetic field.

Which of the following statements is true about the direction of motion of the electron? (a) Electron moves in a circle whose plane is parallel to the x-y plane (b) Electron moves in a circle whose plane is parallel to y-z plane (c) Electron moves faster in the negative y-direction (d) Electron changes direction and begins to move in the positive x-direction

14. When $^{238}_{92}U$ decays by emitting an alpha particle, the daughter nuclide is __________ (a) $^{234}_{90}Th$ (b) $^{234}_{92}U$ (c) $^{238}_{93}Np$ (d) $^{238}_{91}Pa$
15. In a thermos-flask cork is placed between the evacuated inner cylinder and the outer jacket to reduce heat loss or gain due to ______ process (a) combustion (b) convection (c) radiation (d) conduction
16. The process of ejecting electrons from a body by heating it is referred to as (a) Thermionic emission (b) Photoelectric effect (c) Electron conduction (d) Thermal conduction
17. An object is 5cm away from the pole of a convex lens whose radius of curvature is 30cm. What is the nature of the object's image? (a) Real and magnified (b) Real and diminished (c) Virtual and magnified (d) Virtual and diminished
18. When white light passes through a glass prism, the colours of light coming out arranged in the order of decreasing angle of deviation are (a) Red, orange, yellow (b) Violet, green, red (c) Yellow, red, blue (d) Blue, orange, green
19. Calculate the effective resistance between terminals A and B in the circuit section shown in Fig. 2.



(a) 3.0Ω (b) 21.0Ω (c) 2.0 (d) 4.3Ω

20. Calculate the equivalent capacitance between terminals A and B in the circuit section shown in Fig. 3.

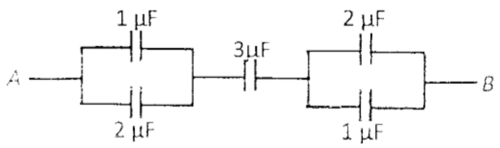


Fig. 3

(a) 1.0μF (b) 4.3μF (c) 6.0μF (d) 9.0μF

21. A broadcasting station transmits at a frequency of 89.5MHz. What capacitance together with an inductance of 30μH is needed to tune to this station? (a) 4.16nF (b) 2.68kF (c) 4.16pF (d) 2.98μF
22. Both proton and electron carry charges whose magnitude equal 1.6×10^{-19}C. Calculate the electrostatic potential between the electron and the proton in a hydrogen atom when the separation between them is 0.5×10^{-10}m ($\varepsilon_0 = 8.85\times10^{-12}N.m^2/C^2$) (a) 1.84×10^{3}J (b) 2.1×10^{-7}J (c) 9.2×10^{-8}J (d) 4.6×10^{-18}J

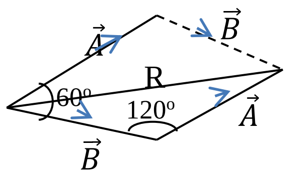
2015 OAU Post UTME: Physics Solution

1. In simple harmonic motion, the velocity is maximum when the body is passing through the equilibrium position while it is minimum at the extreme ends.
The acceleration, on the other hand, is maximum at the extreme end and minimum as the body passes through the equilibrium point.
The correct option is A
2. m = 1.5kg, F = 32N, e = 0.35m and A = 0.2m
The force constant of the spring can be found using Hooke's law:
F = Ke
32N = K × 0.35m
$K = \frac{32N}{0.35m}$ = 91.429Nm^{-1}
The maximum kinetic energy equals the total energy of the spring = $\frac{1}{2}KA^2$
$= \frac{1}{2}\times 91.429 \times 0.2^2$
= 1.83J
Alternatively, the maximum kinetic energy can be obtained using the mass and the maximum velocity.
V_{max}= WA
But the angular velocity, W is not given
$$W = \sqrt{\frac{K}{M}} = \sqrt{\frac{91.429}{1.5}} = 7.807radS^{-1}$$
∴ V_{max} = 7.807 × 0.2 = 1.561mS^{-1}
$K.E_{max} = \frac{1}{2}MV_{max}^2 = \frac{1}{2} \times 1.5 \times 1.561^2$
= 1.83J as above
The correct option is C
3. The magnitude of the average velocity is given by $\frac{displacement_1 + displacement_2 \ldots}{time_1 + time_2 \ldots}$
Displacement$_1$ => 3km = 3000m
Displacement$_2$ => 4km = 4000m
Time$_1$ => 4mins => 240s
Time$_2$ => 3mins => 180s
$$Average\ velocity = \frac{3000m + 4000m}{240s + 180s}$$
$= \frac{7000m}{420s}$ => 16.667ms^{-1}≈ 16.7ms^{-1}
The correct option is D
4. From Newton's second law of motion, F = ma
It follows that $a = \frac{F}{m}$
Since all the materials have the same mass and force of the same magnitude is applied on them, they accelerate at the same rate. It has nothing to do with the material of the object in motion.
The correct option is D
5. Since the two masses are connected by a string, they have the same acceleration.
Then,
T – Fr = 5a……………. (i) (from F = ma)
80 – T = 8a………………. (ii)
Adding equations (i) and (ii), we have:

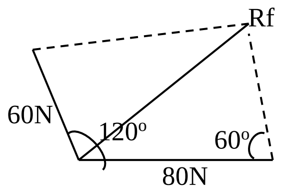
T – Fr + 80 – T = 5a + 8a
∴ 80 – Fr = 13a
But a = 4.5ms^{-2}, Then
80 – Fr = 13(4.5)
Fr = 80 – 13(4.5)
Fr = 80 – 58.5
Fr = 21.5N

6. The force of attraction between two masses is given by $F = \frac{GM_1M_2}{r^2}$
G is the gravitational constant, m_1 and m_2 are the respective masses while r represents their distance apart.
The forces of attraction of the various bodies can be found as shown below:
Between body 1 and 2,
$F_2 = \frac{G \times M \times 2M}{0.5^2}$ = 8GM^2
Between body 1 and 3,
$F_3 = \frac{G \times M \times 3M}{0.75^2}$ = 5.333GM^2
Between body 1 and 4,
$F_4 = \frac{G \times M \times 5M}{0.9^2}$ = 6.173GM^2
The third body, 3Mkg exerts the least gravitational force of attraction on the first body.
The correct option is B

7. The linear coefficient of expansion, $\propto = \frac{\Delta l}{l\Delta\theta}$
Where Δl and $\Delta\theta$ represent change in length and change in temperature respectively. l represents the original length. For the two metals given, aluminium and iron, we have
$\propto_{Al} = \frac{\Delta l_{Al}}{l_{Al}\Delta\theta_{Al}}, \propto_{Fe} = \frac{\Delta l_{Fe}}{l_{Fe}\Delta\theta_{Fe}}$
Then, $\Delta l_{Al} = \alpha_{Al} l_{Al}\Delta\theta_{Al}, \Delta l_{Fe} = \alpha_{Fe} l_{Fe}\Delta\theta_{Fe}$
The original length aluminium is 0.24 times the length of iron, i.e. l_{Al}= 0.24l_{Fe}; then
$$\Delta l_{Al} = \propto_{Al} \times 0.24 l_{Fe} \times \Delta\theta_{Al}$$
$$\Delta l_{Al} : \Delta l_{Fe} = \frac{0.24 \propto_{Al} l_{Fe}\, \Delta\theta_{Al}}{\propto_{Fe} l_{Fe}\, \Delta\theta_{Fe}}$$
$$= \frac{0.24 \times 23 \times 10^{-6} \times 60}{11 \times 10^{-6} \times 45}$$
= 0.6691 : 1 = 1 : 1.49
The correct option is B

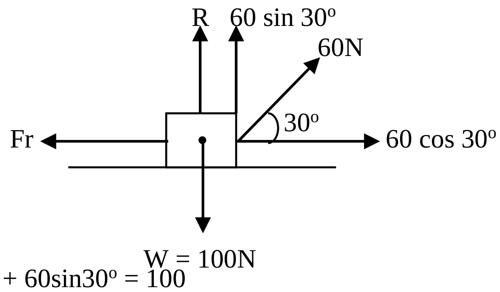
8.


R^2 = $\vec{A}^2 + \vec{B}^2 - 2(\vec{A}.\vec{B})\, cos120^o$
18^2 = 15^2 + $\vec{B}^2$ + 15$\vec{B}$
Then
$$\vec{B}^2 + 15\vec{B} - 99 = 0$$
Solving quadratically, $\vec{B}$ = 4.96N
The correct option is B

9. Let us find the resultant force acting on the object


Rf^2 = 60^2 + 80^2 – 2(60 × 80) cos 60
= 3600 + 6400 – 4800
= 5200,
$Rf = \sqrt{5200}$ = 72.11N
Applying F = ma
72.11 = 5 × a,
$a = \frac{72.11}{5}$ = 14.4ms^{-2}
The correct option is A

10. Black and dull bodies are the best absorbers and the best emitters of heat. A black and dull body thus loses heat faster than any other body.
The correct option is A

11.


R + 60sin30° = 100
R = 100 – 60sin30° = 70N
Fr = μR = 0.15 × 70 => 10.5N
Resultant horizontal force = 60cos30° – Fr
= (51.962 – 10.5)N
= 41.462N
F = ma
41.462 = 10 × a
$a = \frac{41.462}{10}$ = 4.1462ms^{-2}
= 4.15ms^{-2}
The correct option is B

12. Time taken to reach the cliff is half the time taken for the to – and – fro journey
$\therefore t = \left(\frac{1.2}{3}\right)_s$ = 0.6s, speed = 340ms^{-1}
Distance = speed × time
= 340ms^{-1}× 0.6s =>**204m**
Alternatively, One can employ the formula
$x = \frac{vt}{2}$where$x$ = distance from the cliff, v = velocity of sound and t = time to receive echo.
$x = \left(\frac{340 \times 1.2}{2}\right) m$ = 204m
The correct option is B

13. A charged particle moving in a magnetic field experiences a force which is given by F = qvBsinθ
Where q is the charge, v is the velocity of the charged particle while B represents the magnetic flux. The direction is usually found using Fleming's left hand rule.
If the charged particle is initially moving perpendicularly to the magnetic field, it is caused to move in a circular or spiral path.
The correct option is A

14. Let the daughter nuclide be ${}^a_bX$; then
$${}^{238}_{92}U \rightarrow {}^a_bX + {}^4_2He$$
238 = a + 4, a = 238 – 4 = 234
92 = b + 2, b = 92 – 2 = 90
Therefore,
${}^a_bX = {}^{234}_{90}X = {}^{234}_{90}Th$

The correct option is A

15. In a thermoflask, the silvery wall reduces heat loss or gain due to radiation. The vacuum reduces heat transfer due to convection and conduction. The stopper reduces heat exchange due to convection and evaporation while the cork does reduce heat loss or gain due to conduction.

The correct option is D

16. Thermionic emission is a process whereby electrons are ejected from the surface of a heated metal. It is different from photoelectric emission where ejection of electrons is due to illumination with sufficient light.

The correct option is A

17. Object distance, u = 5cm

$F = \frac{r}{2} => \frac{30cm}{2}$ => 15cm

Using the len's formula:

$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ $\qquad \frac{1}{15} = \frac{1}{5} + \frac{1}{v}$

$\frac{1}{v} = \frac{1}{15} - \frac{1}{5} = \frac{1-3}{5} => \frac{-2}{15}$

$\frac{1}{v} = \frac{-2}{15}$

-2v = 15, $v = \frac{15}{-2}$ => -7.5cm

Since v is negative the image is virtual magnification, $m = \left|\frac{v}{u}\right| = \frac{7.5}{5} = 1.5$

Since m is greater than 1, the image is magnified. It is therefore virtual and magnified.

The correct option is C

18. The increasing order of the angles of deviation of the components of white light when the light is made to pass through a glass prism is:

Red → Orange → Yellow → Green → Blue → Indigo → Violet (i.e. ROYGBIV)

If the decreasing order is demanded, one is to simply reverse it. By this, we have VIBGYOR.With the colours selected in this question, we have: Violet > Green > Red

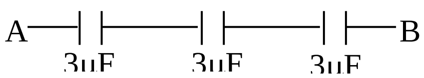
The correct option is B

19. Let us first determine the effective resistance from the combination of the two parallel 6Ω resistors: $\frac{1}{R_E} = \frac{1}{6} + \frac{1}{6}, R_{\varepsilon_1} = 3$

This is then combined with 3Ω serially, then $R_{\varepsilon_2} = R_{\varepsilon_1} + 3 = 3 + 3 = 6\Omega$. This resultant value is in parallel with the last 6Ω resistor. Thus, $\frac{1}{R_{\varepsilon_3}} = \frac{1}{6} + \frac{1}{R_{\varepsilon_2}} = \frac{1}{6} + \frac{1}{6}, R_{\varepsilon_3}$ = 3Ω

The correct option is A

20. For each of the parallel combinations of 1μF and 2μF, the effective capacitance is 3μF. Then we have



This now constitutes a series combination of three 3μf capacitors. Thus

$\frac{1}{C_E} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$

$= \frac{1}{3} + \frac{1}{3} + \frac{1}{3}$

$\frac{1}{C_E} = \frac{1+1+1}{3} = \frac{3}{3}, C_E = 1\mu F$

The correct option is A

21. Resonance occurs at 89.5MHz, then

$X_L = X_C, 2\pi f L = \frac{1}{2\pi f C}$

$2\pi f L \,.\, 2\pi f C = 1, 4\pi^2 f^2 LC = 1$

Therefore,

$C = \frac{1}{4\pi^2 f^2 L} = \frac{1}{4 \times 3.142^2 \times (89.5 \times 10^6)^2 \times 30 \times 10^{-6}}$

$= \frac{1}{9.489 \times 10^{12}} => 1.05 \times 10^{-13} F$

No correct option

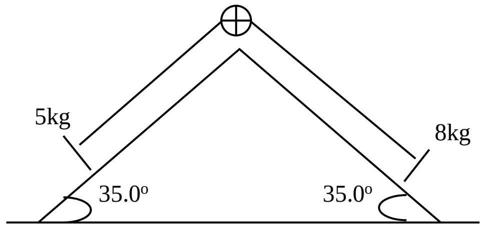
22. Electric potential at a particular point is defined as the workdone to bring a unit positive charge from infinity to that point. Electric potential, $v = \frac{a}{4\pi\varepsilon_0 r}$

Therefore, $v = \frac{1.6\times10^{-19}}{4 \times 3.142 \times 8.85\times10^{-12} \times 0.5\times10^{-10}}$

= 28.77J

PHYSICS QUESTION 2014

1. Which of the following statements is not true about the properties of pressure in a liquid? Pressure (a) at any point in a liquid is at right angle in all directions (b) is the same at all points on the same horizontal plane in a liquid (c) decreases with height and independent of shape and volume of the container (d) is dependent on the shape and volume of the container
2. Which of the following is not a Newton's law of motion? (a) The time rate of change of linear momentum is directly proportional to the external force applied and it takes place in the direction of the force. (b) In the absence of external forces, an object at rest remains at rest and an object in motion continue in motion with a constant velocity (c) If two objects interact the force, $\bar{F}_{12}$ exerted by object 1 on object 2 is equal in magnitude and opposite in direction to the force, $\bar{F}_{12}$ (d) the force required to stretched or compress a spring is directly proportional to the extension or compression, provided that the elastic limit is not exceed.
3. A force of 10N is applied to a spring of elastic spring constant of 200N/m. Calculate the energy stored in the spring (a) 2.50 (b) 0.25J (c) 400J (d) 40.0J
4. Which of the following is not a self-luminous object? (a) glow-worm (b) star (c) moon (d) sun
5. Which of the following is not an application of Total Internal Reflection? (a) Mirage (b) Binoculars (c) Optical Fibers (d) Driving mirror
6. The reading on ADE temperature scale at the ice melting point is $40^{=}$A and $80^{=}$A at the steam point. Calculate the reading on the Celsius scale equivalent to $50^{=}$A (a) $25^{=}$C (b) $40^{=}$C (c) $45^{=}$C (d) $60^{=}$C
7. Calculate the heat required to convert 10g of ice at -10ºC to water at 50ºC. The specific heat capacity of ice and water are 2100J/kgK and 4200J/kgK respectively. The latent heat of Fusion of ice is $3.4\times10^5 JK^{-1}$ (a) 2.10KJ (b) 4.20KJ (c) 3.21KJ (d) 5.71KJ

8. 

Calculate the acceleration of the system (in terms of the acceleration due to gravity, g), when it is released (a) 0.38g (b) 0.13g (c) 0.15g (d) 0.39g

9. A train starting from rest accelerates at the rate of 6 m/s^2 for 20 seconds to attain a constant speed and it further travelled for another 20 seconds and decelerates at rate of 3m/s^2 for 20 seconds. Calculates the total distance (in Kilometre) traveled by the train (a) 4.0km (b) 4.4km (c) 4.8km (d) 3.6km

10. Anomalous behaviour of water refers to (a) Boiling of water at 100°C (b) Freezing of water 0°C (c) Contraction of water when it is heated between 0°C and 4°C (d) Evaporation of water at ambient temperature

11. Which of the following statements is true about collision event? (a) In elastic collision, both linear momentum and kinetic energy are conserved (b) In elastic collision, linear momentum is conserved but the kinetic energy is not conserved. (c) In elastic collision, both linear momentum and kinetic energy are conserved. (d) In inelastic collision, kinetic energy is conserved but the linear momentum is not conserved

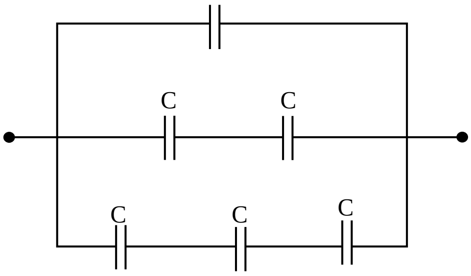
12. Which of the following is the function of a.c n-junction semiconductor device (a) It transforms a direct voltage to AC voltage (b) It steps up an AC voltage (c) It steps down an AC voltage (d) It transforms AC voltage to direct voltage

13. Five 2Ω resistors were connected in two ways, in series and parallel. What is the ratio of the series equivalent resistance to parallel equivalent resistance? (a) 1:50 (b) 2:5 (c) 1:25 (d) 5:6

14. The surface tension of water can be reduced by adding the following except (a) detergent (b) oil (c) grease (d) sand

15. Which of the following is not an eye defect? (a) Astigmatism (b) Hypemetropia (c) Preshyopia (d) Malaopia

16. Calculate the energy stored in a capacitor of capacitance 60μF when voltage of 220V is applied to its terminals (a) 2.0 Joule (b) 1.0 Joule (c) 3.0 Joule (d) 4.0 Joule

17. 

Calculate the equivalent capacitance between the terminals (a) 6C/11 (b) C/6 (c) 17C/6 (d) 6C

18. A material of threshold frequency 4.5×10^{15}Hz, was bombarded with Photons of frequency 8.0×10^{15}Hz. What is the kinetic energy of the emitted photoelectrons (in eV)? [h = 6.60×10^{-34}Js, 1eV = 1.6×10^{-19}J] (a) 2.81eV (b) 2.19eV (c) 5.00eV (d) None of the above

19. Which of the following combinations of factors do not affect evaporation? (i) Temperature (ii) Nature of liquid exposed (iii) Impurities (iv) Pressure (v) Humidity (vi) Draught (vii) Linear expansivity (viii) electronegativity (a) (vii) and (viii) (b) (i), (ii), (iii) (c) (iii), (iv), (v), (vi) (d) (i), (ii), (iii), (iv), (v)

20. A solid is said to sublime if it changes from (a) solid to liquid state (b) solid to molten state (c) solid to gaseous state (d) solid to solid state

21. Sound wave of velocity 340m/s is reflected from an obstacle 5 seconds after leaving the source. Calculate the distance between the source and the obstacle (a) 340m (b) 680m (c) 1020m (d) 850m

22. Which of the following is not true of a force experienced by a current carrying conductor placed in a magnetic field? The force is proportional to (a) the length of the conductor (b) the magnetic field strength (c) the density of the conductor (d) the current in the conductor

2014 PHYSICS SOLUTION

1. The characteristic of pressure in liquids can be summarize as follow
   - Pressure in a liquid increase in direct proportion to the depth of the liquid.
   - Pressure at all points in the same level within a liquid in same
   - Pressure in different liquids at the same depth varies directly with density.
   - Pressure does not depend on the shape and volume of the container

   Option D is correct

2. Note the Newton's laws are basically three, Option A – which says that the time rate of change of linear momentum is directly proportional to the external force applied is referring to second law i.e. $F = \frac{M(V-U)}{t}$

   Option B – refers to first law of motion which states that every object continue in its state of rest or constant motion unless acted upon by an external force.

   Option C – refers to third law of motion which state that to every action there is equal but opposite reaction.

   Option D – is not part of Newton's law rather it is Hooke's Law of elasticity. **Therefore Option D is the correct answer**

3. F = 10N K = 200N/m

Energy stored in a string = $E = \frac{1}{2}ke^2$ or $E = \frac{1}{2}$Fe But e = ?

Recall that F = Ke. The $e = \frac{F}{K} = \frac{10}{200}$ = 0.05m

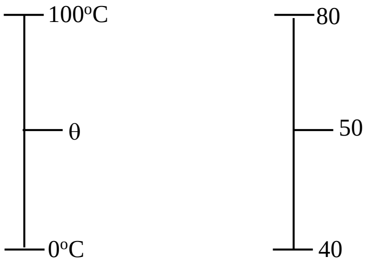
$E = \frac{1}{2}Fe = \frac{1}{2} \times 10 \times 0.05$ = 0.25J

Option B

4. Luminous objects are objects that can produce light on their own.
Moon is a non-luminous object
Option C is correct

5. Mirage, optical fibers, binoculars, field of view of fish under water are all applications to total internal reflection but driving mirrors are convex mirror (diverging).
Option D

6. For a Celsius or centigrade scale – The lower fixed point or (temperature of melting ice) is 0^oC while the upper fixed point or temperature of the steam is 100^oC
For a Kelvin scale – upper point is 373K while lower point = 273K
For a Fahrenheit scale – upper point is 212^oF while lower point 32^oF



By comparing $\frac{\theta - 0^oC}{100^oC - 0^oC} = \frac{50-40}{80-40}$

$\frac{\theta}{100} = \frac{10}{40}$

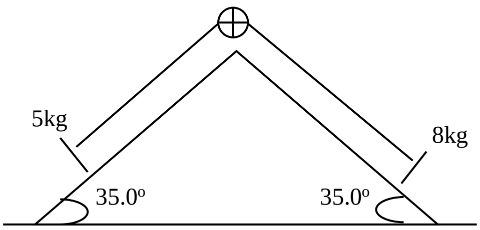
40θ = 1000

$\theta = \frac{1000}{40}$ = 25^oC

Option A is correct

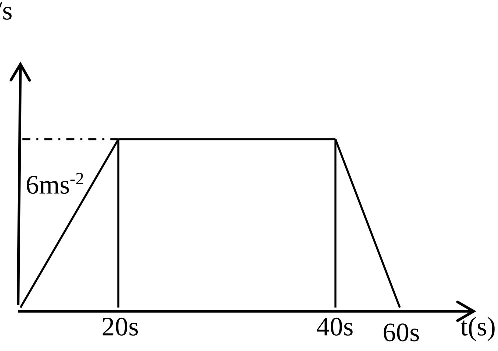
7. Heat required = McΔθ + ML_f + McΔθ
M = 10g = 0.01kg
H = 0.01 × 2100 × (0 – (-10)) + $0.01 \times 3.4 \times 10^5$ + 0.01 × 4200 × 50
H = 210 + 3400 + 2100
= 5710J or 5.71KJ
Option D

8.



Note that the motion of the system (i.e. acceleration of the body) will move toward the heavier mass i.e. the system move towards the lower mass for the 8kgsin35 – T = 8a
For the 5kg mass T – 5g Sin 35 = 5a
But T = T
Then 8g Sin 35 – 89 = 59 + 5g Sin 35
459g – 8a = 5a + 8a
1.72g = 13a
$a = \frac{1.72g}{13}$ = 0.13g
The acceleration is 0.13g in terms of g
Option B is correct

9. Draw out the velocity time graph of the motion and also note that the area of the slope the graph is the total distance covered



v = u + at = 0 + 6(20) = 120m/s
One can calculate the shape of the trapezium as the total distance covered
A = $\frac{1}{2}$(a + b) h
A = $\frac{1}{2}$(20 + 60) × 120
A = $\frac{1}{2}$ × 80 × 120
A = 4800m = 4.8km
Option C is correct

10. Anomalous behaviour of water refers to contraction of water when it is heated between a temperature of 0^oC and 4^oC. It is anomalous because normally when substance are heated, they should expand (i.e. in increase in volume) in the case of liquids and contracts when cooled but this is not so when water is heated between 0^oC and 4^oC in which the volume of water decreases and later increase like any other liquid beyond 4^oC.
Option C

11. **Option A**
Elastic collision – both momentum and kinetic energy are conserved but inelastic collision, momentum is conserved but not kinetic energy.

12. The P-N junction diode can be used as a rectifier and can transform a.c to d.c voltages
Option D

13. Five 2Ω resistors (in series)
N.B when resistors are connected in series, we add them up to get their equivalent resistance
i.e. $R_T = R_1 + R_2 + R_3 \dots R_n$
But we find their reciprocal sum when resistors are arranged in parallel
e.g. $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \dots \frac{1}{R_n}$
Also note that same current pass through the resistors when connected in series and that same voltage pass through it when connected in parallel
Seri I Par V (back to the question)
For series R_T = 2 + 2 + 2 + 2 + 2 = 10Ω

For parallel $\frac{1}{R_T} = \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{5}{2}$

$R = \frac{2}{5}\Omega$

Series: parallel = $10 : \frac{2}{5} = 50 : 2 = 25 : 1$

i.e. multiply $10 : \frac{2}{5}$ by 5 each

None of the answers is correct

14. All except grease can reduce surface

Option C

15. Note that Astigmatism, Hyper metropia and presbyopia are all eye defects except malaopia.

Astigmatism is a eye defect of a lens such that light rays coming from a point do not meet a focal point so that the image is blurred

Hypermetropia is long sight – A disorder of the vision where the eye focuses images behind the retina instead of on it. Therefore distant objects can be seen better than near objects

Presbyopia – In ability of the eye to fosuc due to ageing

Option D is correct as Malaopia is not a eye defect

16. $C = 50\mu F$ $v = 220V$

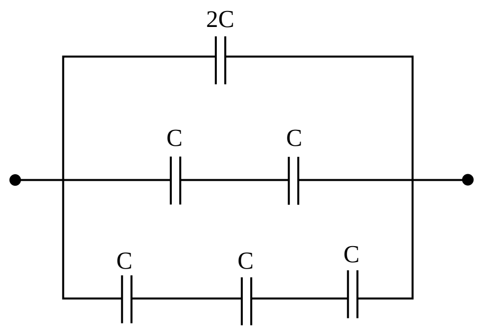
$C = 50 \times 10^{-6}F$ $v = 220V$

Energy stored in a capacitor $E = \frac{1}{2}CV^2$

$E = \frac{1}{2} \times 50 \times 10^{-6} \times 220^2$

$E = 1.21J$

17.



To find the equivalent capacitance, is the reverse of that of resistance

When capacitors are in series (find their reciprocal sum i.e. $\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} ... \frac{1}{C_n}$

But when capacitors are in parallel (we just ad then up)

$\therefore$ For the 2nd line $\frac{1}{C_T} = \frac{1}{C} + \frac{1}{C} = \frac{2}{C}$

$C_t = \frac{2}{C}$

For the 3rd line $\frac{1}{C_T} = \frac{1}{C} + \frac{1}{C} + \frac{1}{C} = \frac{3}{C}$

$C_T = \frac{C}{3}$

$\therefore$ 2C, $\frac{C}{2}$ and $\frac{C}{3}$ are new in parallel

$C_T = \frac{2C}{1} = \frac{C}{2} + \frac{C}{3}$

$\frac{12C+3C+2C}{6} = \frac{17C}{6}$

Option C

18. Threshold frequency, $F_0 = 4.5 \times 10^{15}$Hz

Frequency of incident light = F = 8.0×10^{15}Hz

K.E = ?

N.B $E = \phi + K.E$

But E = hF (energy of incident light)

ϕ = work function or threshold energy

$\phi = hF_o$

$\therefore hF = hF_o + K.E$

$K.E = hF - hF_o$

$= h(F – F_o)$

$K.E = 6.6\times10^{-34} (8\times10^{15} – 4.5\times10^{15})$

$= 6.6\times10^{-34} \times 3.5\times10^{15}$

$= 2.31\times10^{-18}J$

Converting 2.31×10^{-18} J to eV ($1eV = 1.6\times10^{-19}J$)

$= \frac{2.31 \times 10^{-18}}{1.6 \times 10^{-19}}eV = 14.44eV$

None of the above

Options D

19. Evaporation is a process where a liquid turns spontaneously into vapour below its boiling point. Option A is correct as linear expansivity and electronegativity do not affect evaporation

20. A solid is said to sublime if H changes from solid to gaseous state without passing through the liquid state (sublimation)

Option C

21. V = 340m/s t = 5s x = ?

Velocity of echo = $\frac{2x}{t}$ where x = distance

$340 = \frac{2x}{5}$

$2x = 340 \times 5$

$x = \frac{340 \times 5}{2} = 850$

Option D

22. Force experienced by a current carrying conductor placed in a magnetic field. The force F = Bli Sin θ or liB Sin θ

The force is proportional to magnetic field B length of the conductor and the current in the conductor but no proportional to the density of the conductor

Option C

PHYSICS 2013 QUESTION

1. Adeoye moves a distance of 4.0km from a point A on a bearing of N 30° E to a point B and then a distance of 3.0 km on a bearing of S 60° E to a point, C. Calculate Adeoye's resultant displacement from point, A.

A. 10km, N60° E B.5km, N61° E C.3km, 530° E D.4km, 560°E

2. Which of the following statements is true?

A. The unit of mass is Newton (N) B. Weight of an object is a scalar quantity C. The weight of an object varies from one place to another D. The dimensions of weight are $M^{-2}LT^{-2}$

3. Which of the following is not an example of rotational motion?

A. rotation of electric fan blades B. movement of car wheels C. rotation of the earth about its axis D. movement of a loaded spring about its equilibrium position

4. An object of mass, 5 kg placed on an inclined plane (which is at an angle of 30° to the horizontal) is attached to a 10 kg mass through a

pulley, with the 10 kg hanging vertically. Calculate the acceleration of the mass-system in terms of the acceleration due to gravity, g, if there is no friction between the 5 kg mass and the plane.

A. $\frac{2}{5}$g B. $\frac{3}{5}$g C. $\frac{1}{2}$g D.$\frac{3}{4}$g

5. Which of the following is not true about the mechanical energy of a system in a conservative field?
A. total energy is zero B. total energy is the sum of the kinetic energy and the potential energy. C. total energy is equal to the maximum value of kinetic energy D. total energy is equal to the maximum value of potential energy

6. Surface tension is the
A. pressure per unit length on either side of the imaginary line drawn on the liquid surface at rest. B. force per unit length on either side of the imaginary line drawn on the liquid surface at rest. C. current per unit length on either side of the imaginary line drawn on the liquid surface at rest. D. area per unit length on either side of the imaginary line drawn on the liquid surface at rest.

7. A machine has an efficiency of 60%. If the machine applied a force of 2000N to overcome a load of 5000 N, calculate the velocity ratio of the machine.
A 2.4 B. 3.3 C. 4.2 D. 5.5

8. Which of the following statements is true about the hydrostatic pressure?
(i) Pressure increased with height (ii) Pressure is independent of the shape and volume of the vessel (iii) Pressure is the same at all points on the same horizontal p[lane in a fluid (iv) Pressure is independent of the surface area in contact.
A. (i), (ii), (iii) and (iv) B. (i), (ii) and (iii) only C. (ii), (iii) and (iv) only D. (i), (ii) and (iv) only

9. The resistance of a platinum resistance thermomester is 160.5Ω at steam point and 60.5Ω at the melting point of ice. Calculate the resistance of the thermometer at 70ºC
A. 160.5Ω B. 165.5Ω C. 130.5Ω D. 170.5Ω

10. The wall separating a Bakery Oven and its environment is of h, 10m; breadth, 10m; and thickness, 25m. If the rate of heat exchange between the Oven and its environment is 1000 watt and the temperature of the environment is 27ºC, calculate the temperature of the Oven, given that the Coefficient of thermal conductivity of the wall is $0.054Wm^{-1}K^{-1}$.
A. 27.3ºC B. -40.2ºC C. 40.2ºC D. 73.3ºC

11. Which of the following is not an application of expansion of solids?
A. Rivets B. Bimetal strips C. fitting of wheels on rims in Railway Coaches D. Regelation

12. Which of the following is true about melting point of a liquid?
A. The presence of dissolved impurities increases the melting of a pure solid. B. An increase in pressure decreases the melting point of a sub stance that contracts in volume on freezing, more than the one that expands in volume. C. The melting point is not the same as the solidification of a substance. D. The presence of the dissolved impurities does not change the melting point of a pure solid.

13. A boy preparing to have his bath mixed 50kg of water at a temperature of 80ºC with 70kg of water at a temperature of 20ºC. What is the temperature of the water mixture?
A. 45ºC B. 75ºC C. 65ºC D. 35ºC

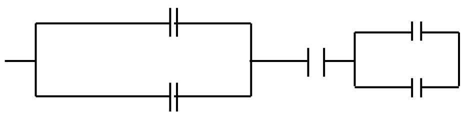
14. Which of the following is not a property of sound wave?
A. Reflection B. Diffraction C. Polarization D. Refraction

15. A pin at the bottom of a beaker filled with water appeared to be elevated when viewed from the top of the beaker. Calculate the displacement of the pin from the bottom of the beaker, if the beaker is filled to 8.0 cm height and the refractive index of water is 4/3.
A. 6.0 cm B. 2.0cm C. 3.0 cm D. 4,0 cm

16. A boy stands at a distance, x from a wall. When he shouted, the echo was heard 2 seconds later. Calculate the distance from the wall, given that the speed of light is 330 m/s.
A. 500m B. 340m C. 250m D. 495m

17. In a 60º prism of refractive index, 1.5, calculate the angle of minimum deviation when light is refracted through the prism.
A. 40.2º B. 37.5º C. 37.2º D. 40.5º

18. Calculate the resultant capacitance of the capacitor network, if each capacitor has a capacitance of 2 μF.



2.0μF B. 1.0μF C. 2.5μF D. 2.3μF

19. A cell of e.m.f 4.0V is connected in series to two resistors 2Ωand 4Ω, which are connected in parallel. Calculate the current which flows through the 4Ω resistor.
A. 1.0 A B. 2.0 A C. 3.0 A D. 4.0 A

20. A step-down transformer is energized by a 220 V a.c supply and supplied a current of 10 A to the secondary winding. Calculate the current which flows through the primary winding ii the ratio of the primary winding to secondary winding is 10:3.
A. 10A B.3A C.4A D.5A

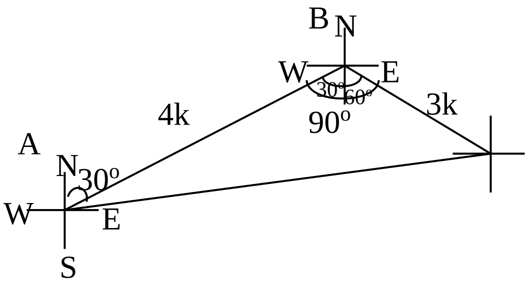
21. All the following properties are characteristics of X-rays except
A. they have short wavelength B. they have no charge C. they are electromagnetic in nature D. they do not affect photographic plates

22. ${}^{238}_{92}U \rightarrow {}^{234}_{90}Th + X$
What particle is emitted in radioactive decay process shown above?

A. β particle B. X-ray C. α particle D. γ ray

PHYSICS 2013 SOLUTION

1.



A right angle triangle

$R^2 = 3^3 + 4^2$

$R^2 = 9 + 16$

R = $\sqrt{25}$ 5km "**B**"

(is the only option with 5km)

2. Force = Newton

Weight is a scalar quantity

Option **"C"** is also correct

Ans. "D"

W = mg

= kg × ms^{-2}

= $kgms^{-2}$ = MLT^{-2}

Not $M^{-2}LT^{-2}$

3. **"D"** The movement of a loaded spring about its equilibrium position is an example of oscillatory motion (to and fro)

4. $a = \left(\frac{m_2 - m_1 sin\theta}{m_2 + m_1}\right) g$

$a = \left(\frac{10-5sin30}{10+5}\right) 10$

$a = \left(\frac{10-2.5}{15}\right) 10$

$a = \left(\frac{7.5}{15}\right) g$

$a = \frac{1}{2} g$

5. **"A"** Option A is incorrect. It would have been right if it were the change in the potential and kinetic energy summation is equal to Zero. All others are correct.

6. **"B"**$S.T = \frac{F}{L}$ Unit = $\frac{N}{m}$

7. $E = \frac{M.A}{V.R} \times \frac{100}{1}$

$M.A = \frac{Load}{Effort} = \frac{5000}{2000}$ = 2.5

$60 = \frac{2.5}{V.R} \times 100$

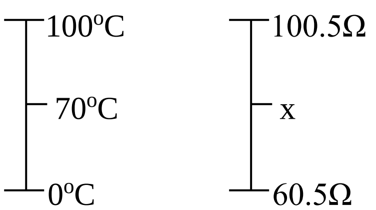
60V.R = 2.5 × 100

V.R = $\frac{2.5 \times 100}{60}$

V.R = 4.1667 = 4.2

8. **"A"**

9.



$\frac{100-70}{70-0} = \frac{160.5-x}{x-60.50\Omega}$

$\frac{34}{76} = \frac{160.5-x}{x-60.50\Omega}$

3(x – 60.5) = 7(160.5 – x)

3x – 181.5 = 1123.5 – 7x

3x + 7x = 181.5 + 1123.5

10x = 1305

$x = \frac{1305}{10}$ = 130.5Ω **"C"**

10. $K = \frac{Qx}{tA\Delta T}$

$\frac{Q}{t} = \frac{KA\Delta T}{x}$

$\frac{Q}{t} = KA\frac{(T_2 - T_1)}{x}$

$2500 = \frac{0.054 \times 10 \times 10\ (T_2 - 27)}{25}$

250 = 5.4(T_2 – 27)

11. **"D"** Regelation which is also known as antifreezing is not an example of solids.

12. **"C"** All others are incorrect. For option B, an increase in pressure increase boiling point would have been correct.

13. $M_H C\Delta\Theta_H = m_C C\Delta\Theta_C$

50 × (80 – x) = 70(x – 20)

4000 – 50x = 70x - 1400

70x + 50x = 4000 + 1400

120 = 5400

$x = \frac{5400}{120}$

$x = 45^o$ **"A"**

14. **"C"** Polarization is an exclusive property of transverse wave.

15. $\eta = \frac{R.D}{A.D}$

R.D = Real depth

A.D = Apparent depth

$d = \frac{A.D}{\eta - 1}$

$\eta = \frac{R.D}{R.D - d}$

ηR.D – ηd = R.D

ηd = ηR.D – R.D

ηd = R.D(η – 1)

$d = \frac{R.D(\eta-1)}{\eta}$

$= \frac{8\left(\frac{4}{3} - 1\right)}{4/3}$

$= \frac{8\left(\frac{1}{3}\right)}{4/3}$

$= \frac{8}{3} \times \frac{3}{4}$

= 2cm **"B"**

16. $v = \frac{2x}{t}$

$330 = \frac{2x}{2}$

660 = 2x

x = 330m (No answer)

But 340 is close **"B"**

17. $\eta = \frac{\sin\left(\frac{A+dm}{2}\right)}{sin\frac{A}{2}}$

$1.5 = \frac{\sin\left(\frac{60+dm}{2}\right)}{sin\frac{60}{2}}$

$1.5 = \frac{\sin\left(\frac{60+dm}{2}\right)}{sin30}$

$1.5 \times 0.5 = \sin\left(\frac{60+dm}{2}\right)$

$0.75 = \sin\left(\frac{60+dm}{2}\right)$

$sin^{-1}0.75 = \frac{60+dm}{2}$

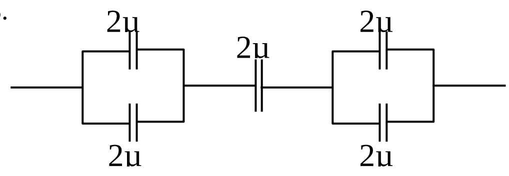
2 × 48.59 = 60 + dm

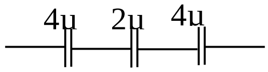
97.18 = 60 + dm

dm = 97.18 – 60

= 37.5 **"B"**

18.





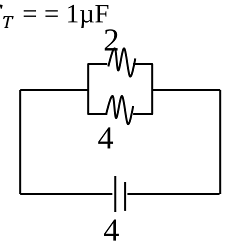
$\frac{1}{C_T} = \frac{1}{4} + \frac{1}{2} + \frac{1}{4}$

$\frac{1}{C_T} = \frac{1+2+1}{4}$

$\frac{1}{C_T} = \frac{4}{4}$

C_T = = 1μF **"B"**

19.



$I = \frac{V}{R_4}$ (current through the 4Ω resistor)

$= \frac{4}{4}$ = 1A **"A"**

20. $\frac{N_S}{N_P} = \frac{I_P}{I_S}$

$\frac{N_P}{N_S} = \frac{I_S}{I_P}$

$\frac{10}{3} = \frac{10}{I_P}$

$10I_p = 30$

$I_p = \frac{30}{10}$

I_p = 3A **"B"**

21. **"D"** All others options are correct. Take note that → frequency increases

R I V U X G – gamma ray

Wavelength decreases ←

22. ${}^{238}_{92}U \rightarrow {}^{234}_{90}Th + {}^{A}_{Z}X$

$238 = 234 + A$

A = 238 – 234

A = 4

92 = 90 + Z

Z = 92 – 90

A = 2

:. ${}^{4}_{2}\alpha$ or ${}^{4}_{2}He$ **"C"**

PHYSICS 2012 QUESTIONS

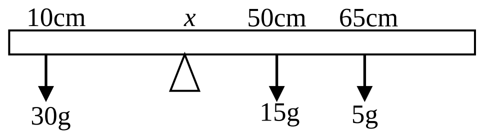
1. A uniform meter rule AB has a mass 15g. A 30g mass is suspended at the 10.0cm mark, and another 5g mass is suspended at the 65.0cm mark. Calculate the position of the fulcrum that will keep the meter rule balanced horizontally. (a) 50.0cm (b) 32.0cm (c) 27.5cm (d) 17.9cm
2. A rectangular block measures 40cm × 25cm × 5cm and is made of a material of density 7800 kg/m^3. Calculate the pressure the block exerts on the floor when it stands on the smallest of its surfaces (a) 3.12 × 10^{3}N/m^2 (b) 6.50 × 10^{3}N/m^2 (c) 1.95 × 10^{4}N/m^2 (d) 3.12 × 10^{4}N/m^2
3. A ship sinks to the bottom of a 250m deep lake. The atmospheric pressure over the lake is 1.03 × 10^5 Pa. Taking the density of water in the lake to be 1000 kg/m^3, calculate the pressure exerted on the boat. (Acceleration due to gravity = 10m/s^2) (a) 2.60 × 10^{6}Pa (b) 2.50 × 10^6 Pa (c) 2.60 × 10^{5}Pa (d) 1.03 × 10^{5}Pa
4. Three knives made of steel, plastic and flat wood respectively are placed on a table for an equal amount of time. The steel knife feels coldest to touch because (a) the steel knife has the lowest temperature (b) the plastic and wooden knives have absorbed more heat from the environment than the steel knife. (c) both wooden and plastic knives have lower densities than the steel knife (d) the steel knife conducts heal faster form the knives than the wooden and plastic knives
5. Two plane mirrors are inclined at angle 45^0 one to another. A ray of light has incident angle 20^0 at the surface of the first mirror. The reflected ray is then incident on the second mirror. Calculate the angle of reflection at the second mirror. (a) 65^0 (b) 45^0 (c) 25^0 (d) 20^0
6. When the length of the string of a simple pendulum is L its period is 0.5π seconds. The period when the length is increased to 4L, will be (a) 0.5π seconds (b) π seconds (c) 2πseconds (d) 4π seconds
7. The coefficient of linear expansion of aluminum is 23 × 10^{-6} K^{-1}. If the volume of a pot made with aluminum at temperature F0 is V0, what will be the change in temperature resulting in a decrease of 0.20% in volume of the pot? (a) -87^{0}C (b) 187^{0}C (c) 125^{0}C (d) 29^{0}C
8. A resonance tube is 40 cm long. The second resonance is heard when the tube is three-quarter full. What is the frequency of the tuning fork placed near the mouth of the tube? (velocity of sound in air is 334 m/s) (a) 2511 Hz (b) 1670 Hz (c) 835 Hz (d) 345 Hz
9. A coin is at the bottom of a bucket filled with a liquid whose refractive index is 1.35. The coin appears to be 120 cm below the surface of the

liquid. Calculate the depth of the liquid. (a) 16.2 cm (b) 13.4 cm (c) 8.9 cm (d) 5.4 cm

10. A car accelerates at 5.0 m/s^2 for 6s, it travels at the speed attained for 20s, and comes to rest after another 4.0s. Calculate the average velocity of the car during the motion. (a) 25.0m/s (b) 20.0m/s (c) 1.25m/s (d) 0.16m/s
11. Which of the following quantities is equal to the area under a velocity-time graph? (a) acceleration (b) distance travelled (c) average velocity of motion (d) total time taken
12. A 25 N lama pulls a 20 body up a 30^0 inclined plane. If the force is parallel to the plane and the body moves up the plane at constant velocity, calculate the magnitude of the frictional force between the body and the plane 1g = 10m/s^2. (a) 35N (b) 25N (c) 20N (d) 15N
13. Which of the following features is used to minimize heat loss due to conduction in a thermo-flask? (a) the space between the two walls of the vacuum flask is evacuated (b) the vacuum flask is separated from the out wall with corks (c) the surfaces of the vacuum flask are silvered (d) the inner and outer walls of the flask are made of steel
14. Which of the following quantities is a scalar quantity? (a) electric field (b) coulomb force (c) electric potential(d) acceleration due to gravity
15. The process by which a solid changes directly to vapour is called (a)evaporation (b) fusion (c) condensation (d) sublimation
16. A 3.0 cm object is placed 12.0 cm in front of a bi-convex lens of local length 8.0 cm. calculate the height of the image of the object. (a) 3.0 cm (b) 6.0 cm (c) 12.0 cm (d) 24.0 cm
17. Five 100-Watt bulbs are put on for 45 days during which the home-owner is on vacation. If 1 KW-hour of electricity costs ₦7.50, how much does it cost the home-owner? (a) ₦168.75 (b) ₦90.00 (c) ₦4050.00 (d) ₦810.00
18. To convert an a.c. generator to a d.c. generator, one needs to (a) remove the brush touching the slip rings (b) laminate the armature (c) replace the permanent magnets with soft iron-core armature (d) replace the slip rings with split rings.
19. To use a milli-ammeter to measure current up to 10 A, what connection needs to be made? (a) a small resistance must be connected in series with the milli-ammeter (b) a small resistance must be connected in parallel with the milli-ammeter (c) A high resistance must be connected in parallel with the milli-ammeter (d) the milli-ammeter must be disconnected from the circuit.
20. Two resistors A and B are made of the same material. The radius of A is three times that of B, and the length of A is half of B. The ratio of the resistance of A to that of B is (a) $\frac{3}{2}$ (b) $\frac{2}{3}$ (c) $\frac{2}{9}$ (d) $\frac{9}{2}$
21. Two 2 μF capacitors are connected in parallel. The combination is connected in series with a 6 μF capacitor. What is the equivalent capacitance for the combination? (a) 10.0μF (b) 8.0 μF (c) 1.5 μF (d) 2.4 μF
22. A student afraid that the substance near him is radioactive places his lecture note between him and the substance. If truly the substance is radioactive, which of the following radiations can be notebook shield him from? (a) gamma rays (b) neutrons (c) alpha particles (d) energetic beta rays

PHYSICS 2012 QUESTIONS

1.



Clockwise moment = Anticlockwise moment
$15(50 - x) + 5(65 - x) = 30(x - 10)$
$750 - 15x + 325 - 5x = 30x - 300$
$50x = 1375$
$x = \left(\frac{1375}{50}\right)$cm = 27.5cm
The correct option is C

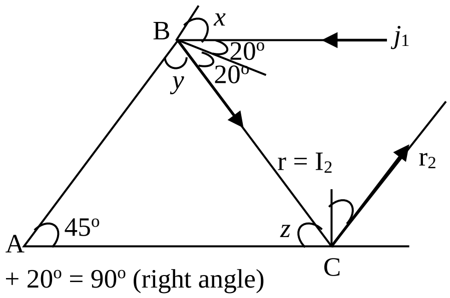
2. Volume of block = 5000cm^3(40cm × 25cm × 5cm)
= $5000\times10^{-6}m^3$
= $5\times10^{-3}m^3$
Density = 7800kgm^{-3}
Mass = Density × Volume
= 7800kgm^{-3} × $5\times10^{-3}m^3$
= 39kg
$$W = mg$$
= (39 × 10) = 390N
With the dimension 40cm × 25cm × 5cm, the smallest surface has an area of 25cm × 5cm
= 125cm^2
= $125\times10^{-4}m^2$
= $1.25\times10^{-2}m^2$
$$Pressure = \frac{Force}{Area} = \frac{390N}{1.25\times10^{-2}m^2}$$
= 31200Nm^{-2}
= $3.12\times10^4Nm^{-2}$
The correct option is D

3. Pressure on the boat = pressure due to water + atmospheric pressure
$P_{H_2O} = h\rho g$ => 250 × 1000 × 10
= 2.5×10^6Pa
$P_{atm} = 1.03\times10^5$Pa
Pressure on the boat = 2.5×10^6Pa + 1.03×10^5Pa
= 2603000Pa
= 2.6×10^6Pa
The correct option is A

4. Steel feels coldest to touch because it is the best conductor of the three materials. It conducts heat

faster from the fingers than the wooden and plastic materials.

The correct option is D

5. 

$x + 20^o = 90^o$ (right angle)
$x = 70^o$
in a similar way,
$y + 20^o = 90^o$, $y = 90^o - 20^o = 70^o$
fromΔABC
$45^o + y + z = 180^o$
$z = 180^o - 45^o - 70^o = 65^o$
but $z + I_2 = 90^o$ (right angle)
Then, $I_2 = 90^o - z = 90^o - 65^o = 25^o$
Since the angle of incidence equals that of reflection, r_2 is also 25^o

The correct option is C

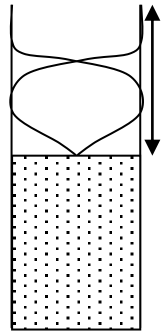
6. $\frac{T_1}{\sqrt{L_1}} = \frac{T_2}{\sqrt{L_2}}$
$T_1 = 0.5\pi \; seconds$
$L_1 = L$
$L_2 = 4L$
$T_2 = ?$
Then,
$\frac{0.5\pi}{\sqrt{L}} = \frac{T_2}{\sqrt{4L}}$
$T_2 = \frac{0.5\pi \times \sqrt{4L}}{\sqrt{L}} = \frac{0.5\pi \times \sqrt{4} \times \sqrt{L}}{\sqrt{L}}$
$= 0.5\pi \; \times \sqrt{4}$
$= 0.5\pi \; \times 2$
Hence, $T_2 = \pi$seconds

The correct option is B

7. $\alpha = 23\times10^{-6}$
$y = 3\alpha = 3\times23\times10^{-6}k^{-1} = 69\times10^{-6}k^{-1} = 6.9\times10^{-5}k^{-1}$
$y = \frac{V_2 - V_1}{V_1\Delta\theta}$
Let $V_1 = V$
$V_2 = V - \frac{0.2}{100}V$ (Since there is a decrease of 0.2%
$= 0.998V$
Then,
$6.9 \times 10^{-5} = \frac{0.998 - V}{V\Delta\theta} = \frac{-0002}{\Delta\theta}$
$\Delta\theta = \frac{-0.002}{6.9\times10^{-5}}$
$= -28.98^oc = -29^oC$

The correct option is D

8. Since the tube is three quarter full, it is one-quarter empty!
Therefore, length of air column = $\frac{1}{4} \times 40cm$ = 10cm
For the second resonance,



$L = \frac{3}{4}\lambda$

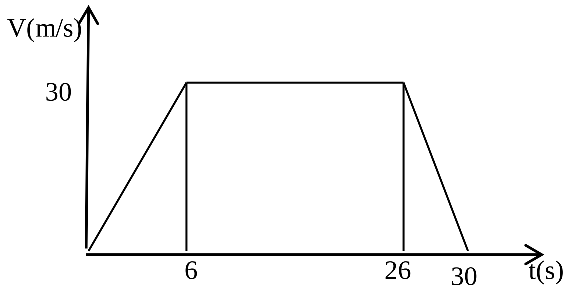
Then, $\lambda = \frac{4}{3}L = \frac{4}{3} \times 10cm$ => 13.333cm => 0.1333m
$Frequency = \frac{Velocity\ (V)}{Wavelenght(\lambda)} = \frac{334m/s}{0.1333m}$ = 2505Hz
Alternatively,
$F = \frac{3V}{4L}$
V = 334m/s, L = 10cm = 0.1m
$F = \frac{3(334)}{4(0.1)}$ = 2505Hz

The correct option is A

9. Refractive index = $\frac{Real\ depth}{Apparent\ depth}$
$1.35 = \frac{Real\ depth}{12.0cm}$
Real depth = 1.35 × 12.0cm => 16.2cm

The correct option is A

10. Average velocity is the ratio of the total distance covered to the total time-taken:
Let us obtain the total distance from a velocity – time graph



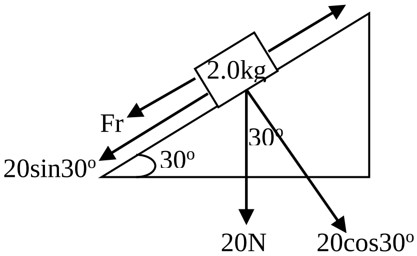
Distance covered = Area of the trapezium
$= \frac{1}{2}(a + b)h$
$= \frac{1}{2}(20 + 30)30$
$= 750m$
Total time-taken = 30s
Average Velocity = $\frac{750m}{30s}$
= 25m/s

The correct option is A

11. The slope of a velocity-time graph gives the acceleration while the area under the graph gives the distance covered

The correct option is B

12.



Since the body moves up the plane at constant velocity, the resulting force on it equals zero i.e. the force up the plane equals the sum of the forces down the plane

25 = Fr + 20sin30º

25 = Fr + 10

Fr = (25 – 10)N = 15N

The correct option is D

13. Several features help in minimizing heat loss in a thermoflask
    (i) The cork reduces heat loss by conduction and evaporation
    (ii) The silvered-walls reduce heat loss by radiation.
    (iii) The vaccum reduces heat loss by conduction and convection

    The correct option is A

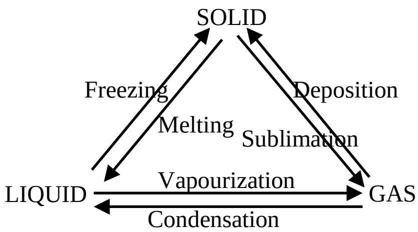
14. Scalar quantities are those that have magnitude but no direction while vector quantities have both magnitude and direction.

    Electric field → vector
    Coulomb force → vector
    Electric potential → scalar
    Acceleration due to gravity → vector

    The correct option is C

15. The process by which a solid changes into vapour without passing through the liquid state is known as sublimation.

    The chart below shows the various forms of change of state and their names.



The correct option is D

16. Object distance, u = 12cm

    Image distance, v = ?

    Focal length, f = 8cm

$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

$\frac{1}{8} = \frac{1}{12} - \frac{1}{v}$

$\frac{1}{v} = \frac{1}{8} - \frac{1}{12} => \frac{3-2}{24} = \frac{1}{24}$

V = 24cm

Magnification, $m = \frac{v}{u}$

Also, $m = \frac{Height\ of\ image, H_I}{Height\ of\ object, H_0}$

Then,

$\frac{v}{u} = \frac{H_I}{H_0}$

$\frac{24cm}{12cm} = \frac{H_I}{3cm}$

$H_I = \frac{24cm \times 3cm}{12cm} => 6cm$

The correct option is B

17. Power = 100W

    For five bulbs, power = 5 × 100W = 500W

    Time = 45days = 45 × 24hrs = 1080hrs

    Total power in KW = $\left(\frac{500}{1000}\right) KW$ = 0.5KW

    Electricity consumed = (0.5 × 1080)KW – hour = 540KWh

    Since 1KWh = ₦7.50, 540KWh will cost ₦(7.50 × 540) = ₦4050

    The correct option is C

18. An a.c. generator can be converted into a d.c. generator by replacing the splip rings with a split ring.

    The correct option is D

19. To measure up to 10A, an ammeter is required. We therefore need to convert the milli-ammeter to an ammeter, a low resistance known as the shunt must be connected with it in parallel

    The correct option is B

20. $R = \frac{Pl}{A}$

    For resistor A

$R_A = \frac{Pl_A}{A_A}$

For resistor B

$R_B = \frac{Pl_B}{A_B}$

Where P = resistivity, which is the same for the two resistors.

$l_A = \frac{l_B}{2}$, Also, radius of A is three times that of B

i.e. $r_A = 3r_B$

$A_A = \pi r_A{}^2 = \pi(3r_B)^2 = 9\pi r_B{}^2$

$A_B = \pi r_B{}^2$

Then,

$R_A = \frac{P \cdot \frac{l_B}{2}}{9\pi r_B{}^2}, R_B = \frac{\rho l_B}{\pi r_B{}^2}$

$\frac{R_A}{R_B} = \frac{P \cdot \frac{l_B}{2}}{9\pi r_B{}^2} \div \frac{\rho l_B}{\pi r_B{}^2}$

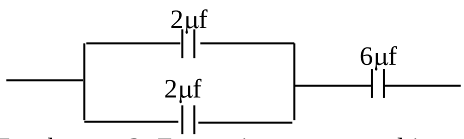
$= \frac{P \cdot \frac{l_B}{2}}{9\pi r_B{}^2} \times \frac{\pi r_B{}^2}{\rho l_B}$

$= \frac{\frac{1}{2}}{9} => \frac{0.5}{9} = \frac{1}{18}$

= 1 : 18

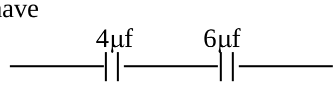
No suitable option

21.



For the two 2μF capacitors connected in parallel, total capacitance = 2μF + 2μF = 4μF

Then we have



For the arrangement in series,

$\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2}$

$\frac{1}{C_T} = \frac{1}{4} + \frac{1}{6} = \frac{6+4}{24} = \frac{10}{24}$

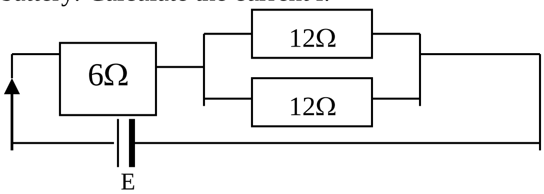
$\frac{1}{C_T} = \left(\frac{24}{10}\right)\mu F$ = 2.4μF

The correct option is D

22. Alpha particles can be stopped by paper. Beta requires thin aluminum foil while it takes a lead block to stop gamma rays.
The correct option is C

PHYSICS 2011

1. A body of mass m slides down an inclined plane with a constant velocity. If the angle of the incline is θ, the coefficient of kinetic friction between the body and the plane is A. cot θ B. cosθ C. tan θ D. sin θ
2. The density of sea water is 1030 kg/m^3. What is the pressure at a depth of 80 m below sea surface? Atmospheric pressure is 1.013×10^{5}Pa and acceleration due to gravity is 10 m/s^2. A. 9.25 × 10^5 Pa B.8.24 × 10^5 Pa C. 7.23 × 10^5 Pa D. 8.34 × 10^9 Pa
3. 6000 J of heat is delivered to 10 g of dry ice at 0^{0}C. What is the final temperature if the container has a heat capacity of 20 J/K? (specific heat of water = 4200 J/Kg.K, latent heat of fusion of ice = 3.33 x 10^5 J/Kg) A.142.9^{0}C B. 63.6^{0}C C. 43.0^{0}C D.0^{0}C
4. A sample of radioactive substance, whose half-life is 16 days, registers 32 decays per second. How long will it take for the rat e of decay to reduce to 2 decays per second? A. 80 days B. 64 days C. 48 days D. 32 days
5. When white light passes through a triangular prism, the emerging rays of light arranged in order of decreasing angle of deviation are A. Red, orange, yellow, green B. blue, green, orange, yellow C. red, green, yellow, blue d. blue, green, yellow, orange
6. The electric field between two parallel plates is E. A particles of mass m and carrying charge q is released at a point half the distance between the plates. The velocity of the particle t seconds after its release is A. qEt/m B. qEt^2/2m C. mqt^2/2E D. mqt/2E
7. The velocity of a 500 kg car moving along a straight road, changes from 12 m/s to 20 m/s in 5 sec. Calculate the average force moving the car. A. 2000 N B. 1600 N C.1200 N D. 800 N
8. When a 2 kg body is at a height 5 m above the floor, its velocity is 4 m/s. What is its total energy at this height? (Acceleration due to gravity = 10 m/s^2) A. 80 J B. 100 J C. 116 J D.180 J
9. An airplane increases it speed from36 km/h to 360 km/h in 20.0 s. How far does it travel while accelerating? A. 4.4 km B. 1.1 km C. 2.3 km D. 1.0 km
10. In the simple circuit shown in fig. 1, E is a 24 V battery. Calculate the current l.



A. 0.1A B. 0.2 A C. 5/6 A D. 2.0 A
11. An object is said to be in simple harmonic motion (SHM) if A. the acceleration is directly proportional to the displacement and is directed toward the equilibrium position of the object. B. the acceleration is inversely proportional to the displacement and directed toward the equilibrium position of the object C. The displacement is directly proportional to the momentum and directed toward the equilibrium position of the object D. the momentum is directly proportional to the displacement and directed toward the equilibrium position of the object
12. A wheel and axle is used to raise a weight of 600 N with an effort of 300 N. If the radii of the wheel and axle are 50 cm and 10 cm respectively, what is the efficiency of the system? A. 40% B. 50% C. 20% D. 10%
13. The term "Viscosity" is used to describe A. surface tension in fluids B. friction in fluids C. surface tension in solids D. moment in solids
14. A particle is in equilibrium under the action of three forces. One force is 40 N towards the west and another is 30 N towards the south. What is the third force acting on the body? A. 40 N, N53^0 E B. 50 N, N37^0 E C. 50 N, N53^0 E D. 40 N, N37^{0}E
15. A hydraulic press works on the principle of transmission of A. force B. energy C. volume D. pressure
16. A cigarette lighter in a car is a resistor that, when activated, is connected across the 12-V battery. If the lighter dissipates 33 W of power, find the resistance it delivers to the lighter. A. 9.90Ω B. 6.60Ω C.4.36ΩD. 17.50 Ω
17. A p-n junction can act as A. an amplifier B. a rectifier C. an inductor D. a capacitor
18. Which of the following is not a thermometer? A. thermocoupleB. pyrometer C. hydrometerD. platinum resistance thermometer
19. Dry air of column of length 10 cm is trapped by a pellete of mercury of length 15cm, with the open end uppermost. When the capillary tube is inverted the length of the air column increased to 25 cm while that of mercury remained constant. Calculate the atmospheric pressure (in cm of Hg). A. 35 cm Hg B. 15 cm Hg C. 20 cm Hg D. 10 cm Hg
20. Sound waves were sent out form a source and after being reflected from an obstacle were

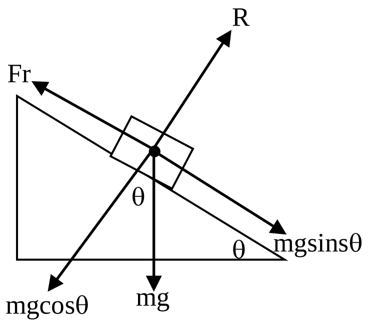
received by a sensor placed beside the source. If the waves were received 10 seconds after they were sent out, calculate the distance between the source and the obstacle. (speed of sound = 330 m/s). A. 990 m B. 660 m C. 1320 m D. 1750 m

21. Which of the following is not true about a chemical cell? A. in primary cells the process through which current is generated is irreversible B. secondary cells can be recharged after they run down by passing a current into the cell in the reverse direction C. positive ions are attracted to the positive electrode where they become neutralized by acquiring electrons. D. primary cells can be recharged

22. Calcium has a work function of 1.9 eV with a wavelength of 150 nm. Calculate the maximum energy of a photo electron emitted. (1 eV = 1.6 × 10^{-19} J, h = 6.6 × 10^{-34} Js). A. 6.35Ev B. 8.25eV C. 14.60eV D. 2.30eV

PHYSICS 2011 SOLUTION

1. Anything we are told that an object is about to move down an inclined plane or it is moving down with a constant velocity, the coefficient of friction, **y = tanθ**, where θ is the angle of friction. Let us consider a body of mass m moving with a constant velocity on a plane inclined at angle θ to the horizontal



Since the object is moving down the plane, the frictional force will be acting upwards. Also, a constant velocity implies that the acceleration is zero.
The resultant force down the plane = mgsinθ - Fr
Force of motion = ma
Equating the two, we have:
mgsinθ - Fr = ma
mgsinθ - Fr = m × 0 = 0
Then,
Fr = mgsinθ, Fr = μR
As can be seen from the diagram, R equals mgcosθ
Then,
Fr = μR = μmgcoθ = mgsinθ
Thus,
$\mu = \frac{mgsin\theta}{mgcos\theta} = \frac{sin\theta}{cos\theta} = tan\theta$
$\boldsymbol{\mu = tan\theta}$
The correct option is C

2. $\rho = 1030$kgm^{-3}, h = 80m, P_{atm} = 1.013×10^5Pa, g = 10m/s
The total pressure at the depth equals the sum of the pressure due to water and the atmospheric pressure.
$P_{H_2O} = h\rho g = 80m\ \times 1030kgm^{-3}\ \times 10m/s^2$
= 824000Nm^{-2}
= 8.24×10^5Nm^{-2} = 8.24×10^2Pa
Total Pressure = $P_{H_2O} +\ P_{atm}$
= (8.24×10^5 1.03×10^5)Pa
= 925300Pa
= 9.253×10^5Pa
The correct option is A

3. The heat supplied will melt the ice and then raise the temperature of the container and the water.
Heat to melt the ice at 0° = ml_f
m = mass, l_f = specific latent heat of fusion of ice.
Heat to raise the temperature of the water formed from 0° to $\theta = mc\theta$
Heat to raise the temperature of the container from 0°C to θ = Hθ, where H = heat capacity of the container.
Thus,
6000J = $Ml_f + mc\theta + H\theta$
= (0.01 × 3.33×10^5) + (0.01×4200×θ) + (20θ)
= 3330 + 42θ + 20θ
= 3330 + 62θ
62θ = 600 – 3330
= 2670
$\theta = \frac{2670}{62}$ = 43.06°C
The correct option is C

4. As time passes, the amount of a radioactive material present decreases. Also, the activity decreases with time. At time, t = 0, let the activity be A_0. The activity is halved for each half-life.

Number of days	Activity	
0	A_0	= 32/s
16	$\frac{A_0}{2}$	= 16/s
32	$\frac{A_0}{4}$	= 8/s
48	$\frac{A_0}{8}$	= 4/s
64	$\frac{A_0}{16}$	= 2/s

Hence, it takes 64days for the activity to reduce to two decays per second.
The correct option is B

5. When white light passes through a triangular prism, it is separated into different colours. This is known as dispersion. The seven colours are red, orange, yellow, green, blue, indigo and violet. This arrangement is according to the angle of deviation. The most deviated is violet while the least is red.
Violet > indigo > blue > green > yellow > orange > red
The correct option is D

6. Therefore, f on the charge due to the field is given by $f = Eq$

Also, the force of motion, F = ma

Then,

$Ma = Eq$ $\qquad a = \frac{v}{t}$

Thus, $\frac{mv}{t} = Eq$

Then, $Mv = qEt$

$V = \frac{qEt}{m}$

The correct option is A

7. Acceleration, $a = \frac{v-u}{t} = \frac{(20-12)}{5s}$m/s = 1.6m/s

$F = ma$

= (500 × 1.6)N

= 800N

The correct option is D

8. Total energy = Potential energy + kinetic energy

$P.E = mgh$

$= (2 \times 10 \times 5)$

= 100J

$K.E = \frac{1}{2}mv^2$

Total energy = 100J + 16J = 116J

The correct option is C

9. Initial velocity, u = 36km/h

Final velocity, v = 360km/h

Time taken, t = 20s = $\left(\frac{20}{3600}\right)$hr = 0.00555hr

$a = \frac{v-u}{t} => \frac{\frac{360km}{h} - 36km/h}{0.00555h}$

= 58378.378km/h^2

(Note that the acceleration is expressed in km/h^2 as opposed to the conventional unit – m/s^2. This is because distance is employed in lalometre while tune is in hr).

From the third equation of motion: $V^2 = u^2 + 2ax$

$s = \frac{v^2 - u^2}{2a} = \frac{360^2 - 36^2}{2(58378.378)}$ = 1.1km

The correct option is B

10. Let us calculate the combined resistance.

For the two parallel resistors, the effective resistance R is given by $\frac{1}{R} = \frac{1}{12} + \frac{1}{12}$

$\frac{1}{R} = \frac{2}{12}, 2R = 12, R = \frac{12}{2}$ 6Ω = 12Ω

$I = \frac{V}{R_T} = \left(\frac{24}{12}\right)$A

The correct option is D

11. An object is said to undergo simple harmonic motion if its acceleration is directly proportional to the displacement from a fixed point and also directed toward the fixed point.

Some examples of simple harmonic motion are:

(i) The motion of the diving board in a swimming pool,

(ii) The beating of the heart,

(iii) The motion of a loaded spiral spring

(iv) The motion of the prongs of a sounding turning fork and

(v) The motion of a loaded test-tube which is vertical in a liquid.

The correct option is A

12. $E = \frac{M.A}{V.R} \times 100\%$

Let us obtain the M.A and V.R and then operate as written above to get the efficiency.

The mechanical advantage, $M.A = \frac{Load}{Effort}$

$= \frac{600N}{300N}$ = 2

The velocity ratio, V.R of a wheel and axle

$= \frac{Radius\ of\ the\ wheel}{Radius\ of\ the\ axle} = \frac{50cm}{10cm} = 5$

Then,

$E = \frac{2}{5} \times 100\%$ = 40%

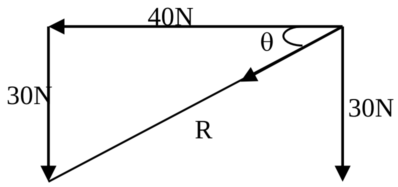
The correct option is A

13. Viscosity is the name given to the resistance in fluids against motion. It is the friction in fluids.

The correct option is B

14. This is mainly to test our knowledge on the relationship between the resultant and the equilibrant of a set of forces.

The two are equal in magnitude but opposite in direction. Let us simply get the resultant and then reverse the direction to get the equilibrant.



Using Pythagora's theorem

$R^2 = 40^2 + 30^2 = 2500$

$R = \sqrt{2500}$ = 50N

$Tan\theta = \frac{30}{40}$ = 0.75

$\theta = tan^{-1}0.75$ = 36.869° = 37°

The angle with the vertical = 90° – 37° = 53°

Hence, the direction is S53°W.

The equilibrant has a value of 50N. The direction is N53°E (opposite that of the resultant as already explained)

The correct option is C

15. A hydraulic press works on the principle of transmission of pressure. The principle is known as Pascal's principle of transmissibility of pressure. It states that when pressure is applied to an enclosed liquid, it is transmitted equally in all directions throughout the body of the liquid.

The correct option is D

16. $P = \frac{V^2}{R}$

Then, $R = \frac{V^2}{P} = \frac{12^2}{33}$ = 4.36Ω

The correct option is C

17. A p – n junction can be used as a rectifier. Rectification involves the conversion of AC to DC.

The correct option is B

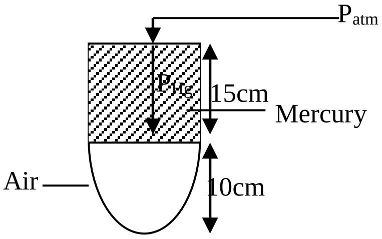
18. All but the hydrometer are type of thermometer.

A hydrometer is an instrument used for measuring the density and the relative density of a liquid.

The correct option is C

19. Let the cross-sectional area of the tube be A.

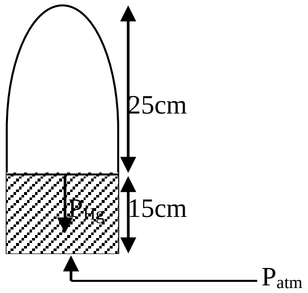
When the open end is uppermost, the total Pressure is the atmospheric pressure plus the pressure dur to the mercury



Volume of air = A × h = A × 0.1 = 01A

$P_{total} = P_{atm} + 15cm$

When the tube is inverted, the total pressure is the atmospheric pressure minus the pressure due to the mercury



Volume of air = A × h = A × 0.25 = 0.25A

$P_{total} = P_{atm} – 15cm$

Applying Boyle's law

$P_1V_1 = P_2V_2$

$(P_{atm} + 15)0.1A = (P_{atm} – 15)0.25A$

$0.1P_{atm} + 15 = 0.25P_{atm} – 3.75$

$0.25P_{atm} – 0.1P_{atm} = 1.5 + 3.75$

$0.15P_{atm} = 5.25$

$P_{atm} = \frac{5.25}{0.15}$ = 35cmHg

The correct option is A

20. For echo, the speed of the sound, $v = \frac{2x}{t}$

Where t = time to receive echo and x = distance

Thus,

$330m/s = \frac{2x}{10s}$

$2x = 3300m$

$x = \left(\frac{3300}{2}\right)$m

= 1650m

21. The process through which current is generated in a primary cell is not reversible while that of a secondary cell is reversible. This is why a primary cell cannot be recharged while a secondary cell can.

The correct option is D

22. Work function = 1.9eV

Energy = $\frac{hc}{\lambda} = \frac{6.6\times10^{-34}\times3\times10^{8}}{150\times10^{-9}}$

= 1.32×10^{-18}J

Let us convert this to eV since the work function is in eV.

Thus,

$1.32\times10^{-18}J = \frac{1eV}{1.6\times10^{-19}J} \times 1.3\times10^{-18}$J

= 8.25eV

$E = W_0 + K.E$

Thus, the kinetic energy of the photo electron emitted = $E - W_0$

= 8.25eV – 1.9eV

= 6.35eV

The correct option is A

PHYSICS 2010

1. Which of the following phenomena cannot be explained by the molecular theory of matter? A. evaporation B. expansion C. conduction D. radiation
2. The most likely measurement of length of an object using a vernier caliper is A. 3.0 cm B. 3.3 cm C. 3.33 cm D. 3.333 cm
3. If 21 g of alcohol of density 0.7 gcm^{-3} is mixed with 10 g of water, what would be the density of the resulting mixture? A. 780 gcm^{-3} B. 0.78 gcm^{-3} C. 30 gcm^{-3} D. 10gcm^{-3}
4. For a particle having an x coordinate that varies in time according to the express ion $x = 4t – 2t^2$. The instantaneous velocity for of the particle at t = 2.5 s is A. 12 m/s B. 6 m/s C. 0 m/s D. 10 m/s
5. A long-jumper leaves the ground at an angle of 20^0 above the horizontal and at a speed of 11 m/s. How far does it jumps in the horizontal direction? A. 0.384m B. 7.94m C. 8.45m D. 0m
6. A mass of 0.5 kg is attached to one end of a helical spring and produces an extension of 2.5 cm. The mass now set into vertical oscillation of amplitude 10mm. The period of oscillation is (g = 10 m/s^2) A. 0.33 s B. 100 s C. 200 s D. 280 s
7. A boat is passing under a bridge. The deck of the boat is 15m below the bridge. A small package is to be dropped from the bridge onto the deck of the boat when the boat is 25m form just below the drop point. What (boat) speed is necessary to have the package land in the boat? (g = 9.8 m/s^2) A. 17 m/s B. 14 m/s C. 1.7 m/s D. 4.9 m/s
8. An 0.60kg rubber stopper is whirled in a horizontal circle of 0.80m radius at a rate of 3.0 revolutions per second. What is the tension in the string? A, 14 N B. 80 N C. 170 N D. 24 N
9. An automobile is traveling at 60km/hr. Calculate the angular velocity of the 0.35m radius wheels. A. 16.67 rad/s B. 47.6 rad/s C. 21 rad/s D. 171.4 rad/s
10. An air bubble at the bottom of a lake has a volume of $20cm^3$, pressure of 4.9Pa, and temperature 4^0C. The bubble rises to the surface where the temperature is 20^0C and the pressure 1.0Pa. Find the volume as the bubble reaches the surface. (take 1 atm = 1.0 x $10^5N/m^2$) A. $124cm^3$ B. $319cm^3$C. $60cm^3$ D. $104cm^3$

11. A gas at constant pressure of 4.0×10^5 Pa is cooled so that its volume decreases form $1.6m^3$ to $1.2m^3$, What work is performed by the gas? A, 6.4×10^5 J B. 3.2×10^5 J C. 1.6×10^5 J D. 0.4 $\times 10^5$ J
12. Highly polished silvery surfaces are:A. poor absorbers but good emitter of radiation B. good absorbers and good emitters of radiation C. poor emitters but good reflectors of radiation D. poor emitters and poor reflectors of radiation
13. An 0.040kg string 0.80m long is stretched and vibrated in a fundamental mode with a frequency of 40 Hz. What is the speed (of propagation) of the wave and the tension in the string? A. 64 m/s B. 340 m/s C. 32 m/s D. 128 m/s
14. What is the total power output of a source with intensity 0.050 W/m^2 at a distance of 3.0m from the source? A,112W B.5.6W C. 15W D. 30W
15. The superposition of two or more waves to produce a maximum or zero effect at a point is known as A.reflection B.refractionC. diffraction D. interference
16. The acceleration due to gravity A. increases with increasing altitude B. decreases with increasing altitude C. increases with increase with increase in the square of the altitude D. is not affected by the altitude
17. Which of the following statements are correct of nuclear fission? During the process I. energy is released II. More neutrons are released than those that cause fission III. Small nuclei merge into large nuclei IV. There is a loss on mass
18. Which of the following statements is not true? A. electric field intensity is force per unit charge B. electric potential is a vector C. the S.I. unit of electric field strength is N/C D. electric field intensity is equal to potential gradient
19. Which of the following about electrolysis is false? A. liquid that conduct electricity and are split chemically by the current are electrolyzed B. the current is brought into the electrolyte by the anode C. the current is taken away form the electrolyte by the cathode D. the container which holds the electrolyte and the electrode is the voltmeter
20. Which of the following is not true about properties of x-rays? A. they are not deflected by magnetic or electric field B. they ionized a gas, making it a conductor C. they are massive D. they have high penetrating power
21. A transformer is connected to a 240 V supply. The primary coil has 40 turns, and the secondary is found to be 960 V. What is the ratio of the number of turns of the primary coil to the number of turns of the secondary coil? A. 1 : 4 B. 4 : 1 C. 1 : 6 D. 6 : 122
22. Which of the following is not true about an object that is projected upwards at angle θ. A. the velocity is maximum at the maximum height B. the acceleration along the horizontal direction is zero C. the maximum range (R_{max}) for an object moving with speed u is given by $\frac{u^2}{g}$D. the time it takes to get the maximum height is equal to the time it takes to comes back to the point of projection
23. When three coplanar non-parallel forces are in equilibrium. i. they can be represented in magnitude and direction by the three sides of a triangle taken in order. ii. The liens of action meet at a point iii. The magnitude of any one force equals the magnitude of the resultant of the other two forces. iv. any one force is the equivalent of the other two. A. i and iii only B. i and ii only C. i, ii, iii, and iv D. none of them
24. Which of the following statements is not TRUE about a body performing simple harmonic motion? A. the linear speed is the product of the angular speed and the radius or amplitude B. the linear acceleration is the product of the square of the angular speed and the displacement. C. frequency is the number of complete revolution per second made by a vibrating body D. the SI unit of amplitude is Hertz (Hz).
25. In the force of gravity on an object of mass m, the gravitational field strength, g, is given by the following equation. A. $g = \sqrt{mF}$ B. $g = mF$ C. $g = m\sqrt{F}$ D. $g = \frac{F}{m}$

PHYSICS 2010 SOLUTION

1. Evaporation, expansion and conduction can be explained using the molecular theory of matter while radiation cannot.
   The correct option is A
2. The accuracy of a metre rule is 0.1cm, that of the vernier caliper is 0.01cm while that of the micrometer screw guage is 0.001cm.
   Accurately, the number of decimal places in the accuracy of each instrument correlates with that of the effective measurement it can make.
   Thus,
   3.0cm and 3.3cm can be read accurately from a metre rule **3.33cm is that of the vernier caliper** while the micro metre screw guage takes 3.333cm
   The correct option is C
3. Mass of alcohol 21g
   Density of alcohol 0.7gcm^{-3}
   Mass of water 10g
   Density of water 1gcm^{-3}
   Can you simply add the two densities together to obtain the overall density?
   Capital NO! Density is not an extrinsic property. Extrinsic properties 9mass, length etc) can just

be added to get the overall value. Being intrinsic, density cannot be so treated.

For a number of substances, the resulting density $\frac{mass_1 + mass_2 + mass_3 \ldots}{Volume_1 + Volume_2 + Volume_3 \ldots}$

$V_{alcohol} = \frac{M_{alcohol}}{\rho_{alcohol}} = \frac{21g}{0.7gcm^{-3}}$ = 30cm^3

$V_{H_2O} = \frac{M_{H_2O}}{\rho_{H_2O}} = \frac{10g}{1gcm^{-3}}$ => 10cm^3

$$resulting\ density = \frac{21g}{30cm^3 + 10cm^3}$$

$= \frac{31g}{40cm^3}$ = 0.775gcm^{-3}

The correct option is B

4. $x = 4t - 2t^2$

The instantaneous velocity equals $\frac{dx}{dt}$

= 4 – 4t

Substituting t = 2.5, we have

= (4 – 10)m/s

= -6m/s (i.e. 6m/s in the opposite direction)

The correct option is B

5. The distance covered in the horizontal direction is the horizontal range which is given by

$R = \frac{U^2 sin2\theta}{g}$

$= \frac{11^2 \times \sin 40^o}{9.8}$

= 7.94m

The correct option is B

6. M = 0.5kg, e = 2.5cm = 0.025m

We can obtain the force constant, k, by using Hooke's law of elasticity

F = Ke, F = W = mg = 0.5 × 10 = 5N

Then,

5N, K × 0.025m

$K = \frac{5N}{0.025m}$ = 200Nm^{-1}

The period = $T = 2\pi\sqrt{\frac{m}{k}}$ = 2 × 3.142$\sqrt{\frac{0.5}{200}}$

= 0.3142s

Alternatively, we can evaluate the angular velocity and then use it to obtain the period

$w = \sqrt{\frac{k}{m}} = \sqrt{\frac{200}{0.5}}$ = 20rads^{-1}

Since $w = \frac{2\pi}{T}$, $T = \frac{2\pi}{w} = \frac{6.284}{20}$ = **0.314s as before**

No suitable option

7. The package in moving under constant acceleration – the acceleration due to gravity. For the package to land on the boat, the boat must neither be too fast nor slow. It must arrive exactly at the point below the deck at the same instant the package arrives. We are to simply calculate the time-taken by the package to travel down and then determine the velocity with which the boat can move so that it takes the same time to get there.

<u>For the package</u>

U = 0, S = 15m, a = g = 9.8m/s^2, t = ?

$S = U + \frac{1}{2}at^2$

$15 = 0(t) + \frac{1}{2} \times 9.8t^2$

$15 = 4.9t^2$, $t^2 = \frac{15}{4.9}$ = 3.0612

$t = \sqrt{3.0612}$ = 1.7496s

<u>For the boat</u>

$t$must be 1.7496s, s = 25m

$Velocity = \frac{25m}{1.7496s}$ = 14.28m/s

The correct option is B

8. $T = F\ \frac{MV^2}{r}$

m = 0.6kg, r = 0.80m

w = 3reV/s = (3 × 2π)rad/s (Since 1reV = 2πrad)

$$v = wr$$

= (18.852 × 0.80)m/s = 15.08m/s

Then,

$T = \frac{0.6 \times 15.08^2}{0.8}$ = 170.55N

The correct option is C

9. V = 60km/hr = $\frac{60 \times 100m}{3600s}$ = 16.67m/s

Since the linear velocity of the wheel equals that of the automobile, the angular velocity of the wheel can be determined with this value

$v = wr, w = \frac{v}{r} = \left(\frac{16.67}{0.35}\right)$rad/s = 47.62rad/s

The correct option B

10. P_1 = 4.9Pa, V_1 = 20cm^3, T_1 = 4^{o}C = 277K, P_2 = 1.0Pa, V_2 = ?, T_2 293K

$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

$V_2 = \frac{P_1V_1T_2}{P_2T_1} = \frac{4.9Pa \times 20cm^3 \times 293K}{1.0Pa \times 277K}$

= 103.66c,^{3}

= 104cm^3

The correct option is D

11. Workdone = PΔV

P = 4.0 ×10^{5}Pa = 4.0×10^{5}Nm^{-2}

ΔV = 1.6m^3 – 1.2m^3 = 0.4m^3

Then, w = (4 × 10^5 × 0.4)J

= 1.6×10^{5}J

The correct option is C

12. Polished surfaces are only good reflectors. They are poor absorbers and poor emitters

The correct option is C

13. $V = \sqrt{\frac{T}{\mu}}$

Where v = velocity of wave, T = tension in the string and μ = mass per unit length

Is the tension, T given? No!

The tension does not refer to the weight of the string. It refers to the tension applied in plucking the string. This is not given in the question

Also, $F = \frac{1}{2L}\sqrt{\frac{T}{\mu}}$

Since $V = \sqrt{\frac{T}{\mu}}$, $F = \frac{1}{2l} \times V = \frac{V}{2l}$

Then,

40Hz = $\frac{V}{2(0.8)m}$

V = [40 × 2(0.8)]m/s

= 64m/s

The correct option is A

14. $I = \frac{P}{4\pi r^2}$

Where I = intensity, P = power, r = distance from the source, I = 0.05W/m^2, r = 3.0m

Thus,

$0.05 = \frac{P}{4 \times 3.142 \times 3.0^2}$

P = (0.05 × 4 × 3.142 × 3.0^2) W

= 5.65W

The correct option is B

15. When two or more waves are superposed, they interact constructively or destructively. This is known as interference.

The correct option is D

16. The higher we go, the smaller the value of the acceleration due to gravity. It decreases as the attitude increases.

The correct option is B

17. During nuclear fission (i) energy is released (ii) more neutrons are released (iii) a large nucleus splits into smaller nuclei (Hence, statement iii given in the question is not correct) and (iv) there is a loss of mass

18. The electric potential is a scalar and not a vector quantity.

The correct option is B

19. The liquids that conduct electricity and are split by the current are called electrolytes.

The anode is the electrode through which current enters and electron leaves the electrolyte.

The cathode is the electrode through which electron enters and current leaves.

The container which holds the electrolyte and the electrodes is known as the **voltameter** and **not voltmeter** as suggested by option D

The correct option is D

20. X-rays are not charged; thus, they are not deflected by electric or magnetic

They are capable of ionizing gases.

They are **not massive**

They have high penetrating power

The correct option is C

21. For a transformer

$\frac{N_P}{N_S} = \frac{V_P}{V_S}$ ……………………….. (i)

Also,

$\frac{N_P}{N_S} = \frac{I_S}{I_P}$ ……………………….(ii)

Please carefully note the difference between the two.

N, V and I refer to the number of turn, voltage and current respectively for the primary and secondary coils. The ratio of the number of turns of the primary coil to that of the secondary coil is $\frac{N_P}{N_S}$.

From equation (i)

$\frac{N_P}{N_S} = \frac{V_P}{V_S} = \frac{240V}{960V}$ = 1 : 4

The correct option is A

22. The velocity of the object is maximum at the point of projection (not at the maximum height). At the maximum height, the velocity is zero.

The correct option is A

23. All the statements are correct

The correct option is C

24. Since amplitude is a displacement, its S.I. unit is the metre (m) and not Hertz (Hz)

The correct option is D

25. The gravitational field strength is the force per unit mass i.e. $g = \frac{F}{m}$

The correct option is D

PHYSICS 2009 QUESTIONS

1. The force with which an object is attracted to the earth is called A. acceleration B. mass C. gravity D. impulse E. weight
2. The refractive index of a liquid is 1.5, if the velocity of light in a vacuum is 3.0 x 10^{8}ms^2, the velocity of light in the liquid is A. 1.5 x 10^{8}ms^{-2} B. 2.0 × 10^{8}ms^{-1} C. 3.1 × 10^{8}ms^{-1} D. 4.5 × 10^{8}ms^{-1}E. 9.0 × 10^{8}ms^{-1}
3. A train has an initial velocity of 44m/s and an acceleration of -4m/s what is its velocity after 10 seconds? A. 2m/s B. 4m/s C. 12m/s D. 12m/s E. 16m/s
4. A man of mass 50kg ascends a flight of stair 5m high in 5 seconds. If acceleration due to gravity is 10ms^{-2}, the power expended is A. 100w B. 300w C. 250w D. 400w E. 500w
5. A machine has a velocity ratio of 5. If it requires a 50kg weight to overcome 200kg weight, the efficiency us A. 4% B. 5% C. 40% D. 50% E. 90%
6. If the force on a charge of 0.2 coulomb in an electric field is 4N, then electric intensity of the field is A. 0.8 B. 0.8N/C C. 20.0N/C D. 4.2N/C E. 20.0C/N
7. The resistance of a wire depends on A. the length of the wire B. the diameter of the wire C. the temperature of the wire D. the resistivity of the wire E. all of the above
8. Which of these is not contained in a dry cell? A. carbon rod B. paste f magnesium dioxide C. paste of ammonium chloride D. zinc case E. copper rod
9. A simple pendulum 0.6m long has a period of 1.5s, what is the period of a similar pendulum 0.4m long in the same direction A. $1.4\sqrt{\frac{2}{3}}$B. $1.5\sqrt{\frac{2}{3}}$ C. $2.25s$ D. $1.00s$ E. $2.00s$
10. A device that converts sound energy into electrical energy is A. the horn of a motor car B. an AC generator C. A microphone D. telephone earpiece E. a loud speaker
11. Radio waves have velocity of 3×10^{8}ms^{-1}, if a radio station sends out a broadcast on a frequency 800KHz, what is the wavelength of the broadcast? A. 375.0m B. 26.70m C. 240.0m D. 37.5, E. 26.7m

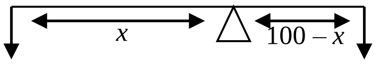
12. Which of these is not a fundamental S.I. unit A. Ampere B. Kelvin C. seconds D. radians
13. If two masses 40g and 60g respectively, are attached firmly to end of a light metre rule, what is the centre of gravity of the system? A. at the mid-point of the metre rule B. 40cm from the lighter mass C. 40cm form the heavier masses D. 60cm from the heavier mass E. indeterminate because the metre rules is light
14. To find the depth of the sea, a ship sends out a second waves and receives an echo after one second. If the velocity of sound in water is 1500m/s, what is the depth of the sea A. 0.75km B. 1.50km C. 2.20km D. 3.00km E. 3.75km
15. What is the neutron in the Uranium isotope$^{238}_{92}X$? A. 92 B. 146 C. 238 D. 330 E. 119
16. The mode of heat transfer which does not require material medium is A. conduction B. radiation C. convection D. propagation
17. The height at which the atmosphere ceases to exist is about 80km, if the atmospheric pressure on the ground level is 760mmHg, the pressure at height of 20km above the ground is A. 380mmHg B. 570mmHg C. 190mmHg D. 480mmHg
18. Which of the following is common to evaporation and boiling, they A. take place at any temperature B. are surface phenomena C. involve change of state D. take place at a definite pressure E. none of the above
19. Which of the following instrument has pure tone? A. Guitar B. vibrating string C. tuning forks D. screen E. horns
20. Four lenses are being considered for use as a microscope object. Which of the following focal length is mot suitable A. -5mm B. +5mm C. 5cm D. +5cm E. -5.5mm
21. The product PV where P is pressure and V is a volume has the same unit as A. force B. power C. energy D. acceleration E. all of the above
22. Two strings of the same length and under the same tension give notes of frequency in the ratio 4 :1. The masses of the strings are in the ratio of A. 2 :1 B. 1 : 2 C. 1 : 4 D. 1 : 7 E. 1 : 16
23. A house hold refrigerator is rated 200 watts. If electricity costs 5k per kwh, what is the cost of operating it for 20 days. A. ₦4.80 B. ₦48.00 C. ₦480.00 D. ₦4800.00 E. ₦240.00
24. The resistance of a 5m uniform wire of cross sectional area of $0.2 \times 10^6 m^2$ is 0.45. What is the resistivity of the material of the wire? A. 1.10×10^5ohms B. 4.25×10^6 ohms C. 2.40×10^7 ohms D. 1.70×10^8 ohms E. 1.40×10^8 ohms
25. When a yellow card is observed through a blue glass, the card would appear as A. black B. green C. red D. white E. purple

PHYSICS 2009 SOLUTION

1. The weight of an object is defined as the earth's pull (a force) on the object
   The correct option is E
2. $n = \frac{Velocity\ of\ light\ in\ vaccum}{Velocity\ of\ light\ in\ medium}$
   $1.5 = \frac{3\times10^8 m/s}{Velocity\ of\ light\ in\ the\ liquid}$
   $Velocity\ in\ the\ liquid = \frac{3\times10^8 m/s}{1.5}$
   $= 2\times10^8$m/s
   The correct option is B
3. u = 44m/s, a = -4m/s^2, t = 10s, v = ?
   $$v = u + at$$
   = 44 + (-4 × 10)
   = 44 – 40
   = 4m/s
   The correct option is B
4. $Power = \frac{Workdone}{Time} = \frac{mgh}{t}$
   $= \left(\frac{50 \times 10 \times 5}{5}\right) W$ = 500W
   The correct option is E
5. V.R = 5, l = 200kg = 2000N
   E = 50kg = 500N
   $M.A = \frac{l}{E} = \frac{2000N}{500N} = 4$
   $Efficiency = \frac{M.A}{V.R} \times 100\%$
   $\frac{4}{5} \times 100\% = 80\%$
   The correct option is E
6. F = 4N, q = 0.2C
   $F = Eq, E = \frac{F}{q} = \frac{4N}{0.2C}$
   $= 20Nc^{-1}$
   The correct option is C
7. The resistance of a wire depends on:
   (i) The resistivity of the material
   (ii) The length of the wire
   (iii) The cross-sectional area of the wire and
   (iv) The temperature
   The correct option is E
8. A copper rod is not a component of a dry cell.
   The correct option is E
9. $\frac{T_1}{\sqrt{L_2}} = \frac{T_2}{\sqrt{L_2}}$
   T_1 = 1.5s, L_1 = 0.6m, T_2 = ? L_2 = 0.4m
   $\frac{1.5}{\sqrt{0.6}} = \frac{T_2}{\sqrt{0.4}}$
   $T_2 = \frac{1.5 \times \sqrt{0.4}}{\sqrt{0.6}} => 1.5\sqrt{\frac{0.4}{0.6}}s$
   $=> 1.5\sqrt{\frac{4}{6}}s$
   $=> 1.5\sqrt{\frac{2}{3}}s$
   The correct option is B
10. A microphone converts sound energy into electrical energy while a loudspeaker converts electrical to sound energy
    The correct option is C
11. V = 3×10^8m/s, F = 800KHz = 8×10^5Hz
    $Frequency = \frac{Velocity}{Wavelenght}$

$Wavelenght = \frac{Velocity}{Frequency} = \left(\frac{3 \times 10^8}{8 \times 10^5}\right)$m
= 375m
The correct option is A

12. Ampere, Kelvin and Seconds are the S.I units of electric current, temperature and time respectively. Since the aforementioned quantities are fundamental quantities, the units are fundamental units. The radian is not a fundamental quantity.
The correct option is D

13.


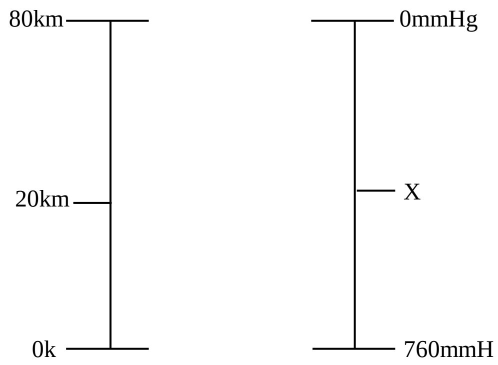
Clockwise moment = Anticlockwise moment
60(100 – x) = 40x
6000 – 60x = 40x
100x = 6000
$x = \frac{6000}{100}$ = 60cm
The centre of gravity is 60cm from the lighter mass or 40cm from the heavier mass
The correct option is C

14. $V = \frac{2x}{t}$
$1500 = \frac{2x}{1}$
$x = \frac{1500}{2}$ => 750m = 0.75km
The correct option is A

15. Atomic number = proton number = 92
Mass number = 238
Mass number = proton number + neutron number
238 = 92 + neutron number
Neutron number = 239 – 92 = 146
The correct option is B

16. Conduction and convection require material medium while radiation does not.
The correct option is B

17. **Method 1**
At the ground level, 80km of air is exerting the given pressure (760mmHg) with this we can evaluate the density of air which can then be used to get the pressure at any height
P = 760mmHg = 10^5Nm^{-2}, h = 80km = 8×10^4m
g = 10m/s^2
$P = h\rho g$
$10^5 = 8\times10^4 \times \rho \times 10$
$\rho = \left(\frac{10^5}{8 \times 10^4 \times 10}\right)$kgm^{-3} => 0.125kgm^{-3}
At 20km above the ground level, the height of air above = 80km – 20km = 60km = 6×10^4m
$P = h\rho g$
= $6 \times 10^4 \times 0.125 \times 10$
= 75,000Nm^{-2}
Since 10^5Nm^{-2} = 760mmHg, 75000Nm^{-2}
$= \left(\frac{760}{10^5} \times 75000\right)$mmHg
= 570mmHg
The correct option is B
Method 2
Since the pressure at the ground level is known and that at the highest level can be inferred (It is zero because no air is above this point), the pressure at any point can be obtained by extrapolation.



$\frac{20-0}{80-20} = \frac{X-760}{0-X}$
$\frac{20}{60} = \frac{X-760}{-X}$
-20X = 60X – 45600
-20X – 60X = -45600
-80X = -45600
$X = \left(\frac{45600}{-80}\right)$mmHg
= 570mmHg as obtained earlier
The correct option is B

18. Both evaporation and boiling refer to a change from liquid to gaseous state. However, there are cogent differences between the two.
Evaporation occurs at all temperature while boiling occurs at a particular temperature known as the boiling point.
Evaporation is a surface phenomenon while boiling occurs throughout the body of a liquid.
Evaporation does not depend on pressure, while boiling occurs when the saturated vapour pressure equals the external pressure.
The correct option is C

19. A sounding tuning fork produces a pure tone.
The correct option is C

20. First of all, it must be remembered that convex lenses are employed in microscopes. The focal length of a convex or converging lens is positive.
Also, it is worthy of notice that the magnifying power is inversely proportional to the focal length of the lens. It is therefore, desired that the focal length is very small.
Then, **+5mm** will be an ideal focal length.
The correct option is B

21. Let us obtain the unit of PV and then check the parameter that has the same unit.
Pressure, P is measured in Nm^{-2}
Volume, V is measured in m^3
Then PV will be measured in Nm^{-2} × m^3 = Nm
Energy = Force × Distance
= N × m = Nm
The correct option is C

22. $F = \frac{1}{2l}\sqrt{\frac{T}{\mu}}$
Since μ is mass per unit length, it can be connected that $F \propto \frac{1}{\sqrt{m}}$

$F = \frac{k}{\sqrt{m}}$, $k = F\sqrt{m}$

$k = F_1\sqrt{m_1} = F_2\sqrt{m_2}$

Then,

$\frac{F_1}{F_2} = \frac{\sqrt{m_2}}{\sqrt{m_1}}$

Ratio of frequency = 4 : 1

$\frac{4}{1} = \frac{\sqrt{m_2}}{\sqrt{m_1}}$

By squaring both sides, we have:

$\frac{4^2}{1} = \frac{m_2}{m_1}$

$\frac{m_2}{m_1} = \frac{16}{1}$

$\frac{m_1}{m_2} = \frac{1}{16}$ = 1 : 16

The correct option is E

23. Power = 200W = 0.2kW
Time = 20days = (20 × 24)h = 480h
Energy consumed = (0.2 × 480)kWh
= 96kWh
Since 1kWh costs 5k, 96kWh will cost (5 × 96)k = 480k =>₦4.80k
The correct option is A

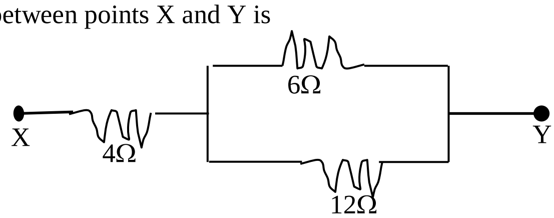
24. $R = \frac{\rho l}{A}$
$\rho = \frac{RA}{l} = \frac{0.45 \times 0.2 \times 10^6}{5} = 1.8 \times 10^4 \Omega m$

25. The colour of an object is the colour it reflects. A green object absorbs other colours and reflects only green. This is why it is seen as green.
When a yellow card is observed through a blue glass, it appears black.
The correct option is A

PHYSICS 2008 QUESTIONS

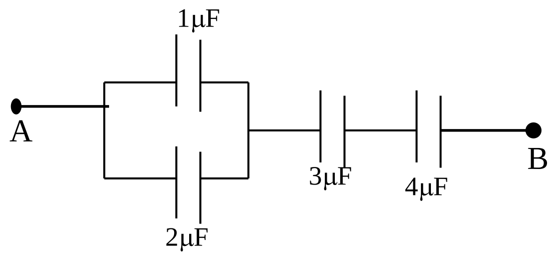
You may find the following useful
Acceleration due to gravity g 9.8m/s
Gas constant R 8.31 J/mol.K
Speed of light in vacuum 3.0×10^8m/s
Plank constant h 6.63×10^{-34} J.s

1. A convex lens of focal length 10.0cm is used to form a real image which is half the size of the object. How far from the object is the image? A. 45cm B. 30cm C. 15cm D. 20cm
2. A diverging lens of focal length 20cm forms an image halve of the size of the object. What is the object distance? A. -11.11cm B. 100cm C. 60cm D. 8.71cm
3. An object of height 3.00cm is placed 10cm in front of a biconvex lens of focal length 15cm,, the image of the object is A. real and 3.00cm tall B. virtual and 9.00 cm tall C. virtual and 3.00cm tall D. real and 9.00cm tall
4. The most suitable type of mirror used for the construction of a searchlight is the: A. concave mirror B. spherical mirror C. convex mirror D. parabolic mirror
5. Light waves and ripples of water are similar because both A. are longitudinal waves B. have the same velocity C. can be diffracted and refracted D. have the same frequency
6. Three 4-Ω resistors were connected in series by 'Tola while Ade connected the same set of resistors in parallel. The ratio of the value obtained by Ade to that obtained by 'Tola is A. 1:2 B. 1:9 C. 1:10 D. 1:5
7. Three resistors are connected as shown in the diagram below. The equivalent resistance between points X and Y is



A. 8.0Ω B. 22.0Ω C. 4.25 ΩD. 3.27Ω

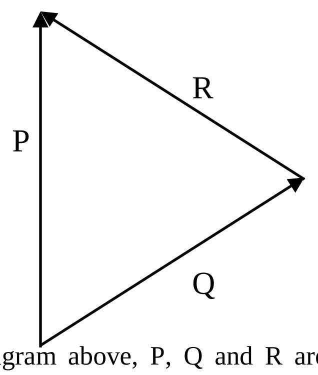
8. A coil of copper wire of N turns is kept rotating between the poles of a permanent magnet such that the magnetic flux linking the coil changes continuously. Which of the following statements is TRUE? A. an e.m.f. is induced in the coil such that when the change of flux is positive the e.m.f is positive, and when the change of flux is negative, the e.m.f is negative B. an em.f is induced in the coil whose magnitude is inversely proportional to both the number of turns in the coil and the rate of change of magnetic flux C. ane.m.f is set up in the permanent magnet which reduces the flux in the coil to zero D. current flows in the coil and an e.m.f. is set up proportional to both the rate of change of the flux and the number of turns.
9. What is the equivalent capacitance between points A and b in the diagram below?



A. 12.0μF B. 1.1μF C. 1.6μF D. 7.7μF

10. The principle of operation of an induction coil is based on A. Ohm's law B. Ampere's law C. Faraday's law D. Coulomb's law
11. The half-life of a radioactive element is 9 days. What fraction of atoms has decayed in 36 days? A. $\frac{1}{16}$ B. $\frac{1}{4}$ C. $\frac{1}{2}$ D. $\frac{15}{16}$
12. Which of the following radiations cannot be deflected by an electric field on magnetic field? (i) α-rays ii. β-rays iii. γ-rays iv. X-ray A. (i) and (ii) only B. (i) and (ii) only C. (iii) and (iv) only D. (i) and (iii) only
13. The equation ${}^{150}_{62}X \longrightarrow {}^{150}_{63}Y + k^{0}_{-1} +$ energy, represents A. α-decay B. β-decay C. γ-decay D. photon emission

14. Calculate the length of a displaced pendulum bob that passes its lowest point twice every second. (g=10ms^{-2}) A. 1.000m B. 0.253m C. 0.450m D. 0.58m
15. A vehicle of mass m is driven by an engine of power P from rest. Find the minimum time it will take to acquire a speed v. A. $\frac{mv^2}{P}$2 B. $\frac{mv^2}{2P}$C. $\frac{mv}{P}$ D. $\frac{mv}{2P}$
16. When a ball rolls on a smooth level ground, the motion of its centre is A. translational B. random C. oscillatory D. rotational
17. A body accelerates uniformly form rest at 6ms^{-2} for 8 seconds and then decelerates uniformly to rest in the next 5 seconds. The magnitude of the deceleration is A. 9.6ms^{-2} B. 48ms^{-2} C. 24.0ms^{-2} D. 39.2ms^{-2}
18. A nail is pulled from a wall with a string tied to the nail. If the string is inclined at an angle of 30^0 to the wall and the tension in the string is 50N, the effective force used in pulling the nail is A. $25N$ B. $25\sqrt{3}N$ C. $50N$ D. $50\sqrt{3}N$
19. A box of mass 40Kg is being dragged along by the rope inclined at 60^0 to the horizontal. The frictional force between the box and the floor is 100N and the tension on the rope is 300N. How much work is done in dragging the box through a distance of 4m? A. 680J B. 200J C. 100J D. 400J
20. A 70kg man ascends a flight of stairs of height 4m in 7s. The power expended by the man is A. 40W B. 100W C. 280W D. 400W
21. Which of the following statements is not true? A. as the slope of an inclined plane increases, the velocity ratio decreases B. the efficiency of an inclined plane decreases as the slope increases C. the effort required to push a given load up an inclined plane increases as the slope increases D. the mechanical advantage of smooth inclined plane depends on the ratio of the length to the height of the plane



22. In the diagram above, P, Q and R are vectors. Which of the options gives the correct relationship between the vectors? A. P = R – Q B. P = R + Q C. P = Q – R d P + R + Q = 0
23. If M and R the mass and radius of the earth respectively and G is the universal gravitational constant, the earth's gravitational potential at an altitude H above the ground level is A. –GM/H B. –GM/R(+H) C. –GM/2H D. –GM/(R-H)
24. The ice and steam points of a thermometer are 20mm and 100mm respectively: A. temperature of 75^{0}C corresponds to Ymm on the thermometer. What is Y? A. 100mm B. 70mm C. 80mm D. 60mm
25. An electric kettle with negligible heat capacity is rated at 2000W, if 2.0Kg of water is put in it, how long will take temperature of water to rise from 20^{0}Cto 100^{0}C? (Specific heat capacity of water = 4200JKG^{-1} K^{-1}) A. 336s B. 240s C. 168s D. 84s
26. A quantity of ice – at 10^{0}C is heated until the temperature of the heating vessel is 90^{0}C. Which of the following constants NOT required to determine the quantity of heat supplied to the vessel? A. specific latent heat of vaporization B. specific heat capacity of ice C. specific latent heat of fusion D. specific heat capacity of water
27. The scent from a jar of perfume opened moves through the air molecules by A. evaporation B. osmosis C. diffusion D. convection

PHYSICS 2008 SOLUTION

1. f = 10cm, $m = \frac{H_I}{H_0} = \frac{v}{u} = 2$
Then, v = 2u
$\frac{1}{f} = \frac{1}{u} + \frac{1}{v} = \frac{1}{u} + \frac{1}{2u} = \frac{2+1}{2u} = \frac{3}{2u}$
$\frac{1}{10} = \frac{3}{2u}$
2u = 30
$u = \frac{30}{2}$ => 15cm
$v = 2u = 30cm$
Distance between the object and the image is v + u = 30cm + 15cm = 45cm
The correct option is A
2. For a diverging lens, f = -ve
F = -20cm, u = 2v (since the image is half the object)
$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$
$\frac{1}{-20} = \frac{1}{2v} + \frac{1}{v}$
$-\frac{1}{20} = \frac{1+2}{2v} = \frac{3}{2v}$
$-2v = 60$
$v = \left(\frac{60}{-2}\right)$cm = -30cm (a virtual image
$u = 2 \times 30$cm = 60cm
The correct option is C
3. f = 15cm, u = 10cm, v = ?
$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$
$\frac{1}{15} = \frac{1}{10} + \frac{1}{v}, \quad \frac{1}{v} = \frac{1}{15} - \frac{1}{10}$
$= \frac{2-3}{30} = -\frac{1}{30}$
$\frac{1}{v} = \frac{-1}{30}$, -v = 30, v = -30cm
Since v is negative, the image is virtual
$m = \frac{v}{u} = \frac{H_I}{H_0}$

$\frac{30}{10} = \frac{H_I}{3}$

$10 \times H_I = 3 \times 30$

$H_I = \left(\frac{3 \times 30}{10}\right)$cm = 9cm

Thus, the image is virtual and 9cm tall

The correct option is B

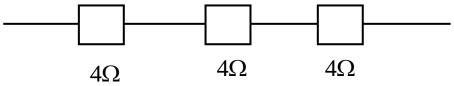
4. A parabolic mirror is the most suitable mirror for constructing a search light.

The correct option is D

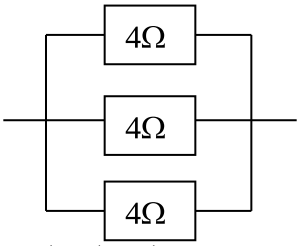
5. Both light waves and water waves can be reflected, refracted and polarized.

The correct option is C

6.



Req = 4Ω + 4Ω + 4Ω = 12Ω



$\frac{1}{Req} = \frac{1}{4} + \frac{1}{4} + \frac{1}{4}$

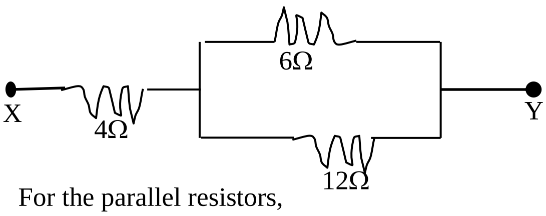
$\frac{1}{Req} = \frac{1+1+1}{4} = \frac{3}{4}$

$Req = \frac{4}{3}$ => 1.333Ω

Ratio of resistance 1.333 : 12 = 1 : 9

The correct option is B

7.



For the parallel resistors,

$\frac{1}{R} = \frac{1}{6} + \frac{1}{12} = \frac{2+1}{12} = \frac{3}{12}$

$R = \frac{12}{3}$ = 4Ω

This is now connected in series with a 4Ω resistor

R_T = 4Ω + 4Ω = 8Ω

The correct option is A

8. Current and emf are set up resultantly. The emf value is proportional to the rate of change of flux and to the number of turns

The correct option is D

9. Unlike for resistors, parallel connection of capacitors give a total capacitance which is the direct sum of the individual capacitances.

The 1μF and 2μF capacitors will give 3μF. This is now in series with another 3μ and a 4μF capacitor

The combined resistance, R_C is given by

$\frac{1}{R_c} = \frac{1}{3} + \frac{1}{3} + \frac{1}{4} = \frac{4+4+3}{12}$ =>$\frac{11}{12}$μF

$\frac{1}{R_c} = \frac{11}{12}$, $R_c = \frac{12}{11}$ => 1.1μF

The correct option is B

10. The principle of operation of an induction coil is based on Faraday's law

The correct option is C

11. Let the original amount of the radioactive element be N

Day	Fraction that remains
0	N
9	$\frac{N}{2}$
18	$\frac{N}{4}$
27	$\frac{N}{8}$
36	$\frac{N}{16}$

After 36 days, $\frac{N}{16}$ remain. Then, the amount that has decayed = $N - \frac{N}{16} = \frac{15}{16}$N

i.e. $\frac{15}{16}$ of the original amount

The correct option is D

12. α - rays and β-rays are positively and negatively charged respectively. Consequently, they are deflected by electric and magnetic field, α-rays are deflected towards the negative pole of an electric field and the south pole of a magnetic field. β-rays are deflected towards the positive pole of an electric field and the north pole of a magnetic field.

Since X-rays and γ-rays are not charged, they are not deflected by those fields.

The correct option is C

13. ${}^{150}_{62}X \rightarrow {}^{150}_{63}Y + {}^{0}_{-1}K$

${}^{0}_{-1}K = {}^{0}_{-1}\beta$ (The values, and not the symbol, matter)

This is a beta decay

The correct option is B

14. Since it passes its lowest point twice every second, its period is I_s

$T = 2\pi\sqrt{\frac{l}{g}}$

$1 = 6.284\sqrt{\frac{l}{10}}$

$1^2 = (6.284)^2 \times \frac{l}{10}$ => $\frac{6.284^2 l}{10}$

$l = \frac{10}{6.284^2} = \frac{10}{39.4887}$ = 0.253m

The correct option is B

15. The energy supplied by the engine equals the kinetic energy of the vehicle

i.e. Power × Time = $\frac{1}{2}mv^2$

$Pt = \frac{mv^2}{2}$

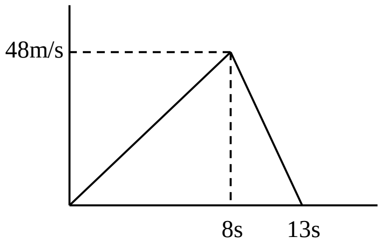
$t = \frac{mv^2}{2P}$

The correct option is B

16. When a ball rolls on a smooth level ground, the motion of its centre is translational while that of the circumference is rotational.

The correct option is A

17. **Method 1**



From the graph, deceleration = $\frac{48m/s - 0m/s}{13s - 8s}$

$= \frac{48m/s}{5s}$ = 9.6m/s^2

The correct option is A

Method 2

We can employ the equation of motion. The final velocity during acceleration here is the initial velocity during deceleration

For acceleration

u = 0, a = 6m/s^2, t = 8s, v = ?

$$v = u + at$$

= 0 + 6(8)

= 48m/s

For deceleration

u = 48m/s, v = 0 (since it comes to rest), t = 5s

$v = u + at$

0 = 48 + 5a

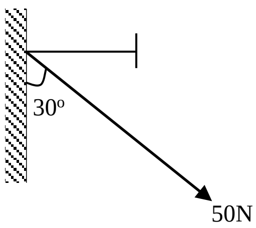
5a = -48

$a = \frac{-48}{5}$ = -9.6m/s^2

Since acceleration and deceleration are opposite, the deceleration is 9.6m/s^2

The correct option is A

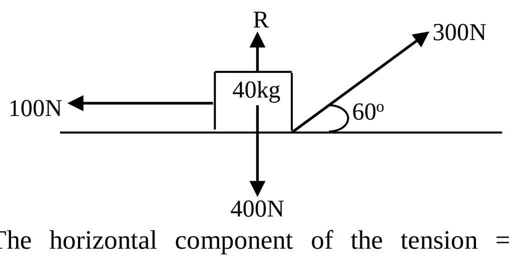
18.



The component of the 50N force pulling the nail is the horizontal. Component which equals 50sin30^0= 25N

The correct option is A

19.



The horizontal component of the tension = 300cos60 = 150N

The vertical component = 300sin60^o = 259.808N

The resulting horizontal force = 150N – 100N = 50N

Workdone = Force × Distance

= 50N × 4m

= 200Nm

= 200J → B

20. $Power = \frac{mgh}{t}$

$= \frac{70 \times 10 \times 4}{7}$

= 400W

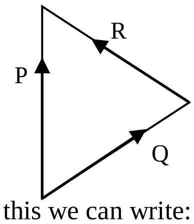
The correct option is D

21. The mechanical advantage of a smooth inclined plane does not depend on the length and height of the plane.

It is the velocity ratio that depends on length and height of plane.

The correct option is D

22.



For this we can write:

(i) P – Q – R = 0 and

(ii) P = Q + R = R + Q

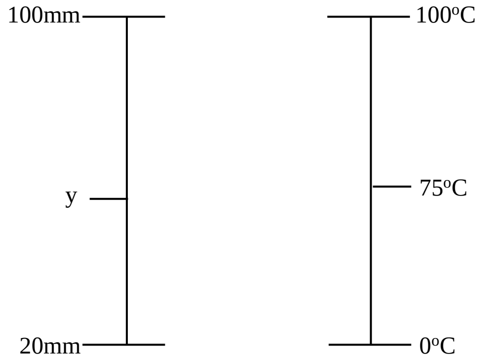
The correct option is B

23. On the surface of the earth, the gravitational potential = $\frac{Gm}{R}$

At a height H above the surface of the earth, the gravitational potential = $\frac{-Gm}{R+H}$

The correct option is B

24.



$\frac{100-y}{100-20} = \frac{100-75}{100-0}$

$\frac{100-y}{80} = \frac{25}{100}$

10000 – 100y = 2000

-100y = 2000 – 10000

-100Y = -8000

$y = \frac{-8000}{-100}$ = 80mm

The correct option is C

25. $IVt = mc\Delta\theta$

Power × time = $mc\Delta\theta$

2000 × t = 2 × 4200 × (100 – 20)

2000t = 672000

$t = \frac{672000}{2000}$ = 336s

The correct option is A

26. To determine the heat demanded to raise the temperature of the ice to 0^{o}C from -10^{o}C, the specific heat capacity of the ice must be known.

To determine the heat required to melt the ice, the specific latent heat of fusion of ice must be known.

To calculate the heat needed to raise the temperature of the water formed from 0ºC to 90ºC, the specific heat capacity of water must be given.

The specific latent heat of vaporization is not needed since the water is not to be boiled off.

The correct option is A

27. The scent of the perfume moves through the air molecules by diffusion

The correct options is C

PHYSICS 2007 QUESTIONS

You may find the following useful:
Acceleration due to gravity g 9.8m/s
Gas constant R 8.31 J/mol.K
Speed of light in vacuum 3.0×10^8m/s
Plank constant h 6.63×10^{-34} J.s

1. What is the dimension of pressure? A. $ML^{-1}T^{-2}$ B. MLT^{-2}C. ML^2T^{-3} D. ML^{-3}
2. Calculate the length of a displaced pendulum bob that passes its lowest point twice every second. ($g=100ms^{-2}$) A. 1.00m B. 0253m C. 0.450m D. 0.58m
3. When a ball rolls on a smooth level ground, the motion of its centre is A. translational B. random C. oscillatory D. rotational
4. A vehicle of mass m is driven by an engine of power P from rest. Fin the minimum time it will take t acquire a speed v. A. $\frac{mv^2}{P}2$ B. $\frac{mv^2}{2P}$C. $\frac{mv}{P}$ D. $\frac{mv}{2P}$
5. A box of mass 40Kg is being dragged along by the rope inclined at 60^0 to the horizontal. The frictional force between the box and the floor is 100N and the tension on the rope is 300N. How much work is done in dragging the box through a distance of 4m? A. 680J B. 200J C. 100J D. 400J
6. A body is projected from the earth's surface with the hope of letting it escape from the earth's gravitational field. What is the minimum escape velocity? (a) $11.3kms^{-1}$ (b) $13.3kms^{-1}$ (c) $12.3kms^{-1}$ (d) $14.3kms^{-1}$ (Earth's radius = 6.4×10^3 km, g = $10ms^{-2}$)
7. A uniform rod PQ of mass 2Kg and length of 1m is pivoted at the end P. If a load of 14N is placed on it at the centre, find the force that should be applied vertically upwards at Q to maintain the rod in equilibrium A. 7N B. 28N C. 68N D. 17N
8. The energy contained in wire when it is extended by 0.02m by a force of 500N is A. 10^4 J B. 10^3 J C. 10 J D. 5J
9. What is the acceleration due to gravity 'g' on the moon, if g is $10ms^{-2}$ on the earth? A. $0.74ms^{-2}$B. $0.1ms^{-2}$ C. $10.0ms^{-2}$ D. $1.67ms^{-2}$
10. A 5kg block is released from rest on a smooth plane inclined at an angle of 30^0 to the horizontal. What is the acceleration down the plane? A. $5.0ms^{-2}$ B. $8.7ms^{-2}$ C. $25.0ms^{-2}$ D. $5.8ms^{-2}$ (g = $10ms^{-2}$)
11. A rectangular metal block of volume $10^{-6}m^3$ at 273K is heated to 573K, if the coefficient of linear expansion is $1.2 \times 10^{-5}K^{-1}$, the percentage change in its volume is A. 1.5% B. 1.1% C. 0.1% D. 0.4%
12. A temperature scale has a lower fixed point of 40mm and an upper fixed point of 200mm. What is the reading on this scale when thermometer reads 60^0C? A. 136.0mm B. 33.3mm C. 96.0mm D. 36.0mm
13. A 500kg car which was initially at rest travelled with an acceleration of $5ms^{-2}$, what is its kinetic energy after 4s A. 2.5×10^3J B. 10^5J C. 5×10^3J D. $\times10^5$J
14. The temperature at which the water vapour in the air saturates the air and begins to condense is A. melting point B. triple point C. dew point D. melting point
15. The period of a simple pendulum will increase by what factor if its inextensible length increased by a factor of four A. 2π B. 4 C. 2 D. ¼
16. An air column 10cm in length is trapped into the sealed end of a capillary tube by a 15cm column of mercury with the tube held vertically. On inverting the tube, the air column becomes 15cm long. What is the atmospheric pressure during the experiment? A. 76cm B. 75cm C. 60cm D. 70cm
17. An electric cell has an internal resistance of 2Ω. A current of 0.5 A was measured when a resistor of resistance 5Ω was connected across it. Determine the electromotive force of the cell. A. 3.5VB. 2.5V C. 4.5V D. 2.35V
18. The speed of light in air is $3 \times 10^8ms^{-1}$. If the refractive index of light from air to water is $\frac{4}{3}$, calculate the speed of light in water. A. $2.25 \times 10^8ms^{-2}$ B. $2.25\times10^8ms^{-1}$ C. $4.00\times10^8ms^{-1}$ D. $4.33\times10^8ms^{-1}$
19. It is known that an atomic nucleus comprises positive charged protons. Which of the following also exists in the nucleus? A. A beta particle B. An alpha particle C. A neutron D. an electron
20. The silver wall of a vacuum flask prevents heat loss due to A. conduction B. convection C. radiation D. diffraction
21. The electromagnetic waves that are sensitive to temperature changes are A.ultra-voilet rays B. gamma-rays C. infra-red rays D. x-rays

22. Under constant tension and constant mass per unit length, the note produced by a plucked string is 500Hz when the length of the string is 0.90m. At what length is the frequency 150Hz? A. 6m B. 3m C. 5m D. 4m

23. Two bodies P and Q are projected on the same horizontal plane, with the same initial speed but at different angles of 30^0 and 60^0 respectively to the horizontal. Neglecting air resistance, what is the ratio of range of P to that of Q? A. $1:1$ B. $1:\sqrt{3}$ C. $\sqrt{3}:1$ D. $1:2$

24. A capacitor of 2.0×10^{-11}F and an inductor are joined in series. The value of the inductance that will give the circuit a resonant frequency of 200KHz is A. $\frac{1}{16}$H B. $\frac{1}{8}$H C. $\frac{1}{64}$H D. $\frac{1}{32}$H

PHYSICS 2007 SOLUTION

1. $Pressure = \frac{Force}{Area}$
$= \frac{Mass \times acceleration}{Area}$
$= \frac{Mass \times \frac{Voleocity}{Time}}{Area}$
$= \frac{Mass \times \frac{\frac{Distance}{time}}{time}}{Area}$
Then, the dimension = $\frac{M \times \frac{\frac{L}{T}}{T}}{L \times L}$
$= \frac{M \times \frac{L}{T^2}}{L^2}$
$= ML^{-1}T^{-2}$
The correct option is A

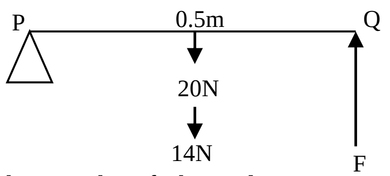
2. This is the same as question 14, 2008
The correct option is B

3. This is the same as question 16, 2008
The correct option is A

4. This is the same as question 15, 2008
The correct option is B

5. This is the same as question 19, 2008
The correct option is B

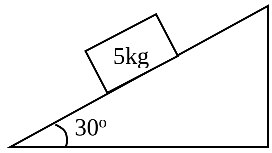
6. Escape velocity, $v = \sqrt{2gR}$
$= \sqrt{2 \times 10 \times 6.4 \times 10^6}$
= 11313.7m/s
= 11.3km/s
The correct option is A

7.


The weight of the rod is concentrated at the centre. Also, a force of 14N is placed at the centre.
Total weight at the centre = 20N + 14N = 34N
Applying the principle of moment
C.M = A.C.M
34N × 0.5m = F × 1m
F = 17N
The correct option is D

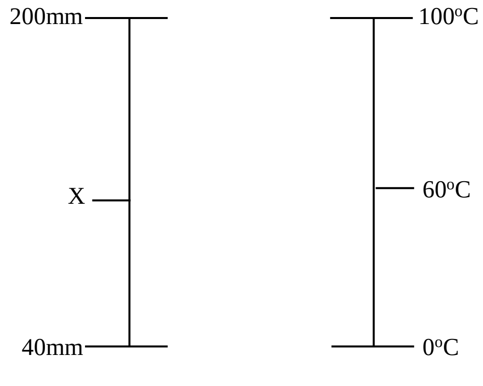
8. Energy stored in an elastic material when given an extension or compression = $\frac{1}{2}Fe = \frac{1}{2}ke^2$
F = 500N, e = 0.02m
$E = \frac{1}{2} \times 500 \times 0.02$
= 5J
The correct option is D

9. The acceleration due to gravity on the moon is one-sixth that on the earth.
Thus $g_{moon} = \frac{1}{6} \times g_{earth} = \frac{1}{6} \times 10 = 1.67$ms^{-2}

10.


Regardless of what the mass is, the acceleration down a smooth plane = gsinθ
= 10sin30
= 5m/s^2
The correct option is A

11. $\alpha = 1.2\times10^{-5}$, $Y = 3\alpha = 3.6\times10^{-5}$
$Y = \frac{\Delta V}{V\Delta\theta}$
The percentage change in volume = $\frac{\Delta V}{V} \times 100\%$
Thus, we can just manipulate the first equation to get the second. Let us do it!
$Y = \frac{\Delta V}{V\Delta\theta}$
$Y = \frac{\Delta V}{V} \times \frac{1}{\Delta\theta}$
Multiplying both sides by Δθ, we have
$Y\Delta\theta = \frac{\Delta V}{V}$
$\frac{\Delta V}{V} \times 100\% = Y\Delta\theta \times 100\%$
$= 3.6\times10^{-5} \times (573 - 273) \times 100\%$
= 1.1%
The correct option is B

12.


$\frac{X-40}{200-40} = \frac{60-0}{100-0}$
$\frac{X-40}{160} = \frac{60}{100}$
100(X – 40) = 160 × 60
X – 40 = 16 × 6 = 96
X = 96 + 40 = 136mm
The correct option is A

13. To know it kinetic energy after 4s, we must know its velocity at that instant
u = 0, a = 5m/s^2, t = 4s, V = ?

$$v = u + at$$
= 0 + 5(4)
= 20m/s
$K.E. = \frac{1}{2}mv^2$
$= \frac{1}{2} \times 500 \times 20^2$
$1 \times 10^5 J$
The correct option is B

14. The dew point is the temperature at which the water vapour in the air saturates the air and begins to condense.
The correct option is C

15. $\frac{T_1}{\sqrt{L_1}} = \frac{T_2}{\sqrt{L_2}}$
$L_2 = 4L_1$
Then,
$\frac{T_1}{\sqrt{L_1}} = \frac{T_2}{\sqrt{4L_1}}$
$T_2 = \frac{T_1 \times \sqrt{4} \times \sqrt{4}}{\sqrt{L_1}} = T_1 \times \sqrt{4} = 2T_1$
Thus, the period will increase by a factor of 2.
The correct option is C

16. Please visit question 19, 2011 for detailed explanation
$10 \times (15 + P_{atm}) = 15 \times (P_{atm} - 15)$
P_{atm} = 75cm of Hg
The correct option is B

17. r = 2Ω, I = 0.5A, R = 5Ω, E =?
$I = \frac{E}{R+r}$
$0.5 = \frac{E}{5+2}$
$E = 0.5(5+2) = 0.5 \times 7 = 3.5V$
The correct option is A

18. ${}_an_w = \frac{Velocity\ of\ light\ in\ air}{Velocity\ of\ light\ in\ water}$
$\frac{4}{3} = \frac{3\times10^8 m/s}{Velolcity\ of\ light\ in\ water}$
$Velocity\ in\ water = \left(\frac{3\times3\times10^8}{4}\right)$ms^{-1}
$= 2.25\times10^8$ms^{-1}
The correct option is B

19. The nucleus of an atom contains protons and neutrons
The correct option is C

20. The silver wall of a vacuum flask prevents heat loss due to radiation.
The correct option is C

21. Infra-red rays are very sensitive to temperature changes.
The correct option is C

22. From the relation $F = \frac{1}{2l}\sqrt{\frac{T}{\mu}}$, it can be connected that $F \propto \frac{1}{L}$
Then, $F = \frac{K}{L}, K = FL = F_1L_1 = F_2L_2$
$F_1 = 500Hz, L_1 = 0.90m, F_2 = 150Hz, L_2 = ?$
$500\text{Hz} \times 0.90\text{m} = 150\text{Hz} \times L_2$
$L_2 = \frac{500Hz \times 0.90m}{150Hz}$ = 3m
The correct option is B

23. $R = \frac{U^2 \sin 2\theta}{g}$
$R_P = \frac{U^2 \sin 60^o}{g}, \ R_Q = \frac{U^2 \sin 120^o}{g}$
$R_P \div R_Q = \frac{U^2 \sin 60^o}{g} \div \frac{U^2 \sin 120^o}{g}$
$= \frac{U^2 \sin 60^o}{g} \times \frac{g}{U^2 \sin 120^o}$
$= \frac{\sin 60^o}{\sin 120^o} = \frac{0.8660}{0.8660} = 1 : 1$
The correct option is A

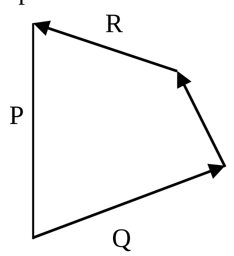
24. At resonance, the capacitive reactance equals the inductive reactance
i.e. $X_c = X_L$
$\frac{1}{2\pi FC} = 2\pi FL$
$L = \frac{1}{2\pi FC} \times \frac{1}{2\pi F}$
$= \frac{1}{4\pi^2F^2C} = \frac{1}{4 \times 3.142^2 \times (2\times10^5)^2 \times 2.0\times10^{-11}}$
$L = \frac{1}{32}H$
The correct option is D

PHYSICS 2006 QUESTIONS

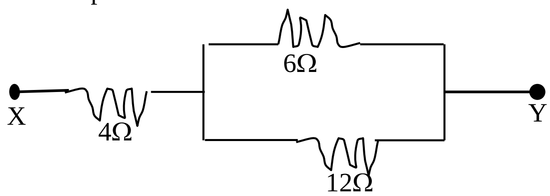
You may find the following useful:
g = 9.8m/s^2
R = 8.31Jmol.K
V = 3.0 × 10^{8}m/s
h = 6.63 × 10^{-3} Js

1. The ice and steam points of a thermometer are 20mm and 100mm respectively: A. temperature of 75^{0}C corresponds to Ymm on the thermometer. What is Y? A. 100mm B. 70mm C. 80mm D. 60mm
2. An electric kettle with negligible heat capacity is rated at 2000W, if 2.0Kg of water is put in it, how long will take temperature of water to rise from 20^0 to 100^0? (Specific heat capacity of water = 4200JKG^{-1} K^{-1}) A. 336s B. 240s C. 168s D. 84s
3. A quantity of ice – 10^{0}C is heated until the temperature of the heating vessel is 90^{0}C. Which of the following constants NOT required to determine the quantity of heat supplied to the vessel? A. specific latent heat of vaporization B. specific heat capacity of ice C. specific latent heat of fusion D. specific heat capacity of water
4. The scent from a jar of perfume opened moves through the air molecules by A. evaporation B. osmosis C. diffusion D. convection
5. A convex lens of focal length 10.0cm is used to form a real image which is half the size of the object. How far from the object is the image? A. 45cm B. 30cm C. 15cm D. 20cm
6. A diverging lens of focal length 20cm forms an image halve of the size of the object. What is the object distance? A. -11.11cm B. 100cm C. 60cm D. 8.71cm
7. An object of height 3.00cm is placed 10cm in front of a biconvex lens of focal length 15cm,, the image of the object is A. real and 3.00cm tall

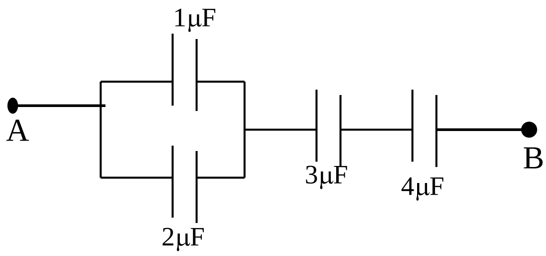
B. virtual and 9.00 cm tall C. virtual and 3.00cm tall D. real and 9.00cm tall

8. The most suitable type of mirror used for the construction of a searchlight is the: A. concave mirror B. spherical mirror C. convex mirror D. parabolic mirror
9. Light waves and ripples of water are similar because both A. are longitudinal waves B. have the same velocity C. can be diffracted and refracted D. have the same frequency
10. A body accelerates uniformly form rest at $6ms^{-2}$ for 8 seconds and then decelerates uniformly to rest in the next 5 seconds. The magnitude of the deceleration is A. $9.6ms^{-2}$ B. $48ms^{-2}$ C. $24.0ms^{-2}$ D. $39.2ms^{-2}$
11. A nail is pulled from a wall with a string tied to the nail. If the string is inclined at an angle of 30^0 to the wall and the tension in the string is 50N, the effective force used in pulling the nail is A. $25N$ B. $25\sqrt{3}N$ C. $50N$ D. $50\sqrt{3}N$
12. A 70kg man ascends a flight of stairs of height 4m in 7s. The power expended by the man is A. 40W B. 100W C. 280W D. 400W
13. Which of the following statements is not true? A. as the slope of an inclined plane increases, the velocity ratio decreases B. the efficiency of an inclined plane decreases as the slope increases C. the effort required to push a given load up an inclined plane increases as the slope increases D. the mechanical advantage of smooth inclined plane depends on the ratio of the length to the height of the plane
14. 

In the diagram above, P, Q and R are vectors. Which of the options gives the correct relationship between the vectors? A. P = R – Q B. P = R + Q C. P = Q–R D. P + R + Q = 0
15. If M and R the mass and radius of the earth respectively and G is the universal gravitational constant, the earth's gravitation potential at an altitude H above the ground level is A. –GM/H B. –GM/R(+H) C. –GM/2H D. –GM/(R-H)
16. The equation $^{150}_{62}X \rightarrow\ ^{150}_{63}Y + k^{0}_{-1}$ + energy, represents A. α-decay B. β-decay C. γ-decay D. photon emission
17. Which of the following radiations cannot be deflected by an electric field or magnetic field? (i) α-rays ii. β-rays iii. γ-rays iv. X-ray A. (i) and (ii) only B. (i) and (ii) only C. (iii) and (iv) only D. (i) and (iii) only
18. The half-life of a radioactive element is 9 days. What fraction of atoms has decayed in 36 days? A. $\frac{1}{16}$ B. $\frac{1}{4}$ C. $\frac{1}{2}$ D. $\frac{15}{16}$
19. Three 4-Ω resistors were connected in series by 'Tola while Ade connected the same set of resistors in parallel. The ratio of the value obtained by Ade to that obtained by 'Tola is A. 1:2 B. 1:9 C. 1:10D. 1:5
20. Three resistors are connected as shown in the diagram below. The equivalent resistance between points X and Y is



A. 8.0Ω B. 22.0Ω C. 4.25 Ω D. 3.27Ω
21. A coil of copper wire of N turns is kept rotating between the poles of a permanent magnet such that the magnetic flux linking the coil changes continuously. Which of the following statements is TRUE? A. an e.m.f. is induced in the coil such that when the change of flux is positive the e.m.f is positive, and when the change of flux is negative, the e.m.f is negative B. an em.f is induced in the coil whose magnitude is inversely proportional to both the number of turns in the coil and the rate of change of magnetic flux C. an e.m.f is set up in the permanent magnet which reduces the flux in the coil to zero D. current flows in the coil and an e.m.f. is set up proportional to both the rate of change of the flux and the number of turns.
22. What is the equivalent capacitance between points A and b in the diagram below?



A.12.0μF B. 1.1μF C. 1.6μFD. 7.7μF
23. The principle of operation of an induction coil is based on A. Ohm's law B. Ampere's law C. Faraday's law D. Coulomb's law

PHYSICS 2006 SOLUTION

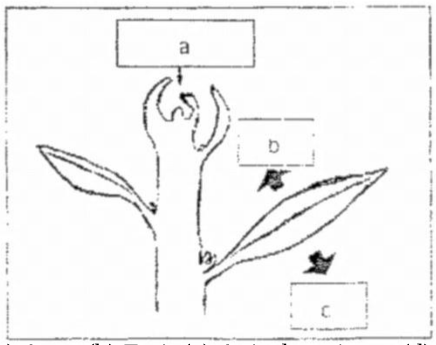
1. This is the same as question 24, 2008
The correct option is C
2. This is the same as question 25, 2008
The correct option is A
3. This is the same as question 26, 2008

The correct option is A

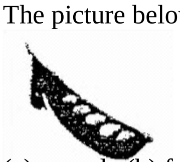
4. This is the same as question 27, 2008
The correct option is C
5. This is the same as question 1, 2008
The correct option is B
6. This is the same as question 2, 2008
The correct option is C
7. This is the same as question 3, 2008
The correct option is B
8. This is the same as question 4, 2008
The correct option is D
9. This is the same as question 5, 2008
The correct option is C
10. This is the same as question 17, 2008
The correct option is A
11. This is the same as question 18, 2008
The correct option is A
12. This is the same as question 20, 2008
The correct option is D
13. This is the same as question 21, 2008
The correct option is B
14. This is the same as question 22, 2008
The correct option is B
15. This is the same as question 23, 2008
The correct option is B
16. This is the same as question 13, 2008
The correct option is B
17. This is the same as question 12, 2008
The correct option is C
18. This is the same as question 11, 2008
The correct option is D
19. This is the same as question 6, 2008
The correct option is B
20. This is the same as question 7, 2008
The correct option is A
21. This is the same as question 8, 2008
The correct option is D
22. This is the same as question 9, 2008
The correct option is B
23. This is the same as question 10, 2008
The correct option is C

2015 OAU Post UTME: Biology Questions

1. Which of these statements about succession is incorrect? (a) Succession is a change of species composition, community structure and function over time and space (b) Succession is usually set in motion by some sort of disturbance (c) Succession is both directional and predictable (d) Succession begins only on a bare ground
2. Example of genetic diseases include any of the following except ______ (a) diabetes (b) Cystic fibrosis (c) Hepatitis (d) Epilepsy
3. In his theory of evolution, Darwin identified _____ as the main cause of natural selection (a) physiological pressure (b) ecological pressure (c) environmental pressure (d) biological pressure
4. One of these is not a unique feature of meiosis (a) synapsis (b) Homologous recombination (c) reduction division (d) cytokinesis
5. One of these organisms is not an autotroph (a) Spirogyra (b) Zea may (c) Rhizobium (d) Mushroom
6. One of these feature is not typical of most animals (a) Heterotrophic (b) Multicellular (c) Sessile (d) Motile at some stage of life cycle
7. The cnidarians use their nematocysts only for _______ (a) Capturing prey (b) Courtship (c) Gas exchange (d) Sensing chemicals
8. In the diagram below, the part labeled (a) is



(a) Apex (b) Fruit (c) Apical meristem (d) Flow
9. The picture below is an example of



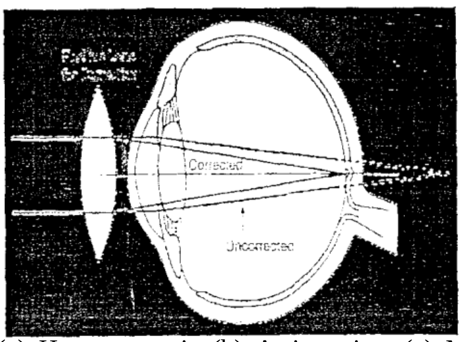
(a) capsule (b) follicle (c) legume (d) schizocarp
10. One of the following is not true (a) Saprophytic nutrition involves feeding on soluble organic material from inorganic substances (b) Symbiosis is a nutritional relationship in which both organisms involved derived benefit (c) A parasite causes injuries to its host in the course of getting its food (d) holozoic mode of nutrition can be seen in animals, carnivorous plants and some protists
11. In which biome would you expect to have the shortest growing season? (a) Tropical rain forest (b) Guinea savanna (c) Sudan savanna (d) Deserts
12. The concept of tropic structure of a community emphasizes (a) The dominant form of vegetation (b) The main predator (c) The feeding relationship within a community (d) The richness of species in the community
13. One of these is an agency responsible for conserving natural resources in Nigeria (a) PDP (b) NCF (c) ACC (d) NEMA
14. Which happens to a tadpole after 45 days old? (a) It becomes fully grown, mouth becomes wider, horny jaw disappears completely (b) It becomes fully grown, mouth becomes wider, tail disappears completely (c) It becomes fully grown, mouth becomes wider, external gill disappears completely (d) It becomes fully

grown, mouth becomes wider, slit disappears partially

15. If a human skin cell with 46 chromosomes divide by mitosis, how many chromosomes will each daughter cell have? (a) 23 (b) 12 (c) 46 (d) 92
16. Effect of air pollution does not include ____ (a) Formation of carboxyl hemoglobin (b) displacement of digested food (c) Lowering plant yield (d) Damage breathing organs
17. The lack of special supportive structures in bryophytes restricts them to one of the following types of growth (a) Lateral growth (b) Upward growth (c) Downward growth (d) Aerial growth
18. Physical observable characteristics of an organism is called _______ (a) Genotype (b) Phenotype (c) Allele (d) Locus
19. In the evolutionary trend, in which phylum do we begin to see a complete digestive tract? (a) Ctenophora (b) Platyhelminthes (c) Nematoda (d) Mollusca
20. Absorption is maximum in the small intestine because of _______ (a) The presence of wili (b) It length (c) Its thin walls (d) All the above
21. Which type of placentation does the diagram below represent?

(a) Parietal (b) Axile (c) Marginal (d) Free-Central

22. In man, abnormality with chromosome 21 often leads to a genetic problem called _____ (a) Respiratory syndrome (b) Carcinoma syndrome (c) Klinefelter (d) Down syndrome
23. The presence of extensive amounts of rough endoplasmic reticulum in a cell is an indication that the cell is involved in ______ (a) Synthesis and metabolism of CH_2O_3 (b) Synthesis and secretion of proteins (c) Synthesis of ATP (d) Contraction
24. The condition in the diagram below is



(a) Hypermetropia (b) Astigmatism (c) Myopia (d) Cataract

25. Which of the following is not present in RNA? (a) Adenine (b) Guanine (c) Cytosine (d) Thymine

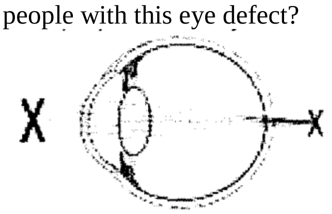
2015 OAU Post UTME: Biology Solution

1. Primary succession begins on an area not previously occupied by vegetation. Secondary succession beings on an area that has been inhabited before. **The correct option is D**
2. Certain diseases are said to be genetic. This is because they are hereditary. Such diseases include sickle cell anaemia, epilepsy, glaucoma, cystic fibrosis, diabetes and phenylketonuria. **The correct option is C**
3. Organisms that tend to survive and reproduce are those whose variations give them competitive advantage over other. They are best adapted to the environment. Darwin, therefore identified **environmental pressure** as the main cause of natural selection. **The correct option is C**
4. Cytokinesis is the division of the cytoplasm of the parent cell into daughter cells. It is not peculiar to meiosis. As it occurs in meiosis, so does it occur in mitosis.**The correct option is D**
5. Mushroom is a fungus. Fungi generally lack chlorophyll and have to depend on other organisms for food, that is, they are not autotrophic. **The correct option is D**
6. To start with,sessility is the inability of an organism to move about. Some animals are sessile, for example the adult organisms in the phylum porifera. However, the vast majority of animals are not **sessile**. **The correct option is C**
7. Nematocysts or cnidocyst are structures used by cnidarians for capturing prey and for defence. **The correct option is A**
8. The part labelled is the apical meristem. The apical meristems are located at the apices of roots and shoots and are directly involved in their elongation. **The correct option is C**
9. The diagram represents a legume (pod). A legume is a simple fruit formed from a superior monocarpous pistil. **The correct option is C**
10. Saprophytic nutrition involves feeding on soluble organic materials from dead animals or plants. **The correct option is A**
11. The shortest growing season is expected from the desert. In a desert, rainfall is very low and humidity may not be up to 25%. **The correct option is D**
12. The tropic structure of a community emphasizes the feeding relationship within the community. **The correct option is C**
13. NEMA: National Emergency Management Agency; NCF: Nigerian **Conservation** Foundation; PDP: If you don't know the full meaning of PDP, you are probably a Chinese; ACC: Accident Compensation Corporation. Now, which of the above deals with **Conservation** of resources? Good! **NCF**. **The correct option is B**
14. 40th – 45th day is the young toad stage. The tadpole changes into a small toad by reabsorbing the tail. **The correct option is B**

15. When a cell undergoes mitotic division, the chromosome number is conserved. There will, therefore, be 46 chromosomes in each daughter cell. **The correct option is C**
16. Displacement of digested food is not an aftermath of air pollution. **The correct option is B**
17. Lateral growth in bryophytes is due to the lack of special supportive structures. **The correct option is A**
18. The physical observable characteristics of an organism is called its phenotype while the genetic constitution of an organism is known as its genotype. **The correct option is B**
19. It is in phylum Nematoda, we begin to see a complete digestive tract. Nematodes have complete digestive system with mouth and anus. The Platyhelminthes have mouth but no anus. **The correct option is C**
20. The integral portion of nutrients are absorbed in the small intestine because of its thin wall and large surface area. There are also very tiny, hair-like projections known as villi which further provide larger surface area for absorption. **The correct option is D**
21. This is parietal placentation because the ovules are attached to the sides of a syncarpous ovary. **The correct option is B**
22. Down syndrome, also known as trisomy 21, is a genetic disorder caused by the presence of all or part of a third copy of chromosome 21. **The correct option is D**
23. The rough endoplasmic reticulum plays a large role in the synthesis of complex proteins and amino acids. **The correct option is B**
24. The diagram illustrates hypermetropia. This is an eye defect in which parallel rays of light are brought to a focus behind the retina. **The correct option is A**
25. It is DNA that has thymine. RNA has Uracil instead. **The correct option is D**

BIOLOGY QUESTION 2014

1. What type of lens is used in the spectacles of the people with this eye defect?



(a) concave lens (b) convex lens (c) all of the option (d) none of the options
2. Which of these components of the phloem has the cytoplasm pushed to the sides while also lacking nucleus? (a) Companion cell (b) Sieve tube elements (c) Sieve plate (d) Parenchyma
3. Which of the following statements about phylum Cnidaria is correct? (a) They are diploblastic animals (b) Body is sac-shaped with three openings (c) They are bilaterally symmetrical (d) They are triploblastic animals with two openings
4. Which of these biological statements is correct? (a) Reflex action are also known as involuntary actions (b) The fore brain consists of the cerebrum and the olfactory lobes (c) The right hemisphere f the brain controls the right half of the body (d) The outer part of the human cerebrum is made up of grey matter
5. The best known plant species that occur in fresh water swamp vegetation is (a) *Rihzophora racemosa* (b) *Acrostichum aureum* (c) *Mitragyna ciliate* (d) *Triplochiton scleroxylon*
6. Bacteria that live in the human intestine assist in the digestion and feed on nutrients the human consumed. This relationship might best be describes (a) Commensalism (b) Ectoparasitism (c) Endoparasitism (d) Mutualism
7. The xylem elements perform the function of transport but they also help to support plants because they (a) are internally located (b) are tubular (c) have rigid thick wall (d) constantly absorb water
8. Biosphere is best described as (a) all parts of the earth where life exists (b) the non – living parts of an ecosystem (c) all component of an ecosystem (d) all the members of a single species in a habitat
9. Muscle fatigue is caused by________ (a) Accumulation of ethanol in the muscle cells (b) accumulation of methanol in the muscle cells (c) accumulation of lactic acid in the muscle cells (d) accumulation of pyruvic acid in the muscle
10. At what stage of the meiotic prophase do the homologous chromosomes attract each other and then pair up? (a) Leptotene (b) Paachytene (c) Zygotene (d) Diplotene
11. Which of the organelles is not directly connected to cell division process? (a) centromere (b) microtubules (c) golgi bodies (d) spindle apparatus
12. During the light dependent reaction (a) glucose is formed (b) carbon IV oxide is fixed (c) NADPH and ATP are synthesized using electron released from water (d) water is split and the electrons produced are used for glucose synthesis
13. Meiosis II is similar to mitosis in that (a) Homologous chromosome synapse (b) Sister chromatids during anaphase (c) The chromosome number is reduced (d) The daughter cells are diploid
14. The mechanism of opening and closing the stomata is associated with the (a) guard cells (b) stoma (c) lenticels (d) air spaces
15. Holozoic is seen in (a) Animals carnivorous plants and protozoans (b) Animals, prostist and carnivorous plants (c) Animals, carnivorous plants and fungi (d) Animals, fungi and prostist

16. The part of the tooth that contains blood vessels and verve fibres is (a) root (b) enamel (c) entine (d) pulp cavity
17. The food substances that are stored in readiness for times of food shortages are (a) carbohydrates (b) fats and oils (c) proteins (d) vitamins
18. Which of these is the role of the liver in digestion? (a) synthesis of lipase (b) secretion of trypsin (c) secretaion of bile and bicarbonate for emulsification of fats (d) Otorego of bile for hydrolyses of starch
19. Respiratory surfaces generally have the following characteristics with exception of (a) it must be thick but permeable (b) it must be moist (c) it must passes a large surface area (d) it must be richly supplied with blood vessels
20. Which of these is not a water soluble vitamin? (a) thiamine (b) riboflavin (c) folic acid (d) calciferol
21. Which of these following biomes is currently being ravaged by desertification? (a) Derived savanna (b) Southern Guinea savanna (c) Northern Guinea savanna (d) Sudan savanna
22. Which of the largest of the middle ear ossicles? (a) anvil (b) stipe (c) hammer (d) incus
23. The process by which plants and animals are modified in structure, physiology and behaviour in order to survive is known as (a) evolution (b) adaptation (c) succession (d) aggregation
24. Which of these plants can withstand extreme dryness? (a) *Cactus* (b) *Raphia palm* (c) *Triplochiton scleroxylon* (d) *Parkia biglobosa*
25. Animals that do not allow their body temperature to vary with the ambient are called? (a) homoeothermic (b) poikilothermic (c) amphibians (d) reptiles

2014 BIOLOGY SOLUTION

1. **A** – With no iota of doubt, you already know that the condition indicated in this diagram is either short sightedness (**Myopia**) or long sightedness (**Hypermetropia**).
   How does one tell one from the other? If the image is formed behind the retina rather than upon it, it is long sightedness. It is short sightedness if the image is formed before the retina (just as what we have here). Short sightedness is corrected by using concave lenses while long sightedness is corrected by using convex lenses.
2. **B –** Sieve tube elements are mainly employed for the transport of sugar throughout the plant. They lack nucleus at maturity and the cytoplasm is pushed to the sides.
3. **A –** In cnidarians, the body wall comprises two layers of cells separated by jelly – like substance known as mesoglea. The possession of two body layers is known as diploblastism.
4. **C –** The brain is made up of two large hemispheres. The right hemisphere controls the left half of the body while the left hemisphere controls the right half of the body.
5. **C –** The best known plant in fresh water swamp forest is *Mitragyna ciliata*
6. **D –** It can be deduced that both the human being and the bacteria benefit from each other. When an association is founded on **mutual** benefit, it is known as **mutualism**.
7. **C –**The possession of rigid thick wall enables xylem elements to perform supportive function addition to transportation.
8. **A –** Biosphere is a collective name for all the parts of the earth where life exists.
9. **C –** During a strenuous exercises the supply of oxygen for body cells becomes inadequate. This makes the organism respire **anaerobically**. In plants, anaerobic respiration produces carbon(iv)oxide and ethanol. In animals, lactic acid is produced instead. It is the accumulation of this acid that lead to muscle fatigue.
10. **C –** It is at the zygotene stage or the meiotic prophase that homologous chromosomes attract each other and their pair up. This involves laying down of a lattice protein between the homologous chromosomes forming a structure known as synaptonemal complex.
11. **C –** The golgi bodies are the sites of synthesis of cell wall polysaccharide and glycoprotein.
12. **C** – During the light dependent reaction water is split into hydrogen ion and hydroxide ion. ATP and NADPH are also produced. What is your take on the statement contained in option D? It is correct that splitting of water occurs but it is off the mark to say that glucose is synthesized there. The synthesis of carbohydrate occurs at the light independent stage..
13. **B –** Just as in mitosis (anaphase), the centromere splits and the sister chromatids (of the poles at the second anaphase of the meiotic division. Please note that difference between the first and the second anaphase of the meiotic division. In the first anaphase, the homologous chromosomes (not the sister chromatids yet) move towards the opposite poles.
14. **A –** The mechanism of opening and closing of the stomata is associated with the guard cell.
15. **B**– Holozoic nutrition entails feeding on solid organic materials synthesized by other organism or other organisms themselves. This mode of nutrition is seen in animals, carnivorous plants and some protists.
16. **D –** The pulp cavity is the part of the tooth that contains blood vessels and nerve fibres.
17. **B –** Fats and oils are used to provide energy and are also stored in readiness for time of good shortage.
18. **C –** Bile, a substance secreted from the liver is used for the emulsification of fats.
19. **A –** The following are the properties good respiratory surface must possess:
    (1) It must be moist
    (2) It should possess a large surface area. This will enable sufficient amount of gases to be exchanged

(3) It must be richly supplied with blood vessels
(4) It must be thin. This is because diffusion becomes inefficient when the distance is greater than 1mm.
(5) It must be permeable, so that gasses can pass through easily.

20. **D –** Water soluble vitamins are **B_1** (thiamine), **B_2** (riboflavin), Nicotinic acid, folic acid and ascorbic acid. Fat soluble vitamins are vitamin **A** (retinol), **K** (phylloquinone) and **D** (caleiferol).
21. **D** – Sudan/Sahelian savanna is a biome that is currently being ravaged by desertification.
22. **C –** The maleus or hammaer is the largest of the three ear ossicles while the stirrup (stapes) is the smallest.
23. **A –** The process by which plants and animals are modified in structure, behaviour and physiology in order to survive is known as **evolution**.
24. **A -** A plant that can withstand extreme dryness must be a desert plant – a xerophyte cactus is an example of desert plants
25. **A -** Animals whose body temperature does not depend on the external environment are said to be homoeothermic while those that allow their body temperature to vary with that of the external environment are said to be poikilothermic.

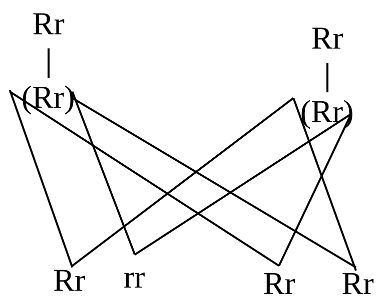
BIOLOGY QUESTION 2013

1. The number of cranial nerves that connect the brain to various parts of the body is
A. 10 pairs B. 11 pairs C. 12 pairs D. 13 pairs
2. The following pair of ions is involved in transmission of impulses by neurons:
A.K and Na ions B.Na and Mgions C. K and Cl ions D. K and Ca ions
3. The probability of producing an heterozygote progeny in a cross between two heterozygote individuals of pea plant is
A. 1/3 B. ¼ C. ½ D.2/3
4. One of these does not protect the body from harmful effect of disease-causing micro-organisms A. Anti toxin B. Phagocytes C. Antigens D. Antibodies.
5. Which of these statements with respect to individual organism is most correct?
A. It refers to animals only B. It refers to either the plants or animals. C. Its ecology can be carried out in zoo only D. It is used in reference to plant only.
6. One of these is not a major Biome in West Africa
A. Rain forest B. Savanna C. Comferous forest D. Mangroove
7. The media of transportation in living things include all but
A. Cytoplasm B. WaterC. Eosin D. Blood and lymph
8. Gaseous exchange through the lungs is called A. cutaneous breathing B. buccal breathing C. pulmonary breathing D. larynxial breathing
9. In saprophytic mode of nutrition
A. organisms feed on insoluble organic material B. decomposition is not possible at allC. nutrient recycling is possible. D. no animal is involved.
10. A macro element which is not directly connected with formation of chlorophyll is
A. nitrogen B. iron C. magnesium D. sulphur
11. During the light dependent reaction
A. glucose is formed B. carbon IV oxide is fixed C. NADPH and ATP are synthesized using electron released from water D. water is split and the electrons produced are used for glucose synthesis.
12. Eutrophication refers to growth of……
A. Bacteria B. Fungi C. Protophytes D. Algae
13. All the following statements are consistent with the concept of trophic structure except
A. At every feeding stage some energy is wasted from the chain B. The nearer the organism to the beginning of a food chain, the greater the available energy of the organism C. The first trophic level is occupied by the autotrophs D. There are few number of organisms at the start of a food chain.
14. All these statements about plants succession are correct except
A. Plant succession is the process of community change at one place over time. B. Plant succession is usually measured over the course of several years to hundred years C. Succession proceeds from pioneer to climax phases D. Succession is often not directional and so difficult to predict
15 Which of the followings is not true about finger punt?
A. It is useful in detecting crime. B. No two individuals have the same finger print. C. It is a heritable character. D. It is environmentally induced.
16. Characteristics of continuous variation include all of the following except
A. Produced by many genes B. Influenced by the environment C. Occurs in a normal distribution curve D. most of the organisms in the population fall at the tail ends of the range
17. Homeostasis is defined as
A. Regulation of both external and internal conditions of organisms B. Maintenance of internal environment of an organism C. Maintenance of internal and external environment of an organism D. Regulation of the chemical environment of an organism
18. Effectors are
A. muscles which work in response to the stimulus received from the motor nerves. B. glands which work in response to the stimulus received from the motor nerves. C. muscles or glands which work in response to the stimulus received from the motor nerves. D. efferent neurons
19. The mechanism of opening and closing the stomata is associated with the
A. guard cells B. stoma C. lenticels D. air spaces
20. The part of the kidney where each tubule begins is called
A. Capsule B. Cortex C. Glomerulus D. Urether

21. The factors for two pairs of contrasting characters areinherited independent of each other. This is
A. Menders first law of inheritanceB. Mendel’s second law of inheritanceC. Mendel’s law of segregation of germinal units D. Mendel’s law of independent pairing of germinal units
22. All of the following green algae are colonial forms except?
A. Gonium B. Volvox,C. Pandorina, D. Anabaena
23. Allele is
A. an alternate form of a geneB. aunit of inheritance C. the position or location of the gene on a chromosome D. number of chromosomes in the gamete
24. One of these statements about sympathetic Nervous system is untrue
A. It stimulates many parts of the body in times of danger B. It stimulates the heart beat C. It functions like the adrenal glands D. It lowers the blood pressure
25. One of the following elements is not associated with leaf chlorosis
A. Nitrogen B. IronC. Calcium D. Magnessium

BIOLOGY SOLUTION 2013

1. **C** – There are 12 cranial nerves which are paired that connects the brain to the other parts of the body. They are; terminal, olfactory, optic, oculomotor, trochlear, trigeminal, abducens, facial, vestibulocochlear, glossopharyngeal, vagus, accessory and hypoglosssai.
2. **A –** The transmission of impulses is been aided by the Na^+ and K^+ ions. At the beginning of impulse transmission, the Na^+ moves into axon causing depolarization, then the K^+ channels open and moves out of the axon. The impulse then travels down the axion to the axion terminal where it signals other neurons.
3. **C –** Let the heterozygote pea plant be Rr



Total F_1 generation = 4
Heterozygote progeny (offspring) = 2
Homozygote progeny (offspring) = 2
:. Heterozygote progeny = $\frac{No\ of\ Heterozygote}{Total\ no\ of\ F_1 generation} = \frac{1}{4} = \frac{1}{2}$

4. **C –** Antigens do not protect the body from infections organism. Antigens are actually foreign substances that invade the body but are been lock up by antibodies and later destroyed by the phagocytes.
5. **B –** Individual organism refers to both plants and animals.
6. **C –** Corniferous forest are mainly located along the coast of northern California.
7. **C –** Eosin is not a medium of transportation in living organism. Eosin is a fluorescent reed dye which is used to stain the cytoplasm, collagn and muscle fibers for examination under the microscope.
8. **C –** Pulmonary breathing is aided by the lungs to take in and release air. Cutaneous breathing is aided by larynxial breathing is a voice box that moves the vocal cords apart and serve breathing.
9. **A –** Saprophytic nutrition involves the feeding on dead or decayed organic matter.
10. **D –** Nitrogen, iron and magnesium are required for chlorophyll formation and synthsis. But sulphur only help in the electron transfer reaction of photosynthesis.
11. **C –** During light dependent reaction, the water splits to form H^+ and O_2. The electron ($NADP^+$) from the photosystem of the plant accepts 2 electrons and react with H^+ in the stroma to form NADPH of which will power smotic synthesis of ATP.
12. **D –** Eutrophication refers to the growth of phytoplankton like algae over and above what would be considered as market. This is caused by the surplus of the key nutrients (nitrogen and phosphorus).
13. **D –** Organism are of much numbers at the start of the food chain but as the chain goes higher, the number of the organisms reduce.
14. **C –** In plant succession, succession proceeds from pioneer phase to climax phase at which the rate of increase in net productivity of the plants is consumed by its own heterotrophs.
15. **C –** Finger prints cannot be inherited. Each person has his/her own unique fingerprint which is formed with the help of DNA.
16. **D –** Continuous variation occurs in a normal distribution curve as average. This variation occurs as a result of genetic factors as well as environmental factors.
17. **B –** Homeostasis is the maintenance of a constant internal environment like temperature, water and oxygen level.
18. **C –** Effectors are muscles, glands or organs capable of responding to a stimulus, especially a nerve impulse.
19. **A** – The mechanism of stomata opening and closing depends on the guard cells. As water enters the guard cells, turgidity increases which pulls the cells causing the opening stomata. The closing stomata has a reverse process of opening.
20. **B –** The kidney is composed of an outer cotex where different kidney tubules begin.
21. **B –** Mendel’s second law of inheritance wwhich is the law of independent assortment states that

separate genes for separate traits are passed independently of one another from parents to offspring.

22. **D –** Volvox, pandorina and gonium are examples of algae that are made similar cells which are joined or massed together but cannot be differentiated from each other which are called colony. But Anabaena is an algae with beadlike cells of which are in solitary forms.
23. **A –** Allele is an alternative form of a gene (one member of a pair).
24. **D –** The sympathetic nervous system accelerate the heart rate, thus, raising the blood pressure. It also help to mobilize the body's nervous system fight or flight response (adrenalin secretion). It is however constantly active at a basic level to maintain homeostasis.
25. **A –** Chlorosis is a yellowing of leaf tissue due to the lack of chlorophyll which are mainly caused by nutrient deficiencies in plants, like deficiency of calcium, iron, magnesium (manganese), copper, phosphorus and zinc.

BIOLOGY 2012

1. Hierarchical organization of living organism is in one of the following orders: (a) atoms, molecules, compound, cells, tissues, organs, systems, organism (b) atoms, molecules, organelles, cells, tissues, organs, systems, organism (c) atoms, elements, molecules, cells, tissues, organs, systems, organism (d) atoms, molecules, elements, cells, tissues, organs, organ systems, organism
2. In a monohybrid cross between round seed and wrinkled seed, given that round is dominant over wrinkled, what is the number of wrinkled seed that would be formed at F_2 if the total number is 7324? (a) 456 (b) 786 (c) 686 (d) 860
3. Steepness of slope generally affect________ (a) rainfall (b) drainage (c) sunlight (d) all of the options
4. Bacteria differ from eukaryotic forms of life in that they: (a) are causes of all infectious diseases (b) have no nuclear membrane (c) reproduce by binary fission (d) have a thick cell-wall
5. Which of the following statements is NOT correct? Hormones (a) are circulated by blood and lymph (b) have their effect on target organs (c) complement nervous co-ordination (d) are not produced in specific glands
6. Sister chromalids are: (a) two identical copies of a single chromosome produced during s-phase (b) pairs of chromosomes (c) points of attachments of centromeres to the chromosomes (d) chromosomes found inc ells of sisters
7. One of the groups of organism below is critical in the entire process of nutrient cycling. (a) aves (b) nematode (c) mammalian (d) fungi
8. If 80 grasshoppers are found in a filed with a total area of $100cm^2$ what is the population density of grasshopper in the field? (a) 0.08 per m^2 (b) 0.8 m^2 (c) 80 m^2 (d) 100 m^2
9. Which of the following is part of the axial skeleton in a mammal? (a) phalange (b) tarsal (c) sacrum (d) patella
10. A major difference between Arachnids and Annelids is that ________ (a) in Annelids, body consists of dissimilar segments unlike in Arachnids (b) In Annelids, body consists of similar segments unlike in Arachnids (c) In Arachnids, the cephalothoraxas is not distinct unlike in Arachnids (d) In Annelids, the cephalothoraxas is distinct unlike in arachnids
11. The process in which the internal environment of an organism is maintained is called (a) co-ordination (b) homeostasis (c) excretion (d) metabolism
12. One of the following organism exhibits a closed and single circulatory system (a) insect (b) earthworm (c) fishes (d) mammals
13. Coelomates are animals with (a) no body cavity (b) true body cavity (c) false body cavity (d) two germ layers
14. This tissue is made up of tracheids, vessels, fibres and parenchyma. What is it? (a) phloem (b) sclerenchyma (c) xylem (d) ground tissue
15. The important processes which bring about recycling of carbon dioxide between the biotic and abiotic components of an ecosystem are all of the following except (a) photosynthesis (b) respiration (c) decay (d) burning of fossil fuels
16. When a cut is made on the trunks of certain trees, the milky fluid excuded is called (a) rubber (b) resin (c) alkaloid (d) latex
17. The system of membrane-lined sacs that forms channels throughout the cytoplasm and whose membrane is continuous with the nuclear membrane is the (a) mitochondrion (b) ribosomes (c) endoplasmic reticulum (d) Golgi apparatus
18. The type of farming which involves raising livestock only is called (a) mixed farming (b) subsistence farming (c) pastoral farming (d) monoculture
19. Fruits that develop without fertilization and are seedless are known as _________(a) parthenocarpic fruits (b) aggregate fruits (c) simple fruits (d) epicarpic fruits
20. Nerve endings are located in which part of the tooth (a) crown (b0 cement (c) pulp cavity (d) gum
21. Which of the following is the hardest material in the body of animals? (a) cartilagve (b) bone (c) enamel (d) dentine

22. Grasping fingers and toes as well as eyes positioned in front of the head are features of (a) cartaceans (b) carnivores (c) rodents (d) primates
23. One of the major differences between DNA and RNA is that_______ (a) DNA is made of ribose sugars and double stranded unlike RNA (b) DNA is made of ribose sugar and single stranded unlike RNA (c) RNA is made of ribose sugar and double stranded like DNA (d) RNA is made of ribose sugar and single stranded unlike DNA
24. One of these statements is true of caryopsis (a) pericarp and seed coat are fused (b) pericarp is free from seed coat (c) pericarp splits open (d) pericarp with a superior ovary
25. A biological species must possess the following characteristics except (a) live only in one place (b) must interbreed (c) must produce fertile offspring (d) the mating between members must be free

BIOLOGY SOLUTION 2012

1. **B –**
2. Incorrect or incomplete question
3. **B –**
4. **B –** Bacteria are prokaryotic cells
5. **D –** Hormones are produced in a specic gland
6. **A** –
7. **D –** Fungi is a decomposer
8. **B** - Population density = $\frac{Total\ No\ of\ organism}{ARea}$
   Total = 80
   Area = 100m^2
   Population Density =$\frac{80}{100} = 0.8$m^2
9. **C** – Sacrum, the axial skeleton include skull, vertebra, shoulder blade
10. **B** –
11. **B** –
12. **C** – Fish
13. **A –**
14. **C** – Xylem, which is a sead soil
15. **C –**
16. **D** –
17. Golgi bodies
18. **C** – Pastoral farming
19. **A–**parthenecarpic
20. **C** – pulp carvity
21. **C** – enamel which covers the crown
22. **D**– Primates
23. **D** –
24. **A** – in caryopsis, the pericarp is fused with the seed
25. **A –**

BIOLOGY 2011

1. Which of these groups of animals rivals mammals in the display of parental care features? A. Birds B. reptiles C. pisces D. none of the above
2. An assemblage of populations of different species which interact through trophic and spatial relationship is best described as a(n) A. city B. community C. ecosystem D. niche
3. In which biome would you expect to have the shortest growing season?
   A. tropical rain forest B. guinea savanna C. Sudan savanna D. deserts
4. Which of the following is an incorrect statement about savanna? A. it occupies about 80% of the lard surface of Nigeria B. it has no woody species C. it is usually burnt annually D. it is a closed or nearly closed cover of grasses
5. Which of the following habitants cannot be used for the study of succession?
   A. abandoned farmland B. a pond C. savanna grassland D. well cultivated farmland
6. In which of these associations is much harm done to one of the partners?
   A. symbiosis B. commensalisms C. parasitism D. mutualism
7. Effect of air pollutant does not include: A. formation of carboxyl hemoglobin B. displacement of digested food C. loweirng plant yield D. damage breathing organs
8. A typical feature of a plant cell is the presence of A. chromosome in nucleus B. cellulose cell wall C. mitochondria D. membrane around the nucleus
9. Non seed plants are found in A. desert and arctic regions only B. all environments C. cold mountain areas and hot springs C. cold mountain areas and hot springs D. tropical and subtropical regions only
10. Which of the following phyla has been found to be the most successful in the animal kingdom?
    A. phylum Annelida B. phylum Arthropoda C. phylum Chordata D. phylum Mollusca
11. In the tropical rainforest, there is little or no litter on the forest floor because of high A. rainfall
    B. temperature C. light intensity D. rate of decomposition
12. Adaptive features of plants to desert conditions include A. thick barks, succulent stems and sunken stomata B. thin barks, succulent stems and sunken stomata
    C. thin barks, air floats on stems and sunken stomata D. air spaces on tissues, adventitious roots and thin barks
13. The distribution of plants in rainforest is governed mainly by A. vegetation
    B. soil types c. amount of rainfall D. rainfall pattern
14. the greatest influence on a stable ecosystem in nature is A. man B. pollution C. animal D. rainfall

15. Which of the following g is the basic unit of classification? A. genus B. species C. phylum
D. kingdom
16. Which two structures are present in a palisade cell but not in a liver? A. cell wall and cytoplasm
B. cell wall and chloroplast C. cell membrane and cytoplasm D. cell membrane and chloroplast
17. Workers in deep mines usually suffer from dehydration because A. water is lost due to evaporation B. water is lost due to defecation C. water is lost in the form of sweat D. water is lost along with salts in the form of sweat
18. Glucose is reabsorbed in the kidney mainly by A. Bowman's capsule B. Loop of Henle C. Proximal convoluted tubule D. distal convoluted tubule
19. The most common substrate of respiration is A. fats B. amino acids C. glucose D. sucrose
20. The rate of heart beat in an adult human being is A. 71 beats per minute B. 72 beats per minute C. 73 beats per minute D. 74 beats per minute
21. One of the following is not true A. saprophytic nutrition involves feeding on soluble organic material from inorganic substances B. symbiosis is a nutritional relationship in which both organism involved drive benefit C. a parasite causes injuries to its host in the course of getting its food D. holozoic mode of nutrition can be seen in animals, carnivorous plants and some protists.
22. _________ is responsible for the direction of growth and development of the organism A. the nucleus B. the DNA C. the neuron D. the RNA
23. The chromosome number in man is A. 46 B. 23 C. 92 D. 58
24. Effectors are A. muscles which work in response to the stimulus received from the motor nerves
B. glands which work in response to the stimulus received from the motor nerves C. muscles or glands which work in response to the stimulus received from the motor nerves D. efferent neurons
25. The following are formed in the bone marrow except A. platelets B. basophils
C. granulocytes D. lymphocytes

BIOLOGY SOLUTION 2011

1. **A** – Birds
2. **B** – Community, It is the functional unit of an ecosystem
3. **D** – Deserts
4. **B** –
5. **C** –
6. **C** – Parasitism
7. **B –**
8. **B** – Cellulosecell wall
9. **C** –
10.
11. **B –** Phylum Arthropoda
12. **A** –
13. **C** –
14. **A** –
15. **B** – Species, It can be defined as organisms which mate to produce a fertile offspring
16. **A** –
17. **D** –
18. **B** –
19. **C –** Glucose
20. **B** – 72beats per minute 16 tiras for and inhalation and exhalation for biston betularia.
21. **A** –
22. **B** – DNA
23. **A** – 46 but we have 23 pair
24. **C** – Effector include muscles and glands
25. **A** – Platelets

BIOLOGY 2010

1. Which of the following is the basic unit of classification of plants and animals? A. genus B. species C. phylum D. kingdom
2. Mendel's first law is known as the law of A. use and disuse B.segregation of genes C. evolution D. independent assortment of genes
3. An interlocking form / pattern of feeding relationship is called A. food chain B. nutrition
C. consumer D. food web
4. The group of animals described as glorified reptiles is A. pisces B. amphibian C. aves D. mammals
5. The anal and dorsal fins of fish are used for: A. steering B. buoyancy C. upward movement D. controlling rolling movement E. downward movement
6. The significance of mitosis includes all of the following except A. genetic stability B. growth C. cell replacement D. degeneration
7. Which of the following is absent in the prophase stage of meiosis? A. leptonem B. zygonem C. pachynem d. triplonema
8. The photosynthetic pigments include: A. chlorophyll and carotenoids B.chloroplasts and cytochromes C. melanin and haemoglobin D. carotenoids and haemoglobin
9. Which of the following produces both hormones and enzymes? A. pancreas B. ileum C. gall bladder D. kidney
10. Of the following, which one lacks chaetae, tentacles and antennae? A. snail B. earthworm C. millipede D. crab E. snake

11. Etilolation is caused by the influence of A. CO_2 B. water C. mineral salts D. HCl E. light
12. Epigeal germination can be found in: A. sorghum B. maize C. millet D. groundnut
13. __________ is not sex-linked A. stunted growth B. river blindness C. haemophilia D. colour blindness
14. The pryrenoid in Spirogyra: A. usually contains starch B. is suspended by cytoplasmic strands C. is mainly used for respiration D. excrete waste product
15. Flower is to the angiosperm as _________is to gymnosperm A. pines B. cords C. cone D. anther
16. Alternation of sexual and asexual method of reproduction is found in ______ A. euglena B. form C. blue green algae D. grasses
17. ________ is not a non-seed plant. A. fern B. conifer C. cycad D. none of the above
18. Which type of association is shown by a fern growing on the stem of oil palm? A. epiphytism B. saprophytism C. commensalism D. symbiosis
19. Which of the following is likely to encourage inbreeding in plants? A. diecious B. protandious
    C. monoecious D. hermaphrodite
20. The biological association that contributes directly to succession in a community is:A. competition B. predation C. parasitism D. commensalisms
21. Grasses recover quickly from bush fires in the savanna because of their _____ A. fibrous B. roots B. succulent stems C. perenating organs D. rapid growth rate
22. The ability of an organism to live successfully in an environment is known as ______ A. resistance B. competition C. succession D. adaptation
23. The community of plants in which the same species occur from year to year is the _______ A. perennial species B. climax species C. pioneer vegetation D. annual species
24. __________ is an autotrophic mode of nutrition A. chemosynthesis B. saprophytism C. parasitism D. symbiosis
25. Which of the following is not an organ? A. leaf B. heart C. kidney D. bone

BIOLOGY SOLUTION 2010

1. **B –** Species

Kola	-	Kingdom
Peter	-	Phylum
Calls	-	Class
Olamide	-	Order
From	-	Family
Girls	-	Genus
School	-	Species

The most exclusive taxa is the kingdom i.e. It contains all living organisms. As we go from kingdom to species, the similarities increases.

2. **B** –
3. **D** – Food web, is a complex feeding relationship
4. **C –** Aves
5. **A** –
6. **D –**
7. **D** – The prophase stage include leptotene, zygotene, pachytene, diplotene and diakiresis
8. **A** –
9. **A** – Pancreas
10. **E** – Snake
11. **E** – Light
12. **D** –
13. **B** – Riverblindness or onchocerciasis is carried by black fly i.e. the vector while the causative agent is *Onchocerca volvulus*
14. **A** – It is made up of protein and it is used to store starch
15. **C** – Cone
16. **C –** Fern
17. **B** – Conifer is a seed plant
18. **A**– epiphytisim
19. **C** – monoecius
20. **A** – competition
21. **C –** perenating organ
22. **D** –
23. **B** – climax community
24. **A –** chemosynthesis, can be defined as the synthesis of food from inorganic source through oxidation
25. **D** – bone

BIOLOGY 2009

1. One of these is not found in urine A. water B. sodium chloride C. nitrogenous compounds D. calcium chloride E. nitrogenous salts
2. An organism with a apir of indistinguishable genes is a A. heterozygote B. hybrid
   C. allelomorph D. homozygote (e) diploid
3. The fruit formed from a single flower having several carpels is called A. multiple fruit B. simple fruit C. aggregate fruit d. dehiscent fruit E. indehiscent fruit
4. The functions of ossicles (malcus, incur and stapes) in the mammalian ear is to A. vibration
   B. regulate pressures C. support of the inner ear D. maintain balance during motion E. secrete oil
5. Joint skeleton is absent in the A. cockroach B. spider C. millipede D. snail E. housefly
6. Mucors and spirogyra can be put in the same group because they A. unicellular B. have spors are dispersed by wind C. repreoduce sexually D. have bodies made up of thallus and filame alternatively

7. The organ through which nourishment and oxygen diffuse into an embryo is called A. Amnion B. chorion C. umbilical cord D. oviducts E. placenta

8. A tapeworm fasten itself to the intestine of its host with A. neck & bicker B. hooks & suckers C. rostellum & suckers D. proglottis E. Rostelluum, hooks, suckers

9. Which of these is false about the piliferous layer of a root? A. has a thin cuticle B. is outermost layer of the cortex C. may bear root hairs D. breaks down with age E. is replaced cock in old roots.

10. Anaerobic respiration in yeast produce A. CO_2& ethanol B. CO_2& H_2O C. CO_2& O_2 D. CO_2& glucose E. ethanol & H_2O

11. Which one of these set of factors is completely abiotic? A. turbidity, tide, salinity, plankton B. pressure, pH, soil, insect C. water, soil, bacteria, salinity D. pH, bamboos, wind, rainfall E. light, altitude, wind, humidity

12. In which of these are flagella and cilia found A. flatworms B. protozoa C. elenterates D. annelids E. nematodes

13. The three important organs that are situated close to the stomach are A. liver, kidney and gall bladder B. pancreas, liver and kidney C. gall bladder, pancreas and spleen D. liver, kidney and spleen E. kidney, gall bladder, liver

14. The plantain reproduces sexually by A. spores B. buds C. fragments D. sucker E. flowers

15. The major functions of swim-bladder in fish A. breathing B. swimming C. diving D. repelling enemy E. buoyancy

16. The part of the central nervous system concerned with answering and examination question is the A. sinalcord B. cerebellum C. orxy D. cerebrum E. medullar oblongata

17. Wind pollinated flowers usually have A. long style B. sticky stigma C. small and short stigma D. rough pollen grains E. short styles and pollen.

18. The ridicule of a bean seedling grows mostly rapidly in the region A. of the root tip B. below the topsoil C. just round the root tip D. just below the root tip E. just above the root tip

19. A key to similarity between nervous and hormonal system is that both A. involve chemical transmission B. have widespread effects C. shed chemicals into the blood stream D. evoke rak response E. eliminate response

20. The bones of the neck on which the skull rest is A. odontoid B. axis C. occipital D. atlas E. patella

21. A child with blood group genotype different from those of both parents and with a mother of genotype OO, can only have a father of genotype A. A B. B C. OO D. AB E. AA

22. A true climax community A. changes from year to year B. persists until the environment change C. is the first stage in plant succession D. consists of tallest trees and small animals E. is in a state of perturbation

23. In a predator food chain involving secondary and tertiary consumers, the organisms become progressively A. smaller b. equal in number C. larger and fewer along the food chain D. parasitized along the food chain as consumers get bigger E. sparse in distribution

24. Which of these is false about the piliferous layer of a root? A. has a thin cuticle B. is outermost layer of the cortex C. may bear root hairs D. breaks down with age E. is replaced cock in old roots.

25. When it is cold the blood vessels of the skin A. dilate to increase blood flow to the skin B. constrict to reduce the amount of flow of blood C. dilate to reduce the amount of blood flow to the skin D. constrict to increase the amount of blood flowing to the skin

BIOLOGY SOLUTION 2009

1. **D** – Calciumchloride, urine contains 95% and 5% urea, minerals, NaCl etc.
2. **D** – homozygote both are expressed in the same form. They are not dominant over each other e.g. TT or tt.
3. **C –** Aggregate fruit
4. **A** – The ear ossicles help to transmit sound in form of vibration
5. **D** – snail lacks jointed skeleton
6. **C** – both reproduce sexually by conjugation
7. **E** – Placenta
8. **B** – hooks and sucker is used for clinging to the wall of the intestine
9. **A** – Piliferous layer or hypodermis lacks cuticle and stomata. It has a trichoblast cells which is used for the production of root hairs, the root hairs absorb water by osmosis
10. **A** – CO_2 and ethanol, in yeast and plant while in animal anaerobic respiration gives CO_2 and lactic acid (2-hydroxyl-propanoic acid).
11. **E** – Abiotic were not living why biotic are leaving
12. **B** – Protozoa, flagella and cilia are used for locomotion
13. **C –** The human stomach is $5dm^3$ the spleen is attached to it while your pancrease lies beside it, the liver is on the toe, and the liver as a duct directly connects to the gall bladder.
14. **D** – Sucker
15. **E** – Swimming bladder is used for buoyancy or to regulate density
16. **D** - Cerebrum , It is the site of intelligence. The most active part is the cerebral cortex, the

higher the convolution, the higher the cerebral capacity.

17. **C** – Small and short stigma
18. **D –**
19. **B** – have wide spread effect. The effect may be short lived like in nervous system, or permanent like the endocrie system.
20. **D** – Atlas, which allows nodding moving movement and it lacks centrum
21. **D –** AB, If we cross OO with AB, It will give AO, AO, BO, BO, in which A and B are dominant over O
22. **B** – It persist and is constant
23. **C** – The organism becomes few and large in size e.g. grass – mouse – snake
Snake is larger than mouse.
24. **A** –
25. **B –** Vasoconstriction during cold and vasodilation during heat. Vasoconstriction is to reduce blood flow so that heat won't be lost.

BIOLOGY 2008

1. Which of these types of skeleton appropriate to the cockroach A. hydrostatic skeleton
B. exoskeleton C. endoskeleton D. cartilaginous skeleton
2. When proteins are broken down they provide A. oxygen B. carbohydrate C. energy D. amino acids
3. The function of lenticels is A. to receive excess water in the plant B. to absorb water from the atmosphere C for gaseous exchange D. to absorb light
4. One of the functions of xylem A. strengthening the stem B. manufacturing food C. conducting manufactured food D. none of the above
5. People suffering from myopia A. can see near objects clearly B. can see far away objects clearly C. cannot see any object clearly D. are colour-blind
6. The cilia in paramecium are used for A. respiration B. locomotion C. protection D. excretion
7. The study of the organisms and the environment of abandoned farmland is the ecosystem of A. A community b. population C. species d. an ecosystem
8. At fertilization A. one chromosomes from the male joins another from the female B. one gene from the male combines with the another from the female C. the male nucleus fuses with the female nucleus D. one set of chromosome combines with another set from the female
9. The neck region of the tapeworm (Taenia spp.) is responsible for the A. the production of eggs (b) the storage of eggs (c) the formation of new segments (d) the development of the suckers
10. Which of the following is characteristics of the animal cell A. presence of chloroplasts
B. possession of a cellulose cell wall C. absence of large vacuoles D. presence of large vacuoles
11. In the life history of Schistosoma (Bilharzra), one of the following is the intermediate host A. man B. snail C. mosquito top larva D. fish
12. The hormone which tones yup the muscles of a person in the time of danger is from the A. thyroid gland b. pancreas C. adrenal gland D. spleen
13. The movement of molecules from a region of higher concentration to one of lower concentration is A. diffusion B. transpiration C. osmosis D. plasmolysis
14. The region of cell division in a root is A. root cap B. endodermis C. xylem D. meristem
15. Which of the following is not an excretory organ? A. lungs B. kidney C. leaf D. large intestine
16. The part of the mammalian brain responsible for maintaining balance is A. medulla oblongata B. cerebellum C. optic lobe D. cerebrum
17. A sugar solution was boiled with fehling's solutions A and B and the colour remain blue. The sugar tested was A. glucose B. maltose C. fructose D. sucrose
18. The blood vessel which carries digested food from the small intestine to the liver is the A. renal vein B. renal artery C. hepatic artery D. hepatic portal vein
19. The maize grain is regarded as a fruit and not a seed because A. it is covered by a sheath of leaves B. the testa and fruit wall fuse after fertilization C. it has both endosperm and cotyledon D. the pericarp and seed coat are separate
20. Identical twins are produced under one of the following conditions A. two ova fertilized at the same time by the sperm B. one ovum fertilized divides to give two embryos C. two ova fertilized by one sperm D. one ovum fertilized by two sperms
21. Partially digested food ready to leave the stomach is called A. chime B. curd C. glycogen
D. paste
22. The particles found in the blood which play an important role in blood clotting are called
A. platelets B. red blood cells C. monocytes D. granulocytes
23. Ina cross between a normal male and a female carrier for haemophiliac is A. 25% B. 50% C. 70% D. 45%
24. Which of the following hormones is produced during fright or when agitated? A. Insulin B. adrenalin C. thuroxine D. pituitary

25. Grasses-Grasshopper-Lizards-Snakes-Hawks. In the food chain, the organisms which are the least in number are A. Grasses B. Hawks C. te Lizards D. Snakes
26. One significant difference between roots and stem is that A. branch root originate in the pericycle while branch stems do not B. stems are always below the ground while roots are always under the ground C. stems are positively geotropic while roots are negatively geotropic D. stems are sometimes used for storage while roots are never so used
27. the arrangements bleow are steps in protein digestion. Which is the correct sequence? A. A-polypeptides, b-protein, c-amino acids, d-peptones A. a-->b-->c-->d B. c-->d-->a-->b C. b-->c-->a-->d D. b-->d-->a>c

BIOLOGY SOLUTION 2008

1. **B** – exoskeleton, this is found outside, hydrostatic skeleton is found in oligochaete (earthworm) while endoskeleton is found inside e.g. All vertebrates
2. **D**– amino acid is the end product of protein, while protein can be deamninated i.e. the breaking down of protein to ammonia which will be converted to urea to make it less toxic and keto acids i.e. carbon containing compound which can be converted to glucose.
3. **A** – To remove excess water from the stem, it is present at the back of old tree as a slit for gaseous exchange
4. **A** – Xylem performs 2 functions (1) transportation of water and minerals salt by diffusion (2) strengthening the stem because it is lignified containing dead cells
5. **A** – can see near object clearly in myopia or short sightedness because the eye ball is too long
6. **B** – locomotion
7. **D** - Organism is living while abandoned farmland environment may contain living and non living things, therefore ecosystem is the study of living and non-living organism
8. **C** – Male nucleus from sperm fuses with female nucleus from egg or ova to form zygote
9. **C** – formation of new segments. The neck region contains immature proglottides while the tail contains mature proglottides which contains egg and break off to form embrayo known as bladder worm or cystercercus.
10. **C** – Absenceof large vacuoles
11. **A** – The intermediate host of schistosoma or a blood fluke is man A… Another name is the primary host, while water snail *Iymna trunculata* is the secondary host, the larva is known as cercaria. The species of schistosoma include (i) *Schistosoma haematibium* (ii) *Schistosoma mansoni* (iii) *Schistosoma japonicom*. It can be contracted through bathing or drinking contaminated water, it affects the vessels in the urinary track and intestine, the symptom is passing blood after urine.
12. **C** - Adrenal gland, precisely *Adrenal medulla*, secreting adrenaline and non – adrenaline. Adrenaline is also known as epinephrine.
13. **A** – diffusion
14. **D** – Meristen
15. **D** – Large intestine is not a excretory organ it only helps to absorb water and to expel so egested product i.e. solid undigested food like faeces.
16. **B** – Cerebellum, maintenance of balancing and posture and for muscular activities
17. **D –** Sucrose, when it is hydrolyse i.e. Addition of mineral acid, it gives glucose and fructose which give brick red when tested with Fehling
18. **D** – Hepaticportal vein which is the shortest, it begins and ends with capillaries. The largest vein is vena – cava anterior or superior vena – cava. Posterior or interior while the largest artery is Aorta.
19. **B** – Maize is a carryopsis dry indehiscent fruit
20. **B** – One egg fertilized by one sperm and divides into two
21. **A** – Chyme It is pasty in nature while chyle is watery.
22. **A** – Platelets
23. **A** –
24. **B** – Adrenaline or epirephrin
25. **B** – Hawles because they are the largest in that food chain
26. **A** –
27. **D –** Protein – peptone – polypeptide – amino acids

BIOLOGY 2007

1. Where in energy produced in a cell? A. nucleus B. lysosomes C. mitochondria D. nucleolus
2. Which of the following organism does not exist as a single free-living cell? A. Amoeba B. Euglena C. Clamydomonas D. Volvox
3. Euglena is an autotrophic organism because it: A. has flagella B. has plant and animal features
C. can manufacture its food D. moves fast
4. In which of the following organism does a single cell perform all function of active movement nutrition, growth, excretion and photosynthesis? A. paramecium B. amoeba C. Euglena D. hydra
5. What is the function of contractile vacuole in paramecium? A. produces enzymes B. gets rid of excreta C. stores and digests food D. gets rid of excess water
6. The ability of organism to maintain a constant internal environment is known as: A. diuresis B. endosmosis C. palsmolysis D. homeostasis

7. Which of the following is the medium of transportation of nutrients within unicellular organism?
A. Lymph B. plasma C. protoplasm D. serum

8. Ina erobic respiration, oxidative phosophorylation takes place in A. cytoplasm B. lysosome
C. mitochondrion d. ribosomes

9. Bryophytes are different from following plants because they A. are simples mall plants B. carry out alternation of generation C. possess small D. possess no vascular tissue

10. In lower plants like mosses, the structure which performs the functions of roots of higher plants is called A. roots hairs B. rhizoids C. hydphae D. roots

11. Which of the following components of an ecosystem has the greatest biomass? A. primary producers B. primary consumers C. secondary consumers D. tertiary consumers

12. The young shoot of a plant is referred to as A. radicle B. plumule C. bud D. branch

13. The name of a bacterium which derives its energy from oxidizing nitrites into nitrates is
A. ntirosomonas B. azotobacter C. ntirobacter D. Escherichia coli

14. Potometer is used to measure A. rate of osmosis B. rate of diffusion C. rate of transpiration D. rate of photosynthesis

15. Meiotic cell division ensures that A. many similar cells are produced B. chromosome number of cells is halved C. cells produced are doubled D. cells produced posses the same chromosome number

16. The stem of young herbaceous plants are kept upright mainly by A. osmotic pressure B. turgidity C. transpiration pull D. root pressure

17. Which of the following tissues is not found in the stem and root of monocotyledons? A. Xylem
B. Cambium C. Pith D. Pericycle

18. Fruit enlargement can be induced by spraying young ovary with A. gibberellins, ethylene and abssisic acid B. auxin, abscisic acid and ethylene C. auxin, cytokinin and gibberellin D. a uxin, kinin and giberelin

19. A dry, indehiscent, winged fruit formed from one carpel is known as A. schizocarp B. caryopsis
C. samara D. nut

20. A fruit which develops without fertilization is described as A. simple B. aggregate C. multiple D. parthenocarpic

21. A dwarf plant can be stimulated to grow to normal height by the application of A. thyroxin
B. giberelin C. insulin D. kinin

22. The condition known as cretinism is caused by the deficiency of A. vitamin A B. Insulin
C. Thyroxin D. Vitamin C

23. The different between viviparous and oviparous animal is A. possession of yoked eggs B. laying and brooding of eggs C. possession of yolkless eggs D. laying of unfertilized egg

24. The following are features of the tropical rainforest except A. loose and moist soil B. short trees growing beneath tall trees C. scanty trees with small elves D. presence of many animals

BIOLOGY SOLUTION 2007

1. **C** – Mitochondria, It is also known as power house, it is where oxidative phisphorylation of glucose take place, It has a double membrane with a folded partition known as cristae, it is foud in active cells like sperm cells while golgi bodies are found in secreting cells. Lysosones helps to engulf dead cells and for transporting materials in and out of the cells which is known as endocytosis and exocytosis respectively,

2. **D**– volvox exist as a colony, a colony is a group of cells joined together by cytoplasmic strands they undergone division of labour

3. **C** – It has chloroplast for photosynthesis

4. **C** – Euglena has both plant and animal like characteristics

5. **D** – it helps to get ride of excess water while the excretion of ammonia is by diffusion not through contractile vacuole

6. **D**– homeostasis, organs involve in homeostasis include (1) kidney (2) liver (3) skin and endocrine glands (hormones)

7. **C** – Protoplasm because of their high surface area to volume ratio therefore diffusion is efficient

8. **C** – mitochondria

9. **D** – Bryophytes lack vascular bundle which include (xylem and phloem) and they undergo alternation of generation in which the gametophyte generation is dominant and haploid but in petridiophytes the sporophyte generation is dominant and diploid.

10. **B** – Rhizoids

11. **A –** Primary producer, biomass is the total wet and dry mass of an organism, the primary producer has the largest biomass during spring and the consumers has the largest biomass during winter.

12. **B** – Plumule, it develops to shoot while radicle develops to root

13. **C** – Nitrobacter converts nitrite to nitrate, you can see from nitrobacter we can get nitrate, because it contains: Nitrate, just use it to decode it while nitrosomonas converts ammonia to nitrite.

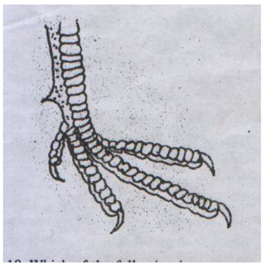
14. **C** – Rate of transpiration can be measured by photometer while osmosis can be measured by osmometer.
15. **B** – The chromosomes number is halve, and it ensures exchange of genetic materials which is known as crossing over
16. **B** – Turgidity makes upright while flaccidity makes it wilt, turgidity is just like an inflated tyre
17. **B** – Cambium is only found in dicot plants which initiate secondary growth by producing vascular cambium and cork cambium
18. **C** – Auxin, cytokinin and gibberellin are positive phyto hormones
19. **C –** Samara
20. **D** – Parthenocarpic, while insect the process by which insects develop without fertilization is known as parthenogenesis
21. **B** – Gibberelin helps to stimulate the growth of dwort plant
22. **C** – Thyroxin, cretinism occurs when there is under secretion of thyroxine before maturity, A cretin is sexually immature while under secretion of thyroxin after maturity is known as dwarfism, A dwarf as a reduced metabolic rate which is known as myexodema.
23. **B** – Layingand brooding of eggs.
24. **C**– Scanty trees with small leaves are characteristics of desert, though savannah has the highest population of animals.

BIOLOGY 2006

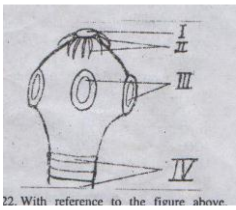
1. Structures found in cells are listed below: i. cell wall ii. Cell membrane iii. Chloroplast iv. Cytoplasm v. nucleus vi. Sap vacuole. Which of these structures are found in both animal cells and plants cells? (a) i, ii and v (b) I, iii and v (c) ii, iii, and v (d) ii, iv, and v
2. Which of the following is not present in the nucleus of a cell? A. chromosomes B. Nucleolus C. Mitochondrion D. Genes
3. An amoeba moving towards a crumb of cake in a pond most likely exhibits A. phototropism B. chemotaxis C. thermotaxis D. mastic movement
4. Which of following cells would most probably contain the greatest number of Golgi bodies A. muscle cell B. secretory cell C. nerve cell D. white blood cell
5. A group of similar cells performing the same function is A. an enzyme b. an organ C. a tissue D. an organelle
6. A plant which grows on another plant without apparent harm to the host plant is called A. a parasite B. a n epiphyte C. a saprophyte D. a predor

$6CO_2 + 6H_2O \xrightarrow{Sunlight} C_6H_{12}O_6 + 6O_2$

7. The oxygen given off during the process in the above equation is derived from A. sunlight B. water C. carbon dioxide D. atmosphere
8. When testing a leaf for starch, why is it first placed in boiling water? A. to extract the chlorophyll B. to remove colour from the leaf C. to dissolve the starch D. to stop chemical reactions
9. The petals of a flower are collectively called A. Calyx B. Capsule C. Carpel D. Corolla
10. Osmosis can be defined as diffusion of: A. water molecules from an area of high concentration to an area of low concentration B. water molecules from a dilute solution to a concentrated solution across a permeable membrane C. water molecules from a concentrated solution to a dilute solution through a semi-permeable membrane D. water molecules from a dilute solution to a concentrated solution through a semi-permeable membrane
11. The ventricles of the mammalian heart are more muscular than the auricles because the A. auricles have smaller capacity B. ventricles are larger in size C. ventricles pump blood to distance organs D. ventricles receive more blood
12. Which of the following statement is not correct about the function of each group of mammalian vertebrae? A. caudal vertebrae support the tail and provide attachment for tail muscles B. Thoracic vertebrae articulate with the ribs C. Lumber vertebrae provide attachment for abdominal muscles D. sacral vertebrae support the skills and allow nodding and rotting movement
13. In the adult toad, gaseous exchange takes place through A. Buccal cavity, skin and spiracle B. Buccal cavity, balder and lungs C. Buccal cavity, skin and lungs D. gills, skin and buccal cavity
14. The brain and the spinal cord constitute the A. autonomic nervous system B. sympathetic nervous system C. somatic nervous system D. central nervous system
15. Which of the following parts of the mammalian brain is involved in taking the decision to run rather than walk B. cerebellum B. medulla oblongata C. midbrain D. cerebrum
16. Which part of the car is responsible for the maintenance of balance? A. cochlea B. tympanic membrane C. Eustachian tube D. semi-circular canals
17. The foot of the bird shown below is strong and has strong claws on its digits. This implies that is bird A. is a scavenger B. is a bird of prey C. uses the foot to supplement wing action D. uses the foot to scratch the soil



18. Which of the following is not a means of conservation A. repracing harvested mature timber trees with their seedlings B. prevention of poaching C. controlling excessive deforestation D. burning of vegetation before cropping
19. One of the following statements is not true of viruses A. they are micro-organism B. they are smaller than bacteria C. they can be seen with an ordinary light microscope D. they cause tobacco disease, polio and smallpox
20. Each of the following is an arthropod except the A. crab B. scorpion C. spider D. snail
21. The largest phylum in the animal kingdom is A. cnidaria B. mollusea c. chordate D. arthropoda



22. With reference to the Figure above, which of these are correct? A. I & II are proglottides and hooks B. I & III are rostellum and suckers C. III & IV are hooks are proglittids D. II & IV are hooks and rostellum
23. The two species of human tapworm canb e distinguished by the presence or absence of A. scolex B. hook C. head D. sucker

BIOLOGY SOLUTION 2006

1. **D** – The structures found in both plants and animal cells include cell membrane or plasma – membrane or plasmalemma, cytoplasm which is the fluid portion and part of the living content of the cells, nucleus which controls cell activities. Other include ribosome, mitochondria, endoplasmic reticulum, golgi bodies and lysosome etc.
2. **C** – The constituents of the nucleus include nuclear membrane nucleus, chromosomes, nucleic acids (RNA and DNA) genes nucleouplasm and nuclear pore
3. **B** – chemotaxis, this is because food contains chemicals, tatic movement is the directional movement of organism towards or away stimulus
4. **B** – Golgi bodies are mostly found in secretory cells such as pancrease because secretes hormones and enzymes. The hormones include insulin and the enzymes include trypsinogen, lipase or steapsin and amylase.
5. **C** – A Tissue is a group of similar cells performing the same functions examples include blood which is made up of 55% plasma and 45% cells 5% dissolved substances.
6. **B** – An epiphyte lives in an aboreal habitat example include fern, they stay there to receive sunlight
7. **B** – Water. During photolysis of water i.e. the breaking down of water to hydroxyl ion and hydrogen ion
   $H_2O \rightarrow OH^- + H^+$
   The hydroxyl ion recombines and form water and liberate oxygen gas
   $4OH^- \rightarrow 2H_2O + O_2 + 4e^-$
8. **D** – To stop chemical action i.e. To kill the protoplasm of the leaf
9. **D** – Corolla
10. **D** – Water molecules from dilute solution to a concentrated solution through a semi-permeable membrane or movement of solutes from low concentration of solute to high concentration of solute through a semi-permeable membrane.
   Note: that osmosis only occur in living cells and water. But osmosis can occur in cellophone which is non-living.
11. **C** – the left ventricles is 3 times the right ventricle because it pumps blood from the heart to other parts of the body which is known as systemic circulation
12. **D** – Sacral supports the pelvis and it join together to form sacrum.
13. **C** – Buccal cavity or mouth skin or cutaneous respiration lungs or pulmonary respiration
14. **D** – Central nervous system. The types of nervous system include (a) central nervous system (brain and spinal cord) (b) peripheral nervous system which include sensory system or afferent (SAN) and motor neurone or efferent (MEN)
15. **D** – Taking the decision to work rather than to run is a voluntary action and most voluntary actions are controlled by cerebrum.
16. **D** - Semi-circularcanals, but if semi-circular canal is not in the option you can choose perilymph because is the fluid that filled the semi-circle canal
17. **B** – The bird is a predator, it has harp claws, strong and curved for grasping prey
18. **D** – Burning exposes the surface of soil and can lead to erosion
19. **C**– Virus can only be seen by an electron microscope, they exist in two state, extracellular and intracellular state
20. **D** – Snail belong to Mollusca and class gastropoda
21. **D** – Arthopoda are largest phylum

22. **B** – That is the structure of *Taeniasolium*, we have 2 species of human tape worm (1) *Taenia solium* and (2) *Taenia saginata*. They both have two hosts. (1) primary host (2) secondary. The primary host or intermediate host of the two species is man while the secondary host are pig and cattle respectively. The only difference between them is that *Taenia solium* has six hooks while *Taeia saginata* lacks hoot.

I	-	rostelum
II	-	hooks
III	-	Sucker
IV	-	immature proglottides

The head together is known as scolex. The hooks and sucker is used for clinging to the wall of the intestine while the proglottides is used for the formation of new segments

23. **B** - Hook
Taenia solium six hook
Taenia saginata no hook
Their embryo is known as cystercercus or bladder worm

OAU POST UTME
GEOGRAPHY QUESTIONS 2015

1. The major source of tropical hardwood for timber production is the (a) Mangrove swamp (b) Coniferous forest (c) Tropical deserts (d) Rain forest
2. A settlement existing primarily for the people to enjoy their leisure is known as a (a) Game town (b) Cultural town (c) Conurbation town (d) Resort town
3. Which of the following is not a rapid type of mass movements? (a) Mudflow (b) Earthflow (c) Solifluction (d) Debris Avalanche
4. Which of the following actions/ features is not true of youthful stage of a river? (a) Later erosion (b) Rapids and waterfalls (c) Interlocking spurs (d) Vertical Erosion
5. Based on Richter Scale, earthquake results in distinct shaking and collapse of less well – constructed buildings from (a) Log – scale 2 (b) Log – scale 3 (c) Log – scale 4 (d) Log – scale 5
6. Which of the following rivers crossed the equator twice? (a) River Congo (b) River Niger (c) River Nile (d) River Zambezi
7. If an international football match is scheduled to start by 6.00pm on Wednesday October 14, 2015 in Location X (Longitude 260ºW) be able to listen or watch the start of the march? (a) 9.40am on Thursday October 13, 2015 (b) 1.00pm on Thursday October 13, 2015 (c) 11.00pm on Tuesday October 13, 2015 (d) 1.20am on Tuesday October 13, 2015
8. Which of the following states of the atmosphere is antagonistic to precipitate formation and pollution dispersal (a) Absolute stability (b) Instability (c) Convective instability (d) Conditional stability
9. Which of the following is a cold Ocean current? (a) Oyashio Current (b) North Atlantic Drift (c) South Equatorial Current (d) North Equatorial Current
10. What is the centigrade equivalent of 158 degree Fahrenheit? (a) 105.56 degree Centigrade (b) 70 degree Centigrade (c) 316.4 degree Centigrade (d) 252. 4 degree centigrade
11. The effects of earth's rotation include the following except (a) Day and Night (b) Time difference from place to place (c) Changes in the altitude to place midday sun (d) deflection of winds and ocean currents
12. Terracing in agriculture is practiced (a) On top of plateaus (b) one the steep slope of escarpments (c) on dry lands (d) on waterlog areas
13. Which of the following terms is associated with glaciation? (a) Zeugen (b) Arete (c) Dolines (d) Mesa
14. Which of the following is not a renewal energy? (a) Wind energy (b) Solar energy (c) hydroelectic power (d) fossil fuel
15. Which of the following tourist centres in Nigeria is the largest? (a) Yankan Game reserve (b) Kainji National Park (c) The Mambila Plateau (d) Obudu cattle reach
16. Which of the following is wrongly paired? (a) Ahanti of Ghana (b) Wolof of Gambia and Senegal (c) Kru of Southern Liberia and Southwest Cote d'Ivoire (d) Mossi of Sierra Leone
17. Which of the following Africa countries has the largest expanse of trophical rain forest? (a) Nigeria (b) Zaire (c) Cote d'Ivoire (d) Ghana
18. Which of the following gases constitutes the least percentage by volume in the earth's (a) Oxygen (b) Ozone (c) Nitrogen (d) Argon
19. In a typical soil profile, the zone of illuviation is also known as (a) A Horizon (b) B Horizon (c) C Horizon (d) D Horizon
20. Which of the following African Countries has the longest coastline? (a) Somalia (b) Angola (c) Kenya (d) Sudan
21. Vauclusian springs are also called (a) Resurgent Streams (b) Perennial Streams (c) Seasonal springs (d) permanent springs
22. In which of the following countries is most extensive area of loess depicts found? (a) China (b) Canada (c) United States (d) United Kingdom
23. Where is the phenomenon of Little Dry Season (August Break) most evident in Nigeria? (a) Southwest (b) Southeast (c) Northwest (d) Northeast
24. Which if the following hydrological centres in Nigeria is most extensive? (a) North central Plateau (b) Eastern Highlands (c) Eastern Highlands (d) Udi Plateau

25. Which of the following trophical cyclones is most violent? (a) Willy nilly (b) Typhon (c) Tornado (d) Hurricane

Geography 2015 Solution

1. D
2. D
3. C
4. A
5.
6. D
7. Wrong Question
8. C
9. C
10. B
11. C
12. B
13. B
14. D
15. B
16. C
17. C
18. D
19. A
20. D
21. A
22. A
23. A
24. A
25. C

GEOGRAPHY QUESTION 2014

1. Which of the following locations in Nigeria has the least mean annual rainfall total? (a) Sokoto (b) Nguru (c) Kano (d) Maiduguri
2. Nigeria's Federal Capital Territory shares boundaries with the following States except (a) Plateau (b) Niger (c) Kogi (d) Kaduna
3. Which of the following is not a major fold mountain of the world? (a) Himalayas (b)Andes (c) Atlas (d) Nakatoa
4. The brief period between sunset and the complete darkness or night is known as (a) dawn (b) solstice (c) twilight (d) night
5. In an international football match is scheduled to start by 2.00pm on Monday 21 September 2014 in location A (longitude 5ºE, at what time and date would people in location B (longitude 295ºW) be able to listen or watch the start of the match? (a) 5.40pm on Sunday 20 September, 2014 (b) 7.00pm on Sunday September, 2014 (c) 10.00am on Tuesday 22 September, 2014 (d) 8.40am on Tuesday 22 September, 2014
6. Which of the following is not associated with youthful stage of river? (a) narrow valley (b) rapids and water falls (c) braided channel (d) port holes
7. Which of the following gases constitutes the largest percentage by volume in the earth's atmosphere? (a) oxygen (b) carbon dioxide (c) nitrogen (d) argon
8. Which of the following is not a worm ocean current? (a) North Equatorial current (b) South Equatorial current (c) North Atlantic drift (d) Peruvian current
9. Which of the following is not an element of climate? (a) precipitation (b) evaporation (c) temperature (d) humidity
10. What is the Fahrenheit equivalent of 95 degree Centigrade? (a) 203 degree Fahrenheit (b) 51 degree Fahrenheit (c) 84.78 degree Fahrenheit (d) 139 degree Fahrenheit
11. Which of the following temperate grasslands is wrongly paired with its location? (a) the Downs of Australia (b) Prairies of Canada and USA (c) The Pampas of Argentina (d) the Steppe of South Africa
12. Which of the following is not true of Anticyclone? (a) a region of relative low atmospheric pressure (b) Blowing clockwise in the northern hemisphere (c) blowing anticlockwise in the southern hemisphere (d) it is also refers to simply as a "HIGH"
13. Which of the following vegetation belts houses the largest concentration of domesticated livestock in (a) Sahel (b)Sudan savanna (c) Montane grassland (d)Forest
14. Which of the following is the cause of the deflection of all freely-moving bodies (e.g. wind and ocean current) to the right of their paths in the northern hemisphere and left in the southern hemisphere? (a) the differences id air pressure (b) the revolution of the earth (c) the rotation of the earth (d) the big size of the earth
15. All of the following are measures of combating soil erosion except (a) terracing (b) cover crops (c) mining (d) contour ploughing
16. The State of equilibrium reached by the vegetation of an area when it is left undisturbed for a long period of time is called (a) trophic level (b) ecosystem 9c) climax (d) phyletic origin
17. Which of the following is not a prominent process of wind erosion in the desert? (a) deflation (b) abrasion (c) attrition (d) solution
18. In which of the following latitudes would you expect more than 24 hours of continuous daylight or darkness at a particular time to the year? (a) 70ºN and S (b) 50ºN and S (c) 45ºN and S (d) 23.5ºN and S
19. One of the major mineral deposits found in Jos Plateau is (a) gold (b) coal (c) lignite (d) columbite
20. "Transhumance" is (a) rearing of animal which involves seasonal movement of animals up and down hill slopes in search of pasture (b) animal rearing which involves movement of farmers from the coast inland (c) animal rearing which involves movement of farmers from desert to the coast (d) animal rearing in which farmers stay permanently over a location

21. The movement of people from a village to a farm settlement may be described as (a)rural-urban migration (b) rural-rural migration (c) suburban-rural migration (d) urban-urban migration
22. In which of the following rock types is petroleum mainly found? (a) igneous (b) sedimentary (c) volcanic (d) metamorphic
23. Under which of the following atmospheric conditions is pollution forced to spread horizontally near the earth's surface than rising vertically trough the atmosphere? (a) convectively unstable atmosphere (b) wen the environmental lapse rate greatly exceeds the adiabatic lapse rate (c) absolutely unstable atmosphere (d) temperature inversion
24. The world's richest fishing ground are found (a) on the continental shelves (b) on the deep sea platforms (c) in oceanic deep (d) in river estuaries
25. Which of the following African cities is situated near the confluence of rivers? Niamey (b) Cairo (c) Port-Harcourt (d) Khartoum

GEOGRAPHY SOLUTION 2014

1. **D -** Maiduguri
2. **A –** These are states that share boundary with Abuja (i) Kaduna at North (ii) Niger at West (iii) Nassarawa at East (iv) Kogi at South.
   Note: Plateau shares boundaries with Nassarawa, Kaduna, Bauchi, Gombe, Taraba and Benue
3. **D –** Examples of fold mountain are Atlas mountain in Morocco (North Africa), Scandinavia mountain, Tienshan mountain in China, Himalayas (Southeast Asia), The Andes (South America), Atlai mountain (Mongoria) and Alps.
   Krakatoa is an example of volcanic dome shaped mountain
4. **C –** Dawon refers to the brief period between sunrise and full daylight while Twilight refers to the brief period between sunset and complete darkness.
5. **C –**
   5^oE 295^oW
   Location A Location B
   2:00pm X

   Longitude difference = $5^oE + 295^oW = 300^o$
   Converting this to time, using 15^o = 1hr = 300^o
   $= X = \frac{300}{15}$ = 20hrs
   Since location B is West of location A, time in location A will be 2:00pm × 20hrs = 22:00hrs
   22.00 – 12.00hrs = 10am on Tuesday
6. **C -** Braided channel
7. **A -** Oxygen
8. **D -** Peruvian current
9. **B** - Evaporation
10. **A –** These should be noted:
   $^oC = \frac{5}{5} \times {^oF} - 32$
   $95^o = \frac{5}{9} \times {^oF} - 32$
   $95^o = \frac{5^oF - 160^o}{9}$
   $5^oF - 160^o = 855^o$
   $5^oF = 1015^o$
   $^oF = \frac{1015^o}{5^o}$
   ^oF = 203 degree
   OR
   $^oC = \frac{5}{9} \times {^o F} - 32 = \frac{^oF - 32}{1.8}$
   $^oF = (1.8 \times {^oC}) + 32^o$
   $^oF = (18 \times 95) + 32$
   $171^o + 32^o$
   $^oF = 203^o$
11. **D -** The Steppe of South Africa
12. **A -** A region of relative low atmospheric pressure
13. **C -** Montane grassland
14. **C -** The rotation of the earth
15. **C -** Mining
16. **C -** Ecosystem
17. **D** - Solution
18. **A -** 70^oN and S
19. **D -** Columbite
20. **A -** Rearing of animal which involves seasonal movement of animals up and down hill slopes in search of pasture
21. **B -** Rural-rural migration (c) suburban-rural migration (d) urban-urban migration
22. **B -** Sedimentary
23. **B -** When the environmental lapse rate greatly exceeds the adiabatic lapse rate
24. **A -** On the continental shelves
25. **D -** Khartoum

GEOGRAPHY QUESTIONS

1. The second Equinox in any year takes place during the month of
   A. March. B. June C. September D. December
2. Which one of the following countries in Africa would be person NOT pass over by crow fly (distance) from Abidjan to Cairo?
   A. Niger B. Libya C. Bukina Faso D. Chad
3. The main work of a river in its torrent state is
   A. Widening its valley B. Down cutting C. Deposition D. Bifurcation
4. Which of the following locations in Nigeria has the highest mean annual rainfall total?
   A. Benin City B. Port – Harcourt C. Calabar D. Ondo
5. The world's longest river is
   A. River Amazon B. River Mississippi C. River Nile D. River Chang Jiang

6. Which of the following is a feature of a rejuvenated river?
A. Incised members B. Braided chanel C. Delta D. Levees
7. Which of the following is not a thermometric scale?
A. Celsius B. Kelvin C. Fahrenheit D. Octas
8. Which of the following is not a dorm of condensation?
A. Snow B. Rime C. Cloud D. Fog
9. The surface of discontinuity between the earth's crust and the mantle is known as
A. Lithosphere B. Barysphere C. Mohocivic discontinuity D. Gutenburg discontinuity
10. Where is Sahel Savanna vegetation belt found in Nigeria?
A. Northwest B. Northeast C. Both Northwest and Northeast D. From the middle belt to the north
11. The largest soil group, found in the temperate grasslands, having a deep, black, nutrient – rich A – horizon, a compact B – horizon and a zone of calcium carbonate accumulation is called a
A. Chemozem B. Pedocal C. Podsol D. Chestnut

12. Drought – tolerant plants are known as
A. Epiphytes B. Hydrophytes C. Drouphytes D. Xerophytes
13. The Major disadvantage of the River Nile as a trade route is that
A. It is too long B. It is too shallow C. It flows during the wet season only D. It has several cataracts
14. The earth rotates from
A. Southeast to Southwest B. Southwest to Southeast C. West to East D. East to West
15. The Tropical maritime air mass attains its maximum incursion over West African hinterlands
A. When the sun is overhead on the Tropic of Capricorn B. When the sun in overhead on the Tropic of Cancer C. When the sun is overhead on the Equator D. When the sun is overhead on both the Tropic of Cancer and Tropic of Capricorn
16. A line joining places of equal salinity is known as
A. Isohaline B. Isoneph C. Isohyet D. Isohel
17. Which of the following continents has the largest area tropical type of climate?
A. Asia B. South America C. North America D. Africa
18. One hour's difference in mean solar time represents what angular difference in longitude?
A. 15º B. 23.5º C. 90º D. 66.5º
19. An extended urban area, typically consisting of several towns merging with the suburbs of one or more cities can best be referred to as
A. Conurbation B. Urban decay C. Growth pole D. Urban inertia
20. Which of the following terms is not associated with desert – type topography?
A. Zuegen B. Yardang C, Mesa D, Uvala
21. The major characteristics of a kerst scenery include
A. Excessive overland flow B. Extensive and long surface flow C. An underground network of caves and streams D. Frequent surface drainage
22. Which of the following landforms originates from Aeolian erosion?
A. Deflation hollow B. Sand dune C. Playa D. Wadi
23. The cheapest means of transport for a long distance travel is by
A. Air B. Rail C. Water D. Road
24. A traveller crossing the International Date Line from America to Asia at 1.00p.m. on Saturday, July 13th, 2013, would have to change his watch to 1.00 p.m. on
A. Sunday, July 14th 2013 B. Friday, July 12th 2013 C. Saturday, July 13th 2013 D. Monday, July 15th, 2013
25. The shaduf method of irrigation was first practiced in
A. Sudan B. Ghana C. Egypt D. Mali

GEOGRAPHY SOLUTION

1. **"C"** - September
2. **"C"** - Bukina Faso
3. **"B"** - Down cutting
4. **"B"** - Port – Harcourt
5. **"C"** - River Nile
6. **"A"** - Incised members
7. **"D"** - Octas
8. **"C"** - Cloud
9. **"C"** - Mohocivic discontinuity
10. **"B"** - Northeast
11. **"A"** - Chemozem
12. **"D"** - Xerophytes
13. **"D"** - It has several cataracts
14. **"C"** - West to East
15. **"C"** - When the sun is overhead on the Equator
16. **"A"** - Isohaline
17. **"D"** - Africa
18. **"A"** - 15º
19. **"A"** - Conurbation
20. **"D"** - Uvala
21. **"C"** - An underground network of caves and streams
22. **"A"** - Deflation hollow
23. **"C"** - Water
24. **"A"** - Sunday, July 14th 2013
25. **"A"** –Sudan

GEOGRAPHY 2012

1. Which of the following is not a rapid type of mass movement? (a) solifluction
(b) rockfall (c) mudflow (d) debris slide
2. A front in which warm air is overtaken and lifted off the ground by cold air in a depression is known as (a) cold front (b)

warm front (c) mixed front (d) occluded front

3. The earth's atmosphere is said to be stable (a) when the environmental lapse rate is greater than the adiabatic lapse rate (b) when the adiabatic lapse rate is greater than the environmental lapse rate (c) when the environmental lapse rate is equal to the adiabatic lapse rate (d) when the environmental lapse rate is not equal to the adiabatic lapse rate.
4. Environmental lapse rate is (a) the rate of temperature changes of an air parcel undergoing vertical displacement (b) the rate of temperature decrease of an air parcel undergoing horizontal displacement (c) the actual rate of decrease of temperature with increase in altitude at a given place at a given moment (d) the rate of horizontal temperature gradient of an air parcel
5. What is the Centigrade equivalent of 95^0 Fahrenheit? (a) 139 ^{0}C (b) 37.2 ^{0}C (c) 35 ^{0}C (d) 3.6 ^{0}C
6. Which one of the following states in Nigeria would a person NOT pass over by crow fly (direct distance) from Ibadan to Makurdi? (a) Osun (b) Ondo (c) Kogi (d) Enugu
7. The latitude which marks the limits of the overhead sun's apparent movement is (a) 0^0 (b) 66.5^0 (c) 32.5^0 (d) 23.5^0
8. Which of the following features is not commonly associated with a river at the floodplain stage? (a) meanders (b) Levees (c) braided channel (d) knickpoint
9. The world's richest fishing ground are found (a) on the continental shelves (b) in oceanic deeps (c) in big and fast-flowing river (d) in river estuaries
10. A deflation hollow is produced by (a) (a) River erosion (b) Water action in a limestone area (c) Wind erosion in deserts (d) wave erosion on the coast
11. Akosombo dam is on the River (a) Niger (b) Nile (c) Congo (d) Volta
12. Which of the following is not characteristic of the international Date Line? (a) The line is approximately along the 180^0 meridian (b) The line has a zigzag shape (c) local time is the same on either side of the line (d) a traveler gains a day when crossing the line from the west to the east
13. Which of the following African cities is situated near the confluence of rivers? (a) Niamey (b) Freetown (c) Cairo (d) Khartoum
14. Ferrel's law states that winds deflect to the (a) left in the northern hemisphere and to the right in the southern hemisphere (b) right in the northern hemisphere and to the left in the southern hemisphere (c) right in both hemispheres (d) left in both hemispheres
15. Given an environmental lapse rate of 0.650C per 100 metres, a place with sea level temperature of 400C and 2500 meter above the sea level will approximately have a temperature of (a) 31.25^0C (b) 23.75^0C (c) 32.5^0C (d) 28.25^0C
16. At the summer solstice (June 21), which of the following latitudes will have the shortest night? (a) 30^0N (b) 30 ^{0}S (c) 50 ^{0}N (d) 50 ^{0}S
17. Which of the following landforms results from wind deposition? (a) Playa (b) Barchan (c) Bajada (d) Fan
18. Which of the following farming practices can be used to check soil erosion? (a) contour ploughing (b) ploughing of land upslope (c) bush burning (d) shifting cultivation
19. Aeolian erosion refers to the work of (a) Plants (b) Wind (c) Ice (d) Running water
20. The difference in the readings on the dry and wet bulb thermometers used to determine (a) relative humidity (b) temperature range (c) evaporation (d) transpiration
21. Which of the following scales should show the greatest amount of detail on a map?
(a) 1:50,000 (b) 1:500,000 (c) 1:20,000 (d) 1:200,000
22. Sandstone is metamorphosed into (a) Slate (b) Quartzite (c) Schist (d) Graphite
23. In humid areas, farmers add lime to the soil to (a) Reduce acidity (b) Act as fertilizer (c) Facilitate the absorption of nutrients (d) encourage the farmers
24. In the hydrological cycle, the transfer of water from the earth's surface to the atmosphere is by (a) evaporation only (b) transpiration only (c) evapotranspiration
(d) condensation
25. In which of the following countries was the shaduf method of irrigation first practiced? (a) Niger (b) Egypt (c) Sudan (d) Ghana

SOLUTION GEOGRAPHY 2012

1.	A	2.	D	3.	4.		
	C	5.	C	6.	D		
7.	B	8.	A	9.	A		
10.	C	11.	D	12.	C		
13.	D	14.	B	15.	B		
16.	C	17.	B	18.	A		
19.	B	20.	A	21.	C		
22.	B	23.	A	24.	C		
25.	B						

AGRICULTURAL SCIENCE QUESTIONS

1. Which of the following is the reason for carrying out quarantine programmes in agricultural? (a) Improving the genetic qualities of planta and animals (b) Ensuring that importers obtain valid license (c) Adapting plants and animals to new areas (d) Screening and isolating imported disease stock

2. One of the following is not an agricultural programe in Nigeria (a) Operation feed the Nation (b) Directorate of Good, Road and Rural Infrasturcure (c) Farm Settlemetn Scheme (d) Green Revolution
3. The Ginnery Industry uses which of the following raw materials? (a) pulp wood (b) oil seeds (c) sorghum (d) cotton
4. Land for wildlife (a) must not be good for agriculture and forestry (b) should be a land sparse with population (c) must be a plain, hill or vally (d) all of the above
5. The humpless breed of cattle which also possesses short horn is (a0 Gudali (b) White Fulani (c) Red Bororo (d) Muturu
6. Which of the husbandry practices is not carried out on pig farming? (a) Castration of excess male (b) Application of iron supplements (c) numbering by ear notching or tattooing (d) Shearing of hairs of the skin
7. In the prevention of yam beetles, the setts should be treated with the following except (a) Aidrin (b) Dieldrin (c) Nuvacron 40 EC (d) Aidrin and Dieldrin
8. Selection of breeding males based on the performance of their female relative is known as (a) Sib selection (b) Individual selection (c) Tandem selection (d) Progeny testing
9. Foot and mouth disease is caused by (a) Parasite (b) Bacteria (c) Virus (d) Fungi
10. The Leghorn and the Cornish breeds of fowl are well noted for their (a) High egg and meat production respectively (b) High meat and egg production respectively (c) Feathers (d) None of the above
11. The mechanical methods of weed control include the following except (a) Hand pulling (b) Flooding (c) Use of parasite (d) Burning
12. Newcastle disease is caused by (a) Virus (b) Bacteria (c) protozoa (d) None of the above
13. The growing of plants under could conditions is known as (a) Etiolation (b) Vernalization (c) Survival (d) Condensation
14. Sources of potassium include the following except (a) Organic manure (b) Oil-palm bunch (c) Mica (d) All of the above
15. One of the following is a symbiotic organism in Nitrogen fixation (a) Clostridium (b) Azotobacter (c) Rhiobium (d) Nitrobacter
16. Excess Nitrogen in the soil leads to (a) stunted growth (b) Leaves tending to drop (c) Encouraging root development (d) Delayed fruiting and ripening of fruits
17. The application of water to sub-soil through perforated pipes is known as (a) flooding (b) surface irrigation (c) subsurface irrigation (d) sprinkler irrigation
18. Farm networth is the difference between 9a) Cost and returns (b) Assests and liabities (c) Profit and loss (d) credit and debit
19. A measure to ameliorate soil acidity is (a) Addition of organic manure (b) Addition of lime (c) Addition of tertilizer (d) Learching
20. The following are breeds of cattle in Nigerian except (a) Holstein (b) White Fulani (c) Sokoto Gudali (d) Muturu
21. Modern genetics has led to the development of (a) transgenic plants (c) genetically engineered plants (c) genetically modified organisms (d) all of the above
22. The main abiotic stresses of maize in Africa are: (a) drought and rust (b) drought and lodging (c) lodging and bores (d) streak and wilting
23. The following are leguminous crops except (a) cowpea (b) groundnuts (c) maize (d) soya beans
24. One of the following is an example of written agricultural extension chennels: (a) TV programmes (b) Focus Group Discussion (c) Radio Broadcasting (d) Handouts

AGRICULTURAL SCIENCE ANSWERS

1. Quarantine involves separating the diseased from the healthy to prevent the disease from spreading. Examples of quarantine able diseases are scabies in sheep, erysipelas in pigs, white rot in onions and potato cyst
   The Correct option is **D**
2. Operation Feed The Nation, Green Revolution and Farm settlement scheme are all agricultural programs.
   The correct option is **B**
3. What first comes to your mind any time you hear the word **"GIN"**? I guess you take it to always mean a strong colorless alcoholic beverage.
   A gin can also mean a machine that separates the seeds from raw cotton fibres. The ginnery industry therefore uses cotton as a raw material
   The correct option is **D**
4. The land to be utilized for wildlife must not be good for agriculture and forestry. It may be a plain, hill or valley. Additionally, there should be sparse population where it is to be located.
   The correct option is **D**
5. The humpless breed of cattle which also possesses short horn is the Muturu. In fact, the word Muturu is a Hausa word for humpless cattle.
   The correct option is **D**
6. Pig husbandry involves processes such as:
   i. Castration of excess males
   ii. Tail docking (to reduce tail biting)
   iii. Teeth clipping (to reduce injuring their mother nipples)
   iv. Ear marking and tattooing (for litter identification)

v. Injection with iron solution (this is because sow's milk is low in Iron)
The correct option is **D**

7. In the prevention of yam beetles, yam setts should be treated with insecticides such as a dieldrin, endosulfan, carbofuran or aldrin. As a matter of fact, aldrin and dieldrin are broad spectrum insecticides which can control other insects such as wood borers, termites, grasshoppers and textile pests.
The correct option is **C**
8. Progeny testing is a test for selective breeding of an individual's genotype by looking at the progeny produced by different mating. It can be used to evaluate characteristics that cannot be measured directly from a particular gender. For example, a bull cannot be assessed for milk production but performance of its female offsprings can be used to determine the use of the animal for the future crosses.
The correct option is **D**
9. Foot – and – mouth disease (**FMD**) is a highly contagious viral infection of all cloven hooved animals such as cattle, sheep, pigs, goats, deer and buffalo. It is characterized by (I) development of fluid – filled lesions (ii) reduced appetite (iii) weight loss (iv) reluctance to stand or move around (v) poor weight gain (vi) reduced milk yield. It is caused by a picornavirus.
The correct option is **C**
10. The leghorn breeds of fowl are excellent layers of white eggs (around 300 per year). They mature quickly but are not large birds. They are therefore preferred for egg- laying. The Cornish breeds, on the other hands, are deceptively big and heavy. They are therefore known for meat production.
The correct option is **A**
11. The use of parasites in weed control is not a mechanical but a biological method.
The correct option is **C**
12. The Newcastle disease is a highly contagious disease in many species of birds. It is characterized by gasping, coughing, depression, inappetence, muscular tremors, drooping wings, twisting of head and neck, paralysis, swelling of the tissues around the eyes and reduced egg production. It is caused by a paramyxovirus.
The correct option is **A**
13. Vernalization is the artificial exposure of plants to low temperature in order to stimulate flowering or to enhance seed production.
The correct option **B**
14. Mica, a mineral containing potassium silicate is a source of potassium. Also, manure, compost and other organic materials are sources of potassium.
The correct option is **B**
15. Rhizobium leguminosarium is a symbiotic organism in nitrogen fixation. It is found in the root nodules of leguminous plants.
The correct option is **C**
16. Too much nitrogen may cause excessive vegetative growth and poorly colored fruit with delayed ripening, aside other effects.
The correct option is **D**
17. Subsurface irrigation (also known as subirrigation) is a method of providing water to plants by raising the water carrying moisture to the root zone by perforated underground pipe
The correct option is **C**
18. Net worth is the difference between what is owned and what is owed. It is the value of all assets minus the total of all liabilities.
The correct option is **B**
19. The most common way to ameliorate soil acidity is to add pulverized limestone to the soil. Magnesium carbonate is also an excellent soil-acid neutralizer.
The correct option is **B**
20. White Fulani, SokotoGudali and Muturu are breeds of cattle in Nigeria. Holstein is not found in Nigeria but in Europe, North America, North Holland, and Friesland and in Northern Germany.
The correct option **A**
21. Development of transgenic plants, genetically, engineered plants and genetically modified organisms are all from modern genetics.
The correct option is **D**
22. The main abiotic stresses of maize in Africa are environment lodging, low nitrogen, drought, salinity and aluminum toxicity.
The correct option is **B**
23. Cowpea, groundnuts and soya beans are leguminous plants while maize is a cereal.
The correct option is **C**
24. The provision of handouts is an example of written agricultural extension channel. The option is **D**

2015 OAU DIRECT ENTRY QUESTIONS
COMMUNICATION AND LANGUAGE SKILLS
SECTION A

1. Mr. Ajibade's colleague **stole his thunder** at the occasion. This means that his colleague…….
   A. Made a lot of noise about Mr. Ajibade's achievements at the occasion
   B. Took the credit for what Mr. Ajibade did at the occasion

C. Boasted about Mr. Ajibade's achievements at the occasion
D. Refused to talk about Mr. Ajibade achievements at the occasion

2. My Uncle, later realized that he was put forward by his party as a stalking horse. This means that my uncle realized …..
   A. His party expected him to perform better than he did
   B. His party knew had no real chance of winning
3. His service to the nation can best be described as a **Yeoman's Service**. This means he did …..
   A. An excellent work
   B. A very hard work
   C. A thankless work
   D. A servant-like work
4. After long illness, finally Bisi **turned up the corner** and now he is completely fine. This means Bisi …..
   A. Went into a comma and was revived
   B. Passed a critical stage
   C. Got expert medical attention
   D. Had a miraculous encounter

Choose from the words or groups of words below each sentence, the word best completes it

5. They set themselves a rather …… task, but they were immensely resourceful.
   A. Scaring
   B. Easy
   C. Daunting
   D. Excruciating
6. As they were going to take their seats in the plane, the attendants were asking for their ….. pass
   A. Travelling
   B. Boarding
   C. Passport
   D. Permit
7. They sold their house for a fair…..
   A. Rate
   B. Cost
   C. Charge
   D. Price
8. I was disappointed when I hear mice ….. in his living room.
   A. Chirping
   B. Screeching
   C. Squeaking
   D. Tweeting
9. The ….. of chains announces to us that the inmates are being let out of the prison for early morning labor in the farm
   A. N
   B. J
   C. B
   D. H
10. In the estate where I live, I am always guided by the principle that **good fences makes good neighbor**. This means…..
    A. My fences must not obstruct my neighbor
    B. I must not be too secluded from my neighbor
    C. I must respect my neighbor's property
    D. I must always be interested in what happens to my neighbor
11. Find the proverb that is closest in meaning to the sentences given below

Become wise after having an unpleasant experience
   A. Look before you leap
   B. A burnt child dreads fire
   C. A stitch in time saves nine
   D. A bird in hand is worth two in the such

Read the following passage carefully and answer the questions that follow, choosing from the options provided.

It was an extremely and event that occurred two days after we had entered on our travels. We were passing through some high grass. Suddenly, we found ourselves in front of a lion which was eating an African Antelope. Romer who happened to child some yards foremost of the three, was so alarmed that he fired at the animal which we had never to do, as it was folly to enrage so powerful a beast when our party was so small.

The lion was lightly wounded. He gave roar that might have been heard for a mile, sprang up on Romer, and with one blow of his paw knocked him off the saddle into the bushes. Our horses which were frightened, wheeled round and fled for the animal was about to attack us. But really he did not make one bound in our direction. We could not pull up until we had gone half a mile and when we did, we saw the lion had torn down the horses which Romer had ridden, and was dragging away the carcass to the right, without any apparent effort on his part. We waited till he was well off and then rode back to the spot where Romer had fallen.

12. Which of the following statement **is not true** of the passage
    A. Romer fell off the horse after the attack by the lion
    B. All the other horses escaped except the one Romer was on
    C. The lion attempted to attack the group, did not succeed
    D. The lion killed the horse Romer was riding
13. Why did Romer fire at the lion?
    A. He was frightened
    B. He understood that the party was small
    C. He was the foremost of the three
    D. He wished to kill the animal
14. The horses fled away because
    A. They heard the roar of the lion
    B. They were frightened by the lion
    C. The lion sprang upon them
    D. The lion sprang upon Romer

15. Why did the group feet they should not fire at the animal
    A. Because they were too close
    B. Because the animal did not attack them
    C. Because they were not many
    D. Because the animal was enraged
16. Why did the group feel they should not fire at the animal
    A. Because they were too close
    B. Because the animal did not attack them
    C. Because they were not many
    D. Because the animal was enraged

Identify the option that is **the exact opposite** of the word in the following sentences:

17. Which of the following means exactly the opposite of **reputable**?
    A. Famous
    B. Notorious
    C. Popular
    D. Renowned
18. If something is not **elastic**, then it is …?
    A. Flexible
    B. Smooth
    C. Fixed
    D. Adaptable
19. Rahman is **laid-back**, while his friend, being the exact opposite, is …
    A. Slow
    B. Restless
    C. Confident
    D. Strong
20. I opted for a **serene** village as opposed to a …. City
    A. Crowded
    B. Large
    C. Rowdy
    D. Dirty
21. Muhammed is very principled unlike his neighbor who is very …..
    A. Disorganized
    B. Jovial
    C. Disciplined
    D. Undisciplined
22. There has always been migration of labor from **distressed** economies to …. Nations.
    A. Industrialized
    B. Troubled
    C. Growing
    D. Developed

Which of the following words **are closest in meaning** to the highlighted words?

23. The insurgents are simply **barbaric**
    A. Uncivilized
    B. Harsh
    C. Unruly
    D. Stubborn
24. I found the story of the politician very **boring**.
    A. Crude
    B. Uninteresting
    C. Short
    D. Snappy
25. Many conference participants **grumbled** about the delay in commencing the ceremony.
    A. Complained
    B. Argued
    C. Talked
    D. Spoke
26. The lawmakers **eventually** reached a consensus on how to allocate the offices
    A. Suddenly
    B. Abruptly
    C. Unexpectedly
    D. Finally
27. I went to the **premier** secondary school in my town
    A. Popular
    B. Biggest
    C. First
    D. Finest
28. The day Bola wedded remains very **memorable** to her
    A. Different
    B. B. important
    C. Interesting
    D. Insignificant

Which of these sentences is correctly punctuated?

29. A. I cannot believe that Ade won the election?
    B. I cannot believe that Ade won the election!
    C. I cannot believe that Ade won the election,
    D. I cannot believe that Ade won the election:
30. A. John, where is my book?
    B. John where is my book?
    C. John; where is my book?
    D. John where is my book?
31. A. What, I cannot fail that course.
    B. What! I cannot fail that course.
    C. What: I cannot fail that course.
    D. What; I cannot fail that course
32. A. These are the things I need: a bag; a towel and an umbrella.
    B. These are the things I need, a bag, a towel and an umbrella.
    C. These are the things I need; a bag. A towel and an umbrella
    D. These are the things I need! A towel and an umbrella
33. A. What exactly is your problem!
    B. What exactly is your problem?
    C. What exactly is your problem.
    D. What exactly is your problem:
34. A. James is friend, a colleague and a neighbor.
    B. james is a friend, a colleague and a neighbor.
    C. James is a friend a colleague and a neighbor
    D. James is a friend; a colleague and a neighbor
35. A. What he said I do not like.
    B. What, he said, I do not like
    C. What he said, I do not like

D. What he said, I do not like

SECTION B

Complete each of the following sentences with the appropriate words from the options provided.

1. I had difficulty locating the book. My mother placed it …..the table
   A. At B. inside C. under D. in
2. The daring robber jumped ….. the fence to escape being arrested.
   A. At B. over C. on D. in
3. What on earth are you explaining ….Him?
   A. With B. by C. to D. on
4. The new principal's notorious …. Coming late to school
   A. In B. on C. for D. about

Which of the options numbered A – D best completes the following sentences?

5. You are Tade's friend …..
   A. Aren't you?
   B. Isn't it?
   C. Won't you?
   D. Weren't you?
6. The students are very determined…
   A. Are they?
   B. Aren't they?
   C. Won't they?
   D. Isn't it?
7. The new administration is not serious…
   A. Isn't he?
   B. Is he?
   C. Isn't it?
   D. Will he?
8. We are not eating tonight…..
   A. Aren't we?
   B. Can we?
   C. Won't we?
   D. Are we?
9. Lagos is very far from Abuja….
   A. Isn't it?
   B. Is it?
   C. Aren't they?
   D. Isn't they?
10. We don't have to travel late….
    A. Don't we?
    B. Do we?
    C. Aren't we?
    D. Won't we?

GENERAL PAPER

SECTION A

1. The current Chief Justice of Nigeria is
   A. Justice Dahiru Mustapha
   B. Justice AlomahMuktar
   C. Justice MohammedUwais
   D. Justice IdrisKutgi
2. Nigeria is bordened in the South by
   A. Gulf of Guinea and the Atlantic Ocean
   B. Gulf of Guinea and the Pacific Ocean
   C. Pacific and Atlantic Ocean
   D. None of the above
3. The bound of the Nigeria Nation is found in
   A. Peace and unity
   B. Freedom, peace and unity
   C. Peace and Justice
   D. Justice and freedom
4. The liberation stadium in Nigeria is in
   A. Abuja FCT
   B. Port Harcourt
   C. Lagos
   D. Kano
5. The last creation of state exercise in Nigeria was done under the regime of
   A. General Murtala Mohammed
   B. General Ibrahim Babangida
   C. General SanniAbacha
   D. General Yakubu Gowon
6. The last Governor General of colonial Nigeria was
   A. Sir John Macpherson
   B. Sir Arthur Richard
   C. Sir James Robertson
   D. Sir Hugh Richard
7. NHIS stands for
   A. National Health Information Scheme
   B. National Housing Integral Scheme
   C. National Health Insurance Scheme
   D. National Housing Informational scheme
8. The last world female football championship was won by
   A. Japan
   B. Germany
   C. United states
   D. Spain
9. Which of the following is not an Economic agency in Nigeria
   A. Niger Delta Development Commission
   B. National Council on Privatization
   C. Federal Inland Revenue Service
   D. Independent Corrupt Practices Commission
10. In football, any foul committed outside the eighteen yards box attracts a
    A. Penalty
    B. Free kick
    C. Red card
    D. None of the above
11. The Nigerian Police went on strike for the first time in
    A. Feb. 2002
    B. Feb. 2004
    C. Feb. 2001
    D. Feb. 2006
12. NIMASA means
    A. Nigerian Maritime Administration and safety Agency
    B. National Institute for Maritime Administration and safety
    C. National Immigration and safety Agency
    D. Nigerian International Marine Administration and safety

13. Which of the following does not belong in the group?
    A. V.O.A
    B. V.O.N
    C. W.H.O
    D. B.B.C
14. The current sovereign of the Vatican City state is
    A. Pope Benedict XVI
    B. Pope Francis
    C. Pope Gregory III
    D. None of the above
15. Lagos is nicknamed "Centre of Excellence" while Yobe is
    A. Jewel of the savannah
    B. The power state
    C. Pride of the Sahel
    D. Home of hospitality
16. The first female senator in Nigeria is
    A. LateefahOkunnu
    B. FunmilayoRansomeKuti
    C. WuraolaEsan
    D. Margaret Ekpo
17. In legal terms, 'Exparte" means
    A. Within the power
    B. Put forward
    C. On behalf of
    D. As evidenced
18. John Adrian Shepherd-Barron invented the
    A. Automated teller machine
    B. Twitter
    C. You tube
    D. Locomotive engine
19. The scientist who invented the thermometer is
    A. George Stephenson
    B. Michael Faraday
    C. Galileo Galilee
    D. Ferdinard Magellan
20. The speed of wind is measured with a
    A. Wind Vane
    B. Barometer
    C. Nanometer
    D. None of the above
21. The law of gravity was discovered by
    A. Michael Faraday
    B. Sir Isaac Newton
    C. Sir James Watt
    D. Galileo Galilee
22. America was discovered in 1492 by
    A. Abraham Lincoln
    B. George Eastman
    C. Thomas Edison
    D. Christopher Columbus
23. The current Secretary General of the United Nation is
    A. Kofi Annan
    B. Banki Moon
    C. Javier Prezez
    D. Boutros Ghali
24. The annual Argungu fishing festival in Nigeria is celebrated in
    A. Taraba State
    B. Sokoto State
    C. Kebbi State
    D. Bauchi State
25. The first to land and walk on the moon is
    A. Neil Armstrong
    B. Yuri Gangarin
    C. Madelein Albright
    D. Ferdinand Magellan
26. The second world war lasted from
    A. 1914 – 1918
    B. 1936 – 1945
    C. 1930 – 1934
    D. 1939 – 1945
27. The largest of the planets
    A. Earth
    B. Pluto
    C. Jupiter
    D. Uranus
28. The world's largest producer of Meat
    A. Nigeria
    B. China
    C. Great Britain
    D. Argentina
29. The largest and deepest sea in the world is the
    A. Red sea
    B. Mediterranean Sea
    C. Sea of Galilee
    D. None of the above
30. The Central Bank of Nigeria commenced operation in
    A. 1959
    B. 1960
    C. 1957
    D. 1964
31. The motto of Nigeria written in the coat of arms is
    A. Peace and Unity
    B. Unity, peace and progress
    C. Unity, Faith, Peace and progress
    D. Unity, Strength, Peace and progress\
32. The capital of Netherlands is
    A. Lisbon
    B. Colombo
    C. Hague
    D. Zurich
33. The maiden edition of the U-17 world cup was won by
    A. Nigeria
    B. Brazil
    C. France
    D. Saudi Arabia
34. The current world footballer of the year is
    A. Lionel Messi
    B. Chritiano Ronaldo
    C. Richado Kaka
    D. Gucho Ronaldinho
35. Stanford Bridge (Chelsea F. C), Anfield ____
    A. Manchester United
    B. Everton F.C.
    C. Sunderland F.C
    D. Liverpool F.C
36. The current ECOWAS chairman is

A. AlassanaQuattara (Cote D'Ivoire)
B. John Kufuor (Ghana)
C. MackySall (Senegal)
D. Paul Biya (Cameroon)

37. The smallest of the continents of the world is
A. Africa
B. Australia
C. South America
D. Antarctica

38. A place where important Government records are kept is called
A. Apiary
B. Archives
C. Library
D. Museum

39. Which of these does not belong to the South East Geo Political Zone in Nigeria
A. Delta State
B. Enugu State
C. Ebonyi State
D. None of the above

40. The first Nigerian to own a radio and television station is
A. Chief M.K.O Abiola
B. Chief BolarinwaAbioro
C. Chief Raymond Dokpesi
D. Dr. Raymond Njoku

SECTION B

1. The longest river in the world is
A. River Zambezi
B. River Niger
C. River Nile
D. River Benue

2. Who amongst the following is a Nobel Peace prize winner?
A. Professor Wole Soyinka
B. Archbishop Desmond Tutu
C. Chinua Achebe
D. Kwame Nkrumah

3. The longest living part of the human body is the
A. Heart
B. Kidney
C. Head
D. Brain

4. The Nigerian appointed Chief justice of Botswana was
A. Justice AdetokunboAdemola
B. Justice AkinolaAguda
C. Justice Mohammed Uwais
D. Justice Fatai Williams

5. The hurricane that ravaged the United states in 2005 was named
A. Hurricane Katrina
B. Hurricane Sylvester
C. Hurricane terror
D. Hurricane Osama

6. The first military coup in Nigeria was led by
A. Gideon Orka
B. Murtala Mohammed
C. BukarSukaDimka
D. Kaduna Nzeogwu

7. Osun state is the " State of living Spring while Abia State is
A. Land of Hope
B. The power state
C. God's own state
D. The people's paradise

8. The title of the traditional ruler of Igala land is
A. MoiFika
B. Moyegeso
C. Ata
D. Amayanabo

9. Which of the following is a domestic airport in Nigeria?
A. Port Harcourt International
B. Maiduguri International
C. Murtala Mohammed International
D. MallamAminu Kano International

10. The "amphibian battalion" is found in the Nigerian
A. Army
B. Navy
C. Air force
D. All of the above

OAU POST-UTME SYLLABUS

EGL 001 – INTRODUCTION TO ENGLISH LANGUAGE I

(a) Introduction to Essay/ Composition

1. Description of a composed text

(i) Unity of theme
(ii) Coherence
(iii) Cohesion

2. Features of Composed text

(i) Focus
(ii) Organization
(iii) Support and elaboration
(iv) Style
(v) Conventions

3. Writing a text

(i) Purpose of writing
(ii) Choosing a topic
(iii) Developing an outline
(iv) Paragraph development
(v) Identifying characteristics of target audience

4. Types of Essay/Composition

(a) Narrative Essay

(i) Form(s)

- Relating personal or imaginative experience;
- Giving a historical account of a person, place, event etc.,
- Narrating a story to support a position (proverb, philosophical view)

(ii) Features

Narration usually involves;
(i) Setting (place, time), (ii) Characters (peo0ple, actors) (iii) Actions, (iv) Use of flashback, (v) Tense usage usually past tense.

(b) Descriptive Essay

(i) Form(s)
Painting in words of persons, objects, scenes, event etc.

(ii) Features

- Emphasis more on showing rather than telling;
- There is always one dominant impression
- Selection of details to support dominant impression
- Reliance on concrete sensory details to support points.

(c) Expository Essay

(i) Form(s)

- Presentation, explanation or expounding of ideas, theories, beliefs;
- Presentation of a proposal on an issue, topic;
- Presentation of an analysis of something, event, issue;
- Presentation of commentary on an issue;
- Writing of an editorial
- Comparing and contrasting two things or event

(ii) Features

- Support ideas ordered logically and linked clearly;
- Use of a thesis (focus/controlling idea to establish control of content;
- Ideas receive adequate support to make them clear;
- Skilled use of language.

(d) Argumentative Essay

(i) Form(s)

- Arguments on both sides of an issue (open)
- Arguments on one side of an issue (closed)

(ii) Features

- Logical presentation of ideas, viewpoints etc;
- Providing sufficient information to support or rebut a position,
- Careful selection of what to present;
- Anticipating counter arguments;

B. Grammar I

1. **The Sentence:** Definitions using various criteria semantic, structure and orthography, with many examples of sentence.
2. **Structural Classification of Sentence:** simple sentence, complex sentence, compound sentence, compound complex sentence, multiple-complex sentence.
3. **Functional Classification of Sentence, namely:** Statements, commands, questions and exclamations.
4. **Basic simple sentence and various sentence parts**
   - Explication of basic simple sentences
   - Sentence part e.g., subject, verb, direct object, indirect object, and subject complement and object complement.
5. **Elements on the grammatical rank scale:** sentence, clause, phrase/group, word and morpheme.
6. **Formation of non-simple sentences**, i.e. variants of the basic simple sentence, formed through the process of *movement, deletion and insertion*. e.g. negative sentence, passive sentence, polar question, *wh* – question etc.
7. **Formation of Non-simple sentences** combination of simple sentences or main clauses to form larger sentences e.g. compound sentence, complex sentence, compound complex sentence, multiple-sentence and multiple-complex sentence.
8. **Structural types of clauses**
   - Noun clause
   - Adjectival clause
   - Adverbial clause

C. Word Formation, Collocation and The English Dictionary

1. **The nature of Language**
2. **Vocabulary Development**
3. **World formation Processes**
   a. Clipping
   b. Blending
   c. Acronym

d. Borrowing
e. conversion

4. **Word Types**
   - Simple, complex, compounds
   a. Morphemes
   b. Lexical Morphemes
   c. Grammatical Morphemes
   d. Idioms
5. **Collection**
   a. Fixed
   b. Non-fixed
6. **The Dictionary**
   a. Types
   b. Functions
   c. Merits
   d. Demerits

D. The Study of Meaning in English

1. **What is Meaning**
   - Not easily definable; a simple term
   - Ambiguous, vague, polysemous
2. **Types of Meaning**
   a. Linguistic Meaning, Phonological, Orthographic, Morphological, Syntactic, Lexical
   b. Contextual Meaning
   Verbal, Cultural, Situational, Stylistic:
   - Meaning defined by theme/subject matter
   - Meaning defined by tenor
   - Meaning defined by mode
   c. Concepts of Meaning
   References, Sense, Componential Analysis, Concept, Denotative and Connotative Meaning, Cognitive, Interpersonal, Textual Meaning, Ambiguity v. Vagueness, Nominalism v. Realism.

E. Common Errors

1. **Introduction: Reasons for common errors in students' writing**
2. **Common Spelling Error**
   - Wrong association of spelling with pronunciation\.
   - Spelling mistakes attributable to poor pronunciation.
   - Words commonly mispelt
   - Unnecessary duplication of letters
   - Writing compound words as separate words
   - British and American spellings
   - Some spelling rules
3. **Errors in choice of Lexical Items**
   - Words with similar sounds but different spellings and meanings
   - Words with same spellings but different meanings
   - Confusion in the use of verbs and nouns
   - Errors in the choice and ordering of adjectives
   - Errors in plural formation
   - Errors in the use of prepositions
4. **Common Errors in Grammar**
   - Common errors in the use of nouns
   - Common errors in the use of verbs
   - Common errors in the use of adverbs and adjectives
   - Common errors in the use of pronouns
   - Common errors in the use of tense
   - Common of tenses in conditional sentence

The Concept of Nigeria English at the Levels of Phonology, Lexis and Syntax.

EGL 002 INTRODUCTION TO ENGLISH LANGUAGE II

A. Essay/Composition

1. **Report Writing**
   a. The Narrative Report
   (i) Topic, audience, situation
   (ii) Features of narration
   (iii) Emphasis on fact
   (iv) Accuracy of information: persons, time, place
   b. The Project Report (based on empirical data)
   (i) Choosing a topic,
   (ii) Developing site of data collection;
   (iii) Data collection
   (iv) Data collection
   (v) Data analysis
2. **Speech Writing**
   - The topic, audience and situation
   - The contact: direct communication with audience
3. **Letter Writing**
   Form(s)
   - Formal letter, also called business letter (to business establishments, government, institution etc);
   - Informal letter, also called friendly or personal letter (to parents, relations, friends etc.
   - Features
   - Presentation, in case of a fonnal letter opening and closure, introduction and body of letter; use of language is formal.
   - Presentation, in case of informal letter introduction, body, conclusion and language not as in case ofa formal letter.
   - Topics in narrative, descriptive and expository types of composition form the content bases for letter writing: can be narrative, descriptive or expository depending upon topic and purpose of writing.

B. Comprehension and Summary

1. **Reading**
   a. Efficient reading,
   b. Academic reading,
   c. Learner centred reading difficulties,
   d. Text-centred reading difficulties.
2. **Mental Processes during Reading**
   a. Flexibility in reading,
   b. Determining reading speed,
   c. SO3R strategy

3. **Comprehension**
    a. Identifying the main ideas,
    b. Topic Sentence,
    c. Identifying supportive details.
4. **Techniques Authors use to Develop Main Ideas**

a. Definition, b. Repetition, c. Examples and Illustrations, d. Justification.

5. **Summarization Skills**

A. Definition, b. Application

6. **Formation ofAnswers to Examination Question**

C. Grammar II
   a. Revision of First Semester course (EGL 001).
   b. The English Verbal Group
   c. Tense, aspect and modality
   d. Concord in the English language
   e. The English nominal group
   f. The Adverbial Group
   g. Sentence Connection in English
   h. Revision

D. The Sound Pattern ofEnglish
   1. The Vocal Organs
   2. The Segmental Sounds: Consonants, Vowels
   3. The English Syllable
   4. Word and Sentence Stress
   5. Intonation in English

E. Punctuation In English

Use of capital letters, the full stop, the comma, the semi-colon, the colon, the inverted comma, the question mark, the dash and the hyphen.

MATHEMATICS

Objectives

In recent times, the gap between the Secondary School Mathematics syllabus and First Year University mathematics has widened. This has led to very poor performance of students in their First Year Mathematics in the University.

The Pre-Degree Mathematics syllabus is designed to bridge this gap and to provide a solid basic foundation necessary for an average student to cope with First Year mathematics in the university. The syllabus is specifically designed for students offering courses in the areas of Science, Social Sciences, Agriculture and Engineering. It is hoped that a good coverage of the syllabuswill remove the missing link and provide adequate pre-requisite to the first year undergraduate university syllabus in mathematics.

COURSE OUTLINE

MTH 001 - FUNDAMENTALS IN MATHENIATICS 1:

Numbers and Basic Arithmetic Operations Integer, real, rational and natural numbers. Number Bases Addition, subtraction, multiplication and division in different bases. (Bases 2-12). Conversion from one base to another. Fractions, Decimals and Approximations. Rates, Ratio and Proportions.

Factorisation of Polynomial Expression

Quadratic equations and Perfect Squares. Simultaneous equations in two and three unknowns. Elementary properties of quadratic functions. Functions of roots of quadratic equations. Factor and Remainder Theorems. Polynomials. Variations and Inequalities. Sequences and Series Arithmetic and Geometric Progressions. Graphs of linear, quadratic and Cubic functions.

MTH 002 FUNDAM ENTALS IN MATHEMATICSI1I

Euclidean Geometry Lines, triangles and polygons. Circles Arcs chords, segments, sectors of circles. Theorems on circles. Measuration Perimeter, Areas and Volumes of solid figures: triangles, rectangles, circles, cylinder, cone, pyramid and prism. Coordinate Geometry; Rectangular and Cartesian coordinates; Mid Point, gradient, distance between two points, equations of a line in the forms $y = mx + c$, $ax + by + c = o$. division of line internally and externally into ratios. Parallel and perpendicular lines. Basic Trigonometry; Sine, Cosine and tangent of an angle. Sine and Cosine formulae. Pythagoras Theorem.

Differentiation; Application of derivatives to rates of change, maxima and minima. Integration of algebraic, trigonometric and exponential functions. Application of integration to areas and volumes.

Introduction to Statistics and Probability; Methods of collection and representation of data. Measures of location - (Mean, median, mode) and measures of dispersion - mean deviation, quartile deviation and standard deviation.

Probability - addition and multiplication of conditional probability. Permutation and combination - application to probability.

CHE 001 – CHEMISTRY I

Fundamental Concepts in Physical Chemistry

1. **Separation of Mixture and Purification or Chemical Substances**
   (a) Pure and impure substances. Melting and boiling points should be mentioned as criteria for purity of chemical substances. Elements, compounds and mixtures. Definition of (i) an element (ii) a compound should be known.
   (b) Production of a mixture by mixing sulphur powder and iron fillings in any proportion; Production of a compound e.g. FeS by mixing iron fillings and sulphur powder in the ratio 28:16 and later heating. Discussion of the differences between a mixture and a compound should be known.
   (c) Separation processes: e.g. by evaporation, simple and fractional distillation, sublimation, filtration, crystallization, precipitation and chromatography.

2. **Atomic Structure, Periodic Table and Nuclear Chemistry**
   (a) Short account of Dalton's atomic theory; Crooke's and J.J. Thompson's experiment. Outline of Ratherford's description of alpha particle scattering and deductions there from.
   (b) Atomic number/proton number; number of neutrons; isotopes, atomic mass. Calculation of relative atomic mass from various isotopic masses. Relative atomic mass and relative molecular mass based on carbon 12 scale.
   Quantum Number, Electronic Energy Levels. Experimental evidence and interpretation of line spectra (quantitative treatment only). Arrangement of electrons in the main and sub-energy levels; shapes of s and p – orbitals only. Rules and principles of filling-in electrons – Aufbau Principles, Hund's Rule of Maximum Multiplicity and Pauli Exclusion Principle. Abbreviated and detailed electronic configuration in terms of s, p and d orbitals from hydrogen to zinc. Mention should be made of the old system e.g. 2, 8, 1 for sodium.

   Periodic Chemistry
   Nuclear Chemistry
   (i) Types of nature of radiations.
   Distinction between ordinary reactions and nuclear reactions. Charges, relative mass and penetrating power of radiations. Balancing of simple nuclear equations.
   (ii) Half-life as a measure of stability of nucleus. Quantative treatment of half-life.
   (iii) Nuclear reactions, Fission and Fusion in nuclear reactions. Natural and artificial radioactivity. Detection of radioation by Gieger-Muler counter or cloud chamber. Generation of electricity; atomic bomb.
   (iv) Effects and application of:
   Carbon dating, use of radioactivity in agriculture, medicine, industry and research.
   Periodicity of the elements: Periodic Law: Trend in periodic properties down the group and across a period. Electronic configurations leading to group and periodic classifications. Periodic properties for the first 18 elements: atomic size, ionic size, ionization energy; electron affinity, electronegativity.

3. **Bonding and Shapes of Molecules**
   Combining power, electrovalency and covalency; The electronic configuration of elements and their tendency to attain the noble gas structure. Electronic structure of molecules; bonding, metallic – bonding and coordinate bonding. Vanderwall's forces. Hybridization of orbitals limited to Sp^3, Sp^2 and sp. Shapes of simple molecules; Linear (H_2, O_2, Cl_2, HCl and CO_2); pyramidal (NH_3); tetrahedral (CH_4); angular (H_2O). Properties of covalent compounds compared with those of electrovalent compounds.

4. **Stoichiometry and Chemical Ractions**
   (a) Symbols; Formulae and Equations, Calculations involving formulae and equations will be required. Mass and volume relationships in chemical reactions and the stoichiometry of such reactions as: (i) precipation, (ii) evolution of gases (iii) displacement of metals (iv) formation of reducation of metal oxides.
   Emperical and molecular formulae; chemical equation
   Law of chemical combination:
   Law of conservation of mass;
   law of constant composition;
   Law of multiple proportion
   (b) Amount of substance and mole ratios
      (i) Mass and volume measurements;
      (ii) The mole as a unit of measurement; avogadro constant as the number of atoms in 12.00g of carbon – 12;
      (iii) Molar quantities and their uses;
      (iv) Mole of electgrons, atoms, moledules, formula units etc.

   Use of mole ratios in determining stoichiometry of chemical reactions. Simple calculations to determine number of entities, amount of substance, mass, concentration, volume and other quantities.

5. **Kinetic Theory of Matter:**
   (a) Postulates in the kinetic theory of matter. Application of the theory to explain the:
      (i) Nature of solids, liquids and gases
      (ii) Changes of state of matter.
      Change of state of magger should be explained in terms of movement of particles. It should be emphasized that randomness decreases (and orderliness increases) from geaseous state of liquid state to solid state.
      (iii) Diffusion of gases. Diffusion should be associated with Graham's Law. Ammonia and hydrochloric acid gases traveling at a meeting point shoul be used to explain (in terms of simple calculations) diffusion.
   (b) The Gas Laws
   Charles, Boyle's, Dalton's, Graham's, Avogadro's Laws and the ideal gas equation. Qualitative explanation of each of the gas laws using the kinetic model. Mathematical relations based on the laws. Molar volume of a gas = $22.4dm^3$ at s.t.p. The general gas equation: $\frac{PV}{T} = K$

6. **Energy and Energy Changes**
   (a) Energy changes accompanying physical and chemical changes. Simple calculations involving chemical reactions. Dissolution of substance in/or reactions with water e.g. Na, NaOH, K, NH_4Cl, positive ΔH or energy absorbed for endothermic reactions. Negative ΔH or energy released for exothermic reactions. Simple calculations involving ΔH based on energy profile diagram will be required. Types of heat changes: Heat of

combination, formation, dilution, reaction, neutralization etc; Hess's Law of constant heat, heat summation should be included.

(b) Entropy as order-disorder phenomon, simple illustrations like mixing of gases and dissolution of salt are required.
$\Delta G^o = 0$ as criterion for equilibrium

(c) Spontaneity of reaction
ΔG^o greater or less than zero as criteria for non-spontaneity and spontaneity respectively.

7. Acids, Bases and Salts

(i) Definitions, general characteristics, preparation, and properties of acids, bases and salts. Arrhenius, Bronsted Lowry and Lewis concept of acids and bases.

(ii) Acids, bases and salts as electrolytes. Weak and strong electrolytes. Evidendce from conductivity and enthalpy of neutralization.

(iii) Hydrolysis of salts. Quanlitative explanation of hydrolysis. Behavior of some salts (e.g. NH_4Cl, $AlCl_{3,}$ Na_2CO_3, $NaHCO_3$, CH_3COON_4) in water.

(iv) pH knowledge of pH scale, pH as a measure of acity and alkanility, ionic product of water. Simple calculation of pH.

(v) Acid-base indicators. Indicators as week organic acids or bases. Colour of indicator at any pH depends on the relative amount of acid and basic forms.

(vi) Acid-base titrations; knowledge of how acid-base indicators work in titrations. Choice of indicators in acid base titrations, standard solutions, molar solutions, molarity/volume relationship in a closed system (i.e. $M \times V =$ constant). Primary standards. Calculations involved in acid/base titrations.

(vii) Deliquencence, hydroscopy and effluorence. Water of crystallization.

8. Oxidation and Reduction Reactions.

(a) Oxidation and reducation processes
Oxidation and reduction in terms of
- (i) Addition and removal of oxygen and hydrogen
- (ii) Loos and gain of electrons
- (iii) Change in oxidation number/state

(b) Ocidation states
- (i) Fundamental definition of oxidation state
- (ii) Deduction of oxidation state of oxygen in O_2H_2O and H_2O_2
- (iii) Deduction of oxidation state of elements in HCl

(c) Oxidising and reducing agents:
Definition of oxidizing and reducing agents in terms of
- (i) Addition and removal of oxygen and hydrogen
- (ii) Loss and gain of electrons
- (iii) Test for oxidizing and reducing agents

(d) Redox equations:
Balancing redox equation by
- (i) Ion, electron or change in oxidation state method
- (ii) Electronic half equation method

9. Rates of Reactions and Chemical Equilibria

(a) Rate of a reaction
Give the definition of reaction rate
- (i) Factors affecting rates:
Physical state, concentration of reactants, temperature catalyst and medium. For gaseous systems, pressure may be used as concentration term. In studying the factors, the following example, amongst others can be used. The reaction between HCl and $Na_2S_2O_3$, HCl and marble in lump and in powdered forms; The decomposition of H_2O_2 or $KClO_3$ in presence and absence of MnO_2.
- (ii) Theory of reaction rates, collision theory and activation energy theory to be treated qualitatively only. Factor influencing collision (i.e. temperature and concentration). Effective collision; Activation energy and enthalpy change. Qualitative treatment of Arrhenius' Law to be noted. Effect of Light on some reactions (e.g. halogenations of alkanes) to noted.

(b) Chemical Equilibria
- (i) General Principles: Reversible reactions; i.e. dynamic equilibrium. The equilibrium constant K must be treated qualitatively.
- (ii) Le Chatelier's Principles: prediction of the effects of external influence of concentration, temperature and pressure changes on equilibrium system.

CHM 002 Chemistry II: Fundamental Concepts in Organic/Inorganic Chemistry.

A. ORGANIC

1. Introduction and nomenclature

(a) Meaning of the term "organic" and the unique nature ofcarbon

(b) Characteristics of organic compounds; hybridization in carbon sp^3 sp^2 and sp; sigma and pie bonds carbon/carbon single, double and triple bonds

(c) Functional group and the homologoun series; types of organic reactions

(d) Nomenclature (naming) ofaliphatic families.

2. Isolation, Purification, Characterization, Isomerisms and Double bond equivalence

(a) Pure solids and pure liquids, separation techniques include detai1s of column and thin layer chromatographies, mobile phase, stationery phase, elution Rf values.

(b) Qualitative and quantitative analyses for carbon and hydrogen;

(c) Isomerism; definition, types of isomerism with various examples including optical isomerism double bond equivalence.

3. The Alkane Hydrocarbon (Paraffin)

(a) General formula; methods of preparation, special method preparing methane nomenclature; physical properties; chemica properties; combustion, molecular formula of gaseo3r hydrocarbons; substitution reaction-chlorination; uses of alkanes
(b) (b) Petroleum, fractional distillations important fractions from fractional distillation; petrol, knocking, octane rating; calculation ofoctane value; cracking.

4. Alkenes olefins
(a) General formula, functional group, nomenclature, methods of preparation
(b) Physical properties, nucleophilic and electrophilic reagents, Addition reactions addition of H,, Br,, HBr, Bromine water, conc. H_2SO_4, Ozone, H_2O and (O) (i.e. hydroxylation)
(c) Test for unsaturation, phlymerisation, combustion 2HRS

5. Alkynes.
(a) General formula, functional group, general method of preparation, special method of preparing ethyne.
(b) Physical properties, chemical properties, combustion; addition reactions addition of Br,, H,, HBr, Water, polymerization, polyvinyl chloride; Reaction ofethyme with $KMnO_4$
(c) Metallic derivatives Copper and silver ethynides sodium ethynides and its uses in preparing higher alkynes.2HRS

6. Introduction to Aromatic Chemistry
(a) Benzene and aromaticity, the kekule and Divar structures
(b) Naming of benzene derivatives
(c) Physical properties of benzene, Industrial sources of benzene from petroleum and from coal tar.
(d) Reactions of benzene addition reactions and substitution reactions.2HRS

7. Haloalkanes
(a) Monohalo alkanes -General formula; some typical members; classification as primary, secondary and tertiary; nomenclature; physical properties methods of
(b) Di-and Tri-haloalkanes; nomenclature and isomerism; Preparations, chemical properties; uses.

8. Alkanols
(a) General formula; few members of the homologous series; nomenclature; laboratory preparation; manufacture from starch
(b) Physical properties, unexpected high boiling point; chemical properties as a compound containing OH group and as an acid, oxidation distinguishing between primary, secondary and tertiary alkanols; uses of alkanols; examples of dihydric and trihydric alkanols.

9. AlkanalsandAlkanones
(a) General formula; the carbonyl group; nomenclature; preparation from alkanols; and from calcium salts of carboxylic acids. Addition reaction with HCN; condensation reactions; oxidation reactions; and reduction; distinguishing between alkanals and alkanones Fehlings Test and Tollen’s Test; the lodoform Test

10. Monocarbocylic (Alkanoic) acid
(a) General formula; functional group, few members of the series and nomenclature; Resonance in carboxylic acids and the presence of a fake carbonyl group.
(b) Laboratory preparation of carboxylic acids, reactions with PCl, with alkanols/W, with electropocitive metals.

11. vatives ofcarboxylic acids
(a) These derivatives are (a) acid chlorides (b) acid anhydrides (c) acid esters and (d) acid amides. The treatment of each should be based on (i) General formula (ii) functional group (iii) method ofpreparation (iv) chemical reactions.
(b) Fats and oils as alkanoates, saponification; distinction between fats and oils; distinction between soaps and detergents.

12. Amines and amino acids
(a) Amines as primary, secondary and tertiary; nomenclature as derivatives of alkanes. Preparations; Reactions with inorganic acids.
(b) Amino acids as bifunctional compounds; nomenclature; reactions of the functional groups; existence as dipolar salt or switterion.

13. Natural and Synthetic polymers
(a) Definitions, of Addition and condensation polymers; plastics and resins; thermoplastic and thermosetting polymers. Properties of polymers. Natural rubber.
(b) Natural polymers; carbohydrates formula, properties and uses; classification as monosaccharides, disaccharides and polysaccharides; reducing and non-reducing sugars using glucose fructose, sucrose/maltose and starch/cellulose as examples; hydrolysis of sucrose and starch.
(c) Proteins as polymers of amino acids molecules linked by the peptide or amide linkage; hydrolysis; uses in living systems.
(d) Synthetic polymers; classification and preparation based on the monomers and co-polymers.

B. INORGANIC CHEMISTRY

Introduction

Metals and non-metals; definition; chemical and physical characteristics.

NON-METALS AND THEIR COMPOUNDS

Hydrogen:

Laboratory preparation, properties and uses. Production from water gas and cracking of petroleum fractions will be expected. Test for hydrogen.

Oxygen

- Laboratory and Industrial preparation.
- Properties and uses;
- Binary compounds of oxygen; Acidic oxides, basic oxides, amphoteric oxides and neutral oxides.
- Air; The usual gaseous constituents; nitrogen, oxygen, water vapour (Argon and Neon) Proportion of oxygen in the air e.g. by burning phosphorus or by using alkaline pyrogallol.
- Air and combustion; structure of the flame Bunsen flame; candle flame; luminous and non-luminous flame.

Water and Solution

- Composition of water. Test for water. Atmospheric gases dissolved in water and their biological significance.
- Water as a solvent.
- Hardness of water; causes and methods of removing it. Advantages and disadvantages ofhard water and soft water. Comparison ofthe degree ofhardness ofdifferent samples of water.
- Treatments of water for town supply.
- Air and combustion; structure of the flame luminous and non- luminous flame

Halogens:

- Chlorine: Laboratory preparation properties, reactions and uses. Uses of chlorine and its compounds should include the following water sterilization, bleaching, manufacture of HCI, plastics and insecticides. Test for chlorine. Preparation and uses of bleaching powder.
- Uses of halogen compounds. Uses should include silver halides in photography and sodium oxochlorate (I) as a bleaching agent.

(e) Chemistry of carbon compounds

- All allotropes of carbon, their structures, properties and uses. The uses of the allotropes should be correlated with their properties and structure.
- Coal: Different types of coal to include anthracite, peat and lignite. Products obtained from destructive distillation of wood and coal.
- Coke: Gasefication and uses; manufacture of synthetic gases and uses.
- Carbon(IV) oxides and uses. Uses to include use as fire extinguishers Action of heat on trioxocarbonate (IV) salts to be known; Test for CO
- Carbon(lI) oxides: Laboratory preparation, properties including its effect on blood. Sources of carbon(II) oxides to include charcoal fire and exhaust fumes.

(f) Sulphur:

- Allotropes and uses preparation ofallotropes not expected.
- Preparation, properties and uses of Sulphur(IV) oxide. The reaction with alkalis should be included. Trioxosulphate(IV) acid and its salts. Also includedis the effect of acids on salts of Trioxosulphate(IV).
- Tetraoxosulphate(VI) acid.Commercial preparation from contact process; properties and uses. Properties as a dehydrating and oxidizing agent and as dilute acid. Test for Tetraoxosulphate(Vl) ion.
- Hydrogen sulphide; preparation; properties as a weak acid, reducing and precipitating agents. Test for sulphides.

(g) Nitrogen

- Preparation of nitrogen in the laboratory and production from liquid air.
- Ammonia: Laboratory preparation; industrial preparation Haber process; uses of ammonia. Ammonium salts and uses. Oxidation of ammonia to nitrogen(IV) oxide and trioxonitrate (V) acid. Test for NH4 Ammonium chloride, tetraoxophosphate (V) as fertilizer only.
- Trioxonitrate (V) acid: Laboratory preparation from ammonia; properties and uses. Trioxonitrate (V) salts action of heat and uses. Test for trioxonitrate (V) ion.
- Oxides ofnitrogen properties. The nitrogen cycle.

METALSAND THEIR COMPOUNDS

(a) Principles of the extraction of metals by:
  i. Electrolysis.
  ii. The use ofcarbon or carbon (II) oxide.
(b) Alkali metals e.g. sodium.
(i) Sodium hydroxide; production by electrolysis of brine; its action on aluminium, zinc and lead ions, Uses, including precipitation ofmetallic hydroxides.
  i. Sodium trioxocarbonate (IV) and Sodium hydrogen trioxocarbonate (IV) production by solvay process; properties and uses. The use of Na_2CO_2 in the manufacture of glass.
  ii. Sodium chloride, properties and uses.
(c) Alkaline earth metals e.g. calcium. CaO, $CaCO_3$, propertiesand uses. Preparation of CaO, $CaCO_3$ from sea shells. Cement and mortar

(d) Aluminium, purification of bauxite, electrolytic extraction; properties and uses ofaluminium and its compounds.
(e) Tin; Extraction from its ores; properties and uses.
(f) Metals of the first transition series:
(i) electronic configuration. (ii) oxidation states.(iii) complexion formation. (iv) formation of coloured ions.
(g) Iron; Extraction from suiphide and oxide ores; properties and uses. Different ores and their properties. Advantages of steel over iron. Test for Fe and Fe'2 ions.
(h) Copper; Extraction from suiphide and oxide ores; properties and uses of copper salts. Special note should be taken of the preparation and uses of copper(II) tetraoxosulphate(VI); Test for Cu ions.
(i) Alloys; Steel, stainless steel, brass, bronze, type metal, duralumin and soft solder; simple reasons for the use of these alloys in preference to the metals from which they are made.

CHEMISTRY, INDUSTRYAND THE ENVIRONMENT

Chemistry in Industry

(i) Natural resources in a particular country. e.g. Nigeria.

- Chemical industries in a particular country and their corresponding raw materials. Distinction between fine and heavy chemicals.
- Factors that determine sitting of chemical industries.
- Effect ofindustries on the community.
- Extraction of metals e.g. Al, Fe, Au/Sn. Raw materials, processing, main products, by-products recycling.

Pollution
Air, water and soil pollution.
(i) sources, effect and control

- Greenhouse effects and depletion of ozone layer.
- Biodegradable and non-biodegradable pollution.

Biotechnology
Food processing, fermentation including production of kenkey/gari, bread and alcoholic beverages e.g. local gin.

PHY 001 – PHYSICS I

A. Mechanics

1. (a) Fundamental and derived quantities and units
(b) Derived quantities and units

2. Position, distance and displacement
Distinction between distance and displacement

3. Mass and Weight
Distinction between mass and weight

4. Time and measurement of time

5. Fluids at rest
(a) Volume, density and relative density
(b) Pressure in fluids
Concept and definition of pressure, Parcal's principle, application to hydraulic press and car brakes. Dependence of pressure on depth. Atmospheric pressure. Simple barometer.
(c) Equilibrium of bodies
(i) Principle of Archimedes
(ii) Law of Floatation

6. Motion
(a) Types of motion: rectilinear, translational, rotational, circular.
(b) Relative motion.
(c) Frictional forces between two bodies (static and dynamic)
(d) Limitation of friction
(e) Viscosity
(f) Circular motion

7. Speed and Velocity
(a) Displacement
(b) Uniform and non-uniform speed and velocity
(c) Displacement time graph

8. Rectilinear acceleration
(a) Uniform and non-uniform acceleration
(b) Velocity time graph
(c) Constant acceleration, gravitational acceleration

9. Scalar and Vectors quantities
(a) Concept of scalar quantities
(b) Addition of vectors
(c) Resolution of vectors

10. Equilibrium of forces
(a) Principles of moments
(b) Parallelogram law of forces, Triangle of forces
(c) Centre of gravity

11. Simple harmonic motion
(a) Definition of simple harmonic motion (S.H.M) Simple pendulum, spiral spring
(b) Speed and acceleration of S.H.M
(c) Period, frequency and amplitude of S.H.M
(d) Energy of S.H.M
(e) Forced vibration and resource

12. Newton's law of motion
(a) First law of motion
Inertia of rest and motion
(b) Second law of motion
Force, Acceleration, Momentum and Impulse
(c) Third law of motion
Action and reaction Linear momentum and its conservation. Collision of elastic

bodies in a straight line. Recoil of a gun, jet and rocket propulsions.

13. Energy

(a) Forms of energy (potential and kinetic, heat, chemical, electrical, light sound and nuclear).\

(b) Conservation of energy.

14. Work, Energy and Power

(a) Work as a measure of energy

(b) \Energy as capacity to do work.

(c) Gravitation field

(d) Power as rate of doing work

(e) Machines: levers, pulleys, inclined plane

B. Heat Energy

(a) Temperature and its measurement

(b) Effect of heat:

(i) Temperature rise

(ii) Change of State

(iii) Expansion, Change in resistance, linear, area and volume expansion.

(c) Heat transfer conduction, convection and radiation

(d) Gas laws: Boyle's law, Charles's Law

(e) Heat capacity, specific heat capacity, latent heat, melting and boiling point, specific latent, heat of fusion and vaporization. Evaporation and Boiling. Vapour pressure.

(f) Humidity, relative humidity and dew point. Weather and humidity.

C. Waves

1(a) Production of waves: propagation of waves

(b) Speed, frequency and wavelength

(c) Wave form

(d) Relationship between frequency

(e) Wavelength (n), period (T) and velocity (V)

2. Type of Waves

(a) Transvere, longitudinal and stationary waves

(b) Representation of waves

3. Reflection, refraction, diffraction, interference, superposition of waves

D. Sound Waves

(a) Electromagnetic Waves

(b) Sources of sound

(c) Transmission, Speed in Solid, liquid and air,

(d) Echoes, reverberation, pitch, loudness

(e) Vibrations in strings

(f) Resonance boxes, sonometer forced vibration, harmonics and overtones

(g) Vibration of air in pipes: open and closed pipes

PHY 002 PHYSICS II

A. Light

1. Light Waves

(a) Sources of light

(b) Rectilinear propagation of light. Shadows, eclipses, pinhole camera.

(c) Reflection of light at plane surfaces: Plane mirrors, Incline mirrors, Rotation of mirrors.

(d) (d)Curved surfaces: concave and convex mirrors. Laws of reflection, formation of images. Use of mirror formulae, magnification. Determination of focal length. Applications e.g. driving mirrors.

(e) Refraction of light: plane surfaces, rectangular and triangular glass prisms. Laws of reflection. formation of images, real and apparent dept, critical angle, total internal reflection. angle ofdeviation. Minimum.deviation.

(f) Refraction of light at curved surfaces. Formation of images. Lens formulae and magnification. Converging and diverging lens.

(g) Applications: Optical instruments. Simple camera, human eye, projector, simple and compound microscopes, telescopes.
Defects of human eyes and their corrections.

(h) Dispersion of light.

Electricity

Electric Field

Electrostatics

(a) Production of electric charges by friction, induction and contact

(b) Types and distribution of charges. Simple electroscope for detection ofcharges.

(c) Electric field lines of forces

(d) Electric force between point charges Coulomb's law, Electric field, Electric field intensity and electric potential

(e) Capacitance: parallel plate capacitor. Farad(f). Capacitors in series and parallel. Energy stored in charged capacitor.

Current Electricity

(a) Production of electric current from primary and secondary cells.

(b) Potential difference and electric current. Ohm's law. Resistance.

(c) Electric circuit.

(d) Electric energy and power. Heating effect of electric energy and its applications. Conversion of electric energy to mechanical energy; electric motors.

(e) Shunt and multiplier. Conversion of a galvanometer into ammeter on a voltmeter.

Resistivity and conductivity.

Measurement of electric current, potential difference, resistance, e.m.f. and internal resistance, of a cell. Potentiometer, Metre Bridgewheatotone bridge.

Magnetic Field

(a) Properties of magnets. Temporary and permanent magnet.Magnetic flux, magnetic flux density. Magnetic field around apermanent magnet, a current-carrying conductor and a olenoid.
Units: Weber(W), Tesla (T)
(b) Force on current-carrying conductor in a magnetic field
(c) Use ofmagnets, electric bells, telephone earpiece.
(d) Magnetic force on moving charged particle

Electromagnetic Field

(a) Concept of electromagnetic field
(b) Electromagnetic induction: Faraday's law, Lenz's law, motor-generator. Inductance.

5. Simple A.C. Circuits

C. Atomic and Nuclear Physics

1. Structure ofAtom

(a) Models of atom; Thomson, Rutherford, Bohr and electron-cloud models
(b) Energy quantization
(c) Photoelectric effect: Dual nature of light. Work function and threshold frequency. Photoelectric equation.
(d) Thermionic emission: Applications

2. Structure of the Nucleus

(a) Composition of the nucleus: Protons and neutrons. Nucleon
Number (A), proton number (z), neutron number (N).
Nuclides. Isotopes.
(b) Radioactivity.
(c) Nuclear reaction. Fusion and Fission. Binding energy, massdefect. $E=mc^2$.

BIO 001 – BIOLOGY I

A. Variety of Organism

1. Characteristics of Organisms:
(a) Cell structure (plant and animal) and functions of cell components; cell divisions.
(b) Differences between plants and animals.
2. Evolutionary Trends in Organisms as shown by a study of the characteristics of the following groups
(a) Monera (prokaryotes) e.g. bacteria and blue-green algae.
(b) Protista (protozoans and protophyt) e.g. *Amoeba* and *Euglena*.
(c) Fungi e.g. mushrooms and bread mould
(d) Plantae (plants)
(i) Broyophyta (mosses and liverworts)
(ii) Pteridophyta (Ferns)
(iii) Gymnospermae (Conifers) and
(iv) Angiospermae (Flowering plants)
(e) Animalia (animals)
(i) Unicellular animals (protozoa)
(ii) Multicellular animals (invertebrates)
(a) Coelanterata (e.g. Hydra)
(b) Platyhelmintes (flatworms)
(c) Nematoda roundworms)
(d) Annelida (e.g. earthworms)
(e) Arthopoda (e.g. insects, millipedes, ticks)
(f) Mollusca (e.g. Snails)
(iii) Multicellular animals (vertebrates)
- Pisces (cartilaginous and bony fishes)
- Amphibia (e.g. toads and frogs)
- Reptilia (e.g. lizards, snakes and turtles)
- Aves (birds)
- Mammalia (mammals)

B. Forms and Function
1. Nutrition
(a) Modes of nutrition
(i) Autotrophic: photosynthetic: chemosythetic
(ii) Heterotrophic: holozoic, parasitic, symbiotic, saprophyte and carnivorous plants
(b) Plant Nutrition
(i) Photosynthesis: Outline of the process; light and dark reactions; materials and conditions necessary for photosynthesis and evidence of photosynthesis.
(ii) The macro – and micro-nutrients required by plants and effects of mineral deficiencies
(c) Animal Nutrition
(i) Food substances: classes carbohydrates proteins, fats and oils, vitamins, mineral salts and water as component f diet.
(ii) Food tests
(iii) The structure of a tooth
Types of teeth and their functions. The relations of dentition to diet as illustrated by an omnivore, e.g. man, a herbivore, e.g. Sheep and a carnivore e.g. dog.
(iv) The alimentary canal of a mammal.
(v) The process of nutrition: ingestion, digestion, absorption, transport and assimilation of digested food.
(vi) Feeding habits: Modifications and mechanism.
2. Transport
(a) Need for transportation:
(b) Materials for transportation: excretory products, gases, manufactured food, digested food, nutrients and hormones.
(c) Structure of the heart, arteries, veins, capillaries and vascular bundles.
(d) Media of transportation: cytoplasm is cells, cell sap or latex in most plants; body fluid in invertebrates, blood and lymph in vertebrates.
(e) Composition and function of blood and lymph
(f) Mechanism of transportation.
3. Respiration
(a) Outline of the process and its significance
(b) Respiratory organs/surfaces:
- Body surface gills, trachea, lungs stomata and lenticels

(c) The mechanism of gaseous exchange in insects, fish, toads, mammals and plants.
(d) Cell (tissue) respiration acrobic, anacrobic and yeast fermentation.
4. Excretion
(a) The meaning and significance of the process
(b) Types of excretory structures;
 Kidney, stoma, lenticel, flame cell, nephridium, malphigian tubule.
(c) Excretory mechanisms in:
(i) Kidney
(ii) Lungs
(iii) Earthworms
(iv) Insects
(v) skin
(d) excretory products of plants
5. Co-ordination and Control
(a) Need for coordination and control
(i) The components, structure and functions of the central nervous system
(ii) The component and function of the peripheral nervous system
(iii) Mechanism of transmission of impulses
(iv) Reflex action
(b) Sense organs
(i) The skin as sense organ
(ii) Organ of smell
(iii) Organ of taste
(iv) Organ of sight
(v) Organ of hearing
(vi) Structure and function of each organ
(c) Hormonal control
(i) Hormones – sites of secretion and functions
Endocrine glands: pituitary, thyroid, parathyroid, adrenal, pancreas and gonads and their secretions.
(ii) Effects of over secretion and deficiency of different hormones
(iii) Plant hormones; The effect of plant hormones (auxin, gibberellins, cytokinins and ethylene) on growth.
(d) Homeostasis:
(i) Regulation of body temperature
(ii) Smo-regulation and maintenance of acid-base balance

BIO 002 BIOLOGY II A. Ecology
1. Basic concepts of habitat, niche, population, community, ecosystem, biosphere.
2. Components of the ecosystem
Abiotic and Biotic: climatic, physiographic, edaphic, chemical, etc.
3. Methods of measuring abiotic factors.
4. Interactions of plants and animals with their environment.
Examples ofassociation: Symbiosis, parasitism, saprophytism, commensalisms, competition and predation.
(a) Food chain: food webs andtrophic levels.
(b) Energy flow in an ecosystem.
(i) Food/energy relationship in aquatic and terrestrial Environment.
(ii) Pyramid of energy, pyramid of numbers and pyramid of biomass.
5. Biotic Communities
(a) Tropical rain forest
(b) Southern guinea savanna
(c) Northern guinea savanna
(d) Sahel savanna
(e) Desert
(f) Swamp/estuarine
6. Ecology of Population
(a) Ecological Succession:
(i) Structural changes in species composition variety or diversity and increase in numbers.
(ii) General characteristics and outcomes of succession.
(b) Primary Succession
(c) Secondary Succession
7. Pollution and Control
(i) Sources/types, effects and the methods of control
(ii) Water and Soil Pollution
Types, composition and effects of pollutants.
8. Conservation of Natural Resources
- The need and methods for conservation of natural resources water, soil, minerals, forest and wild life.
- Agencies responsible for conservation such as WWF, FEPA, NCF.
- Conservation education.
- Conservation laws.
B. Evolution
1. Variation in Population
(a) Morphological variation in the physical appearance of individuals:
(i) Size (height, weight etc);
(ii) Colour (skin, eye, hair, coat of animals);
(iii) Finger prints
(b) Physiological variations:
(i) Ability to roll tongue;
(ii) Ability to taste phenylthio carbamide (PTC)
(iii) Blood groups
(c) Application of vanation:
(i) Crime detection;
(ii) Blood transfusion;
(iii) Determination of paternity.
2. Biology of Hereditary (Genetics)
(a) Inheritance of characters in organisms.
(i) Hereditary variation: heritable and non-heritable characters;
(ii) Mendel's work in genetics
- Mendelian traits and Mendelian laws.
(iii) Chromosomes: The basis of hereditary
- chromosome structure
- process of transmission of hereditary characters from parent to offspring.
(b) Probability in Genetics
(c) Application o fthe principles of heredity to (i) agriculture
(d) Sex determination and sex-linked characters
(ii) medicine
3. Theories of Evolution
(i) Lamarck's Theory
- Evidence for evolution such as fossil records, comparative anatomy and physiology and embryology.

(ii) Darwin's Theory
- Evolutionary trends in plants and animals such as from simple to Complex structural adaptations and from aquatic to terrestrial organisms.

GEOGRAPHY

Aim and Objectives

The programme in Geography is structured to help students develop a holistic view about their physical, human and environment interrelationships. Most importantly the course aims at developing the students' ability to think logically on issues involving man environmental relations.

Among the specific objectives are (a), to acquire sufficient knowledge about the physical and chemical components.of their surrounding environment and (b), to have adequate knowledge about their human environment through such topical issues in social, economic and cultural geography.

GEO 001: A. Physical Geography

(i) The Solar System:
- The earth as a planet in relation to the sun and other planets, the shape and size ofthe earth.
- Earth's rotation and revolution.
- Latitude, longitude and time.

(ii) The Structure of the Earth: internal and external structures
(iii) Rocks: Types, characteristics, formation and uses
(iv) Major Landform Features: mountains, plateaux, plains; their types, modes of formation and uses.
(v) Oceans: ocean basins, salinity, ocean currents (causes, types and effects).
(vi) Weather and Climate: definitions, elements, factors, basic weather instruments, weather records, importance of weather and climate, effects of climatic elements, climatic classification based on Greek and Koppens methods, major climatic types.
(vii) Soils: definition, types and properties; factors and processes of soil formation; soil profile; importance of soils; effects of human activities on soil.
(viii) Vegetation: Types and characteristics; distribution; factors Influencing their distribution.
(ix) Environmental Resources: human; atmospheric; water; vegetation; and mineral resources.
(x) External Processes: Weathering and mass movement; definition, types and features.
(xi) Internal processes of Landform Development: earthquake, vulcanicity, other tectonic processes e.g. compressional, tensional and faulting; action of underground water.
(xii) Environmental Interaction: land ecosystem, environmental balance and interference with the natural environment.
(xiii) Environmental Hazards: Soil erosion, flooding, drought encroachment, deforestation and pollution; causes, effects and preventive methods.

B. Practical Geography

(i) Map reading and interpretation based on a contoured map of West Africa, especially Nigeria.
(ii) Scale and measurement of distance Directions and bearings
(iii) Conventional symbols, relief representation and drawing of relief profiles
(iv) Map reduction and enlargement Slope measurement and calculation Interpretation oftopographical maps
(v) Graphical representation ofgeographical data
(vi) Elementary surveying: types: chain, prismatic and traverse (open and close): equipment and procedures; advantages and disadvantages of each type.

GEO 002:

A. Human Geography

(i) World Population: patterns and factors of population distribution and movement; population growth: factors and problems.
(ii) Settlement: types; patterns; functions; and factors of settlement location.
(iii) Primary Activities: agriculture; lumbering; quarrying; fishing
(iv) Industry: types; major commodities; factors of industrial location; importance of industries; problems of industrial growth in the developing countries; World Trade Routes, with special emphasis on Trade between Nigeria and the outside world.
(v) Transportation: types (merits and demerits of each mode); transportation and economic development; problems of transportation.

B. Regional Geography of Nigeria

(i) Location, position and size of Nigeria.
(ii) Political divisions
(iii) Physical setting: relief, drainage, soils and climate.
(iv) Population
(v) Minerals and power resources
(vi) Agriculture
(vii) Industry and commerce
(viii) Transportation.
(ix) Geographical Regions ofNigeria: Eastern Highlands, Sokoto Plains, Chad Basin, Niger-Benue Trough, Cross River Basin and Southern Coastlands, each of these geographical regions should be treated under Coastlands, each of these

geographical regions should be treated under the following sub-headings.

i. Physical setting
ii. Peoples and population
iii. Settlement
iv. Resources and economic activities
v. Transportation
vi. Problems of development.

C. Regional Geography of Africa with Particular Emphasis on West Africa

Location, size, and position; political divisions and associated islands; major relief regions; drainage systems; climatic types and their characteristic; vegetation; major minerals; and population; and the following selected topics:

i. Lumbering in Congo Basin.
ii. IrrigationAgriculture in the Nile and Niger Basins.
iii. Bush fallowing in West Africa
iv. Plantation agriculture in East Africa.
v. Fruit farming in the Mediterranean region.
vi. Mineral exploitation inAfrica (Copper, Petroleum, Gold etc.)
vii. Major hydro-electricity power projects inAfrica.
viii. International Economic Cooperation in WestAfrica (ECOWAS).